



Module Code & Module Title

CS6P05NP & Final Year Project (Proposal)

Assessment Weightage & Type

Final Report

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Submission Date: 22nd April 2026

Submission Due Date: 22nd April 2026

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1 INTRODUCTION

1.1 Introduction to Topic

In the modern digital age, online education platforms and communities of sharing knowledge are gaining popularity. Good skills that people use like programming, graphic design, language skills, music, photography, and numerous other abilities are not usually taught to them by professional websites due to a lack of opportunity and funds. Meanwhile, there are numerous individuals who are ready to share their expertise but lack a suitable site to exchange knowledge. This brings about a difference between individuals with desires to be educated and those with desires to educate.

The Skill Barter System is meant to resolve this issue by establishing an online system in which users are able to trade skills among themselves. Users do not need to pay money to take courses, rather they can impart a skill they know and acquire a skill they desire with another user. This will form a community of knowledge sharing in which the teachers and even students will be mutually beneficial.

The system is created through a web-based application based on the Laravel framework and PHP, MySQL database, HTML, CSS, and JavaScript. The platform enables its members to create accounts, profiles, add skills, search and find skills, book sessions, take online learning classes, chat with others, post learning materials, rate teachers, and become a premium member with online payment integration.

The primary objective of this project is to establish a friendly, safe, and effective platform to be used to teach and learn online by sharing skills. There is also an admin panel used to control the users, sessions, skills, payments, and activities of the system in general.

1.1.1 Global Context

Online learning platforms like Udemy, Coursera and Skillshare are very popular globally. On these sites, students can acquire different skills remotely with the help of professional teachers. Nevertheless, most of those platforms demand payment to access courses which are not necessarily affordable to all, and mostly not to students in the developing countries.

Additionally, some individuals possess skills that they can impart but lack a platform to pass the information they have. This issue can be addressed by use of a skills exchange platform where people can learn and teach each other without paying high course fees. The world is experiencing an increase in skill-sharing communities due to the high levels of collaboration, learning, and building communities.

The Skill Barter System has a parallel concept except that it is a peer-to-peer learning model that allows users to trade in skills as well as pay to take courses. This increases accessibility and affordability of learning to all.

1.1.2 Nepal Context

Online learning platforms are not yet well-developed in Nepal, and many students use YouTube or free online platforms to acquire new skills. In Nepal, however, there is no appropriate platform where individuals can share skills with others. A lot of students possess such skills as programming, graphic design, video editing, photography, or language skills without a platform to teach and acquire new skills in their turn.

Most of the learning platforms in Nepal are paid platforms, and a significant number of students are unable to afford them. There is also no central platform where the user gets to find the teachers, schedule the learning sessions, interact with one another, exchange learning content and monitor the learning process.

These issues can be addressed by the Skill Barter System, offering the platform of skill exchange and online learning in Nepal specifically. Through this platform, students, freelancers and professionals will contribute their knowledge and get to learn new things as they exchange knowledge among themselves.

1.1.3 Problem Scenario

Learning new skills or obtaining expert services in today's fast-paced digital age often requires financial means. Many people have abilities they can contribute but cannot afford to take advantage of new learning opportunities, especially students, independent contractors, and the jobless. Current platforms lack a suitable framework for skill exchange and primarily concentrate on paid courses, freelance employment, or voluntary work. People lose out on the chance to work together and develop without worrying about money. Furthermore, it is challenging for users to locate compatible partners and uphold transaction transparency in Nepal due to the lack of a centralized system that offers a reliable, equitable barter mechanism for skill exchange.

In the absence of an appropriate skill exchange platform numerous issues arise:

- Individuals seeking the acquisition of new skills cannot easily get access to teachers.
- There are a lot of skilled individuals who cannot get students to teach.
- Educational systems are costly.
- No way of scheduling learning sessions.
- It is hard to communicate between teachers and students.
- File sharing and learning materials do not have a centralized platform.
- Users are unable to follow their learning sessions.
- There is no teacher rating and feedback system.
- Lack of an appropriate online teaching management system.

These issues complicate online learning, which is disorganized and inefficient.

1.1.4 The Project as a Solution

The Skill Barter System offers a practical solution by creating a digital platform where users can register, list their skills, and search for other users willing to exchange them. For example, someone with graphic design skills can trade their services for someone teaching programming. In order to preserve confidence, the platform guarantees safe authentication, mutual agreement verification, and feedback methods. An integrated messaging system allows users to interact with each other, explore postings, and suggest deals. By fostering economic inclusion, community cooperation, and skill development, this project enables people to advance both personally and professionally.

1.1.5 The Scope of The Project

The Skill Barter System seeks to establish a platform through the internet where people can share their skills and learn by interacting with others via online classes. The following modules are the scope of this project:

1) For Student:

- **Create Account and Manage Profile:** Students are able to create an account, log in and edit their profile details such as bio, profile picture, and skills that they wish to acquire.
- **Search Skills and Teachers:** Students are able to browse and search which skills and who can teach the skills.
- **Send Session Requests:** Students are also allowed to request teachers to send session requests that would allow them to learn certain skills.
- **Chat with Teachers:** The students are able to chat with teachers.
- **Download Learning Resources:** Students are able to download learning resources that have been uploaded by teachers.
- **Offer Ratings and Feedback:** The student can rate a teacher and comment after completing a session.
- **Premium Membership:** Students will be able to subscribe to premium membership to get more features.
- **Monitor Learning Progress:** Students are able to check their learning sessions and progress.

2) For teacher:

- **Register and Manage Profile:** The teachers are able to establish and maintain their profile with teaching skills and experience.
- **Add Skills They Can Teach:** Teachers can include skills which they can teach.
- **Accept Session Requests:** Teachers may accept session requests in students.
- **Accept or Reject Requests:** The teachers are able to accept or decline session requests.
- **Manage Learning Sessions:** The use of this feature enables teachers to control the status of the sessions and to indicate that the session is over.
- **Create Learning Resources:** Teachers are able to post learning resources like PDF files, documents, videos, and others.

- **Chat with Students:** Communication between the teachers and students can be made through the chat system.
- **View Ratings and Feedback:** Teachers will be able to view the feedback and ratings provided by students.
- **Premium Membership:** Teachers are able to subscribe to premium membership to have extra features.
- **Analytics Dashboard:** Teachers will be able to see analytics like number of sessions, ratings and students taught.

3) For Administrators

- **Manage Users:** Admin is able to see all users, enable or disable users, change user roles, and delete accounts.
- **Approved Teachers:** Admin has to approve teachers before they can begin teaching.
- **Administrate Skills:** Administrator is able to add, update or remove skills in the system.
- **Manage Session Requests:** Admin has the ability to view all the session requests and modify their status.
- **Control Feedback:** Admin is able to remove unwanted feedback or reviews.
- **Manage Premium Subscriptions:** Admin is able to see and cancel the premium membership.
- **Platform Analytics:** Admin is able to see the system statistics like total users, total teachers, total students, total skills, total sessions, and total revenue.
- **Platform Monitoring:** Admin makes sure that the platform is running and manages system troubles.

In sum, Skill Barter System offers an opportunity to provide students and teachers with a platform to connect, share skills, organize learning sessions, communicate, share resources, and give feedback. The admin handles all the systems and makes sure that the platform runs efficiently and safely.

1.2 Aim and Objectives

Aim:

This project aims to create an online platform that will allow people to quickly and equitably trade their abilities without the need for money. Through secure identification, AI-based matchmaking, gamification, and feedback mechanisms, the system will foster self-learning, communication, and community involvement.

Objectives:

- To create a secure login and registration system for users.
- To implement a skill management module where users can add “skills to teach” and “skills to learn.”
- To design an efficient matching algorithm for skill barter requests.
- To provide scheduling and chat features for session management.
- To develop a responsive interface using HTML, CSS, and JavaScript.
- To integrate a feedback and rating system for user reliability.
- To integrate a payment gateway for Premium Membership.
- To use the Incremental Software Development Model for phased delivery.

1.3 Tools and Resources Required

➤ **Database: MySQL**

XAMPP is an open-source local development environment created by Apache Friends. It includes Apache HTTP Server, MySQL Database, and interpreters for PHP and Perl, making it ideal for web development. MySQL will be used to manage all user data, skill records, and barter requests securely. XAMPP allows for easy setup and local testing before deployment (DSouza, 2024).



Figure 1: Xampp

➤ **Framework: Laravel**

The Model-View-Controller (MVC) design is used by the open-source PHP framework Laravel. It ensures organized and effective code by offering built-in features for database management, routing, and authentication. Laravel will be utilized in this project to handle data movement between the front end and back end, provide the fundamental features of the Skill Barter System, and uphold system security (tutorialspoint, 2024).



Figure 2: Laravel

➤ **Frontend: HTML, CSS, and JavaScript**

JavaScript adds logic and interactivity to improve user experience, CSS improves layout and design, and HTML establishes the webpages' structure. Together, they guarantee that the Skill Barter System is responsive, aesthetically pleasing, and simple to use on desktop and mobile platforms.



figure 3: HTML,CSS & JS

➤ **Backend: PHP**

PHP is a strong programming language that is used to manage backend operations including database communication, barter request processing, and user authentication. It ensures smooth operation between client and server, making the platform secure and dynamic (WillDom, 2023).



figure 3: PHP

➤ **Text Editor: Visual Studio Code**

Visual Studio Code, a free and open-source code editor, supports multiple programming languages. It facilitates the development of the Skill Barter System by providing features including real-time collaboration, debugging tools, and syntax highlighting (UCHENDU, 2021).



figure 4: VS Code

➤ **Documentation: Microsoft Word**

The Skill Barter System will use Microsoft Word to create reports, documentation, and proposals. It ensures that written documents are presented professionally by enabling structured formatting, editing, and integration of figures and tables.



figure 5: MS word 365

➤ **Design Tools: Figma and Draw.io**

System flowcharts, ER diagrams, and UML models will be produced using Draw.io (also called Diagrams.net). Diagramming project designs is made easier by its support for cloud storage and data privacy (Edirisinghe, 2023).



figure 6: Draw.io

2 Structure of The Project

This report is divided into various sections describing the overall development process of the Skill Barter System starting with the original idea, to the final implementation and testing. Different stages of the project are described in each chapter such as research, planning, design, development, testing, and evaluation of the system. This report is structured in the following way.

2.1 Introduction

This chapter provides the general introduction of the project. It describes the idea of the Skill Barter System and the relevance of internet-based learning and platforms of skills exchange in the modern digital world. The chapter also outlines the global scenario and Nepal scenario of online learning platforms and indicate the issues of people who wish to learn and teach skills online.

Additionally, this chapter describes a problem situation and the way the proposed system delivers a solution to such problems. This chapter also outlines the objectives and aims of the project, the scope of the project and constraints of the system. This chapter gives a clear picture as to the reason why the system was developed and the expectations of the system.

2.2 Background

This chapter describes the background research of the project and gives details on the characteristics of the system and technologies involved in the development. It explains the key functional features of Skill Barter System like user management, skill management, session management, online teaching, chat system, file sharing, rating and feedback, premium membership, payment integration and admin panel.

The study and the review of the related articles and journals concerning the online learning platforms, skills exchange systems, and web-based learning management systems are also introduced in this chapter. This chapter also examines existing systems and other platforms and compares the existing systems and the proposed system.

This chapter is meant to demonstrate the research and study that took place before the development of the system as well as to justify the reason behind the proposed system.

2.3 Methodology

This chapter explains the software development process that was adopted to come up with Skill Barter System. Various software development models like Waterfall Model, Agile Model, Scrum Model and Incremental Model are discussed in this chapter. The most appropriate methodology in this project is chosen and described upon the analysis of various methodologies.

Project planning is also a part of this chapter where pre-planning, requirement gathering, system analysis, system design, implementation, testing, and deployment are included. Gantt chart has been used to indicate the project timeline and the Work Breakdown Structure (WBS) has been used to break the project into mini tasks and activities.

This chapter also explains project milestones and deliverables to demonstrate how the project was accomplished in different stages within the specified time.

2.4 Design

This chapter describes the system design and architecture of Skill Barter System. It contains several system diagrams which allow us to comprehend the structure and work of the system.

The figures in this chapter are:

- Use Case Diagram
- Entity Relationship Diagram (ERD).
- Data Dictionary
- Sequence Diagram
- Data Flow Diagram (DFD Level 0 and Level 1)
- Process Diagram
- System Algorithms
- Access Matrix
- Wireframes

Use Case Diagram indicates communication between the system and users. The ER Diagram provides the database structure and the relationships among the various tables within the database. The Sequence Diagram presents the order of interactions of the

system components. The Data Flow Diagram displays the movements of data in the system. Wireframes display layout and design of various pages of the system. The chapter assists in imagining the system prior to implementation.

2.5 Implementation

This chapter explains how the Skill Barter System was implemented. It describes the development of the system with the help of various technologies: Larval, Philippines, MySQL, HTML, CSS and JavaScript.

Various modules are developed in this chapter and include:

- User Module
- Skill Management Module
- Session Management Module
- Online Teaching Module
- Chat Module
- File Upload Module
- Payment Module
- Admin Module

In this chapter, there are also the screenshots of the system interface of the various users who include: unlogged users, logged-in users, teachers, students and admin. This chapter also explains the system architecture and the implementation of the database.

2.6 Sustainability

The chapter describes how the system will sustain and grow in the future. It entails business plan of the system, estimation of costs based on COCOMO model, and business rules that are adhered to in the system. The chapter also has the Software Requirement Specification (SRS) that explains the non-functional requirements, external requirements and functional requirements of the system. Non-functional requirements are of system performance, security, usability and reliability and functional requirements are what the system should do. In this chapter the author describes what the system can be sustained, enhanced, and extended in future.

2.7 Testing

This chapter details the system testing. The system was tested in various forms like unit testing, integration testing, system testing or user acceptance testing.

The chapter has test cases of various system features like:

- User Registration
- Login System
- Profile Management
- Skill Management
- Session Booking
- Chat System
- File Upload
- Rating and Feedback
- Payment Integration
- Admin Panel
- Middleware testing and Security testing.

Table format of the test cases is provided with the information about the test case ID, test description, expected and actual result. Screenshots are also provided to display the testing results. This chapter makes sure that the system is operating well and that it satisfies all system requirements.

2.8 Analysis

The analysis of the system in various analysis techniques is contained in this chapter and they include:

- SWOT Analysis (Strengths, Weaknesses, Opportunities, Threats).
- PEST Analysis (Political, Economic, Social, Technological).
- Risk Assessment

In this chapter, the system is assessed in various standpoints and possible risks and opportunities of the system are determined.

2.9 Conclusion

This is the conclusion of the project. It is a summary of the whole process of project development that includes planning and implementation as well as testing. It also describes the problems encountered in the project development and how the problems were addressed.

Future enhancements and improvements that can be brought to the system in the future are also mentioned in this chapter, which includes the development of a mobile application, integration of video calling, skill matching and recommendation system based on AI.

2.10 References

All the books, research papers, articles, websites, tutorials, and online materials that were utilized in the process of research and development of the project are contained in this chapter.

2.11 Appendix

The appendix also contains other support materials like:

- Survey Results
- Gantt Chart
- Work Breakdown Structure (WBS).
- Additional Diagrams
- System Screenshots
- Code Snippets
- Test Case Tables

This is a section where more information can be gathered as a supporting element of the project documentation.

3 Background

3.1 Features of The Project

Skill Barter System is a web-based application which is meant to enable people to trade skills and knowledge via the internet. The system has a number of features which assist the user to teach, learn, interact, schedule sessions and control learning materials.

The system enables the user to create and register their profiles with the provision of personal data, skills that they can teach and the skills they desire to learn. Users are able to search and request a session with other users according to their skills. Educators are able to accept or decline session requests and plan learning sessions in that way.

The online learning session management is one of the primary characteristics of the system. Before and after sessions teachers and students can schedule lessons, post learning materials and chat. Such files as PDF, notes, and learning resources can also be uploaded.

The system also has a rating and a feedback system whereby students are able to give ratings to teachers after they have attended sessions. This is useful in ensuring quality and trust on the platform.

Another feature of the site is the premium membership system where the user is allowed to upgrade to premium accounts via online payment gateway. Premium users will be offered other features like unlimited session requests, priority matching and search advancement.

The system also has an admin panel without which the admin can handle users, skills, sessions, payments, and system reports.

The key characteristics of the system are:

- User Registration and Login
- Profile Management
- Skill Management (Teach Skills and Learn Skills)
- Skill Search and Matching
- Request and Scheduling of Sessions
- Online Teaching Sessions

- Chat System
- File Upload and Download
- Rating and Feedback System
- Premium Membership
- Payment Integration
- Admin Dashboard
- User Management
- Session Management
- Payment Management
- Reports and Analytics

It will be user friendly, responsive, and secure to ensure that the users can use the platform easily through desktop, tablet, and mobile devices.

3.2 Articles and Journal Review and Study.

The online learning platforms, skill exchange systems, and e-learning management systems were researched before the invention of the Skill Barter System. The work reviewed some articles and journals to gain an insight into the functionality of online learning platforms and the features that such systems require.

Online learning systems enable learners to acquire new skills without necessarily being present in actual physical classes. These are video lectures, learning resources, and learning assignments and communication platforms. Nevertheless, numerous online learning websites are not free, and this restricts accessibility by a number of users.

The skill exchange platform is not the same as the traditional learning platform since through this platform users can exchange skills without necessarily paying money. As an illustration, a user is able to teach graphic design and learn programming in exchange. Such a system encourages teamwork and exchange of knowledge.

Studies also revealed that chat systems, files sharing and time scheduling communication tools are of high significance in the online learning platforms. The rating and feedback systems are also needed to uphold the standards of quality and confidence among the users.

Based on the research, the Skill Barter System was found to have the following significant characteristics:

- User profile management
- Skill listing and searching
- Session scheduling
- Chat system
- File sharing
- Rating and feedback
- Premium features payment system
- Admin management system

The system design and implementation had these features.

3.3 Research and Requirement Analysis

In the research and requirement analysis, several analogous skill-based and learning web sites were studied in detail to have an idea of typical system designs, user experience trends, and functional expectations. The study was primarily concerned with the way these platforms are structured, how they display filtering options, and help the user navigate with ease as he or she goes through skills or learning materials. Focus was directed at the analysis of front-end layout patterns, specifically, the application of organized sections, where filtering and category settings must be located to the one side of the interface, and the actual results of the content are located to the other side. This design strategy was found to be suitable for enhancing usability, refinement of search, and general user experience. A high-level review of the popular modern web technologies was also undertaken in the research in order to have a glimpse of how performance, responsiveness and scaling is established in real systems.

According to the results of this study, system requirements were found and developed. With the help of it, functional requirements, such as skill browsing, filtering, and presentation of content in a structured way, were obtained, as well as non-functional ones, including responsiveness, usability, and security. The UI designs and wireframes developed on the project were also based on these patterns and modified to apply to a skill barter system environment. In this research, practicality, user oriented and industry standards were put in consideration to make the system requirements and design decisions practical.

3.4 Similar Projects

Several existing systems have inspired the development of the Skill Barter System. Platforms like LinkedIn Learning, Fiverr, and Skillshare focus on paid services, while platforms such as Simbi promote non-monetary exchange. However, these platforms often lack localized features suitable for Nepalese users, such as language options, community-based reputation systems, and easy barter matching. The Skill Barter System adopts the best features from these global models and tailors them for the local context.

3.4.1 LinkedIn Learning (<https://www.linkedin.com/learning/>)

Thousands of professionally taught online video courses are available on the well-known site LinkedIn Learning. Professional skill development and career promotion are its main priorities. Users can finish courses and use their LinkedIn profiles to highlight their accomplishments. It prevents users from interacting directly or exchanging skills. On the other hand, SkillXchange emphasizes free, peer-to-peer skill sharing, where users teach and learn from one another at no cost, encouraging collaboration and a sense of community.

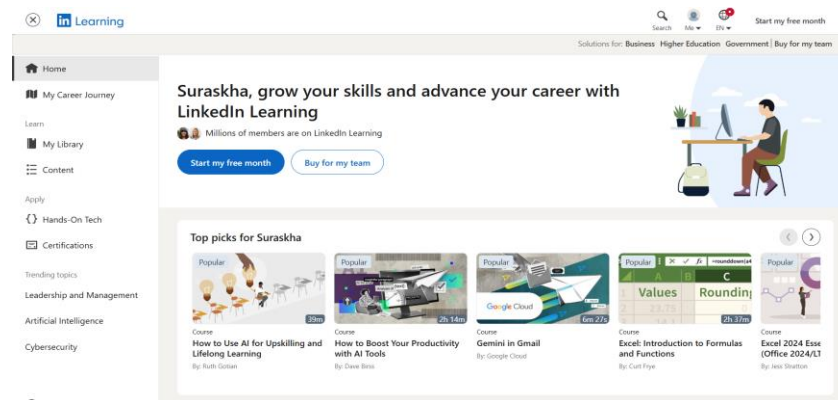


figure 7: LinkedIn Learning's Home Page

Front-end:

- Ember.js (JavaScript framework)
- Core-JS (JavaScript polyfill library)
- Custom UI components (LinkedIn's internal design system)

Back-end:

- Not public

Security:

- HSTS (HTTP Strict Transport Security)
- Cloudflare Bot Management (protection against automation & threats)

CDN:

- Cloudflare

Overview:

LinkedIn Learning uses a highly optimized enterprise-level tech stack designed for speed, security, and heavy user traffic. The presence of Ember.js indicates that the platform uses a single-page application structure, enabling smooth page transitions and fast navigation.

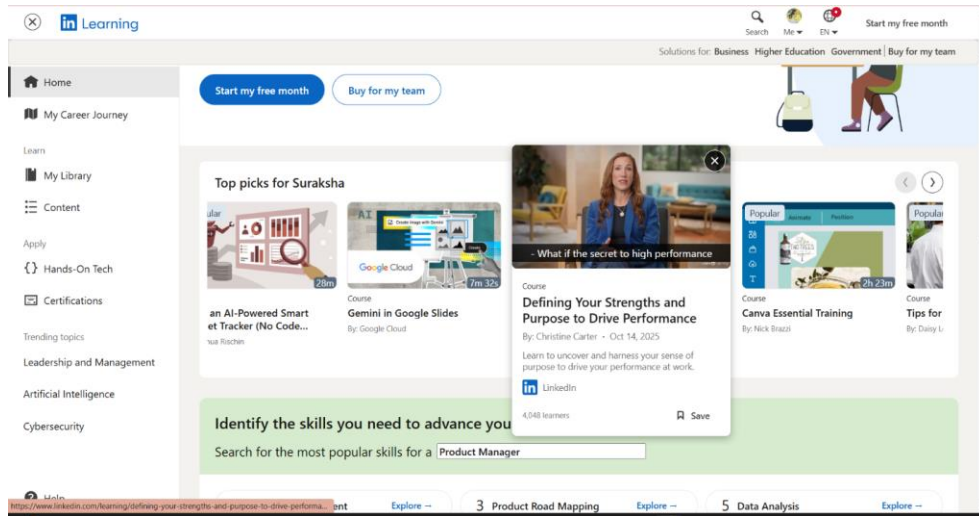


Figure 8: LinkedIn Learning

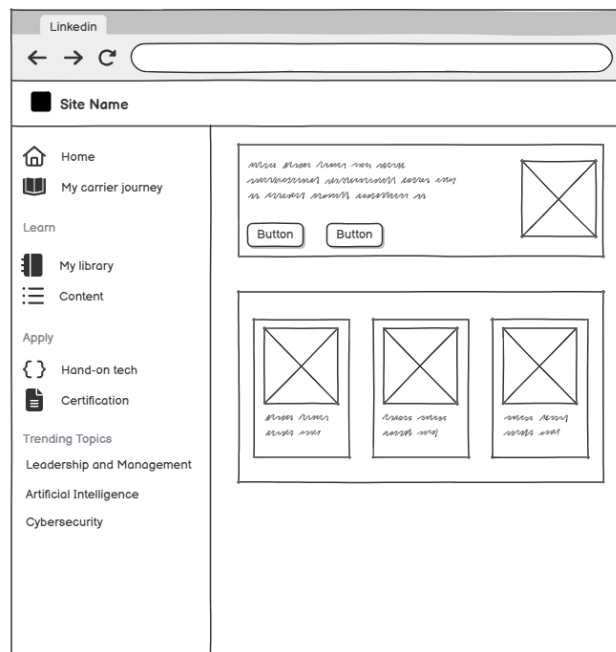


Figure 9: Wireframe of LinkedIn learning

Features I Liked

The platform implements a well-structured card-based listing system. Each course card initially displays a thumbnail and title, while additional metadata becomes visible on hover through a scale transformation. This interaction pattern minimizes visual clutter while ensuring that key information, such as course summary, save action, and learner count, is accessible without leaving the list view.

3.4.2 Skillshare (<https://www.skillshare.com/en/search/classes?query=freelance>)

SkillShare is another global learning platform that allows experts to create and share video lessons in creative fields like design, photography, and writing. For those who cannot afford to pay for their education, access is limited because all classes require a subscription. SkillShare is focused on one-way learning from teachers to students and lacks teamwork or reciprocal communication. The SkillBarter platform, on the other hand, emphasizes two-way learning by enabling each user to act as both a teacher and a learner. It employs an AI-based matching engine to connect people with complementary abilities and a gamified system with badges and points to make learning fun and motivating.

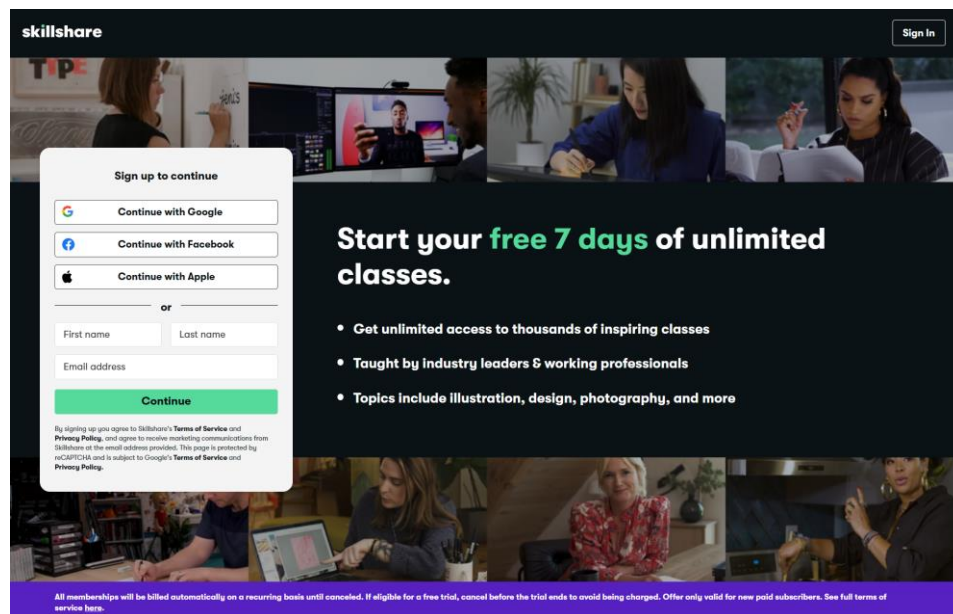


figure 10: Skillshare

Front-end:

- React
- Next.js 12.3.4 (SSR + SSG framework)
- Svelte (used in some modules)
- JSS
- Emotion
- styled components
- Goober (CSS-in-JS library)
- Material UI (UI components)

Back-end:

- PHP (detected through Yii framework)
- Next.js server-side rendering
- AWS (hosting + compute services)

JavaScript Libraries:

- Lodash 4.17.21
- core-js 3.0.1
- jQuery (legacy usage)

Security:

- Cloudflare Bot Management
- HSTS (HTTPS enforcement)

CDN + Server:

- Cloudflare (caching, CDN, security)
- Next.js server environment

Overview:

Skillshare uses a hybrid, modern architecture combining React + Next.js on the front-end with a PHP-based Yii framework on the back-end — showing that the platform is partially migrated to modern infrastructure.

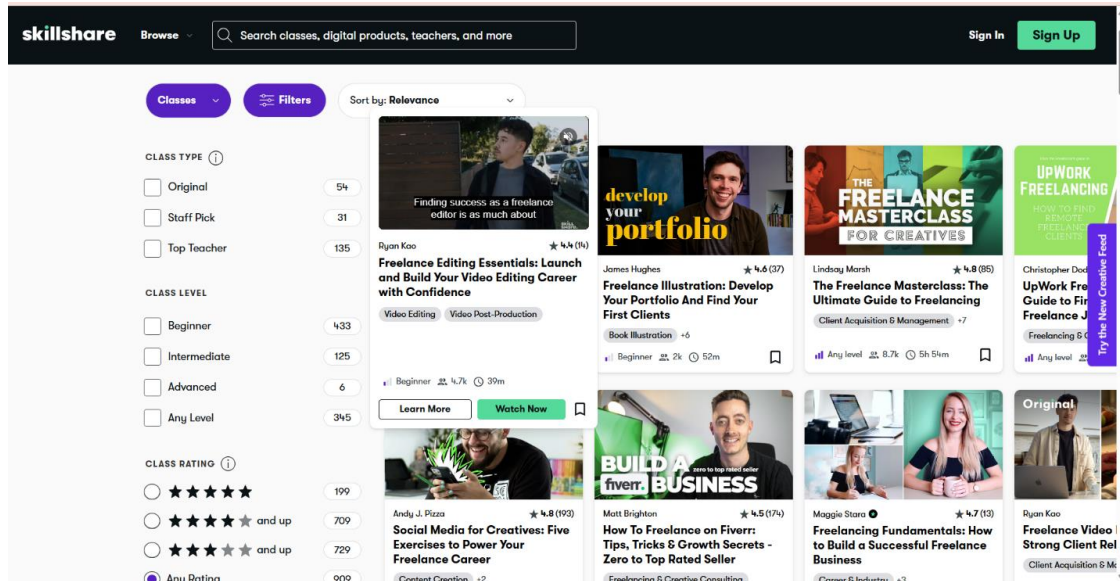


Figure 11: Skill Share

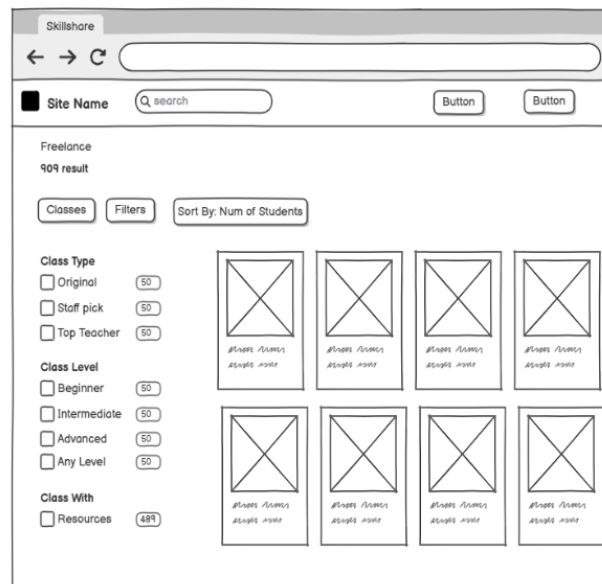


Figure 12: wireframe of Skillshare

Features I liked:

Skillshare provides an efficiently designed browsing layout with filters positioned on the left panel (class type, level, rating) and corresponding course results displayed on the right. This two-column structure enhances discoverability, supports quick refinement of results, and maintains a consistent, visually balanced interface.

3.4.3 Simbi (<https://simbi.com/services/search>)

Simbi is one of the few platforms that allow users to trade services without using money. It works on a credit-based system where users earn credits by offering their skills and spending them to receive services from others. Although Simbi promotes non-monetary exchange, it lacks AI matchmaking, gamification features, and a localized design for users in Nepal. Simbi also does not provide built-in communication or scheduling tools. The SkillXchange system addresses these limitations by introducing an AI skill-matching algorithm, in-app chat and calendar scheduling, and Premium Membership features for users seeking enhanced experience.

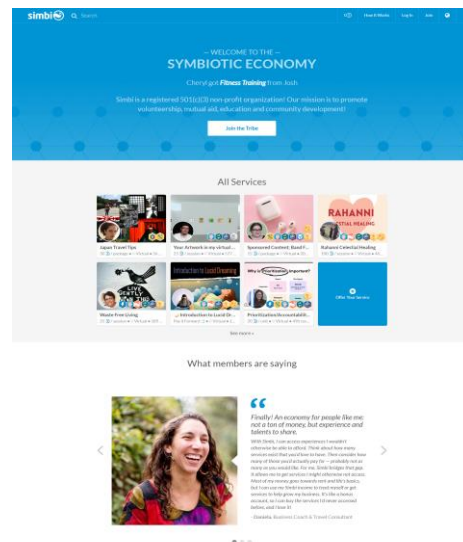


figure 13: Simbi

Front-end:

- Vue.js (primary frontend framework)
- jQuery 4.0.1 (legacy use)
- Lodash 4.17.21
- Hammer.js 2.0.7 (touch gestures)
- core-js 3.32.2 (JS polyfills)
- Bootstrap 3.4.1 (UI layout + components)

Back-end:

- Ruby (primary language)
- Ruby on Rails (main backend framework)
- AWS (hosting infrastructure / PaaS)

Security:

- HSTS (Strict HTTPS Enforcement)

CDN:

- Cloudflare
- cdnjs (JS library delivery)

Overview:

Simbi is built on a classic Ruby on Rails backend paired with a modern Vue.js frontend, providing a balanced architecture that combines reliable backend logic with a responsive, component-based UI.

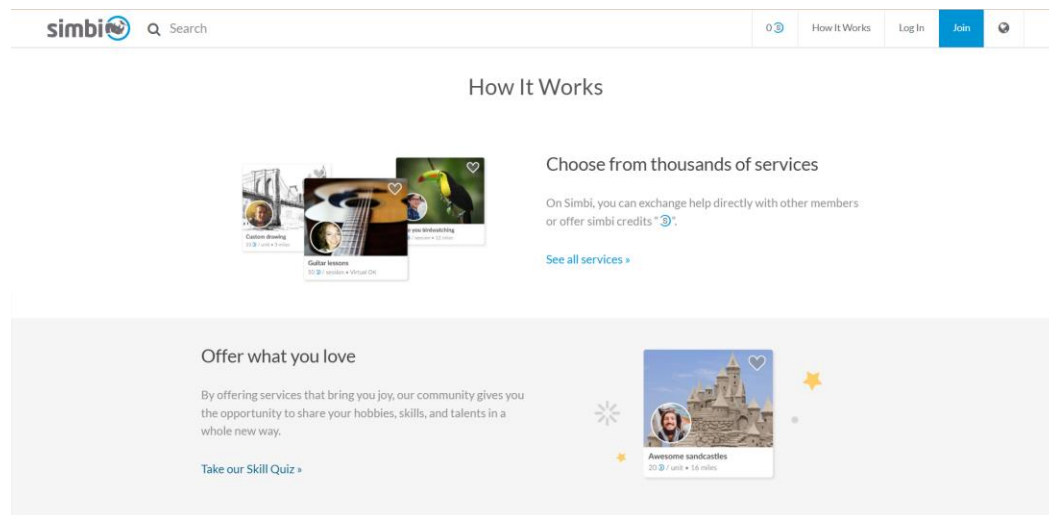


Figure 14: Simbi

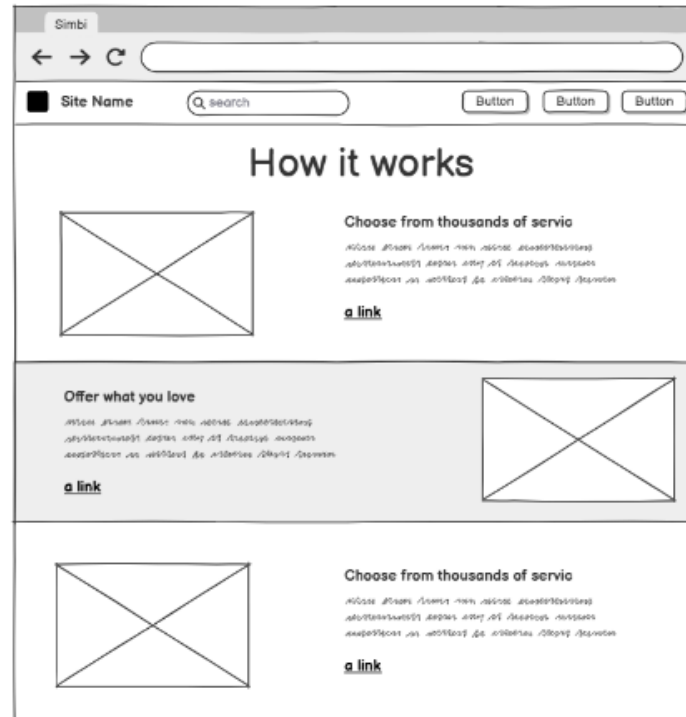


Figure 15: wireframe of Simbi

Featured I liked:

Simbi clearly defines its workflow through a dedicated “How It Works” section supported by contextual links for user guidance. The “Offer What You Love” module is particularly effective as it includes an interactive quiz that helps users identify relevant skills, addressing uncertainty and improving onboarding quality.

3.4.4 PeerList(<https://peerlist.io/scroll>)

Front-end:

- React (primary JavaScript framework for interactive UI)
- Next.js 15.2.3 (modern React-based framework for SSR, SSG, and routing)
- Tailwind CSS (Utility-first CSS framework for fast styling)
- Framer Motion (advanced motion & animation library for UI transitions)
- core-js 3.25.5 (JavaScript polyfills)
- LottieFiles 5.10.2 (animated icons and illustrations)

Back-end:

- Next.js 15.2.3 (also used as backend API routes)
- PHP

Database:

- My SQL

Security:

- HSTS (forces secure HTTPS)
- NextAuth.js (secure authentication framework)
- Google Sign-in (OAuth-based authentication)

Overview:

This platform uses a modern Next.js + React stack combined with Tailwind CSS and Framer Motion for fast, a nimated, and highly responsive UI.

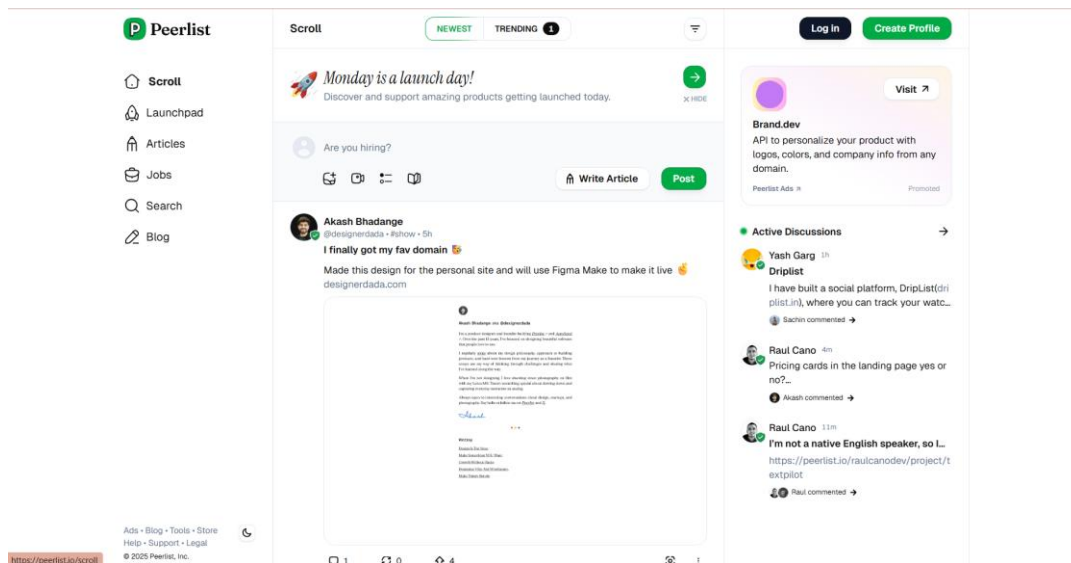


Figure 16: PeerList

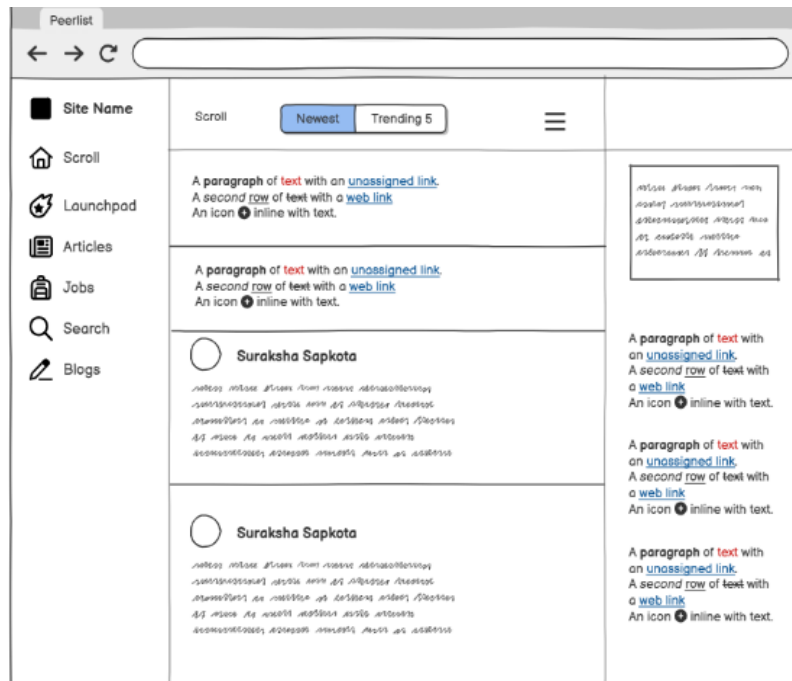


Figure 17: wireframe of peerlist

Featured I liked:

Peerlist has a unique feature where users can share short, useful posts like tips, ideas, and lessons. It also lets people read articles written by other professionals. This makes the platform great for learning and sharing knowledge simply.

3.4.5 Mentorcruise

Front-end:

- React (primary modern JavaScript framework)
- Alpine.js 3.11.1 (lightweight framework used for interactive UI behaviors)
- core-js 3.32.2 (JavaScript polyfills for compatibility)
- jQuery 3.6.0 (legacy script usage for DOM manipulation)

Back-end:

- Django (Python framework) powering dynamic backend logic
- Python as main backend development language
- PHP (used for WordPress-driven sections of the platform)

Security:

- Cloudflare Turnstile (modern captcha alternative for bot protection)
- HSTS (Strict HTTPS for secure communication)
- Cookie Control + Cookiebot 1 (cookie consent + GDPR compliance)
- Cloudflare (DDoS protection, firewall, SSL)

Overview:

This platform uses a hybrid technology architecture, combining:

- Django (Python) for backend application logic
- WordPress (PHP) for CMS content
- React + Alpine.js + HeroUI for responsive UI components
- MySQL for structured data storage

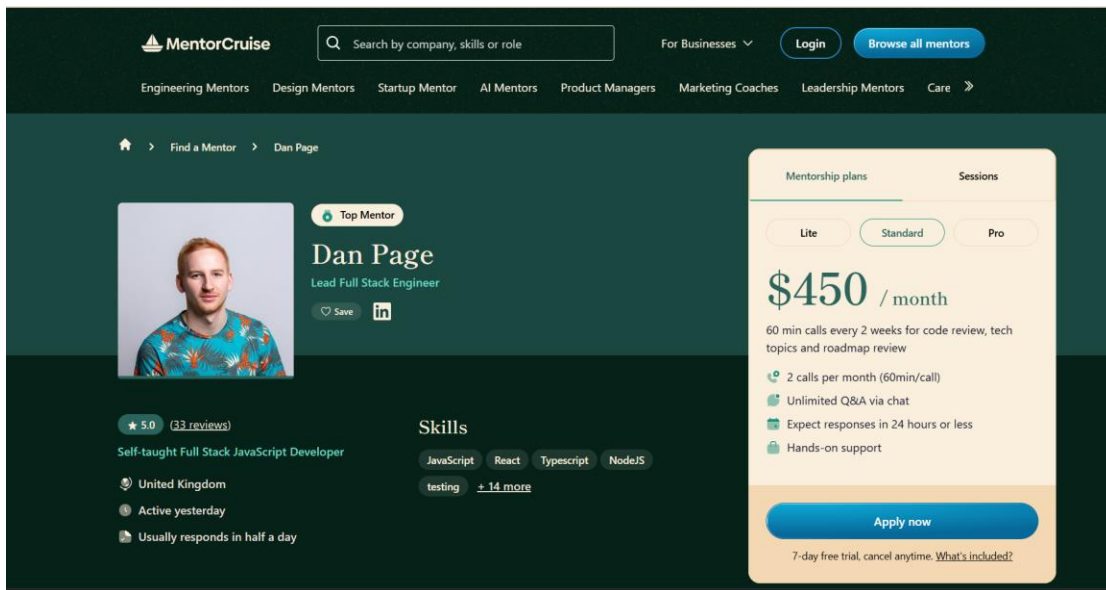


Figure 18: Mentorcruise

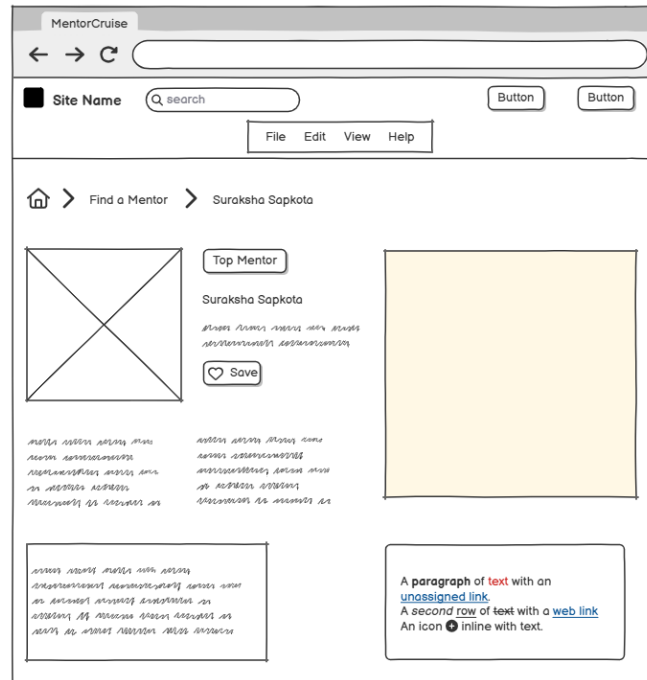


Figure 19: wireframe of mentorcruise

Featured I liked:

The mentor detail page follows a highly optimized information layout, integrating mentor profile images, teachable skills, background description, and user reviews in a single view. This clear hierarchy of information supports efficient decision-making and improves overall discoverability of mentor expertise.

3.5 Journal Research

During the research phase, some academic journals and project-based studies regarding skill exchange, barter systems, and digital credit systems have been considered. The primary aim of this research was to learn more about the way non-monetary and credit-based exchange systems are structured and operated at the backend level.

The studies reviewed pointed to the application of credit, token, or point based systems to achieve equitable value exchange among users. They are systems that keep track of the credits that are earned and those that have been spent; they contain records of the transactions and have the administration level control in place to deter misuse or fraudulence. These kinds of mechanisms contribute to the establishment of trust, accountability and transparency in the platform.

Regarding business and engagement factors, the study revealed that the aspects of gamification, which include points, rewards, and loyalty systems, enhance user engagement and retention. Legally and security-wise, it is worth having records of the transactions and using authentication and access control to handle disputes and protect the data.

Another strength of the journals was the necessity of the presence of clear system actors and use cases, which are normally between users and administrators. Registration, skill management, matching, communication, and monitoring were the core functionalities that were shared by the studies. The study also assisted in determining the major system actors like users, skills, exchange, credits, and messages, which have contributed to the general design and database system of the system.

The journal study was a useful resource in the best practice and served to make sure that the system design is equivalent to the real-life, scalable and secure skill exchange platforms.

3.5.1 SWAP SKILL Platform

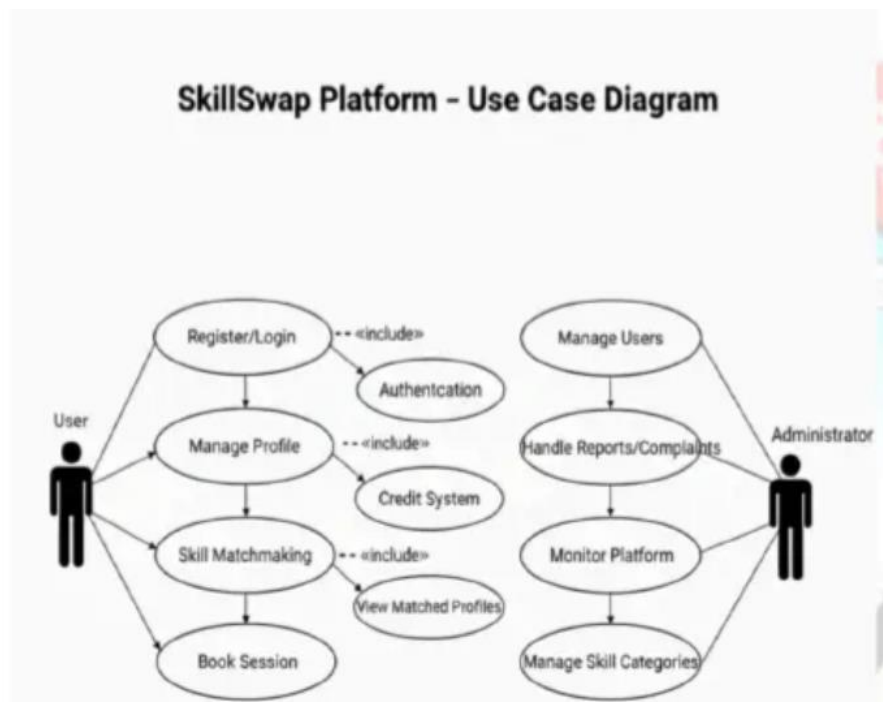
<https://www.scribd.com/document/943225058/Project-File-Swap-Skill>

Backend-Related Financial Components

- Token / credit-based barter engine.
- User account balances (credits earned & spent).
- Credit transfer logic triggered on session completion.
- Transaction log / ledger for every credit transfer.
- Admin-level monitoring for fraudulent or abnormal transfers.
- Feedback/rating storage affecting trust/credit privileges.

Business, Marketing & Legal Reasons for These Attributes

- Business: ensures fair value exchange; prevents freeloading.
- Marketing: tokens/gamification increase engagement and retention.
- Legal: transaction records support dispute resolution and audits.
- Compliance: need to secure financial-related user data and prevent abuse.



Actors include User and Admin, while the use cases include registration, login, skill management, matching, and chat.

3.5.2 Skill_Fusion

https://www.theijire.com/archiver/archives/skill_fusion.pdf

Backend-Related Financial Components

- Digital wallet (store skill-points or credits).
- Points issuance and deduction mechanisms.
- Reward/loyalty module for active contributors.
- Transaction ledger for points transfers.
- Authentication & role-based access for financial endpoints.

Business, Marketing & Legal Reasons for These Attributes

- Business: tracks value exchange and enables the platform economy.
- Marketing: loyalty/points encourage repeat participation.
- Legal: logs needed for disputes, fraud checks, and regulatory traceability.
- Security: authentication protects point balances and transfers.

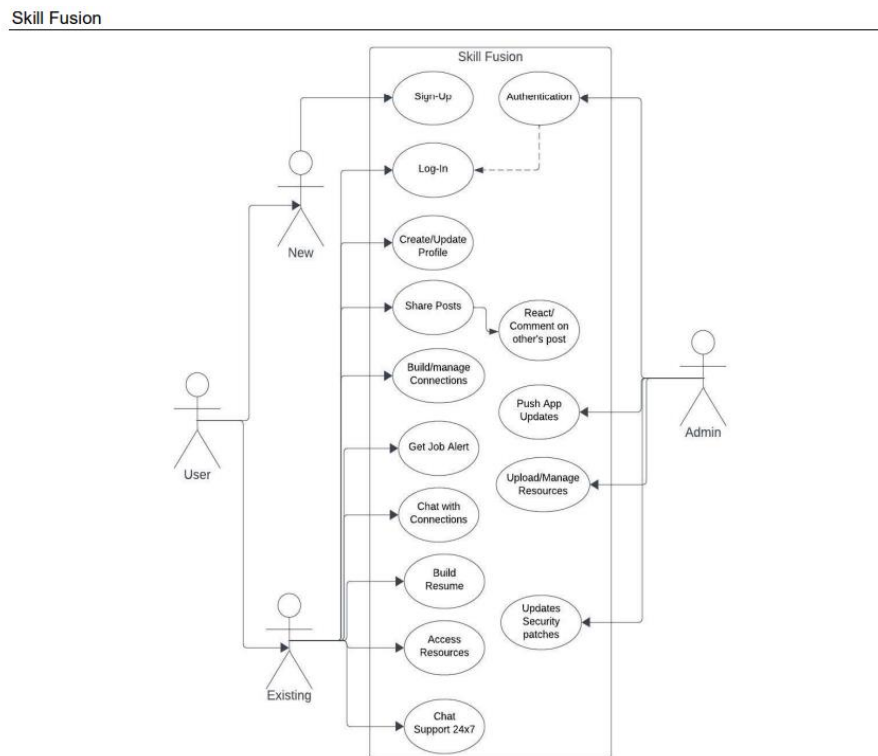


Figure 4: Use Case Diagram

Key Components of Use Case Diagrams:

- **Actors:** Represented as stick figures, actors are entities that interact with the system. They can be users, other systems, or external entities that participate in a use case.
- **Use Cases:** Shown as ovals, use cases outline the precise tasks or operations that the system carries out in reaction to an actor's interaction. Every use case illustrates a unique feature that benefits the actor.
- **System Boundary:** Shown as a rectangle, the system boundary encompasses all of the use cases that the system will address and defines its scope.
- **Relationships:** Associations between actors and use cases, such as generalization (inheritance), inclusion, and extension links, are depicted using lines and arrows, which aid in organizing and refining the interactions.

3.5.3 Micro-payment exchange system

<https://scholarworks.lib.csusb.edu/cgi/viewcontent.cgi?article=3855&context=etd-project&utm>

Backend-Related Financial Components

- E-wallet per user holding e-coins.
- Broker / Issuer module to mint and redeem e-coins.
- Transaction ledger / micro-payment records (sender, receiver, amount, timestamp).
- Secure transfer protocol to prevent double spending.
- Redemption & payout pipeline (convert coins to real currency).
- Audit & reconciliation services for coin issuance/redemption.

Business, Marketing & Legal Reasons for These Attributes

- Business: enables micro-transactions for small-value services/content.
- Marketing: small-cost payments broaden user base and monetization options.
- Legal: precise ledgers required for anti-fraud, AML/KYC where applicable.
- Compliance & privacy: protect users and meet financial regulations when redeeming coins.

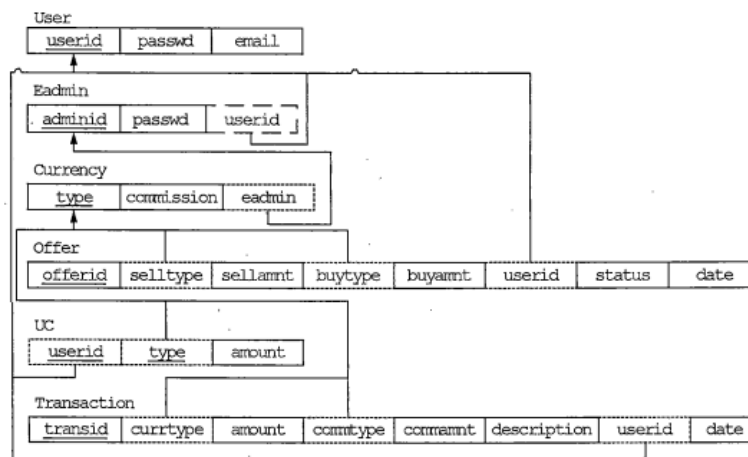


Figure 8. Relation Diagram For the Exchange Service Database

3.5.4 Microservices Architecture for Building a Crypto Freelance Exchange

<https://ceur-ws.org/Vol-4049/paper7.pdf?utm>

Business, Marketing & Legal Reasons for These Attributes

- Marketing Attributes: skill tags, provider reputation.
- Financial Attributes: token balances, conversion rates (token↔points).
- Legal Attributes: verification_status, identity docs reference.
- Analytics Attributes: matching_rate, service latency, success_rate_of_exchanges

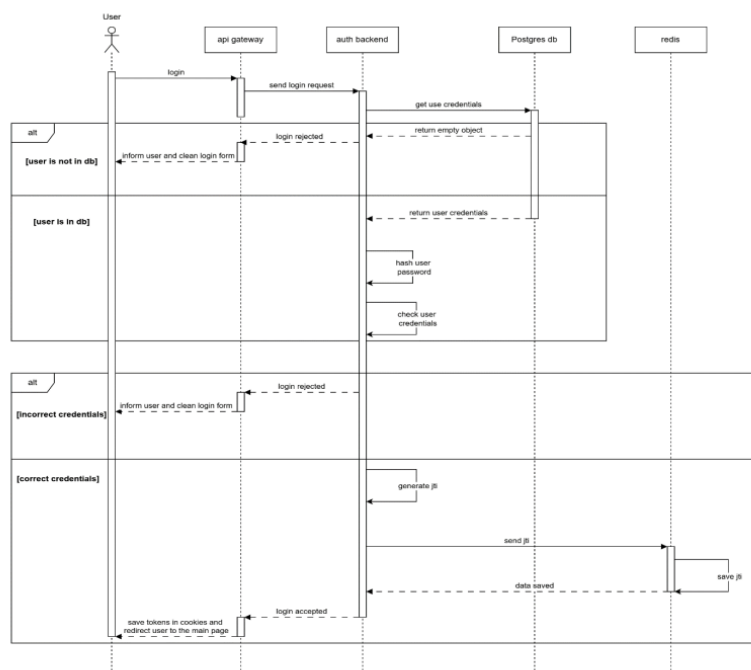


Figure 5: Sequence Diagram of User Authentication Flow

Entities / Attributes

- User Service: user_id, profile, credentials, verification_status
- Skill Service (catalog): skill_id, owner_id, name, category, description, level, metadata
- Match/Exchange Service: match_id, seeker_id, provider_id, matched_skill_ids, match_score, status
- Credit/Token Service: token_id, user_id, balance, earned, spent
- Notification / Messaging Service: message_id, from_id, to_id, content, timestamp

3.6 Comparison

The comparison below highlights the distinctions between similar systems and the proposed skill barter system.

Table 1: Compare my project to another project

Feature	LinkedIn Learning	SkillShare	Simbi	SkillXchange
Skill Exchange	-	-	✓	✓
Monetary Requirement	✓	✓	-	-
Rating & Feedback	✓	✓	✓	✓
Scheduling & Chat	-	-	-	✓
Gamification	-	-	-	✓
Matching	-	-	-	✓
Premium Membership	✓	✓	-	✓
Peer-to-Peer learning	-	-	✓	✓

4 Pre-planning

4.1 METHODOLOGY

The development of the Skill Barter System will follow the Incremental Software Development Methodology. This model divides the project into smaller, manageable increments that are developed, tested, and refined in successive cycles. Each increment adds functionality to the system, ensuring continuous progress and early feedback. This approach suits the project well since it allows flexibility in requirements and iterative improvements based on user testing.

Model Overview

- **Waterfall Model**

The Waterfall Model, introduced by Winston W. Royce in 1970, follows a simple step-by-step development process. The phases include Requirement Analysis, System Design, Implementation, Testing, Deployment, and Maintenance. Each phase is completed before moving to the next one. This model is best for projects where requirements are clearly defined from the beginning and do not change frequently.

Advantages:

- Fixed timeline with well-defined phases.
- Easy to manage since each phase has clear deliverables.
- Good for small projects with predictable requirements.

Disadvantages:

- It is difficult to handle unexpected changes during development.
- Testing happens late, which may increase cost if major errors appear.
- Less flexibility and limited user feedback during early stages.

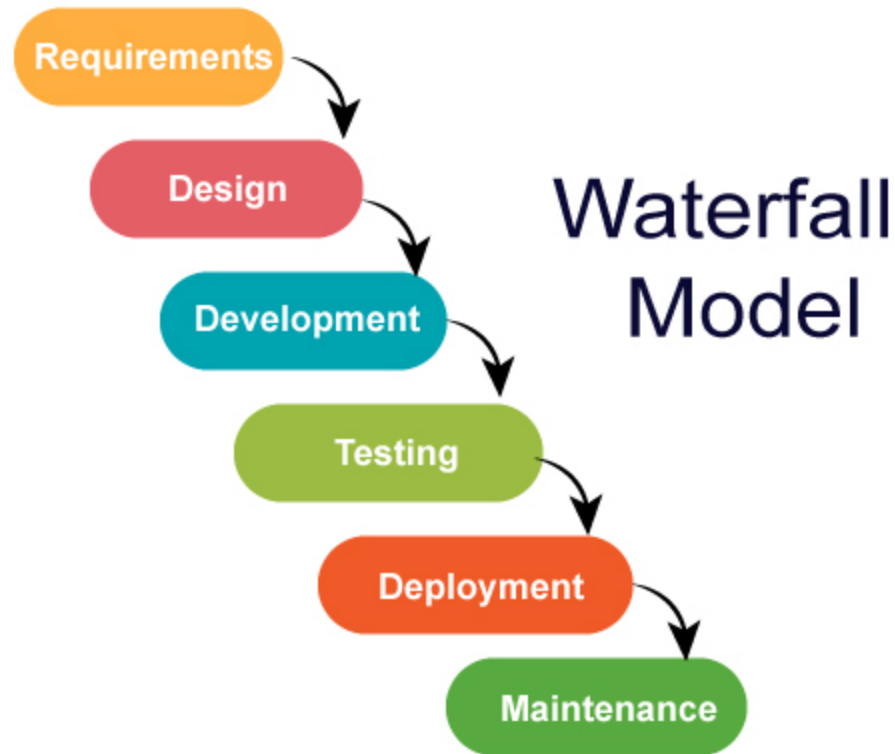


figure 20: Watterfall model

- **Prototype Model**

The Prototype Model focuses on building a quick prototype (or demo version) of the system before the final one is developed. This helps users understand what the system will look like and gives feedback early. The cycle includes Requirement Gathering, Quick Design, Prototype Development, Testing, and Refinement until the system meets user expectations.

Advantage:

- Encourages user involvement and feedback.
- Helps visualize the system early in development.
- Increase customer satisfaction because users see progress.

Disadvantages:

- Increase time and cost due to repeated refinements.
- Create confusion between the prototype and the final system.
- Lead to incomplete documentation.

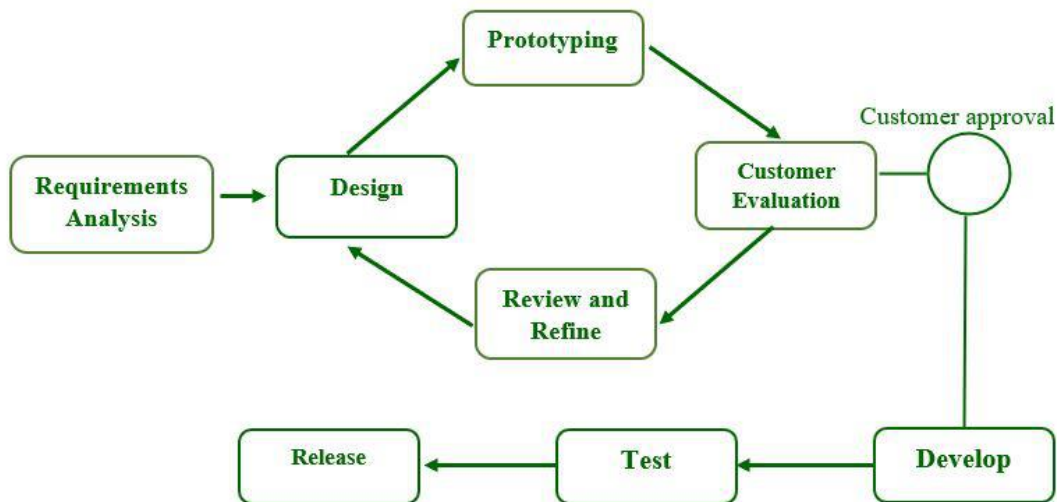


figure 21: Prototype model

- **Spiral Model**

The Spiral Model combines features of both the Waterfall and Prototype models. It is used for large or complex projects that need continuous improvement and risk management. The process repeats in cycles called *spirals*, where each loop includes Planning, Risk Analysis, Engineering, Testing, and Evaluation.

Advantages:

- Focus on risk management and problem-solving.
- Suitable for complex and long-term projects.
- Allows continuous refinement and customer input.

Disadvantages:

- Costly and time-consuming for small projects.
- Hard to manage due to complexity.
- Need experienced developers for risk assessment.

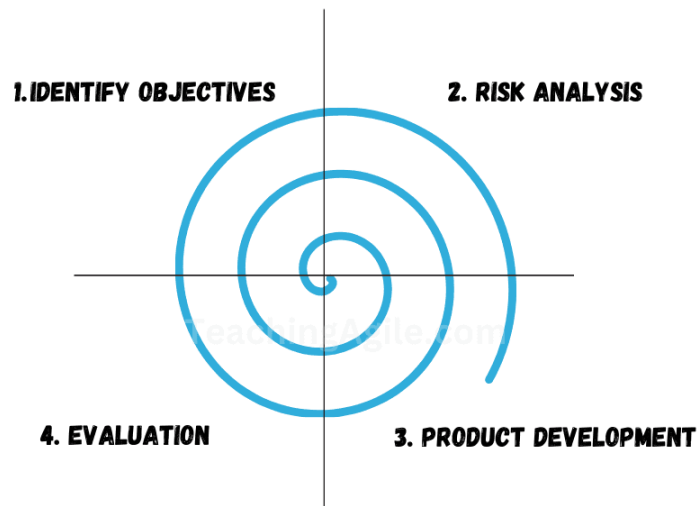
SPIRAL MODEL IN SOFTWARE DEVELOPMENT

figure 22: Spiral model

4.2 Chosen Methodology

The incremental model in software engineering divides a project into smaller modules, or increments, developed and integrated gradually. This iterative approach enables frequent testing, quick feedback, and early detection (Sachan, 2024).

The Incremental Model is a great fit for this project because it breaks the work into smaller steps, making development and integration easier. Each part is tested and improved before moving on, which helps create a more stable and unified system. This method also allows for regular feedback and makes it easier to adjust to changes, ensuring the final product meets user's needs and is strong and reliable.

Advantages of the Incremental Model:

- **Early Delivery:** The basic functions like registration, login and skill management were created during the initial phases and the system was able to generate visible outputs within a short period of time.
- **Flexibility:** The methodology enabled easy change to be made in the process of development. As an illustration, testing and feedback were used to enhance features like session handling and chat system.
- **Risk Reduction:** Mistakes and problems were detected during the initial stages of every increment, and the chances of significant failures in the subsequent stages were minimized.
- **Improved Quality:** Testing of each module was done separately followed by integration, which guaranteed better quality and reliability of the system.
- **Efficient Resource Utilization:** The development was done in small sections that could be handled easily and time and effort were easily allocated.
- **Improved Knowledge of System Requirements:** The requirements were more evident as the project went on, and it was possible to make improvements and refinements in subsequent increments.
- **Simplified Project Management:** Planning, monitoring and controlling the project was easier because the system was broken into smaller increments.
- **Scalability:** New modules like premium membership and payment integration were easily added without interfering with the current modules.

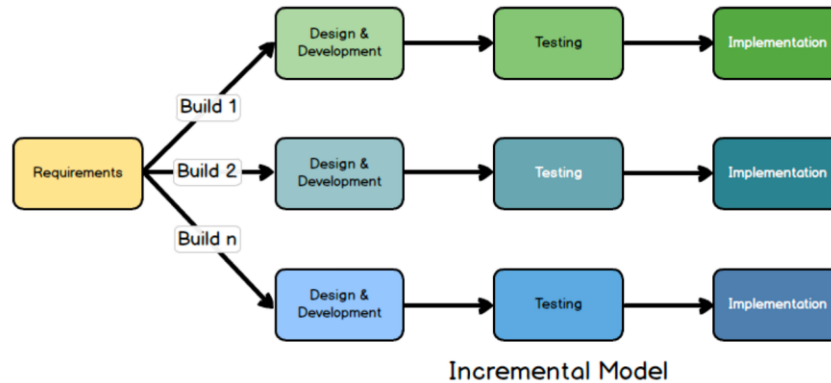


figure 23: Incremental Model

4.3 Incremental Breakdown

The development of the Skill Barter System was divided into multiple increments, where each increment focused on specific functionalities of the system.

- Increment 1 (Project Setup and System Design)

Tasks:

- Requirement Analysis: Research related journals and research papers.
- Comparison of other existing systems.
- System problems and solutions identified.
- Simple diagram drawing (Use Case, Flowchart), Complex diagram development (ERD, DFD, Sequence Diagram)
- Setting up environment (Laravel, database, tools)
- Wireframe design for user interface

- Increment 2 (User Authentication and Profile Management)

Tasks:

- Creation and configuration of databases.
- User registration functionality
- Login system implementation
- Email verification system
- Role management (Student, Teacher, Admin)
- Creation and updating profiles
- Basic dashboard setup

- Increment 3 (Skill Management and Request System)

Tasks:

- Development and management skills.
- Acquire teaching and learning skills.
- Search and skill browsing.
- Implementation of the session request system.
- Accept and reject request feature.
- Request status tracking: Blending of users and skills.

- Increment 4 (Session, Chat and Advanced Features)

Tasks:

- Creation of sessions upon acceptance of request.
- Status of sessions (pending, ongoing, completed)
- Real-time communication through a chat system.
- Resource sharing option (upload and download of files)
- Feedback and rating system.
- Notification system
- Premium membership feature
- Khalti payment integration.

- Increment 5 (Admin Panel, Testing and Finalization)

Tasks:

- Admin dashboard development
- User management (activate, deactivate, delete)
- Teacher approval system
- Admin skill management
- Observation and orders
- System testing and debugging
- Black-box testing
- Integration testing
- Performance testing
- Final documentation preparation

4.4 Work Breakdown Structure (WBS)

The following breakdown represents the major phase of development using the Incremental Model:

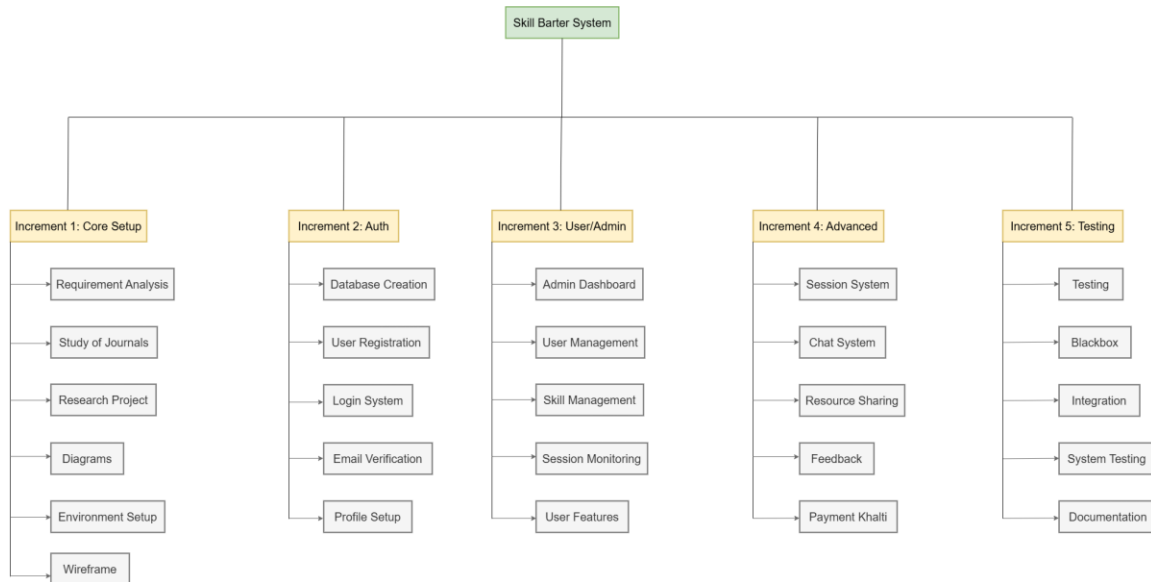


Figure 24: WBS

The Work Breakdown Structure (WBS) of the Skill Barter System displays the way in which the project is broken down into smaller bits and how it is built in stages. It begins with the core setup stage where planning activities such as requirement analysis, research, diagrams, and environment setup is done. It proceeds to authentication, where the database creation, user registration, login, email verification and profile setup are performed. The user and the administration module is developed thereafter with dashboard, user, skill and session monitoring. Then, the more complicated elements like session system, chat, sharing resources, feedback, and payment integration are developed, which are the primary functionalities of the platform. Lastly, testing of the project is undertaken, comprising of various forms of testing and reportages in order to ensure that the system is functioning well. The WBS, in general, demonstrates a clear and systematic manner of executing the project in stages.

4.5 Milestones

Table 2: Milestones

Milestones	Activities	Dates
1	Requirement Analysis	25 July 2025 to 28 July 2025
2	Study of Journals	29 July 2025 to 31 July 2025
3	Research Project	01 Aug 2025 to 03 Aug 2025
4	Diagrams	07 Aug 2025 to 14 Aug 2025
5	Environment Setup	15 Aug 2025 to 18 Aug 2025
6	Wireframes	19 Aug 2025 to 23 Aug 2025
7	Database Creation	24 Aug 2025 to 04 Sep 2025
8	User Registration	04 Sep 2025 to 07 Sep 2025
9	Login System	05 Sep 2025 to 08 Sep 2025
10	Email Verification	07 Sep 2025 to 11 Sep 2025
11	Profile Setup	12 Sep 2025 to 22 Sep 2025
12	Admin Dashboard	21 Sep 2025 to 04 Oct 2025
13	User Management	04 Oct 2025 to 18 Oct 2025
14	Skill Management	18 Oct 2025 to 28 Oct 2025
15	Session Monitoring	29 Oct 2025 to 09 Nov 2025
16	User Features	06 Nov 2025 to 19 Nov 2025
17	Session System	20 Nov 2025 to 13 Dec 2025
18	Chat System	11 Dec 2025 to 30 Dec 2025
19	Resources Sharing	27 Dec 2025 to 17 Jan 2026
20	Feedback	17 Jan 2026 to 27 Jan 2026
21	Payment Khalti	27 Jan 2026 to 13 Feb 2026
22	Testing	13 Feb 2026 to 05 Mar 2026
23	Blackbox	27 Feb 2026 to 29 Mar 2026
24	Integration	28 Mar 2026 to 11 April 2026
25	System testing	09 April 2026 to 19 April 2026
26	Documentation	31 Jan 2026 to 22 April 2026

4.6 Project Gantt Chart

The Gantt Chart below represents the timeline of the Skill Barter System project across the current academic term. Each bar corresponds to the time allocated for major project milestones.

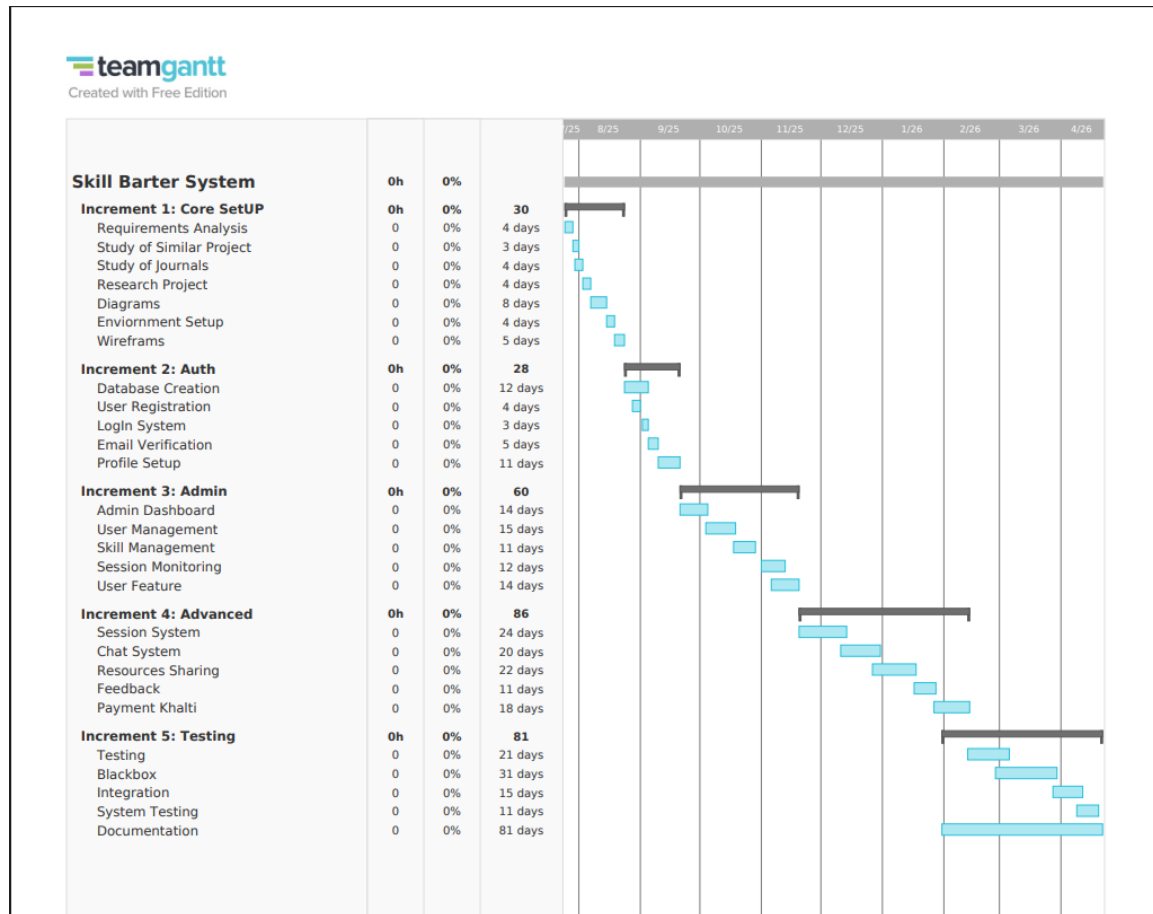


Figure 25: Gantt Chart

The Gantt chart is a development timeline of the Skill Barter System, which illustrates how the project was done in various steps based on incremental approach. The project starts with Increment 1: Core Setup, during which the first activities, including the requirement analysis, research, studying similar projects and journals, creation of diagrams, setting up of the environment, and wireframing, were performed. The primary aim of this stage was to have a solid foundation and how the system needs to be.

Then, increment 2: Authentication was done, with the creation of a database, user registration, the login system, email verification, and profile setup. This stage provides that users will be able to safely enter the system and control their basic information.

Then, increment 3: Admin Module aims to develop the administrative capabilities including the following: the admins dashboard, user management, skill management, session monitoring, and other typical user features. This stage enables the system to be operated and maintained efficiently in the backend.

Next, increment 4: Advanced Features, embodies the key functionalities of the system which are: session handling, chat system, resource sharing, feedback system, and Khalti payment integration. It is the most significant stage when the main idea of the platform skill exchange and interaction are implemented completely.

Lastly, increment 5: Testing involves various kinds of testing which are as follows: functional testing, black-box testing, integration testing, and system testing and documentation. This stage will make sure that the system is stable, without errors and is ready to be deployed.

In general, the Gantt chart clearly indicates that the project was created in stages, with one phase being based on the previous one. This gradual strategy assists in more effective management, fewer risks and the system is well tested and operational at all levels.

5 Design

There are different diagrams on system design and modeling, which were developed to gain insight into the design of the system. The system operation, user interaction with the system, data flow, and database structure are visualized using these charts. UML diagrams composed of Use Case, Class, Sequence and Collaboration were used to model system functionalities, structure and interaction between the different components of a system. To expose data flow within the system, Data Flow Diagrams (DFD) were created. To design the database structure, an Entity Relationship Diagram (ERD) was drawn, and a system flowchart was provided to illustrate the process flow and the logic behind decision making in general. These diagrams helped to make the system design understandable and to implement it on a good basis.

Class Diagram:

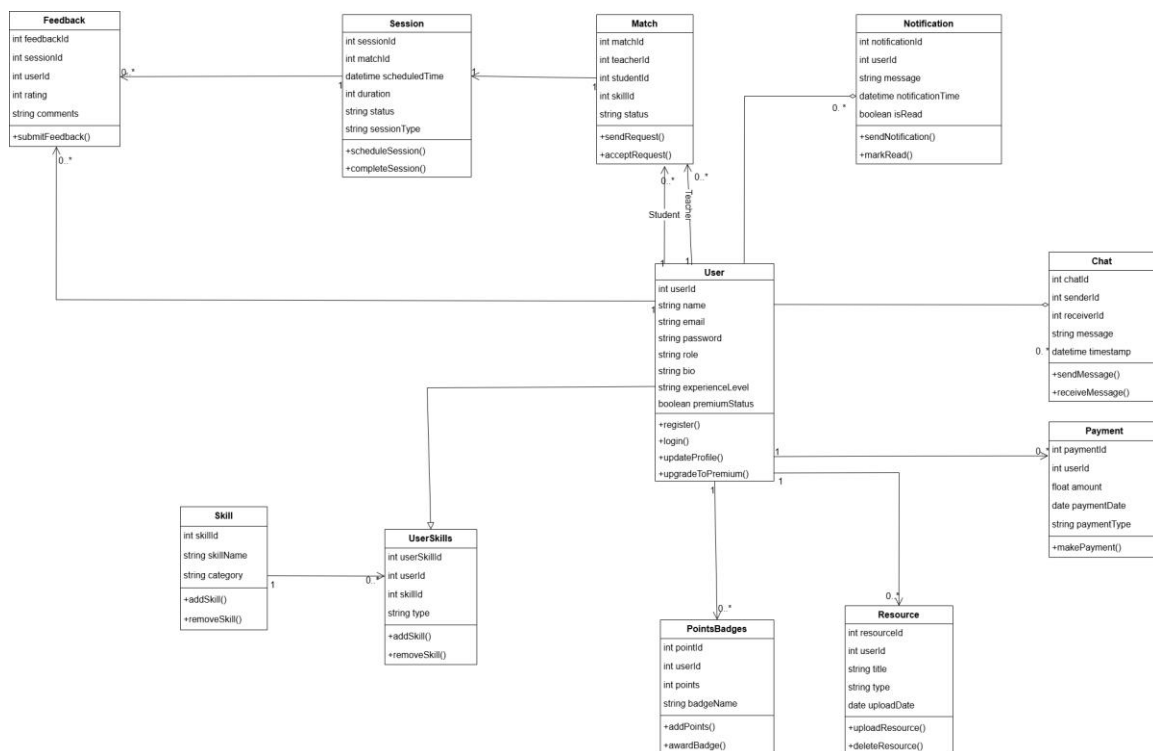


Figure 26: Class Diagram

The class diagram is an object-oriented design of the system that displays the arrangement of functionality into classes. The User class is the central one, but it handles authentication, profile, and premium upgrades. Skills are also handled using the Skill and User Skills classes, where the users are connected to pedagogical roles or to the role of learning. The classes are Match and Session that handle requests and scheduled learning activities and the Chat that facilitates messaging between the users. Resource process uploaded learning content, Payment is used to handle transactions, Notification sends system notification and Feedback receives session ratings. In general, the diagram shows that there is a distinct division of responsibilities, which contributes to scalability and maintainability.

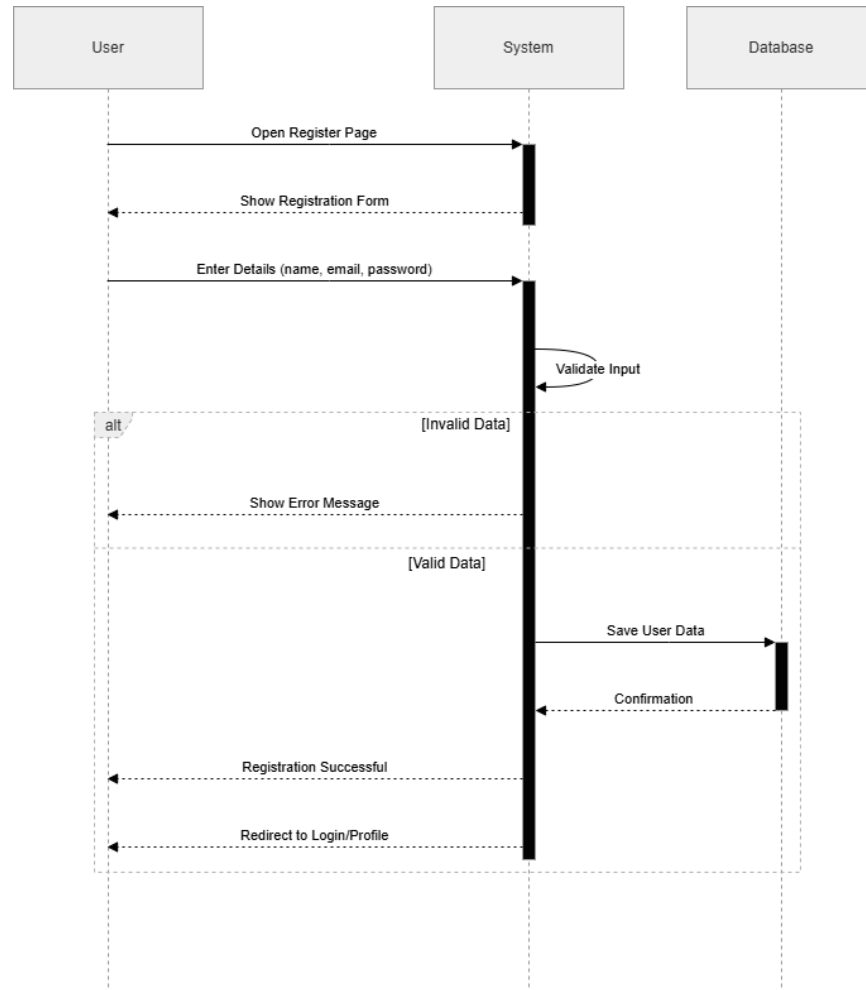
Sequences Diagram:

Figure 27: Sequences Diagram

The sequence diagram is used to show how the user, system and database interact in a step-by-step manner during the registration process. This process starts with the user opening the registration page where the system shows the registration form. The user then proceeds to key in his or her details like name, email and password and submit it to the system. The input data is validated to be correct by the system. When the data is invalid the system will send back the user with an error message. But in case the data is valid, the system continues to store the user information in the database which subsequently provides confirmation. When a successful storage is made, the system informs the user that the registration is successful and redirects him or her to the login page or profile page, which is the end of the registration process.

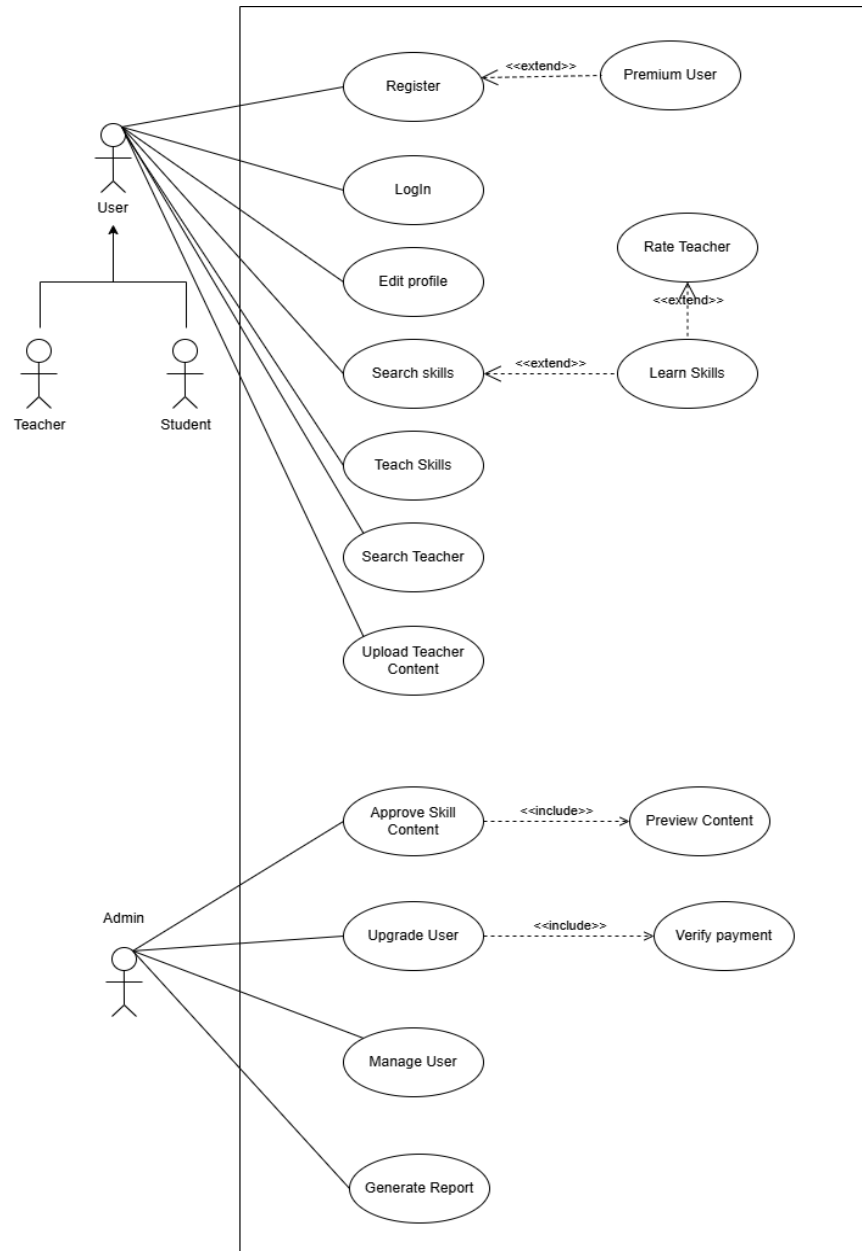
Use Case Diagram:

Figure 28: Use Case Diagram

The use case diagram describes the functional requirements of the Skill Barter System in terms of how various actors interface with the system. User is the primary actor that is further specialized into Student and Teacher who have similar features like registration, login, profile management, and search of skills. There are other applications such as teaching skills and uploading learning content by the teachers and the option of rating teachers after learning is available to the user. The Premium User is presented as an

optional add-on to registration, where one can upgrade to additional features. The admin handles system level activities including skill content approval, user upgrades, payment verifications, user administration and report generation with a well-defined separation of user and administrative duties based on include and extend relationships.

5.1 High Use Case Description (Your System)

Use Case	Actors	Description
Register	User	The user registers an account, which requires him to fill in details of name, email and password. The system authenticates the input and stores user information.
LogIn	User	Users log into the system with credentials. The system authenticates the information and provides access depending on role.
Manage Profile	User	Users have an opportunity to change their profile data, skills, and level of experience.
Search Skills	User	User search for skills or teachers. The system accesses and presents similar results.
Send Request	User	Users make requests of other users to their sessions. The receiver is notified and stored in the system.
Manage Session User	User	Users plan and run sessions. The status of the sessions is monitored by the system.

Provide Feedback	User	Users rate and comment on the session after it is over. Feedback and update of ratings are kept in the system.
Upgrade to Premium	User	A user can upgrade to premium by paying. The payment and subscription are processed by the system.
Make Payment	User	The user provides payment information. The transaction is processed and validated through payment gateway by the system.
Manage Users	Admin	Can deal with user accounts, such as enabling and disabling.
Manage Skills	Admin	The admin adds, updates, or removes the skills offered on the platform.
Manage Feedback	Admin	The admin has access to all the feedback and ratings posted by users.
Manage Dispute	Admin	The admin considers and closes disputes among users.
Manage Payments	Admin	The admin monitors transactions and platform revenue.

5.2 Expanded Use Case: Premium (Payment) Upgrade.

Name: Upgrade to Premium
Actor: User
Description: A user will choose the upgrade option when he or she desires to get the premium features. The user status is verified, and the system redirects the user to the payment gateway. Upon successful payment, the system switches on the premium subscription and provides access to premium features.

User	System
1. The user clicks on the Upgrade to Premium option.	2. The system verifies whether the user is in the present state.
3. The system generates a payment request.	4. The system redirects to the payment gateway.
5. The user is required to key in payment information.	6. The payment is validated and processed by the system.
7. The payment is a success.	8. The system modifies the subscription status.
9. The system shows success message and unlocks premium features.	

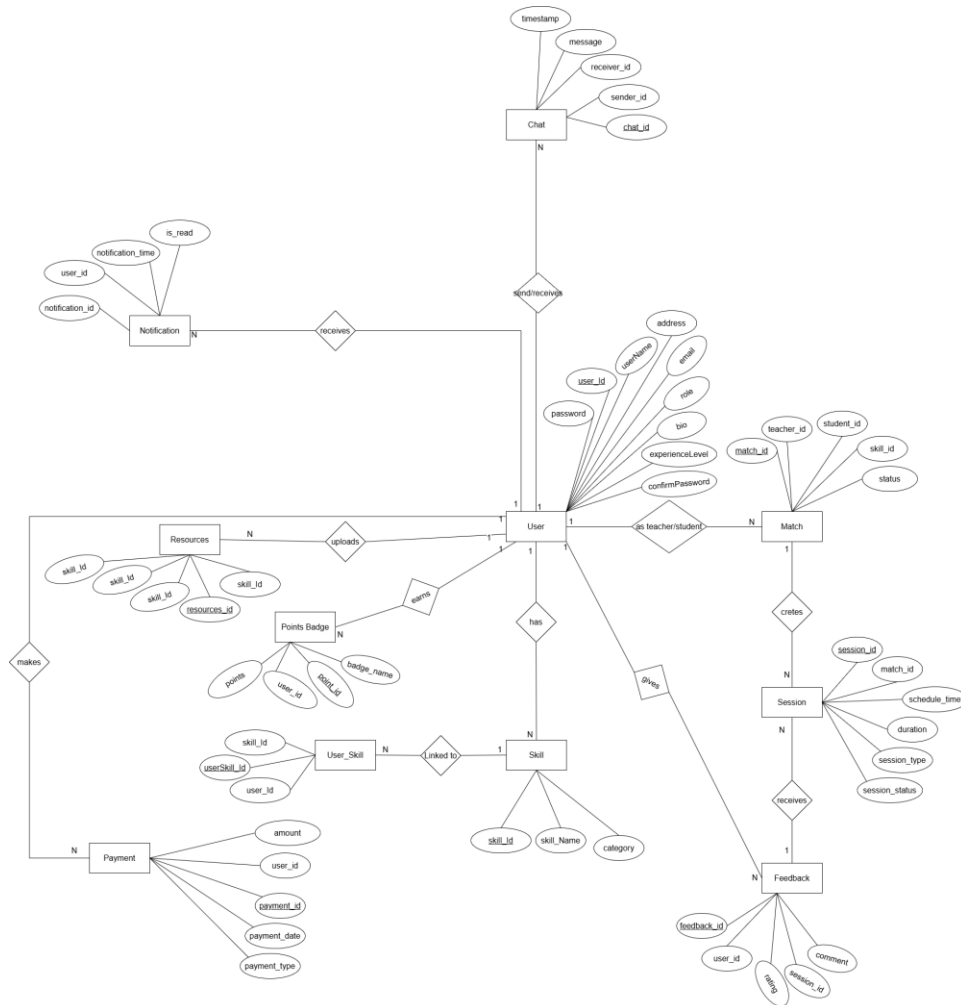
Entity-Relationship Diagram:

Figure 29: ERD

The ERD gives the database format of the Skill Barter System with emphasis on entities, attributes, and their association. The user entity is the central entity that links to the Skills via the User Skill table to handle many-to-many relationships. The Match entity matches students and teachers and may create various Sessions to have scheduled learning. Learning materials that are stored in the entity of Resources are uploaded by teachers and communication among users is facilitated in the entity of Chat. The Premium upgrades and transactions confirmed by the admin are tracked in the Payment entity, the Notifications enable users to see what happens in the system and the Feedback archive the ratings and comments received after the sessions, which creates quality and trust within the platform.

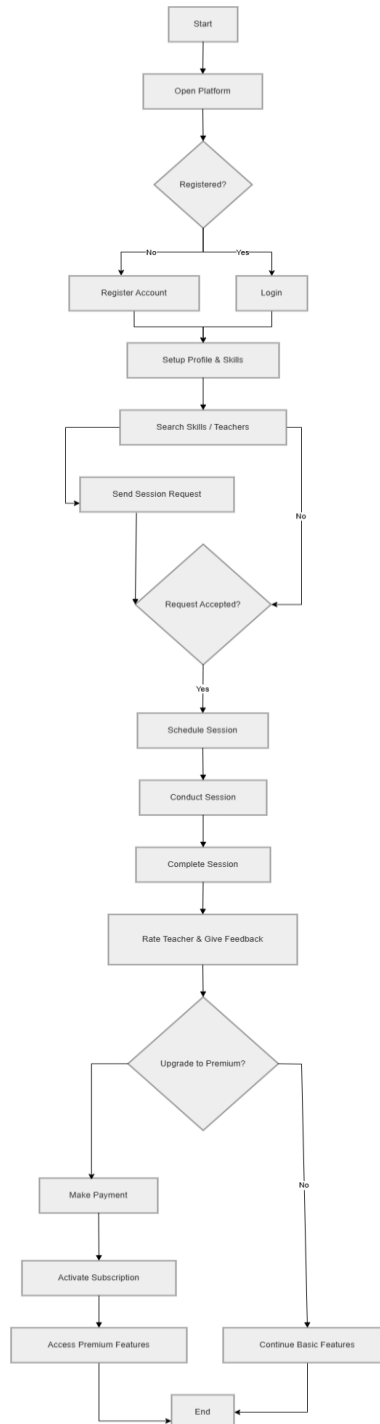
Flowchart:

Figure 30: Flowchart

The flowchart shows the entire workflow of the Skill Exchange platform in detail. It starts when the user opens the platform and verifies their status on whether they have been registered. Otherwise, the user creates an account, otherwise, he/she logs in. Thereafter, the user creates their profile and skills and finds skills or teachers and requests a session. When the request is not accepted, the user will search and resend the request, and when it is accepted, the session is planned. User then provides and completes the session and rates the teacher and gives feedback. The user is then offered the chance to upgrade to a premium plan after this. In case the user wants to upgrade, he or she pays, subscribes to it and accesses premium features. Otherwise, they will keep on using the rudimentary features and the cycle is complete.

DFD 0:

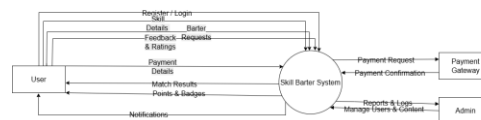


Figure 31: DFD 0

The Level 0 DFD gives an overview of the Skill Barter System on a high level by depicting the flow of data between the external and the entire system. The user engages with the system through registration details, skills, requests, feedback and payment details and the responses include match results, notifications, ratings and rewards. All the payment related information such as payment requests and confirmations is done in the payment gateway making transactions secure. Also, the system uses the admin to communicate with the system to maintain users, resolve conflict, and manage the running of the system. This diagram is more concerned with the data flow than the process steps or time which presents a clear picture of the data flow of information in and out of the system.

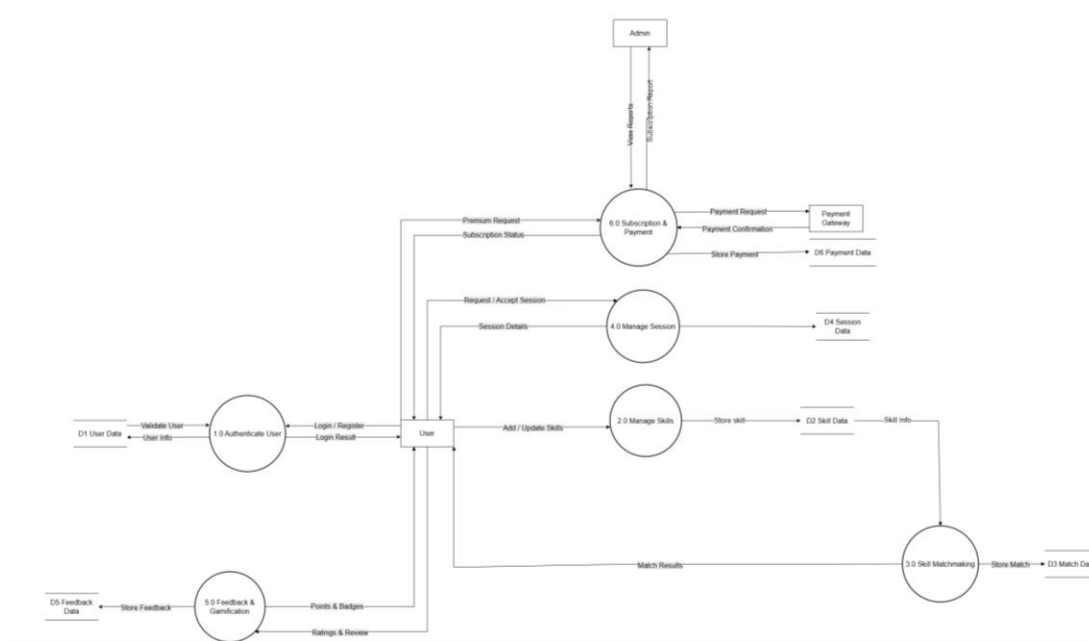
DFD1:

Figure 32: DFD1

5.3 Algorithm

Step 1: Role-based Initiation and Login.

- Start the system
- User chooses role (Student/Teacher/Admin)
- Enter login credentials
- System authenticates user information.
- In case of a valid -go to the respective dashboard.
- In case of invalidity, then show error message.

Step 2: Admin Workflow

- Admin logs into the system and opens the dashboard on the side of the administration.
- Dashboard provides system statistics like users, sessions, skills and revenue.
- Admin can manage:
 - Users
 - Skills
 - Sessions
 - Payments
 - Subscriptions
 - Feedback and disputes
- Admin can view the reports and track system activities.
- Admin is able to answer user problems and control the work of the platform.

Step 3: Teacher Workflow

- Teacher logs in and views teacher dashboard.
- Dashboard shows session requests, sessions to come, and feedback.
- Teacher can:
 - Reject or accept session requests.
 - Book appointment with date and time.
 - Facilitate meeting connection (prior to or throughout course)
 - Conduct and start session.
 - Upload learning materials
 - View student assignments

- Teacher wraps-up and reviews feedback.

Step 4: Student Workflow

- Students enter the system and log in to the student dashboard.
- Skills, teachers and session updates are shown on dashboard.
- Students can:
 - Find and filter by teachers.
 - See teacher profiles and skills.
 - Send session request
 - See session status and notifications.
 - Enter lesson once the teacher begins.
 - Upload assignments
 - View learning materials
- Provide feedback and rating after session.
 - Change password and update profile.

Step 5: Session Workflow

- Student requests teacher to send a session.
- Teacher receives notification
- Request is approved or disapproved by teacher.
- If accepted:
 - Teacher develops session plan (date and time needed)
- Meeting link can be later added.
- Students peruse session information.
- Teacher starts session
- Student joins session
- Session is carried out and done.
- Student gives feedback and assessment.

Step 6: Premium (Payment) Workflow.

- The user chooses premium plan.
- Redirects to Khalti payment gateway.

- User adds payment information and OTP.
- In case of successful payment:
 - updates subscription status.
 - Premium functions are turned on.
- In case of default of payment or cancellation:
 - System displays error message.
 - No subscription update.

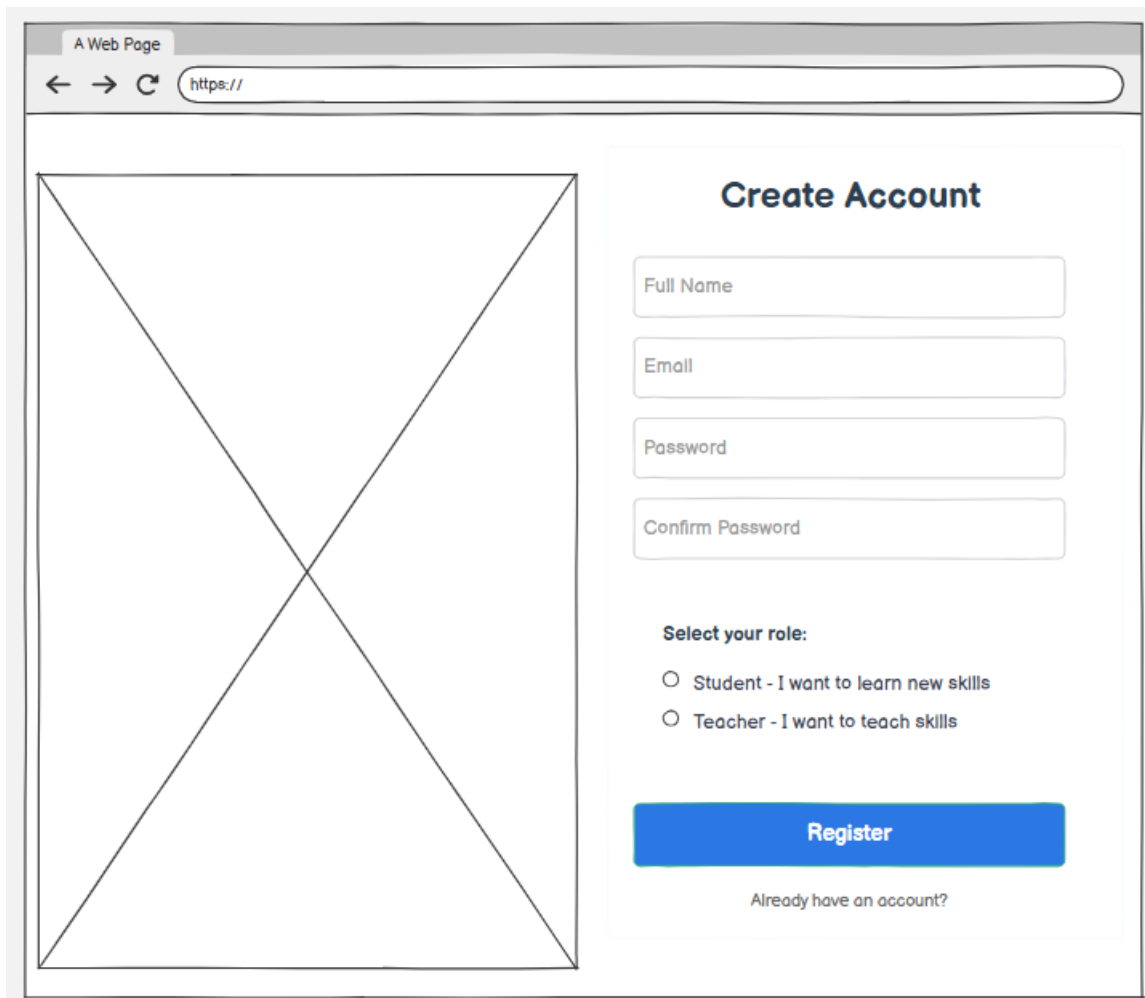
Step 7: Feedback Workflow

- User can rate teacher after completion of the session.
- User adds rating and optional comment.
- System validates input
- Feedback will be stored in database.
- Mean score of teacher is updated.

Step 8: System Exit

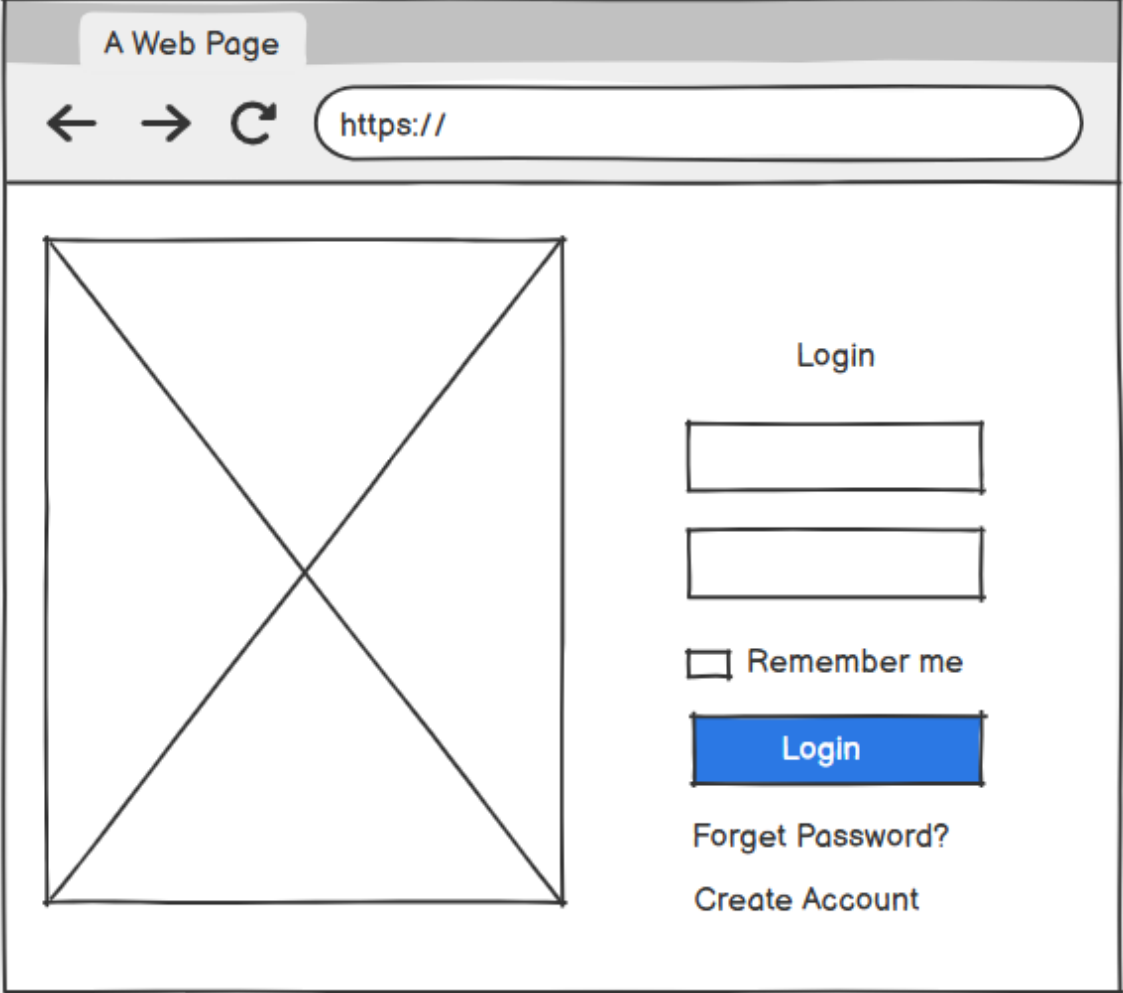
- User leaves the system.
- Session/token is destroyed
- User is redirected to the login page.

5.4 Wireframe



The wireframe shows a web browser window with a tab labeled 'A Web Page' and a URL bar containing 'https://'. The main content area is divided into two sections. On the left is a large rectangular placeholder with a diagonal 'X' across it. On the right is a 'Create Account' form. The form contains four input fields: 'Full Name', 'Email', 'Password', and 'Confirm Password'. Below these fields is a section titled 'Select your role:' with two radio button options: 'Student - I want to learn new skills' and 'Teacher - I want to teach skills'. At the bottom of the form is a blue 'Register' button and a link that says 'Already have an account?'.

Figure 33: Register page's wireframe



The wireframe depicts a web browser window titled "A Web Page". The address bar shows "https://". The main content area is divided into two sections. On the left, there is a large square placeholder with a diagonal 'X' across it. On the right, the text "Login" is centered above two input fields. Below these fields is a checkbox labeled "Remember me". A blue button with the text "Login" is positioned below the checkbox. At the bottom of the right section, there are two links: "Forgot Password?" and "Create Account".

Figure 34: Login page's wireframe

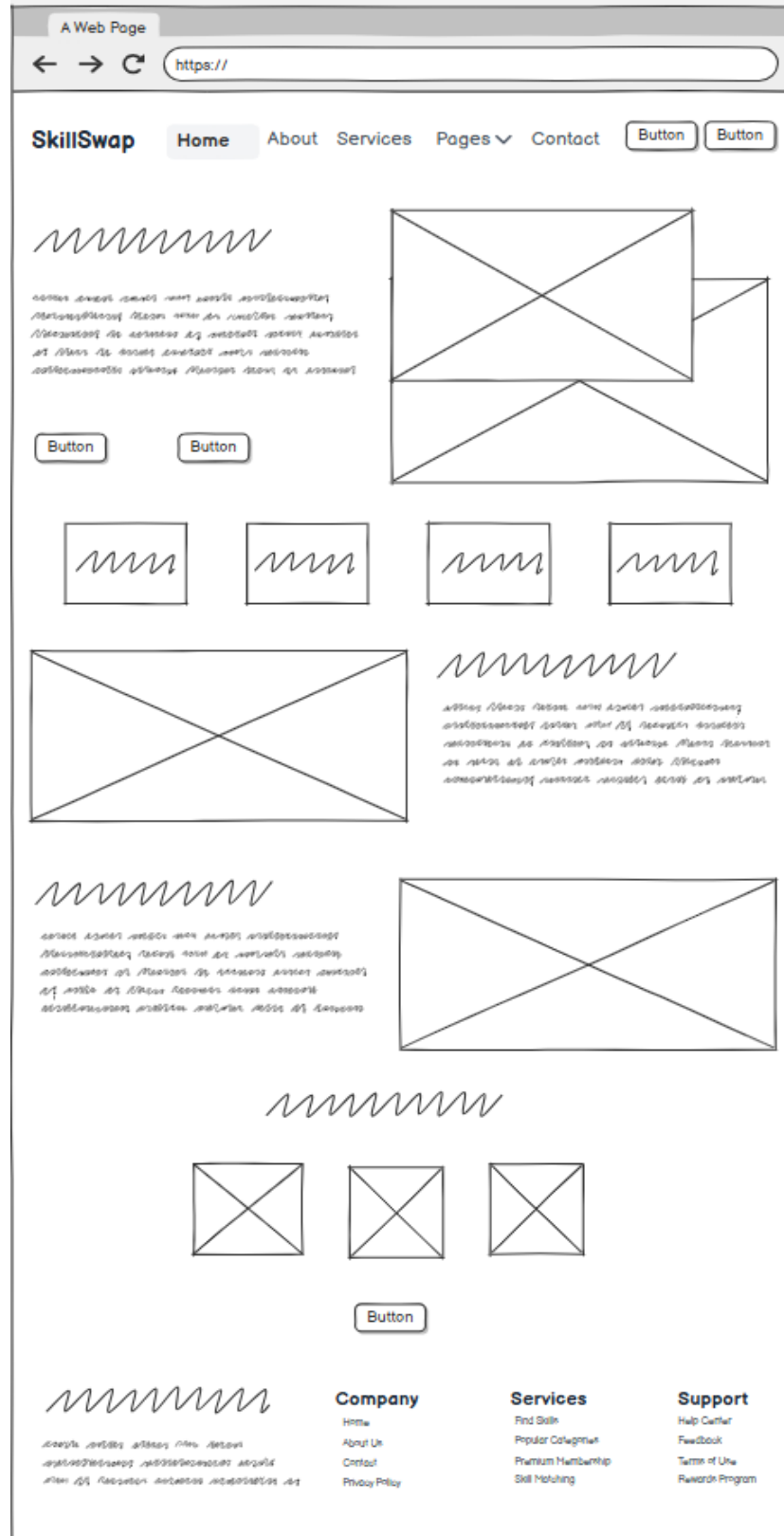


Figure 35: Home page's wireframe

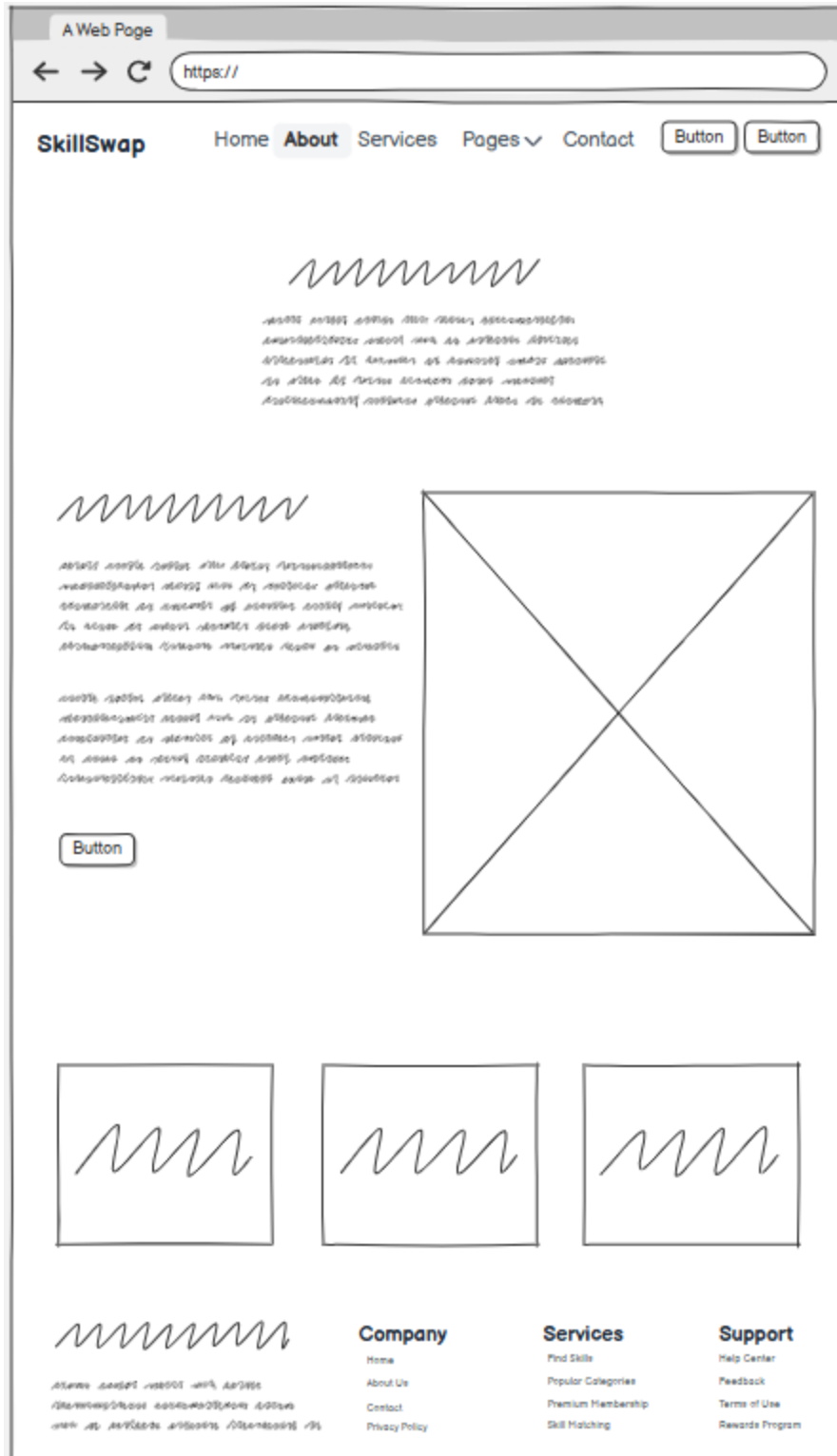


Figure 36: About page's wireframe

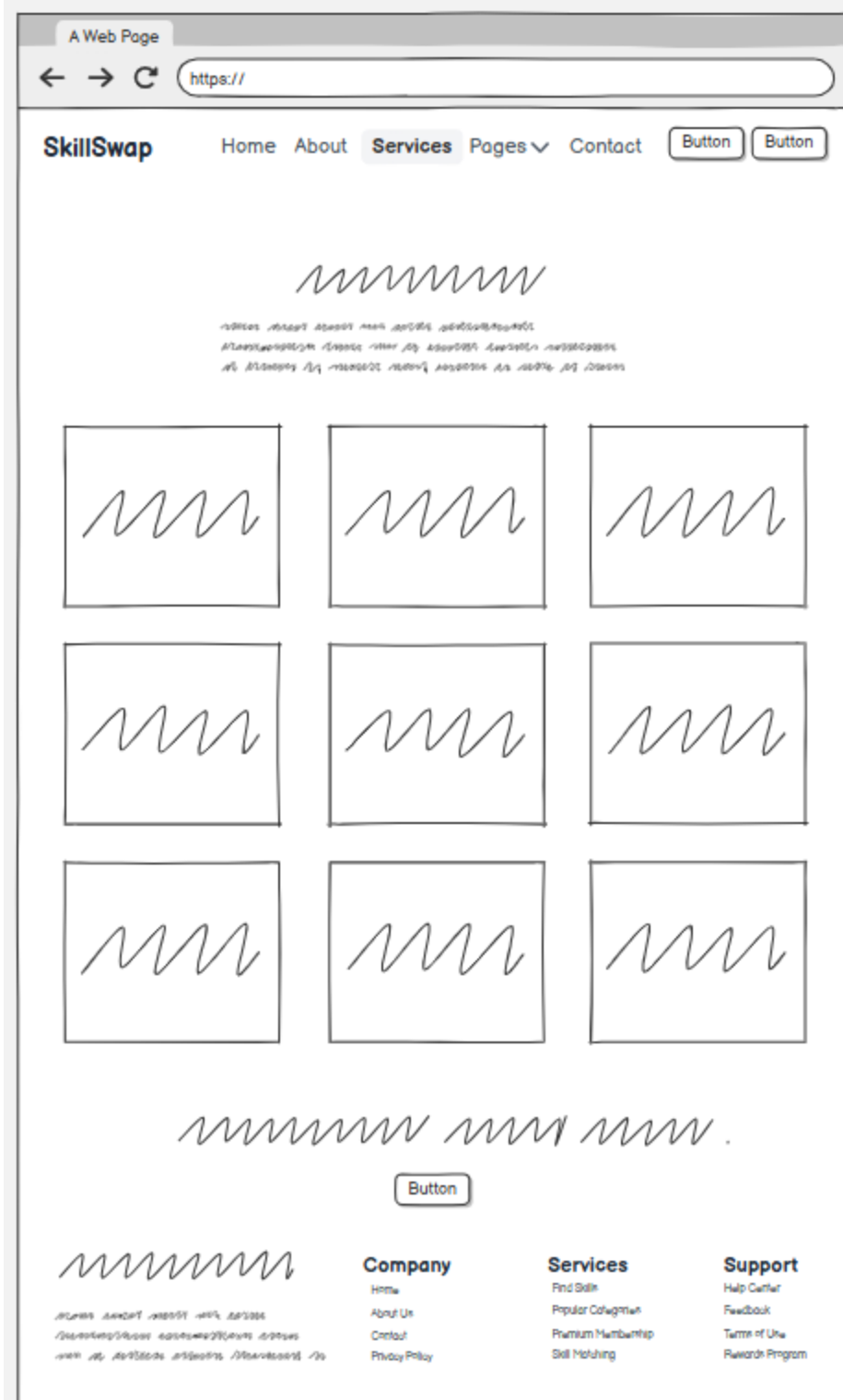


Figure 37: Services page's wireframe

A Web Page
https://

SkillSwap
Home About Services Pages ▾ Contact
Button Button

GET IN TOUCH

Send us a Message

Your Name
Email Address
Subject
Message

Email Us
support@skillswap.com
info@skillswap.com

Call Us
+977 9816681421
+977 981554433

Our Office
Pokhara-17, Chhorepatna
Gandaki, Nepal

Support Hours
Mon - Fri: 9:00 AM - 6:00 PM
Weekend: 10:00 AM - 2:00 PM

Send Message

FAQS

Company
Home
About Us
Contact
Privacy Policy

Services
Find Skills
Popular Categories
Premium Membership
Skill Matching

Support
Help Center
Feedback
Terms of Use
Rewards Program

Figure 38: Contact page's wireframe

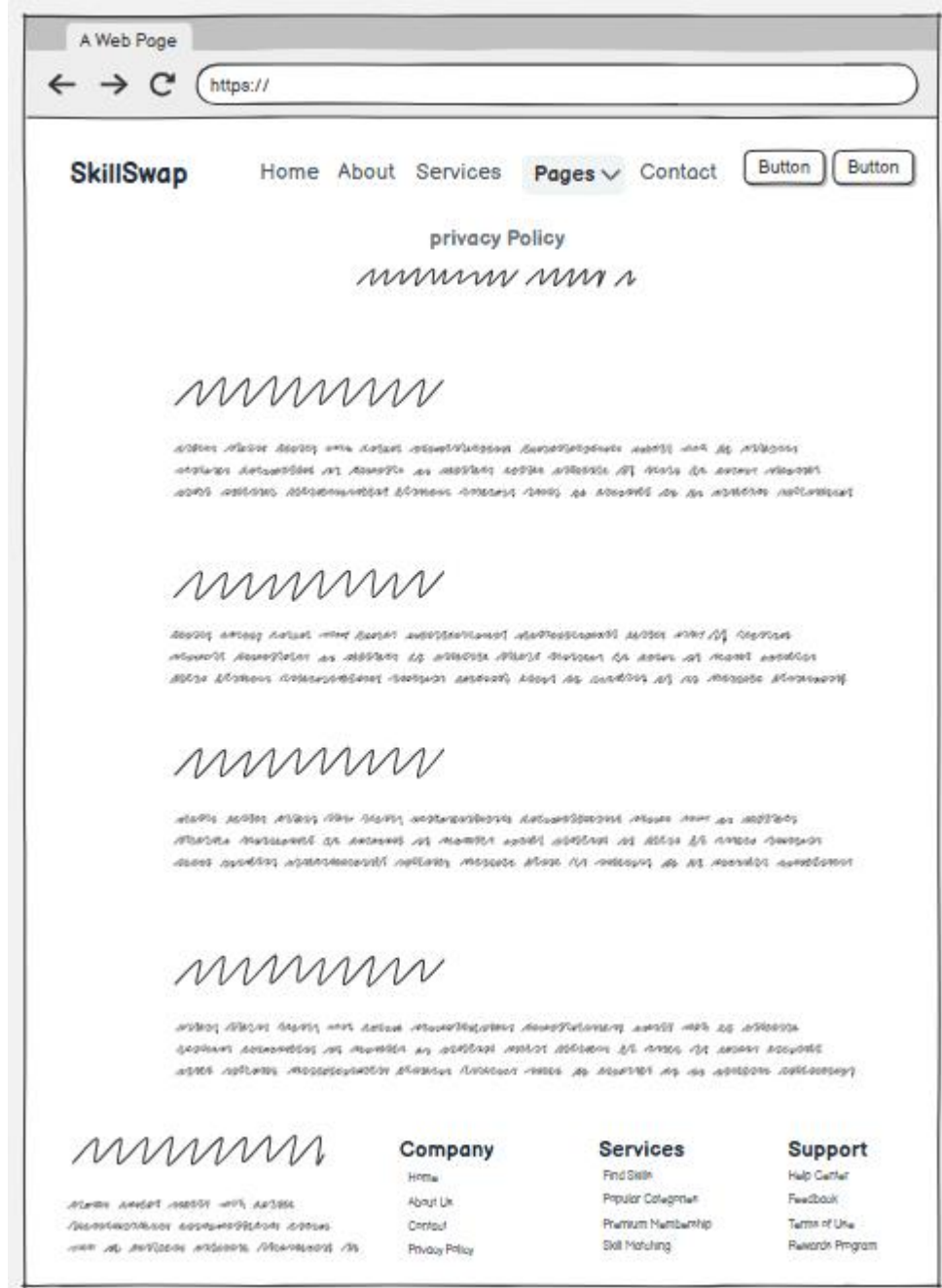


Figure 39: Privacy Policy page's wireframe

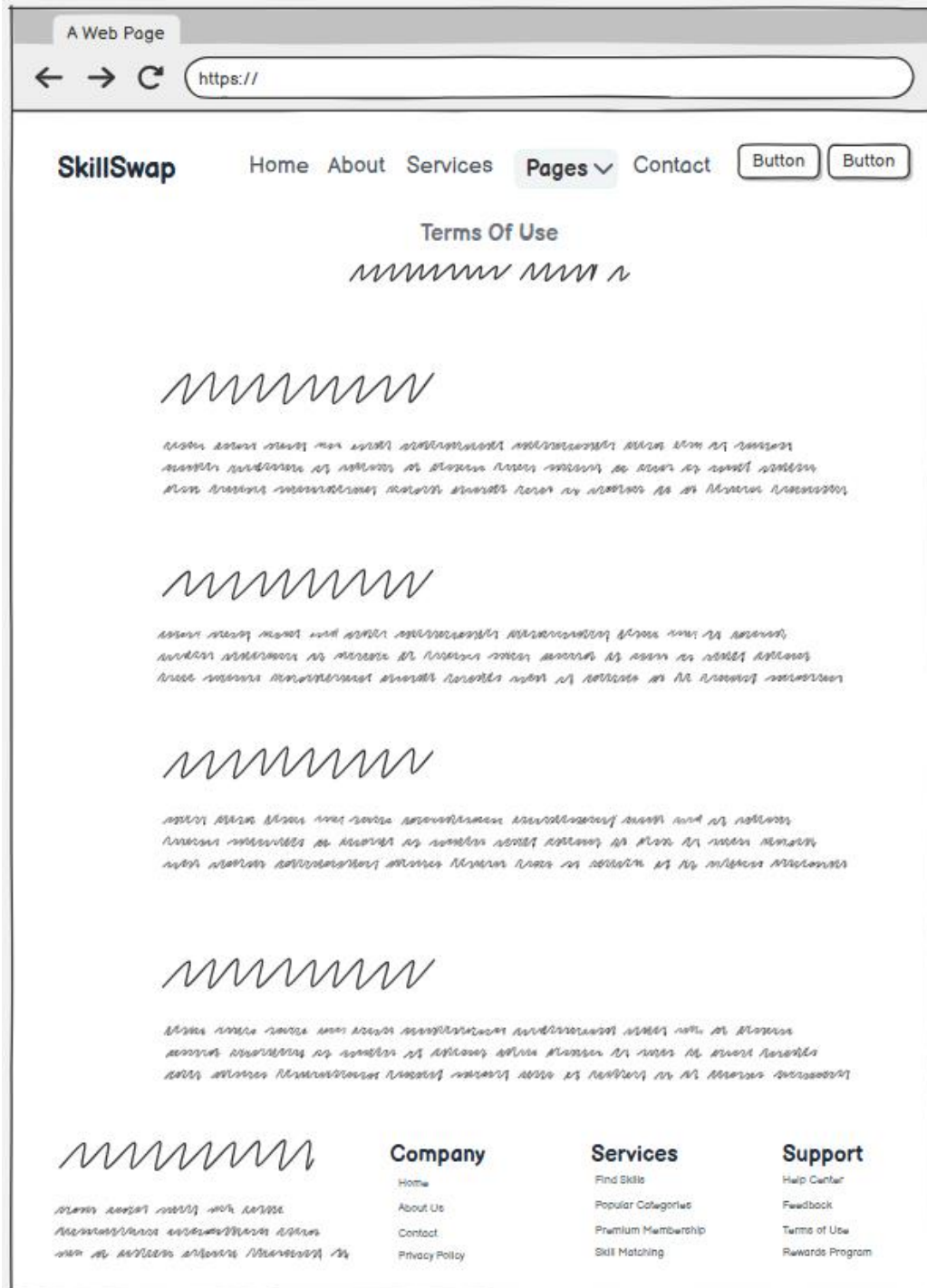


Figure 40: Terms of use page's wireframe



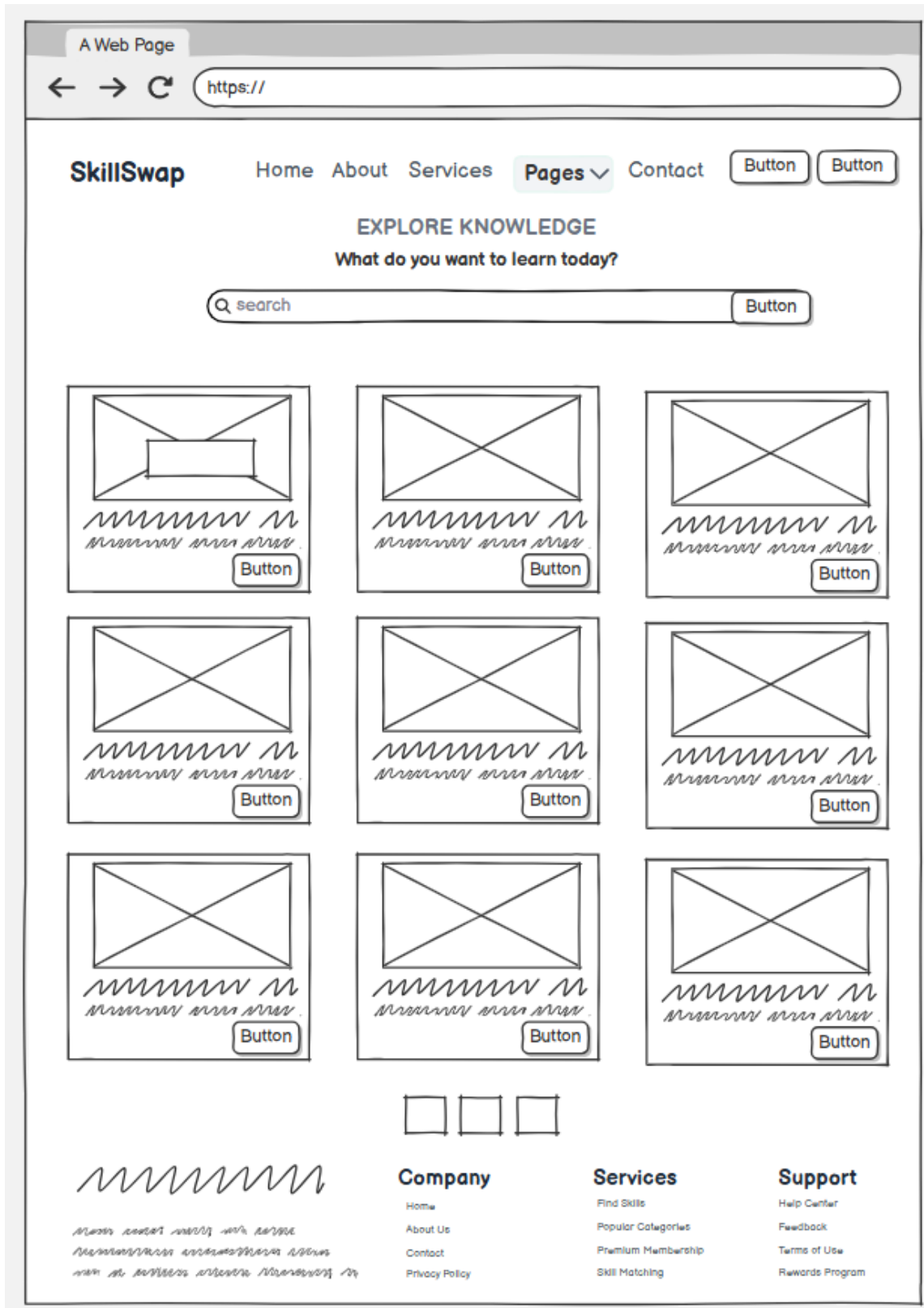


Figure 42: Find-Skill page's wireframe

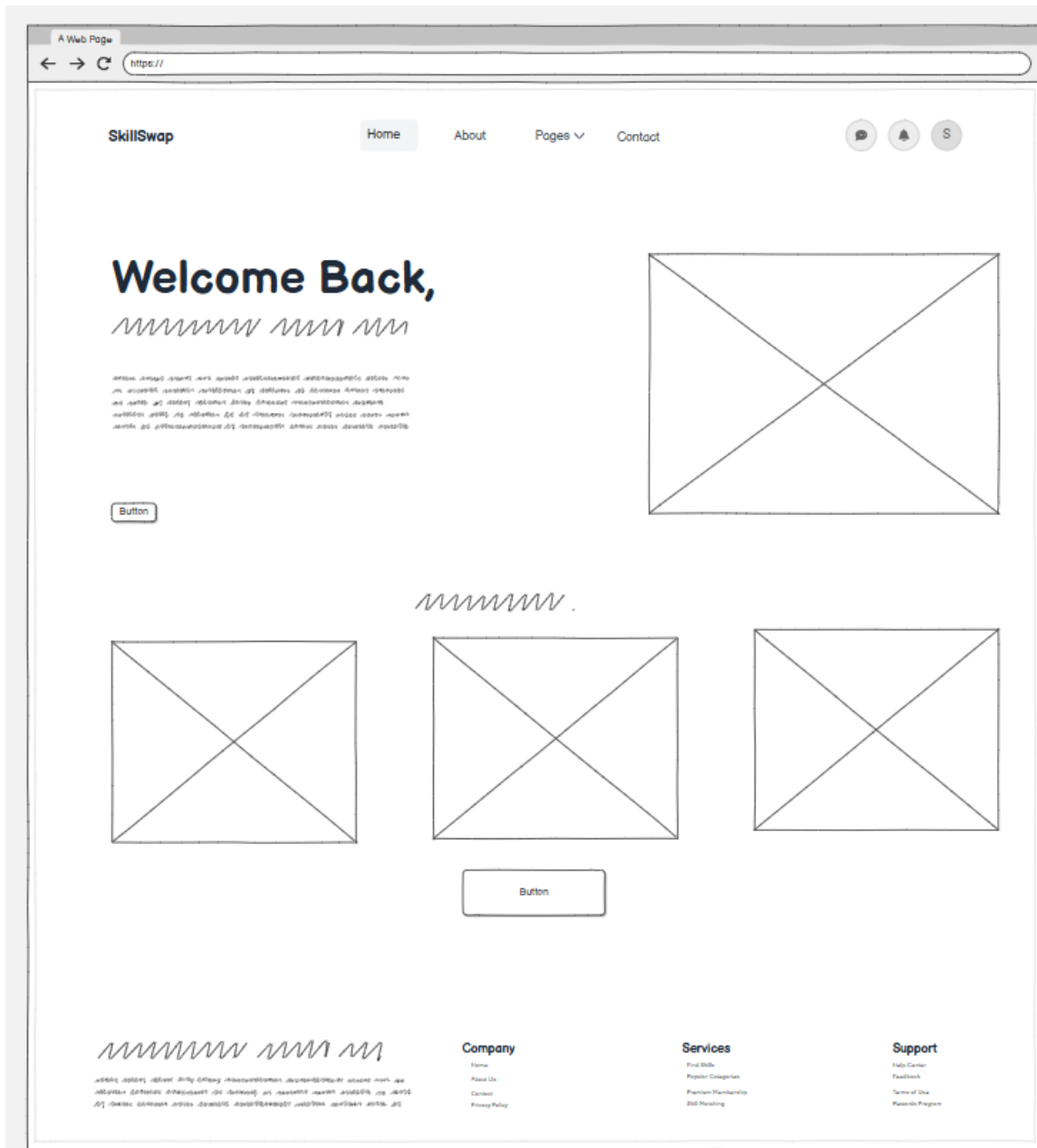


Figure 43: Wireframe of Home page after user login

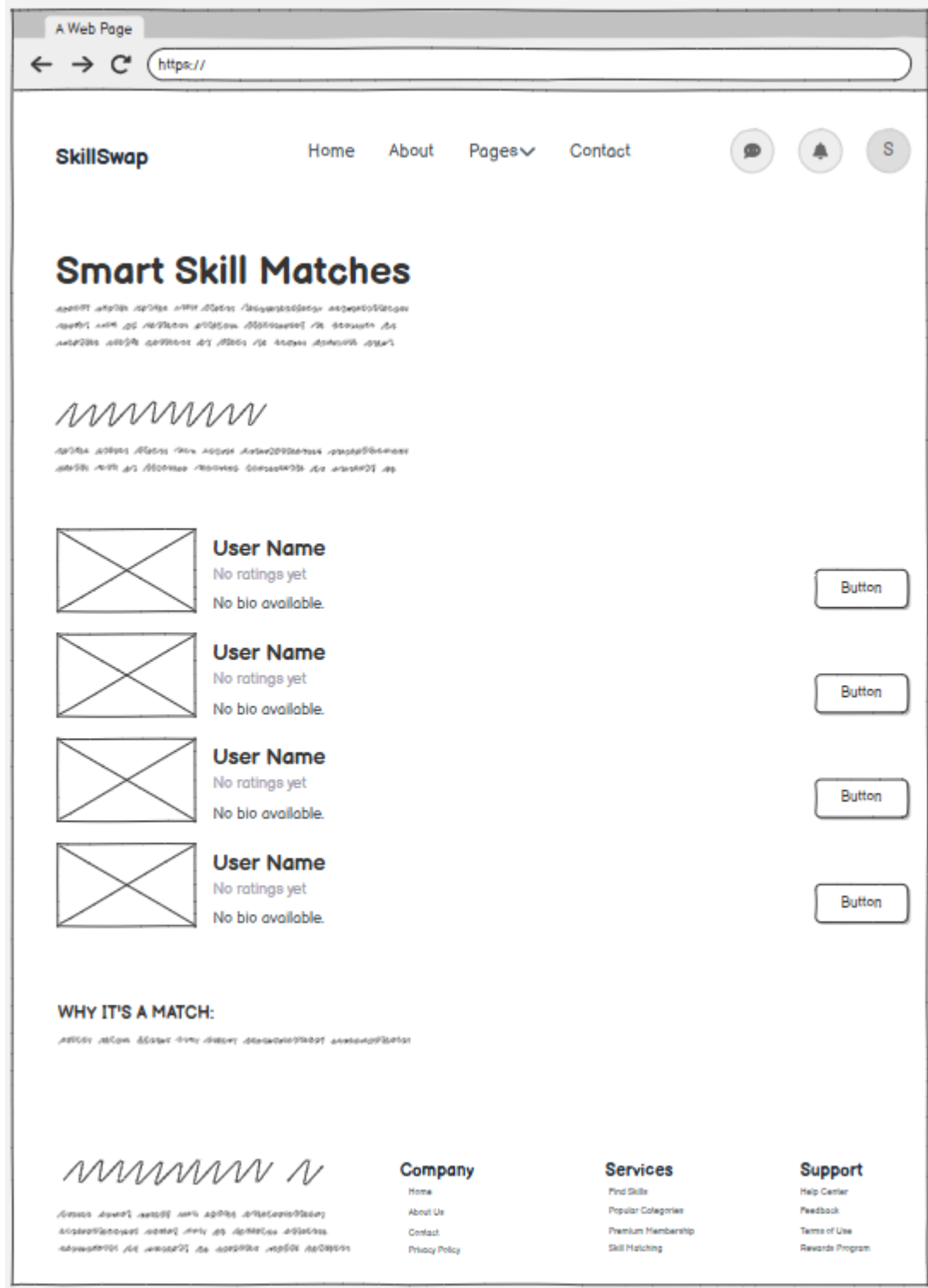


Figure 44: Skill Matching page's wireframe

6 Implementation

6.1 Increment 2 - Database

The name of my database is skillxchange

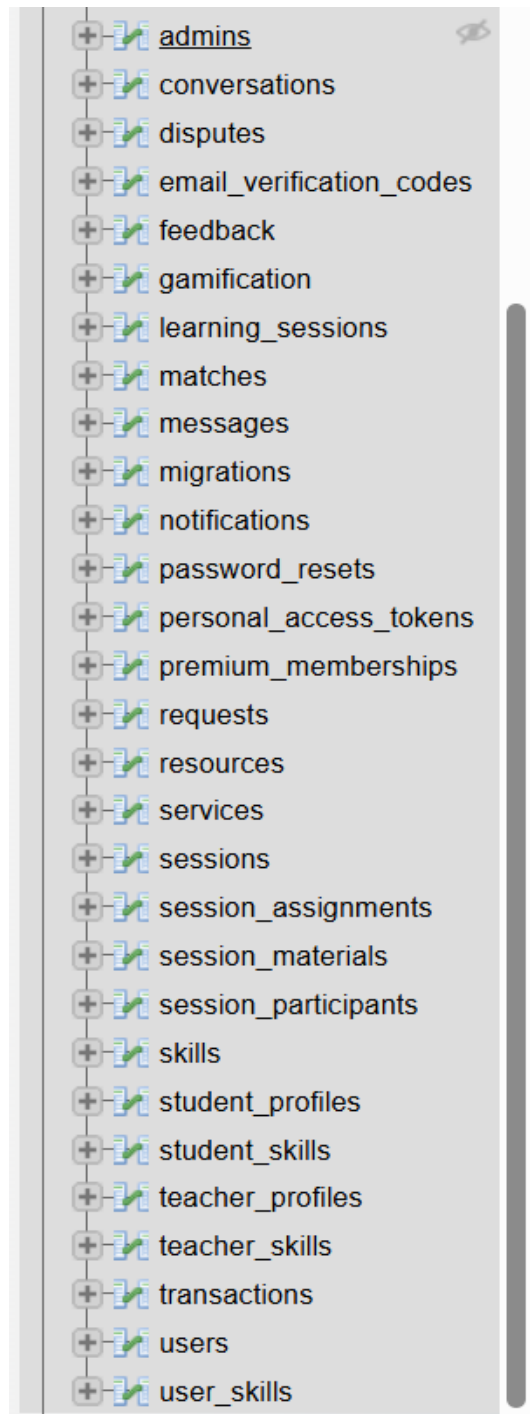


Figure 45: Database table

- Admin table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐ 2	name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 3	email 📧	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 4	password	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 5	remember_token	varchar(100)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 6	created_at	timestamp			Yes	NULL			Change Drop More
☐ 7	updated_at	timestamp			Yes	NULL			Change Drop More

Figure 46: Admin table

- Conversations table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐ 2	user_one_id 📧	bigint(20)		UNSIGNED	No	None			Change Drop More
☐ 3	user_two_id 📧	bigint(20)		UNSIGNED	No	None			Change Drop More
☐ 4	created_at	timestamp			Yes	NULL			Change Drop More
☐ 5	updated_at	timestamp			Yes	NULL			Change Drop More

Figure 47: Conversation table

- Disputes table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐ 2	session_request_id 📧	bigint(20)		UNSIGNED	No	None			Change Drop More
☐ 3	user_id 📧	bigint(20)		UNSIGNED	No	None			Change Drop More
☐ 4	reason	text	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 5	status	enum('open','resolved','refunded','resolved_dismissed')	utf8mb4_unicode_ci		No	open			Change Drop More
☐ 6	admin_notes	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 7	created_at	timestamp			Yes	NULL			Change Drop More
☐ 8	updated_at	timestamp			Yes	NULL			Change Drop More

Figure 48: Disputes table

- Email_verification_codes table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 code	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 used	tinyint(1)			No	0			Change Drop More
☐	5 expires_at	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()	Change Drop More
☐	6 created_at	timestamp			Yes	NULL			Change Drop More
☐	7 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 49: Email verification table

- Feedback table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 target_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 target_type	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 author_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	5 rating	tinyint(4)			No	None			Change Drop More
☐	6 comment	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	7 created_at	timestamp			Yes	NULL			Change Drop More
☐	8 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 50: Feedback table

- Gamification table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 points	int(11)			No	0			Change Drop More
☐	4 badges	longtext	utf8mb4_bin		Yes	NULL			Change Drop More
☐	5 level	int(11)			No	1			Change Drop More
☐	6 created_at	timestamp			Yes	NULL			Change Drop More
☐	7 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 51: Gamification table

- Learning_sessions table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 teacher_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 student_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	4 skill	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	5 start_time	datetime			No	None			Change Drop More
☐	6 end_time	datetime			No	None			Change Drop More
☐	7 meeting_link	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	8 status	enum('scheduled','live','completed','cancelled')	utf8mb4_unicode_ci		No	scheduled			Change Drop More
☐	9 created_at	timestamp			Yes	NULL			Change Drop More
☐	10 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 52: Learning session table

- Matches table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 seeker_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 provider_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	4 user_skill_id	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	5 status	enum('pending','accepted','rejected','completed')	utf8mb4_unicode_ci		No	pending			Change Drop More
☐	6 matched_at	timestamp			Yes	NULL			Change Drop More
☐	7 created_at	timestamp			Yes	NULL			Change Drop More
☐	8 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 53: Match table

- Messages table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 conversation_id	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	3 request_id	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	4 sender_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	5 body	text	utf8mb4_unicode_ci		No	None			Change Drop More
☐	6 created_at	timestamp			Yes	NULL			Change Drop More
☐	7 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 54: Message table

- Migrations table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	int(10)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 migration	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 batch	int(11)			No	None			Change Drop More

Figure 55: Migration table

- Notifications table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	char(36)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	2 type	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 notifiable_type	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 notifiable_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	5 data	text	utf8mb4_unicode_ci		No	None			Change Drop More
☐	6 read_at	timestamp			Yes	NULL			Change Drop More
☐	7 created_at	timestamp			Yes	NULL			Change Drop More
☐	8 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 56: Notification table

- Password_resets table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 email	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	2 token	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 created_at	timestamp			Yes	NULL			Change Drop More

Figure 57: Password reset table

- Personal_access_token table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 tokenable_type	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 tokenable_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	4 name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	5 token	varchar(64)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	6 abilities	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	7 last_used_at	timestamp			Yes	NULL			Change Drop More
☐	8 expires_at	timestamp			Yes	NULL			Change Drop More
☐	9 created_at	timestamp			Yes	NULL			Change Drop More
☐	10 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 58: Personal access token table

- Premium_membership table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 plan	varchar(255)	utf8mb4_unicode_ci		No	monthly			Change Drop More
☐	4 started_at	timestamp			Yes	NULL			Change Drop More
☐	5 expires_at	timestamp			Yes	NULL			Change Drop More
☐	6 status	varchar(255)	utf8mb4_unicode_ci		No	active			Change Drop More
☐	7 created_at	timestamp			Yes	NULL			Change Drop More
☐	8 updated_at	timestamp			Yes	NULL			Change Drop More
☐	9 price	decimal(8,2)			No	0.00			Change Drop More
☐	10 transaction_id	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	11 payment_method	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More

Figure 59: Premium membership table

- Request table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 requester_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 responder_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	4 user_skill_id	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	5 message	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	6 status	enum('open','in_progress','accepted','scheduled')	utf8mb4_unicode_ci		Yes	open			Change Drop More
☐	7 scheduled_at	datetime			Yes	NULL			Change Drop More
☐	8 created_at	timestamp			Yes	NULL			Change Drop More
☐	9 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 60: Request table

- Resources table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	3 title	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	4 description	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	5 category	varchar(100)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	6 file_path	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	7 filename	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	8 mime	varchar(100)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	9 size	int(11)			Yes	NULL			Change Drop More
☐	10 type	varchar(50)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	11 created_at	timestamp			Yes	NULL			Change Drop More
☐	12 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 61: Resources table

- Services table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 title	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 description	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	4 category	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	5 role	varchar(255)	utf8mb4_unicode_ci		No	both			Change Drop More
☐	6 image_path	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	7 teacher_route	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	8 student_route	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	9 created_at	timestamp			Yes	NULL			Change Drop More
☐	10 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 62: Service table

- Sessions table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐ 2	organiser_id 🔑	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐ 3	participant_id 🔑	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐ 4	skill_id 🔑	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐ 5	request_id 🔑	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐ 6	scheduled_at	datetime			Yes	NULL			Change Drop More
☐ 7	status	varchar(255)	utf8mb4_unicode_ci		No	pending			Change Drop More
☐ 8	reschedule_requested	tinyint(1)			No	0			Change Drop More
☐ 9	reschedule_remarks	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 10	created_at	timestamp			Yes	NULL			Change Drop More
☐ 11	updated_at	timestamp			Yes	NULL			Change Drop More
☐ 12	meeting_link	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 13	is_live	tinyint(1)			No	0			Change Drop More
☐ 14	start_time	datetime			Yes	NULL			Change Drop More
☐ 15	end_time	datetime			Yes	NULL			Change Drop More

Figure 63: Session table

- Sessions_assignments table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐ 2	session_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐ 3	student_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐ 4	title	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 5	details	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 6	file_path	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 7	file_url	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 8	created_at	timestamp			Yes	NULL			Change Drop More
☐ 9	updated_at	timestamp			Yes	NULL			Change Drop More

Figure 64: Session assignments table

- Session_material table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐ 2	session_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐ 3	title	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 4	file_path	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 5	file_url	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 6	created_at	timestamp			Yes	NULL			Change Drop More
☐ 7	updated_at	timestamp			Yes	NULL			Change Drop More

Figure 65: Session materials table

- Session_participants table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 session_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 user_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	4 created_at	timestamp			Yes	NULL			Change Drop More
☐	5 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 66: Session participants table

- skills table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 title	varchar(150)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 slug	varchar(180)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 description	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	5 image	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	6 category	varchar(120)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	7 created_at	timestamp			Yes	NULL			Change Drop More
☐	8 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 67: Skill table

- student_profile table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 bio	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	4 level	enum('beginner', 'intermediate', 'advanced')	utf8mb4_unicode_ci		No	beginner			Change Drop More
☐	5 created_at	timestamp			Yes	NULL			Change Drop More
☐	6 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 68: Student profile table

- Students_skill table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 student_profile_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 skill_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	4 created_at	timestamp			Yes	NULL			Change Drop More
☐	5 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 69: Student skill table

- Teacher_profile table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 bio	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	4 experience_years	int(11)			No	0			Change Drop More
☐	5 teaching_style	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	6 rating	decimal(3,2)			No	0.00			Change Drop More
☐	7 created_at	timestamp			Yes	NULL			Change Drop More
☐	8 updated_at	timestamp			Yes	NULL			Change Drop More
☐	9 cv_path	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	10 certificate_path	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	11 citizenship_path	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	12 khalti_id	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More

Figure 70: Teacher profile table

- Teacher_skill table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 skill_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	4 type	varchar(32)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	5 level	enum('beginner', 'intermediate', 'advanced')	utf8mb4_unicode_ci		No	beginner			Change Drop More
☐	6 price	decimal(10,2)			Yes	NULL			Change Drop More
☐	7 location	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	8 created_at	timestamp			Yes	NULL			Change Drop More
☐	9 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 71: Teacher skill table

- Transaction table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 teacher_id	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	4 amount	decimal(10,2)			No	None			Change Drop More
☐	5 admin_share	decimal(10,2)			No	None			Change Drop More
☐	6 teacher_share	decimal(10,2)			No	None			Change Drop More
☐	7 type	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	8 status	varchar(255)	utf8mb4_unicode_ci		No	completed			Change Drop More
☐	9 khalti_pidx	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	10 transaction_id	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	11 created_at	timestamp			Yes	NULL			Change Drop More
☐	12 updated_at	timestamp			Yes	NULL			Change Drop More

Figure 72: Transaction table

- User table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 name	varchar(150)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 email	varchar(200)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 phone	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	5 email_verified_at	timestamp			Yes	NULL			Change Drop More
☐	6 password	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	7 provider	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	8 provider_id	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	9 provider_token	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	10 provider_refresh_token	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	11 oauth_synced_at	timestamp			Yes	NULL			Change Drop More
☐	12 role	varchar(50)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	13 bio	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	14 avatar	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	15 remember_token	varchar(100)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	16 is_active	tinyint(1)			No	1			Change Drop More
☐	17 created_at	timestamp			Yes	NULL			Change Drop More
☐	18 updated_at	timestamp			Yes	NULL			Change Drop More
☐	19 is_teacher_approved	tinyint(1)			No	0			Change Drop More

Figure 73: User table

6.2 Implemented page

6.2.1 Increment 2 Implemented page

a. Guest User's side

This is the 1st page non logged in user is allowed to view

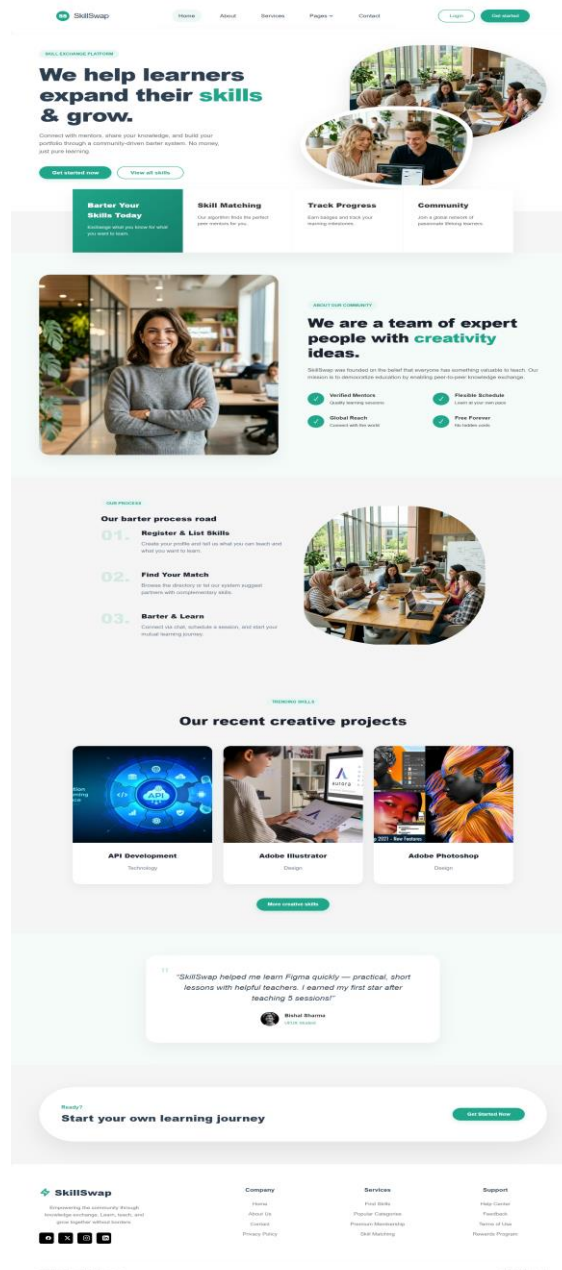


Figure 74: Landing page

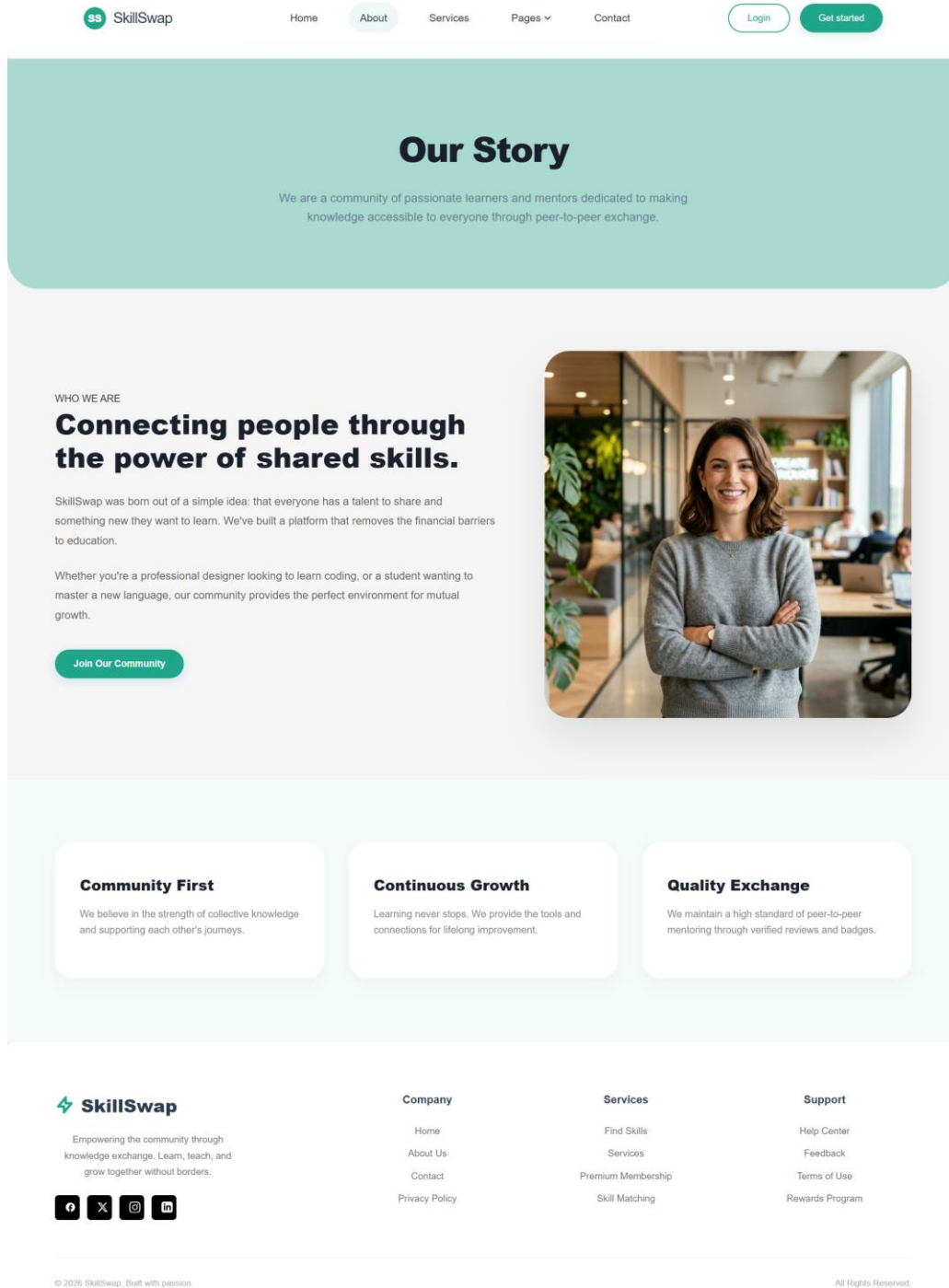


Figure 75: Our story page

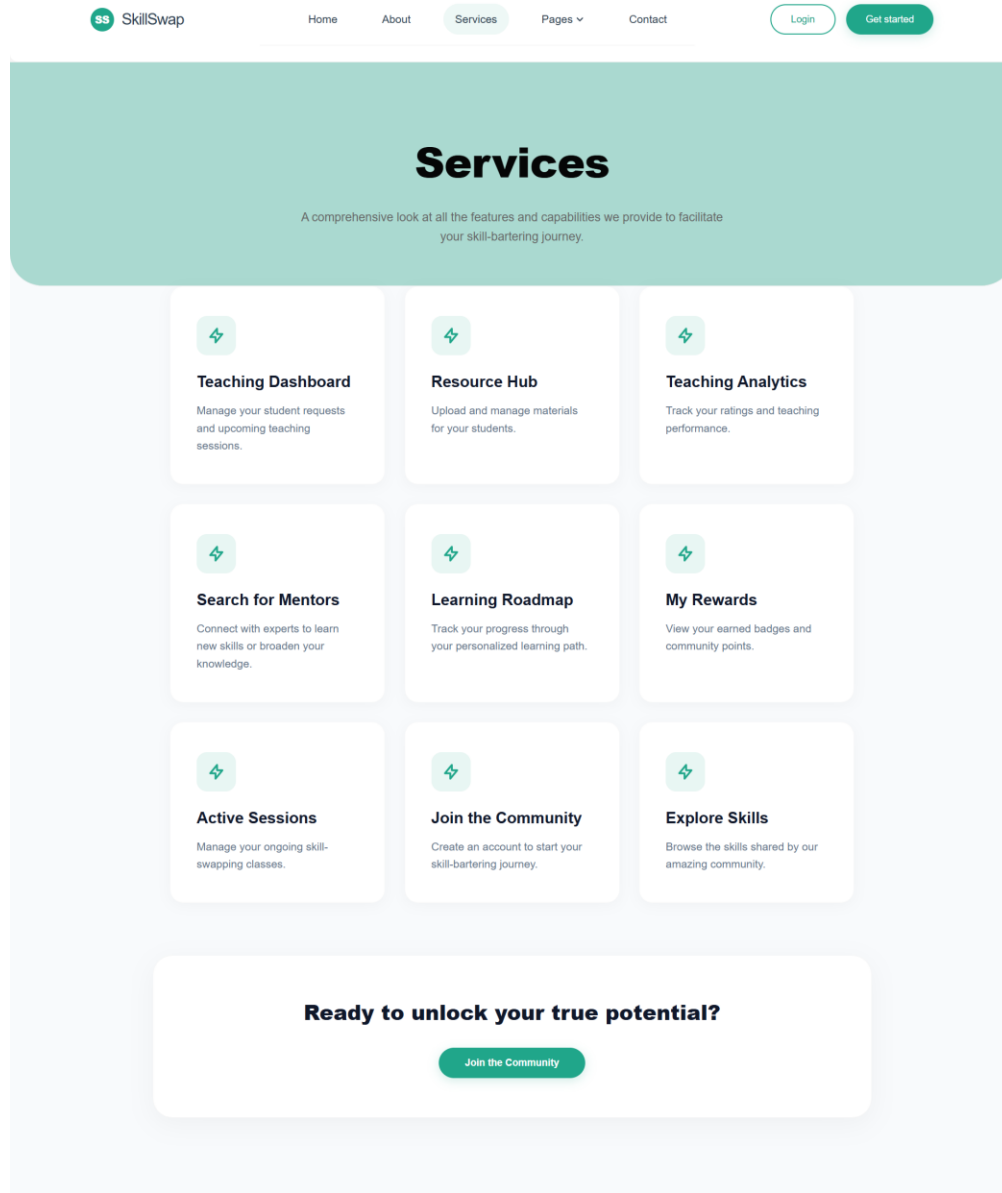


Figure 76: Service page

EXPLORE KNOWLEDGE

What do you want to learn today?

Discover expert mentors and fellow students ready to share their expertise.

Search skills (e.g. Photoshop, Marketing...) All Categories **SEARCH SKILL**

Adobe Illustrator
Vector graphics and illustrations.
24 learners [Details](#)

Adobe Photoshop
Photo editing and manipulation.
12 learners [Details](#)

Affiliate Marketing
Performance-based marketing.
23 learners [Details](#)

AI Fundamentals
Understanding AI systems.
47 learners [Details](#)

API Development
Build and consume REST APIs.
19 learners [Details](#)

Brand Identity Design
Logos and brand systems.
37 learners [Details](#)

Business Analytics
Data-driven decision making.
45 learners [Details](#)

Business Communication
Professional communication.
11 learners [Details](#)

Canva Design
Quick designs using Canva.
48 learners [Details](#)

Content Marketing
Content creation strategies.
22 learners [Details](#)

Creative Writing
Storytelling and writing skills.
39 learners [Details](#)

Critical Thinking
Problem solving techniques.
14 learners [Details](#)

1 2 3 4

SkillSwap

Empowering the community through knowledge exchange. Learn, teach, and grow together without borders.



Company

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Find Skills
Services
Premium Membership
Skill Matching

Support

Help Center
Feedback
Terms of Use
Rewards Program

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Figure 77: Find-skill page

 SkillSwap

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[Login](#) [Get started](#)

GET IN TOUCH

We'd love to hear from you.

Have questions about our platform or want to partner with us? Reach out anytime.

Send us a Message

Your Name

Enter your full name

Email Address

email@example.com

Subject

How can we help?

Message

Your message here...

Send Message

Email Us

support@skillswap.com
info@skillswap.com

Call Us

+977 9816681421
+977 061-554433

Our Office

Pokhara-17, Jyamirkuna
Gandaki, Nepal

Support Hours

Mon - Fri: 9:00 AM - 6:00 PM
Weekend: 10:00 AM - 2:00 PM

FAQS

Common Questions

Is SkillSwap really free?
Yes, our core mission is to enable peer-to-peer barter, meaning you teach to learn. No money involved!

How do I trust a mentor?
We use a rating and review system. Each mentor has a reputation score based on their past sessions.

Can I learn multiple skills?
Absolutely! You can enroll in as many skills as you want as long as you can manage your schedule.

How are sessions conducted?
Sessions can be online (via video call) or in-person at campus, depending on your match's preference.

 **SkillSwap**
Empowering the community through
knowledge exchange. Learn, teach, and
grow together without borders.

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Figure 78: Contact us page

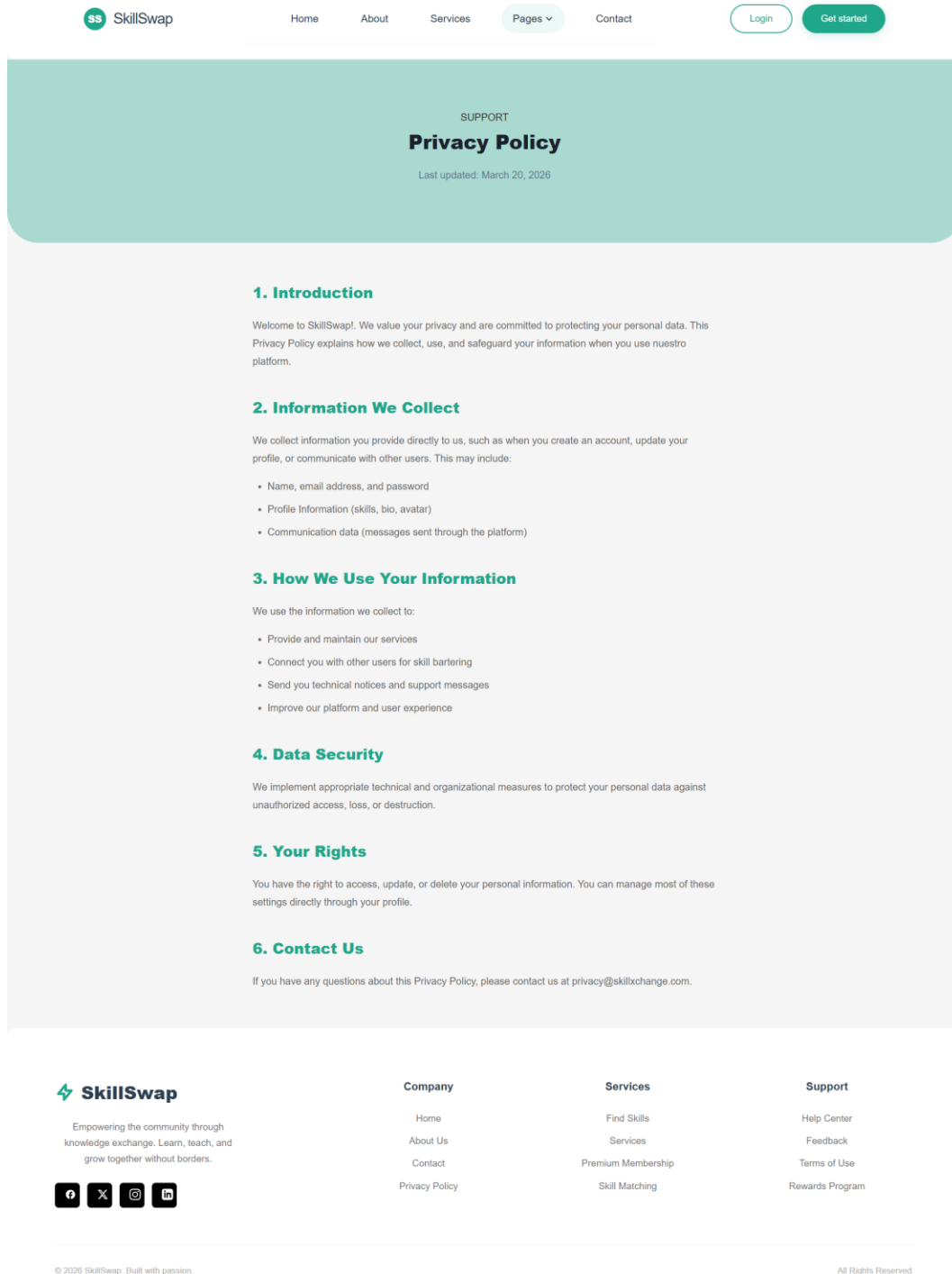


Figure 79: Privacy policy page

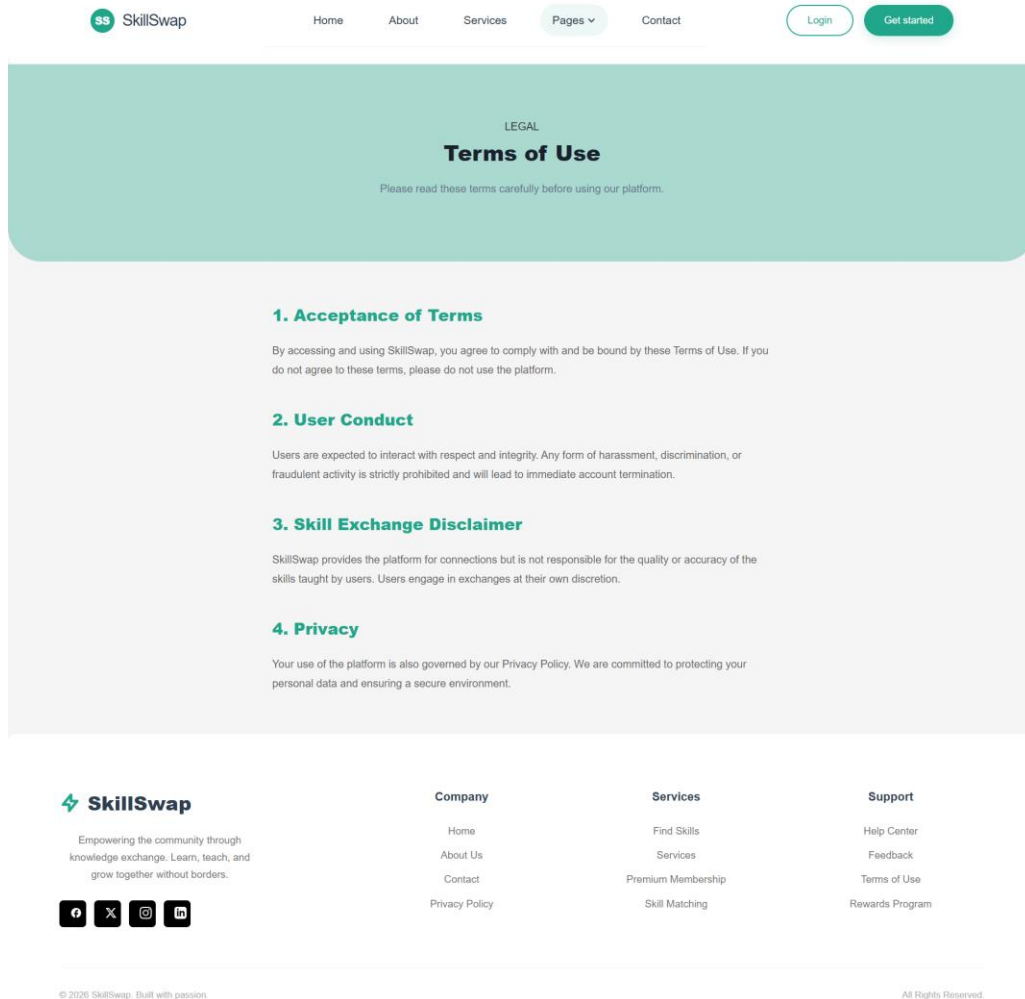


Figure 80: Terms of use page

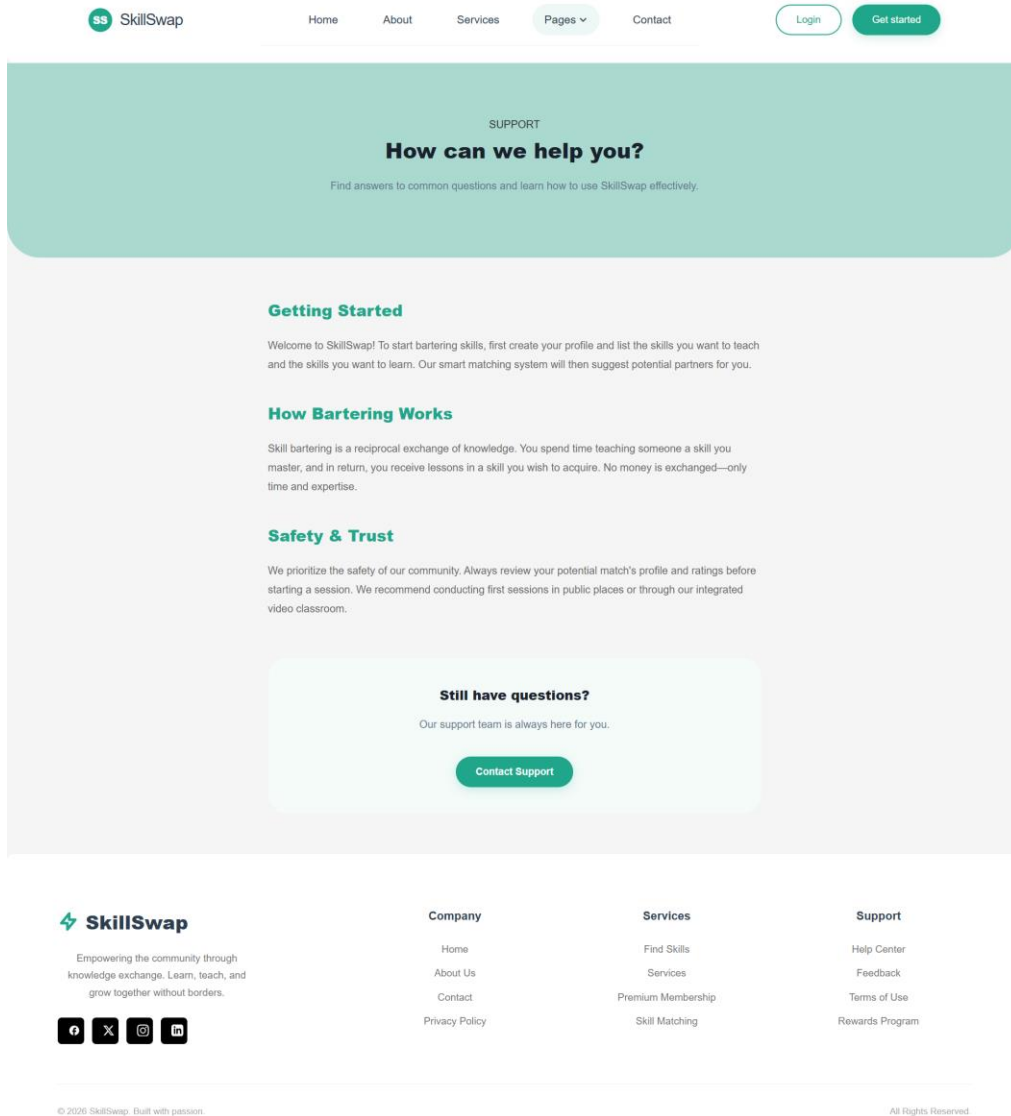
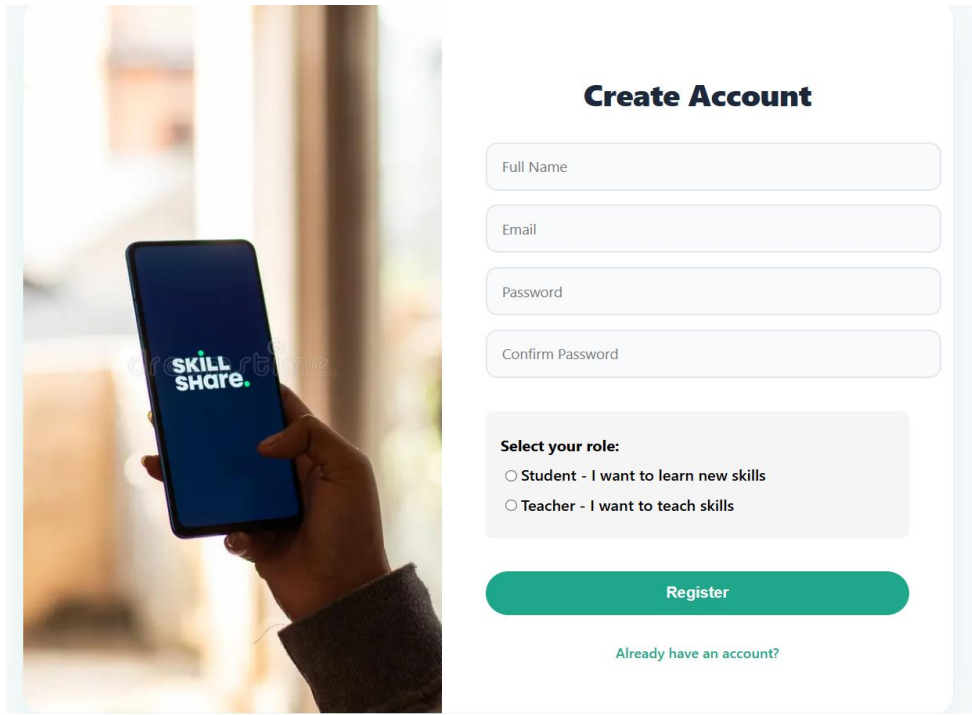


Figure 81: Help center page

User Register Page:

The new user can register from this page.

The image shows a user registration page for 'Skill Share'. On the left, a hand holds a smartphone displaying the 'SKILL SHARE.' logo. On the right, the 'Create Account' form includes input fields for 'Full Name', 'Email', 'Password', and 'Confirm Password'. Below these is a 'Select your role:' section with two radio button options: 'Student - I want to learn new skills' and 'Teacher - I want to teach skills'. A green 'Register' button is at the bottom, with a link 'Already have an account?' below it.

Create Account

Full Name

Email

Password

Confirm Password

Select your role:

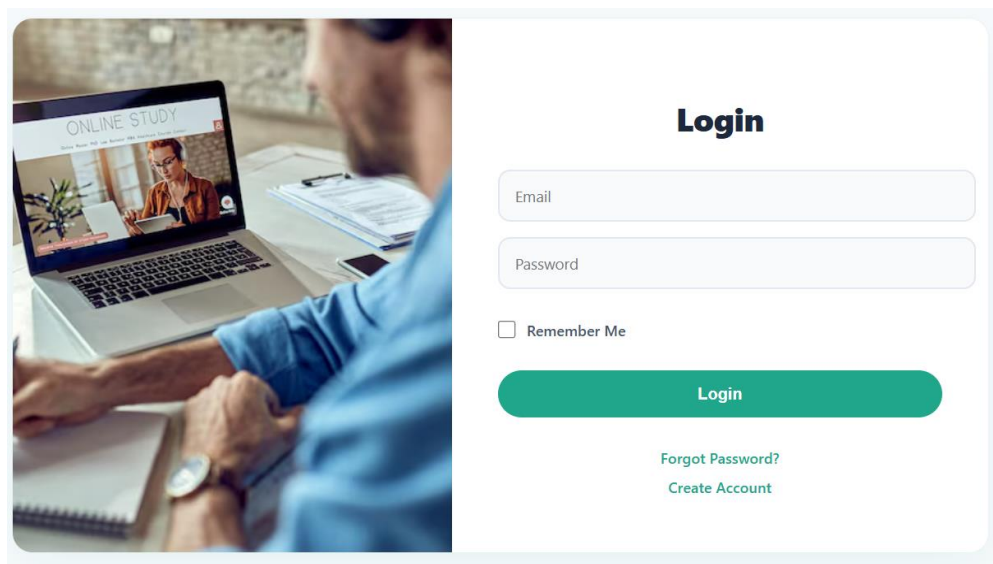
☐ Student - I want to learn new skills

☐ Teacher - I want to teach skills

Register

[Already have an account?](#)

Figure 82: Register page

User Login Page:The image shows a user login page for 'Skill Share'. On the left, a person is seen from the side, working on a laptop that displays an 'ONLINE STUDY' video. On the right, the 'Login' form includes input fields for 'Email' and 'Password', and a checkbox for 'Remember Me'. A green 'Login' button is at the bottom, with links 'Forgot Password?' and 'Create Account' below it.

Login

Email

Password

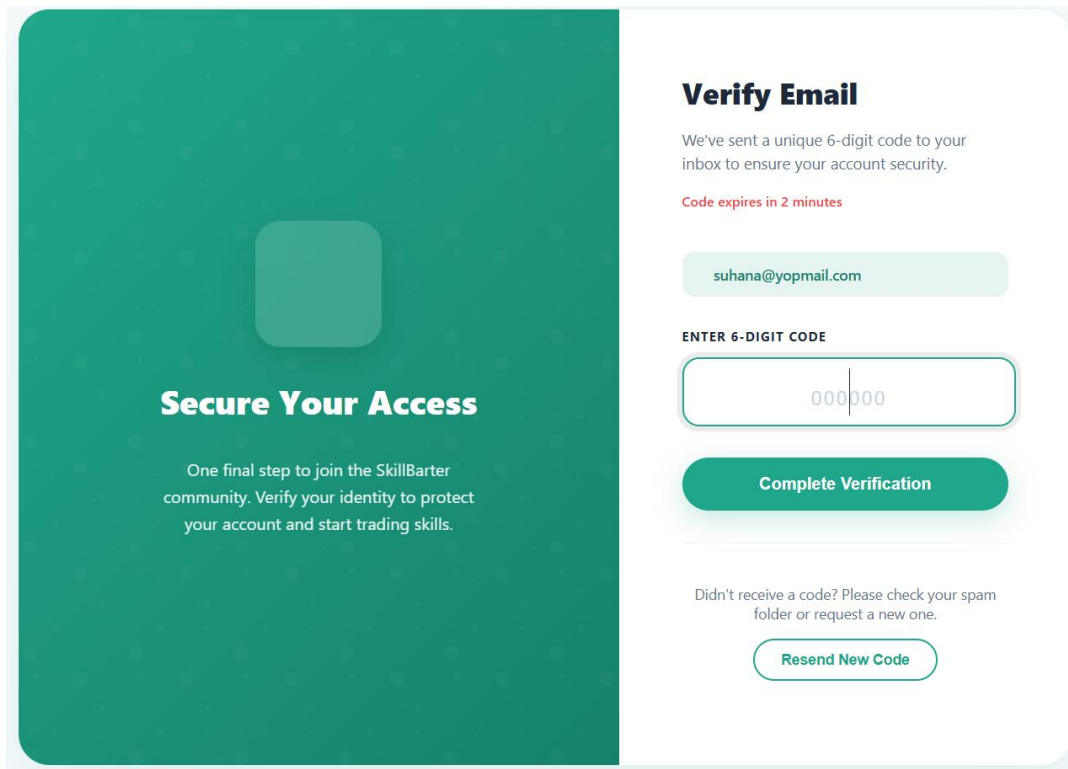
☐ Remember Me

Login

[Forgot Password?](#)

[Create Account](#)

Figure 83: Login Page

Email verification:

The image shows a web interface for email verification. It is split into two main sections. The left section has a dark teal background and contains a large, faint, rounded square icon. Below this icon, the text 'Secure Your Access' is displayed in a bold, white font. Underneath, a smaller line of white text reads: 'One final step to join the SkillBarter community. Verify your identity to protect your account and start trading skills.' The right section has a white background. At the top, it says 'Verify Email' in bold. Below this, a message states: 'We've sent a unique 6-digit code to your inbox to ensure your account security.' A red line of text indicates 'Code expires in 2 minutes'. A light green rounded rectangle displays the email address 'suhana@yopmail.com'. Below this, the text 'ENTER 6-DIGIT CODE' is shown. A white input field with a green border contains the placeholder '000000'. A green button with white text 'Complete Verification' is positioned below the input field. At the bottom of the right section, a message says: 'Didn't receive a code? Please check your spam folder or request a new one.' Below this message is a green button with white text 'Resend New Code'.

Secure Your Access

One final step to join the SkillBarter community. Verify your identity to protect your account and start trading skills.

Verify Email

We've sent a unique 6-digit code to your inbox to ensure your account security.

Code expires in 2 minutes

suhana@yopmail.com

ENTER 6-DIGIT CODE

000000

Complete Verification

Didn't receive a code? Please check your spam folder or request a new one.

Resend New Code

b. Teacher Side User:

This is the home page after user login.

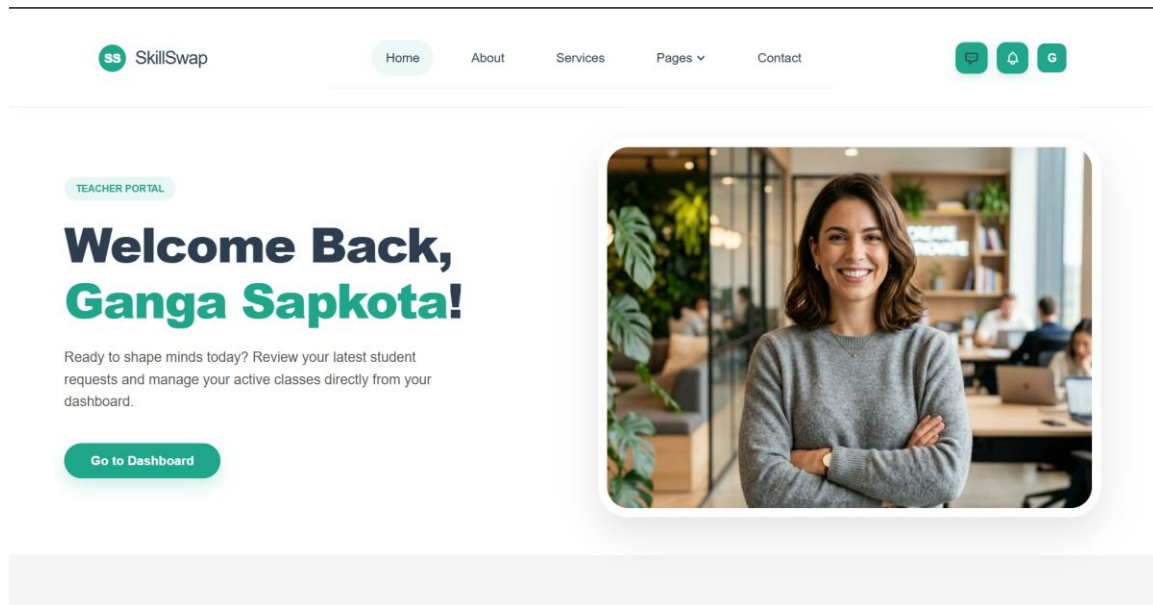


Figure 84: User home page

When a user logs in as a teacher, they need admin approval to access the dashboard.

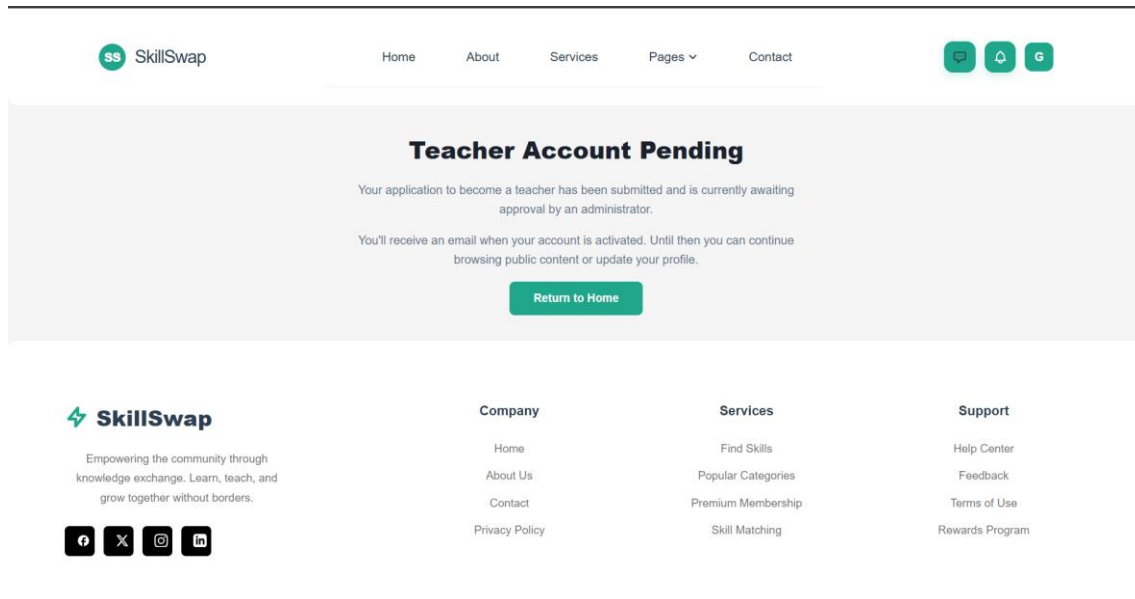


Figure 85: Teacher dashboard before admin approval

The teacher's dashboard after the admin approves access and before adding any skill or session.

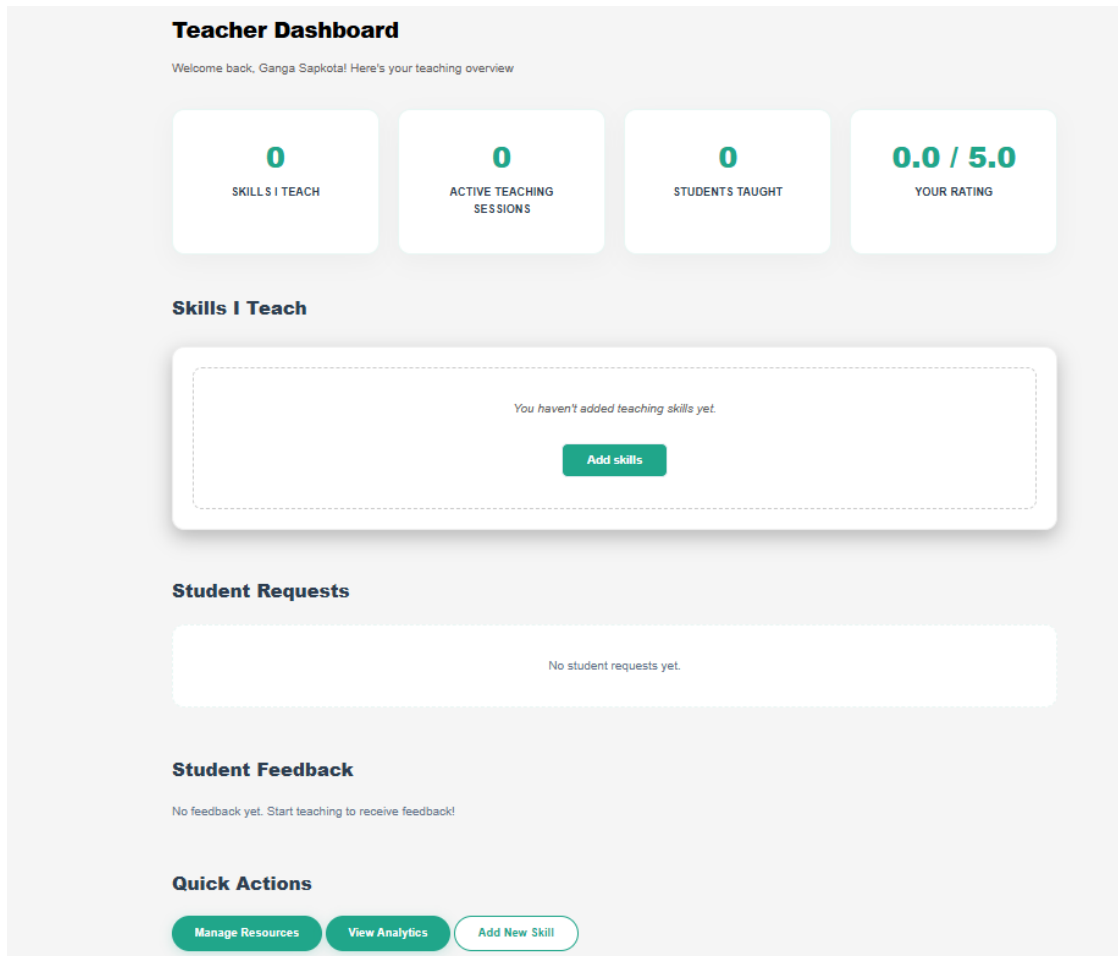


Figure 86: Teacher dashboard before adding skill and session

After adding skills and sessions, this is how the teacher's dashboard looks.

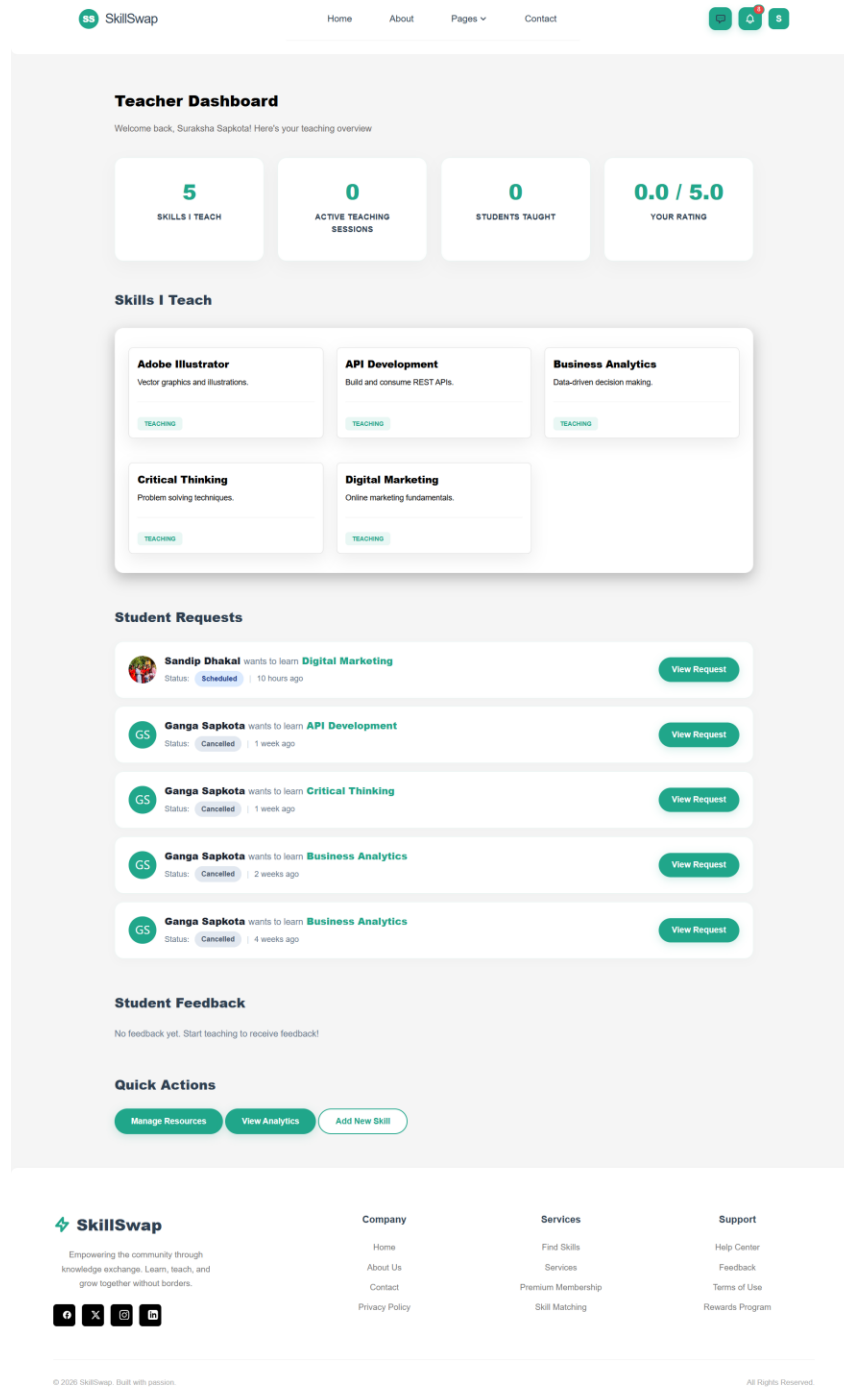


Figure 87: Teacher dashboard after skill or session added

Teachers can add resources for students to help them further.

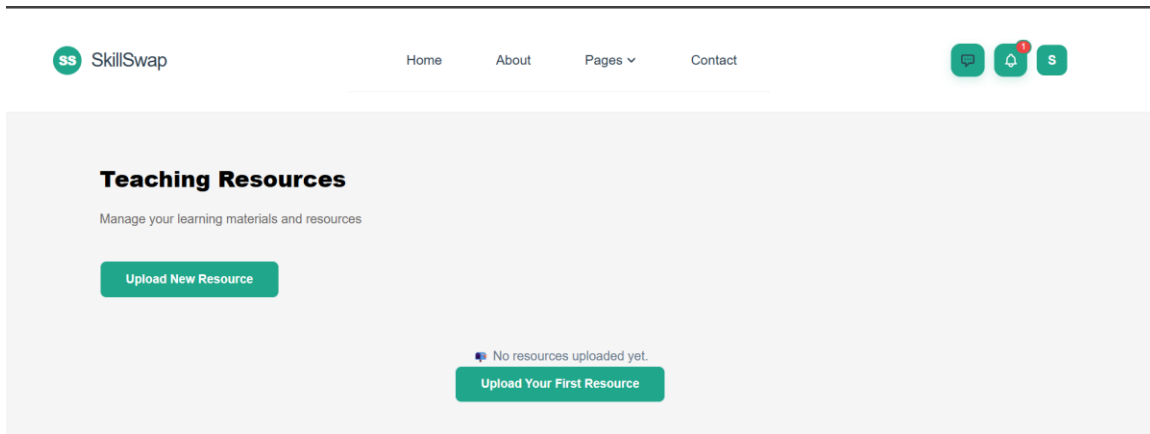


Figure 88: Teacher's resources sharing page

The screenshot displays the 'Upload Teaching Resource' form. The title 'Upload Teaching Resource' is at the top, followed by the instruction 'Share your knowledge materials with students'. The form contains several fields: 'Resource Title *' with the text 'Python Basic Tutorial'; 'Description' with the text 'This is for Testing'; 'Category' with a dropdown menu showing 'Reference'; and 'Upload File * (PDF, DOC, DOCX, TXT max 10MB)' with a file selection area showing 'Choose File' and 'Log1.docx'. At the bottom of the form are two buttons: 'Upload Resource' and 'Cancel'.

Figure 89: Uploading teaching resources

After uploading the resources, that is visible to students when they view the teacher profile.

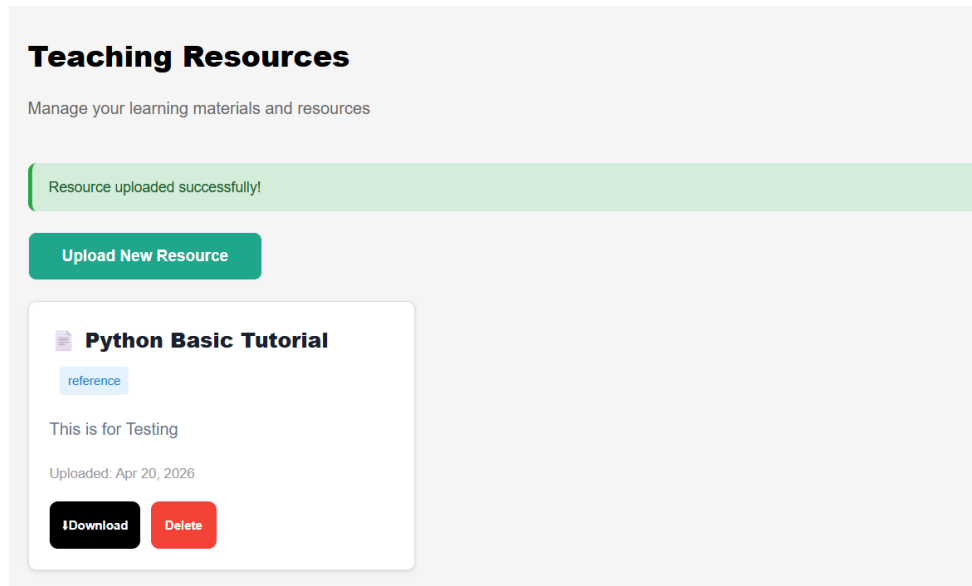


Figure 90: Uploaded resources successfully

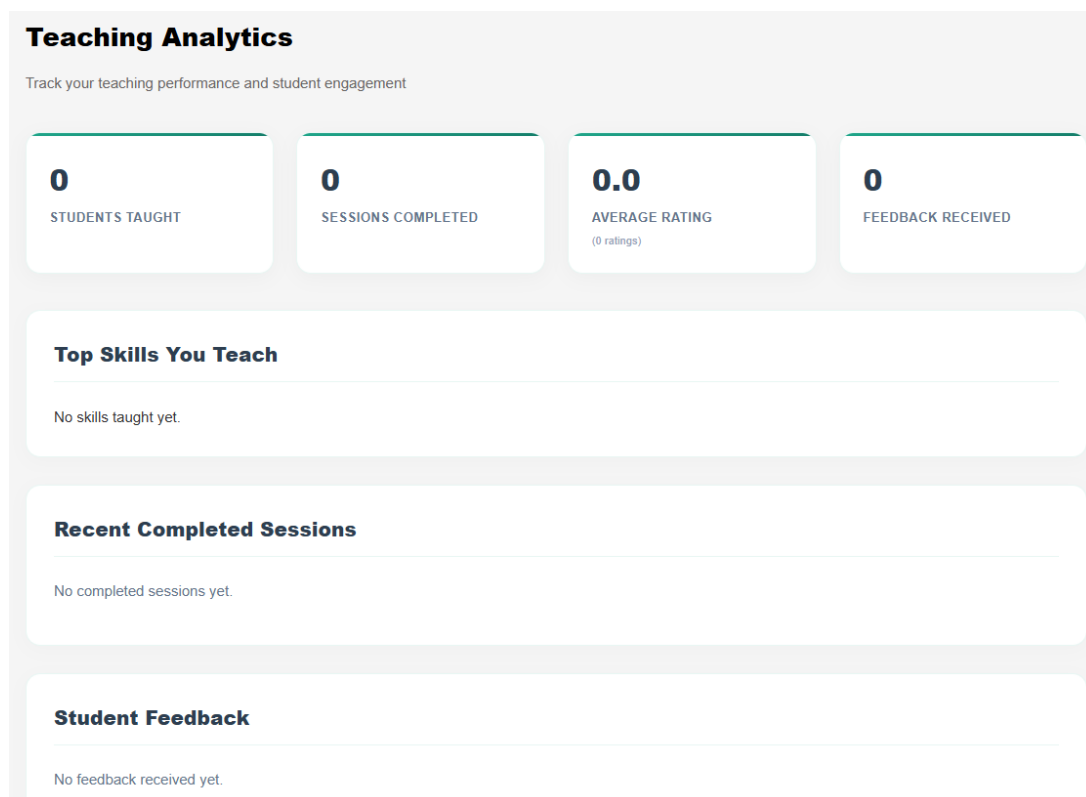


Figure 91: Teaching analytics page

My Skills

Manage the skills you teach

+ Add a Skill

Select Skill *

-- Choose Skill --

Type

Skills I Teach

Level

Beginner

Add Skill

💡 Skills I Teach

Database Design

Intermediate level

Remove

[← Back to Profile](#)

Figure 92: Add new skill page

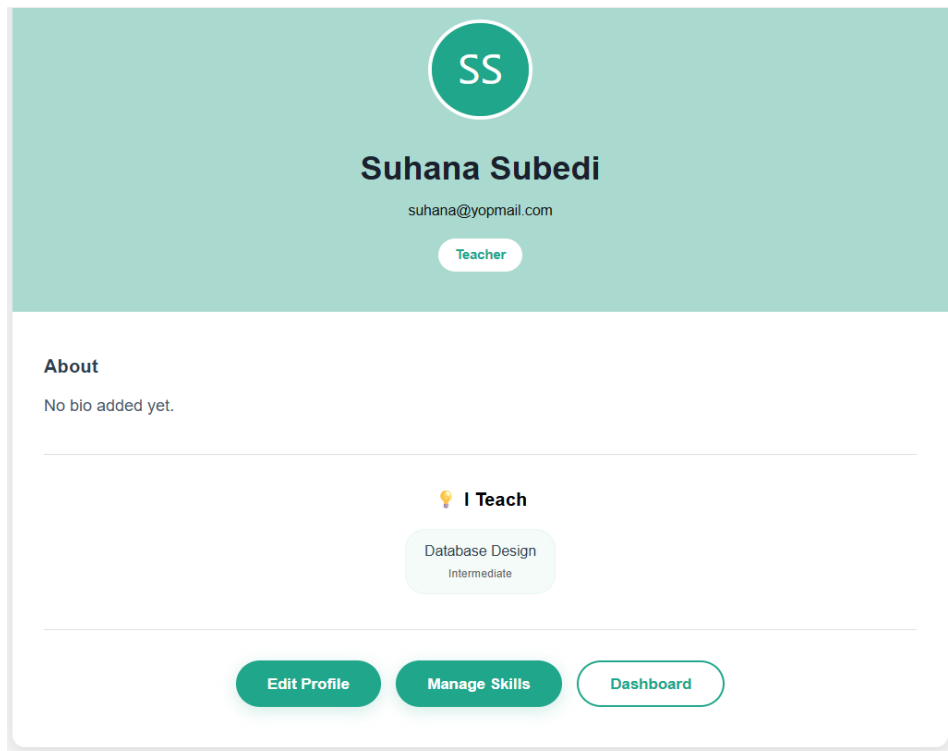


Figure 93: Teacher profile

Teachers get notifications when the student requests them for teaching, and students can find teachers by searching for a mentor and from skill match.

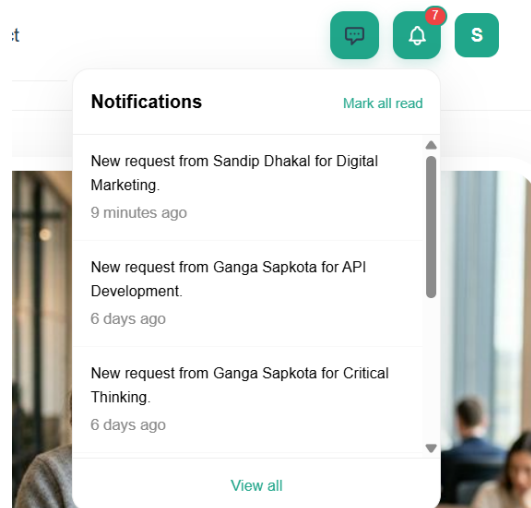


Figure 94: Notification page

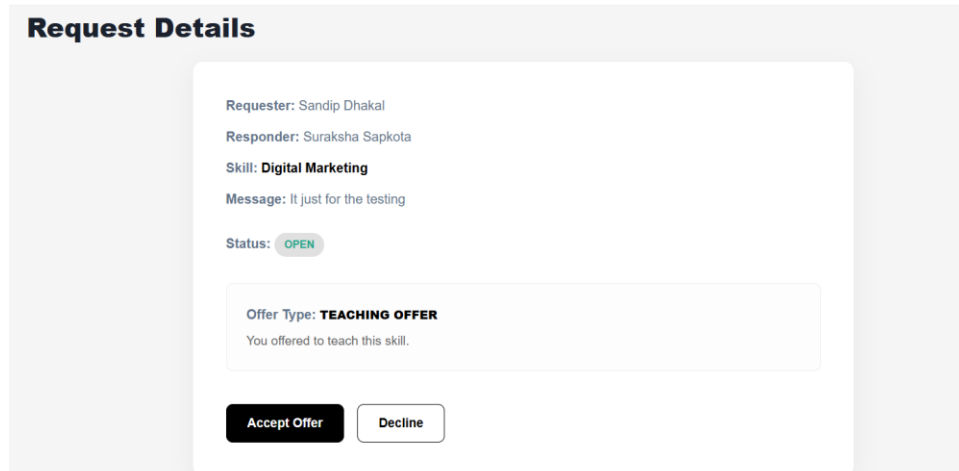


Figure 95: Request details page

Session and notification:

After teacher accept the student's request that redirect to request page where teacher can create a session and after teacher create a session by provide date and meeting link that session created notification goes to student notification bar and also student can request teacher for rescheduling and also teacher get notification of rescheduling and also teacher can update the date and then that's updated date is goes to student notification.

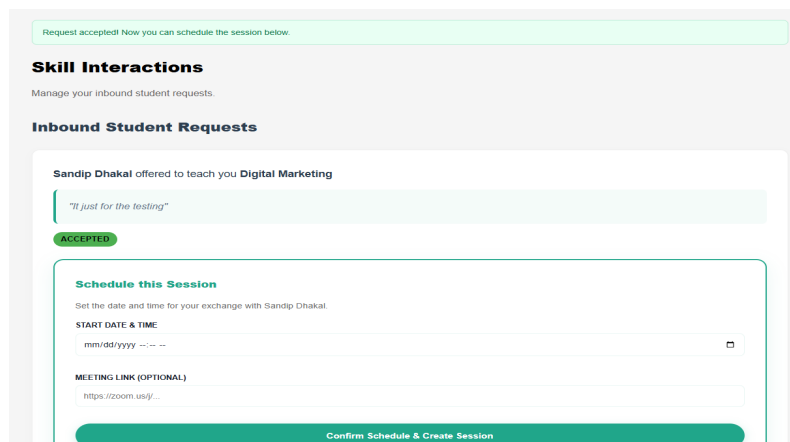


Figure 96: After accepting the request

Sandip Dhakal offered to teach you Digital Marketing

"It just for the testing"

ACCEPTED

Schedule this Session

Set the date and time for your exchange with Sandip Dhakal.

START DATE & TIME

MEETING LINK (OPTIONAL)

Please fill out this field.

Confirm Schedule & Create Session

Figure 97: Creating session

Your Sessions

Active teaching and learning sessions.

Session scheduled successfully and student notified.

MY TEACHING SESSIONS

Digital Marketing Student: Sandip Dhakal SCHEDULED MANAGE CLASS	API Development Student: Ganga Sapkota SCHEDULED MANAGE CLASS	Critical Thinking Student: Ganga Sapkota SCHEDULED MANAGE CLASS
Business Analytics Student: Ganga Sapkota		

Figure 98: After creating a session

After the session, the teacher's classroom looks like this, and also before adding the meeting link

CLASSROOM (TEACHER VIEW)

SCHEDULED

Digital Marketing

TEACHER

Suraksha Sapkota

STUDENTS IN THIS CLASS

No students attached to this session yet.

ADD ANOTHER ACCEPTED STUDENT:

Select Student...

ADD TO CLASS

Session Schedule

CURRENT TIME

April 21, 2028 - 10:26 AM

UPDATE START TIME

04/21/2028 10:25 AM

UPDATE END TIME

mm/dd/yyyy -- --

UPDATE SCHEDULE

Live Class Controls

MEETING LINK (Zoom, Google Meet, etc.)

https://zoom.us/j/...

SAVE

Please save a meeting link first to unlock live class controls.

MATERIALS

No materials shared yet.

UPLOAD NEW MATERIAL

Material Title (e.g. Chapter 1 PDF)

Choose File No file chosen

SECURE UPLOAD

PRACTICE & HOMEWORK

No practice files submitted yet.

MARK COURSE AS COMPLETED

Figure 99: Teacher classroom

Suraksha Sapkota

107

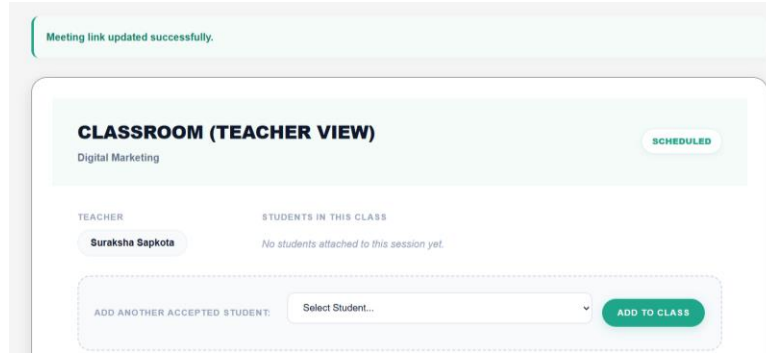


Figure 100: Adding meeting link

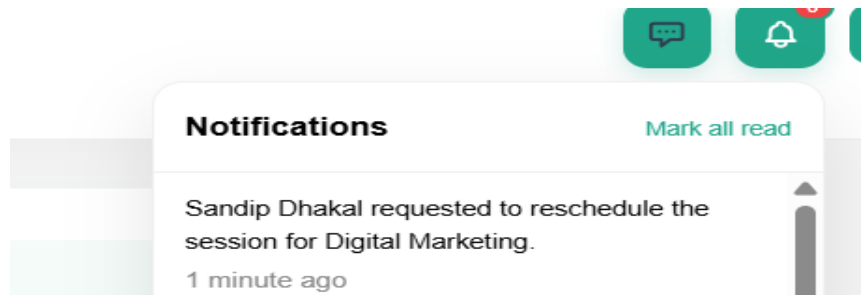


Figure 101: get notification for rescheduling

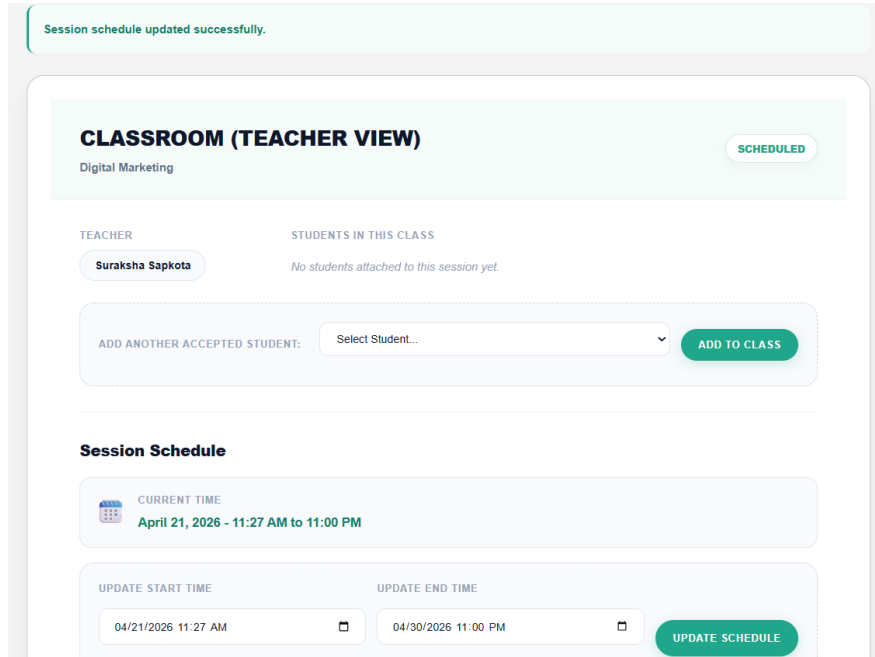


Figure 102: Teacher updated schedule

Live session:

The teacher needs to start the session, and before the teacher starts, the student cannot start the session, and after the teacher starts, the student gets to join the live session box.

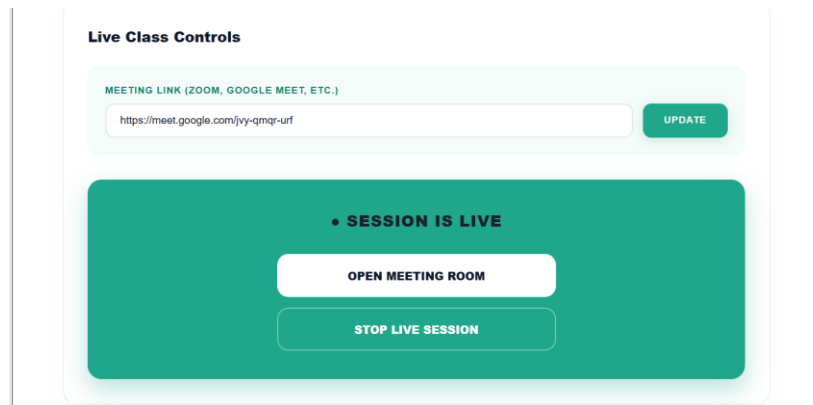


Figure 103: Teacher's Live session box

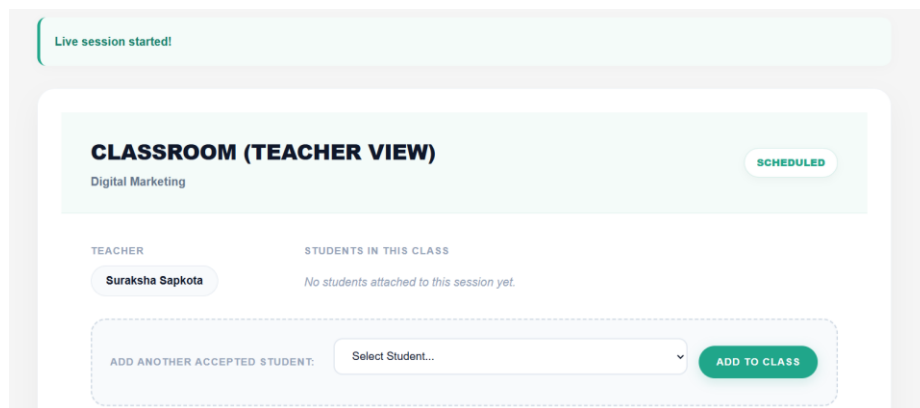


Figure 104: Started session by teacher

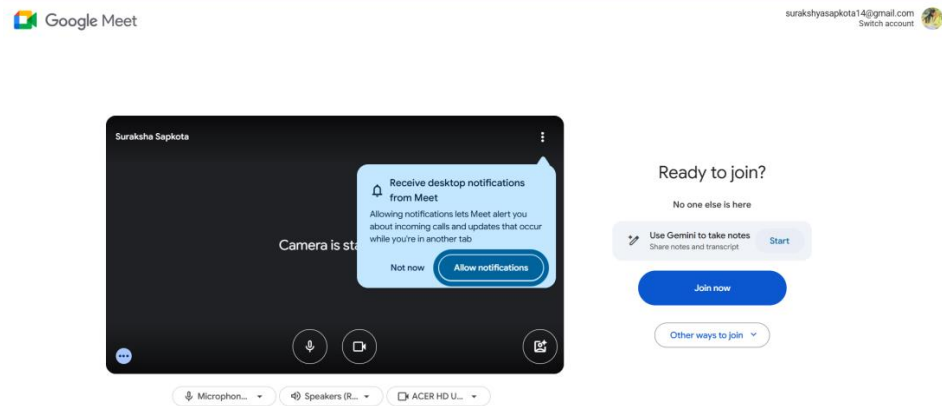


Figure 105: Opened Link

This is how the session is handled and teacher can teach multiple students at the same time if the student has the same skill.

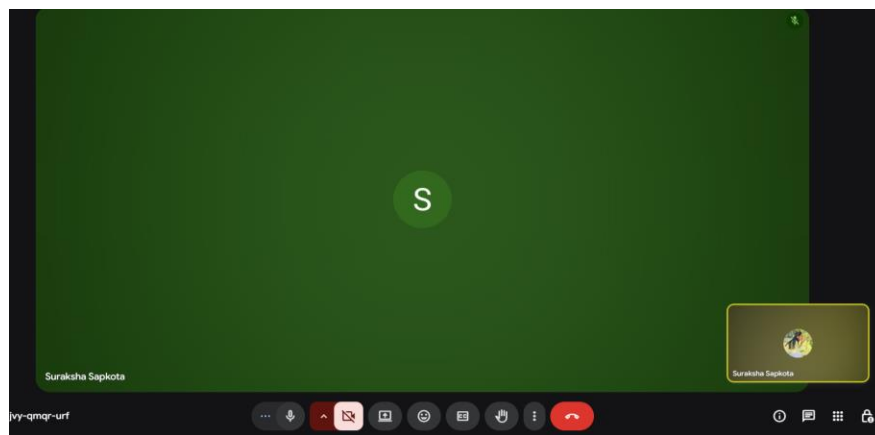


Figure 106: Live session going on

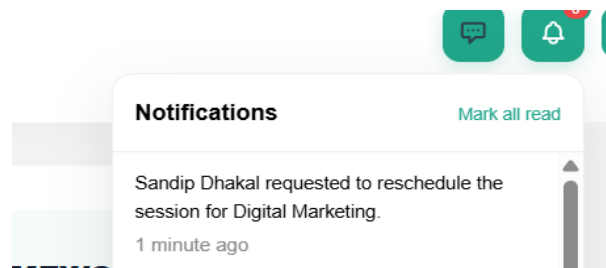


Figure 107: Get notification for reschedule

Session schedule updated successfully.

CLASSROOM (TEACHER VIEW)

Digital Marketing SCHEDULED

TEACHER: Suraksha Sapkota

STUDENTS IN THIS CLASS: No students attached to this session yet.

ADD ANOTHER ACCEPTED STUDENT: ADD TO CLASS

Session Schedule

CURRENT TIME
April 21, 2026 - 11:27 AM to 11:00 PM

UPDATE START TIME

UPDATE END TIME

UPDATE SCHEDULE

Figure 108: Updated schedule successfully

Also, the teacher will be able to upload the materials for the students in the classroom, and that uploaded material will be shown in the students' classroom

MATERIALS

No materials shared yet.

UPLOAD NEW MATERIAL

No file chosen

SECURE UPLOAD

PRACTICE & HOMEWORK

First day

bdutegfjebf jdsfbcisdgf

Submitted: 9 hours ago

@nd day VIEW WORK

now i will upload the file

Submitted: 9 hours ago

Chat System:

In this chat system, the teacher and student will chat with each other, but only the student can send a message to the teacher.

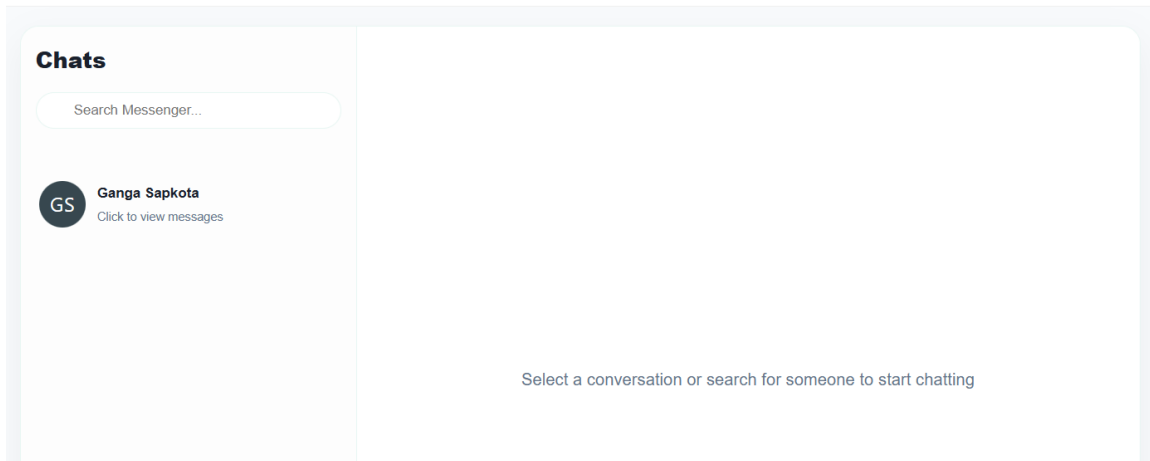


Figure 109: Chat system

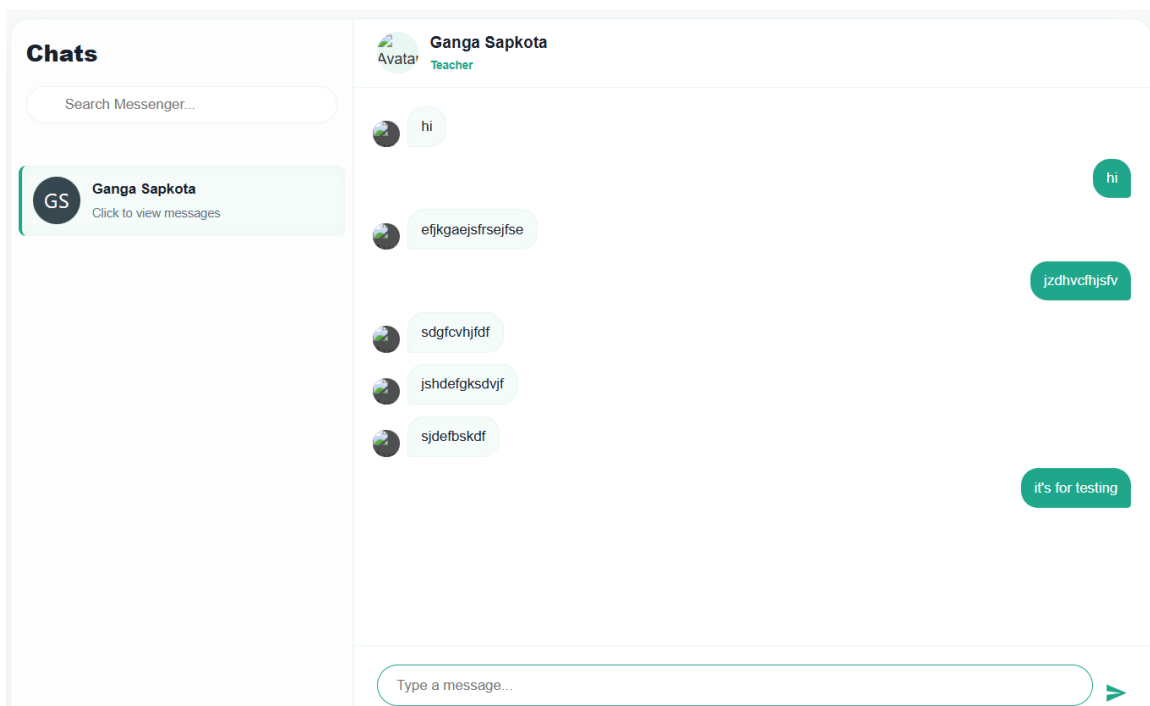


Figure 110: Message

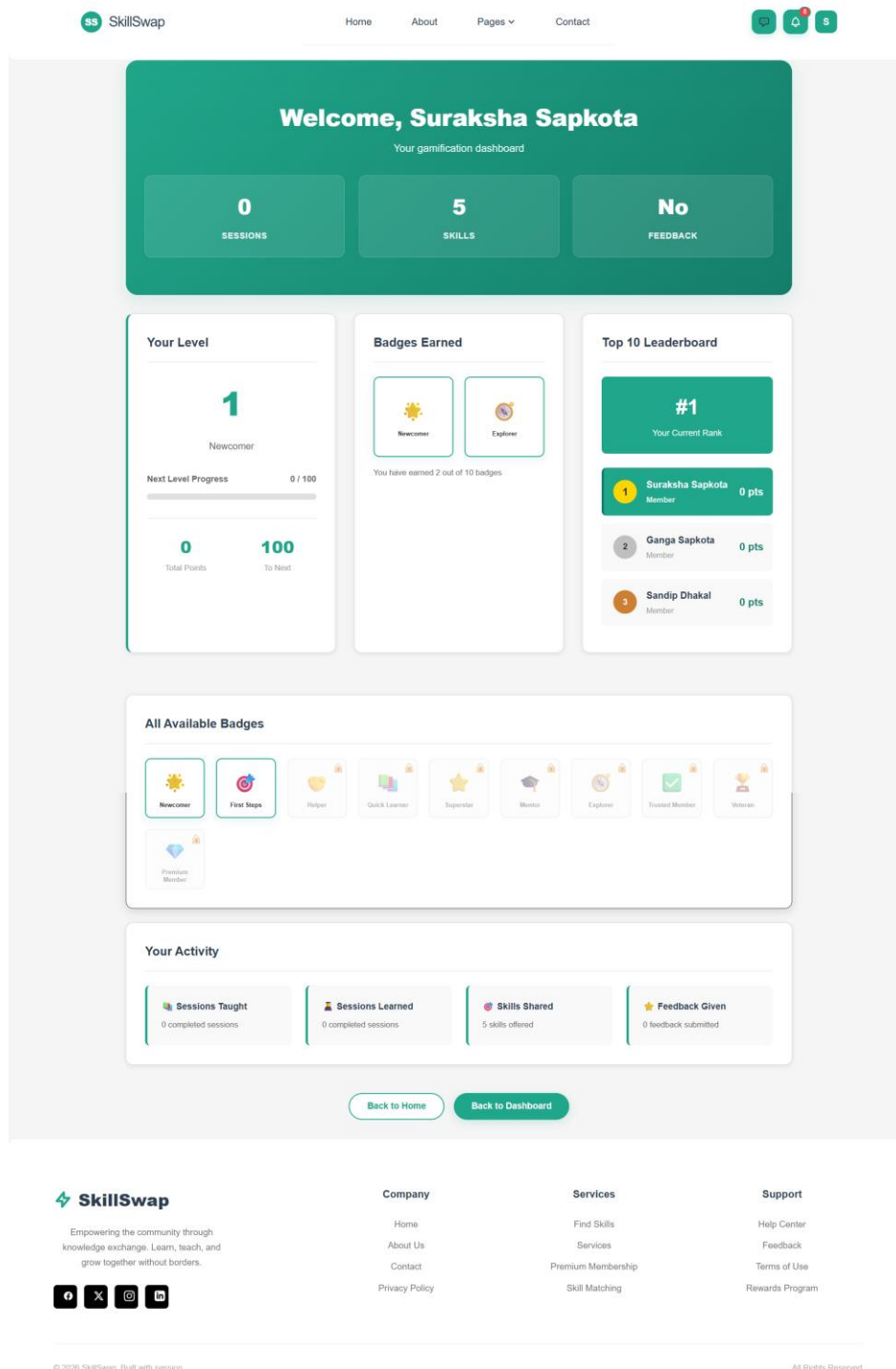
Reward:

Figure 111: Reward page

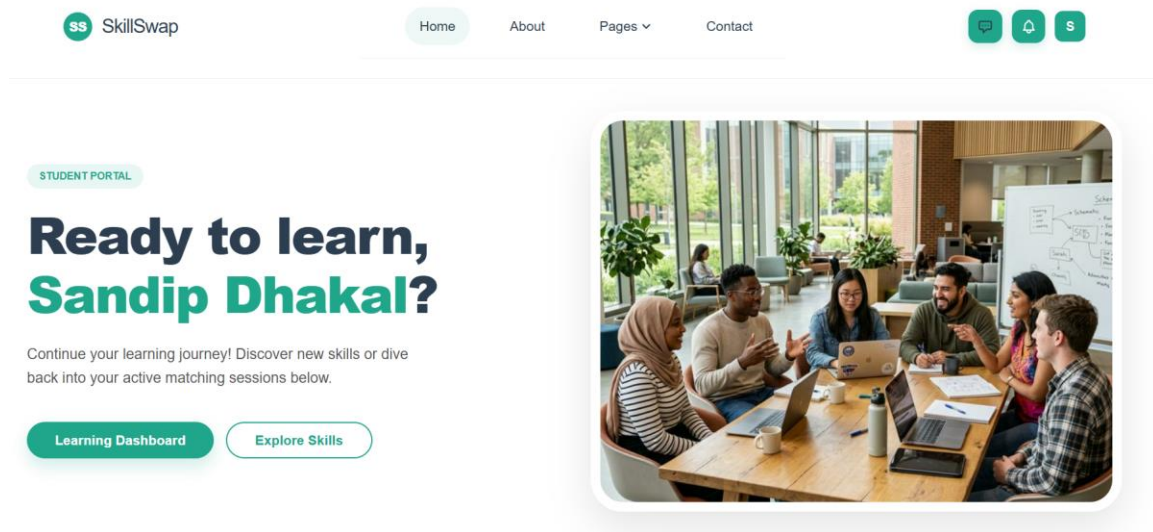
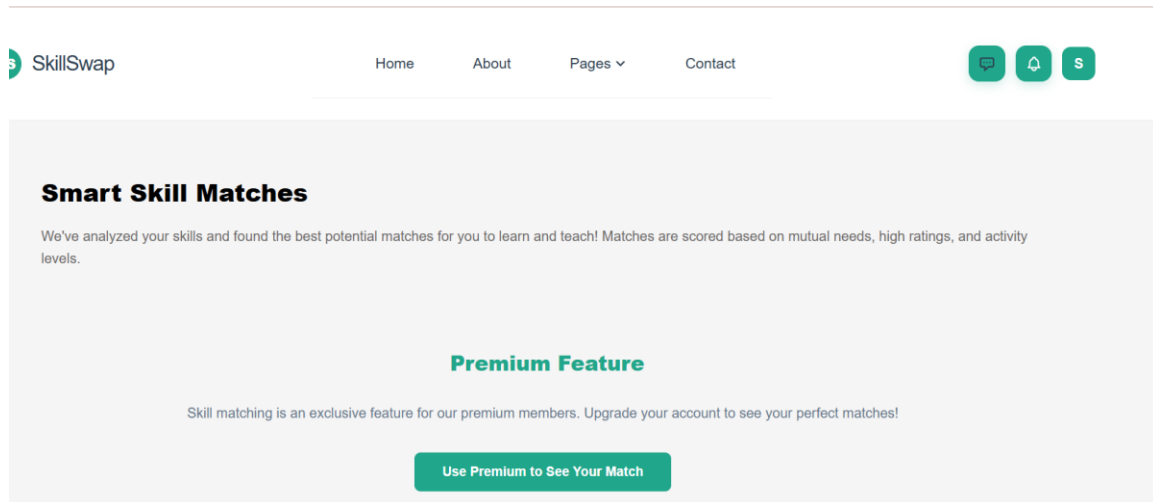
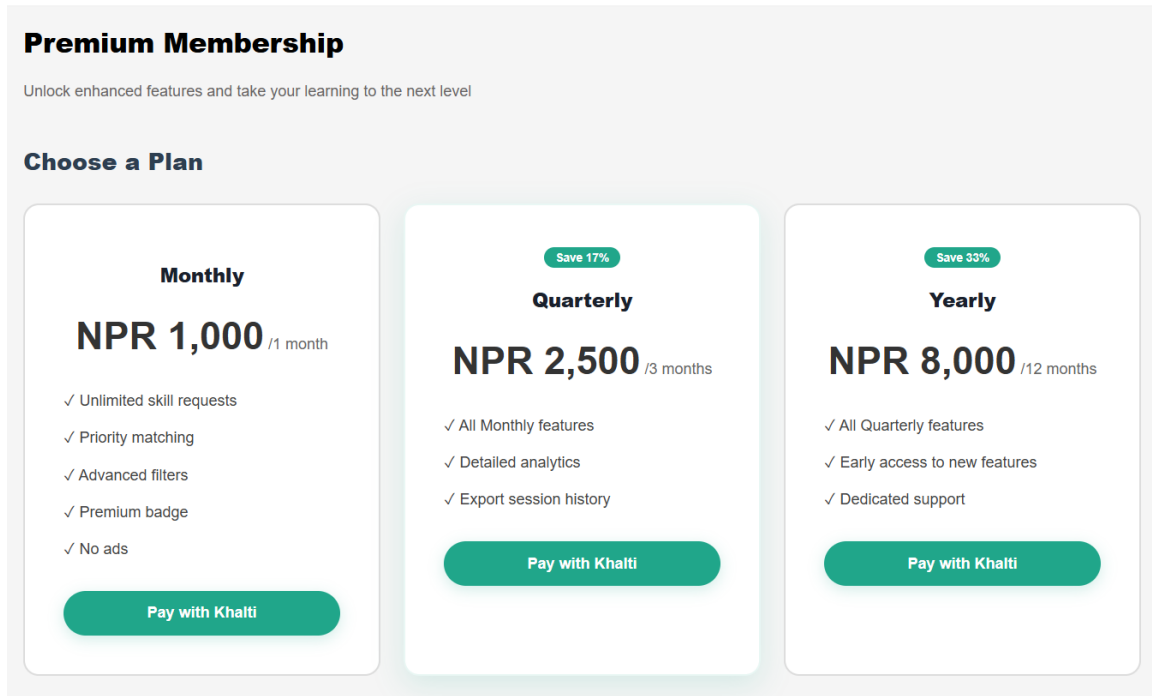
c. Student Side User:

Figure 112: Student User home page

The user can see their matches, but for students, that feature is only available on premium.



Premium:

Premium Membership

Unlock enhanced features and take your learning to the next level

Choose a Plan

Monthly

NPR 1,000 /1 month

- ✓ Unlimited skill requests
- ✓ Priority matching
- ✓ Advanced filters
- ✓ Premium badge
- ✓ No ads

Pay with Khalti

Quarterly

NPR 2,500 /3 months

Save 17%

- ✓ All Monthly features
- ✓ Detailed analytics
- ✓ Export session history

Pay with Khalti

Yearly

NPR 8,000 /12 months

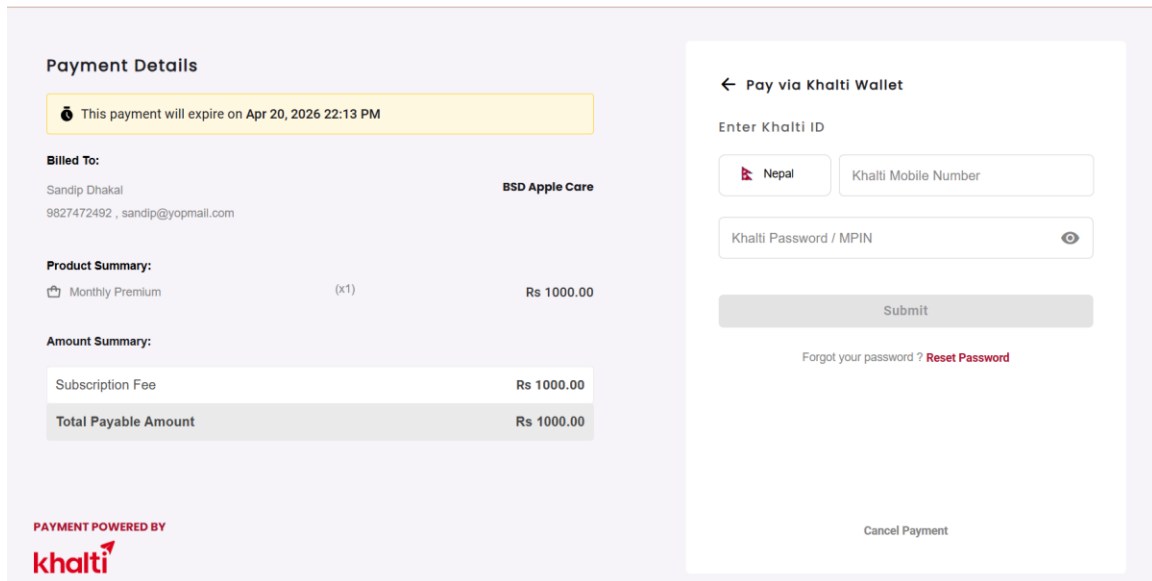
Save 33%

- ✓ All Quarterly features
- ✓ Early access to new features
- ✓ Dedicated support

Pay with Khalti

Figure 113: Premium membership plan

When the user clicks on the plan that redirects to the payment details page.



Payment Details

This payment will expire on Apr 20, 2026 22:13 PM

Billed To:

Sandip Dhakal
9827472492 , sandip@yopmail.com

BSD Apple Care

Product Summary:

Monthly Premium	(x1)	Rs 1000.00
-----------------	------	------------

Amount Summary:

Subscription Fee	Rs 1000.00
Total Payable Amount	Rs 1000.00

PAYMENT POWERED BY
khalti

Pay via Khalti Wallet

Enter Khalti ID

Nepal Khalti Mobile Number

Khalti Password / MPIN

Submit

Forgot your password ? [Reset Password](#)

Cancel Payment

Figure 114: Payments details

Khalti wallet asks for OTP after the user provide there num and MPIN.

The screenshot displays the 'Payment Details' section on the left and the 'Pay via Khalti Wallet' section on the right. The payment details include a yellow warning box stating 'This payment will expire on Apr 20, 2026 22:13 PM'. Below this, the 'Billed To' information shows 'Sandip Dhakal' with contact details. The 'Product Summary' lists 'Monthly Premium (x1)' for 'Rs 1000.00'. The 'Amount Summary' shows a 'Subscription Fee' of 'Rs 1000.00' and a 'Total Payable Amount' of 'Rs 1000.00'. The right side of the interface is titled 'Pay via Khalti Wallet' and includes an 'Enter OTP' section with a 'Nepal' flag selector and a 'Khalti Mobile Number' field containing '9800000005'. Below this is a 'Khalti Password / MPIN' field with a masked input and an eye icon. A section for 'Enter OTP sent to your Mobile Number/Email Address' features six empty input boxes. A large grey button labeled 'Pay Rs. 1000.00' is positioned below the OTP section. At the bottom, there is a link 'Forgot your password ? Reset Password' and a 'Cancel Payment' button.

Payment Details

This payment will expire on Apr 20, 2026 22:13 PM

Billed To:
Sandip Dhakal
9827472492 , sandip@yopmail.com

BSD Apple Care

Product Summary:
Monthly Premium (x1) Rs 1000.00

Amount Summary:
Subscription Fee Rs 1000.00
Total Payable Amount Rs 1000.00

PAYMENT POWERED BY

Pay via Khalti Wallet

Enter OTP

Nepal Khalti Mobile Number 9800000005

Khalti Password / MPIN

Enter OTP sent to your Mobile Number/Email Address

Pay Rs. 1000.00

Forgot your password ? [Reset Password](#)

Cancel Payment

Figure 115: Provide num and MPIN

This screenshot shows the 'Pay via Khalti Wallet' interface with the OTP entry section highlighted. The 'Enter OTP' section includes a 'Nepal' flag selector and a 'Khalti Mobile Number' field with the value '9800000005'. Below this is a 'Khalti Password / MPIN' field with a masked input and an eye icon. The 'Enter OTP sent to your Mobile Number/Email Address' section features six input boxes, with the last box containing the number '4'. A large red button labeled 'Pay Rs. 1000.00' is positioned below the OTP section. At the bottom, there is a link 'Forgot your password ? Reset Password'.

Pay via Khalti Wallet

Enter OTP

Nepal Khalti Mobile Number 9800000005

Khalti Password / MPIN

Enter OTP sent to your Mobile Number/Email Address

9 8 7 6 5 4

Pay Rs. 1000.00

Forgot your password ? [Reset Password](#)

Figure 116: Providing OTP

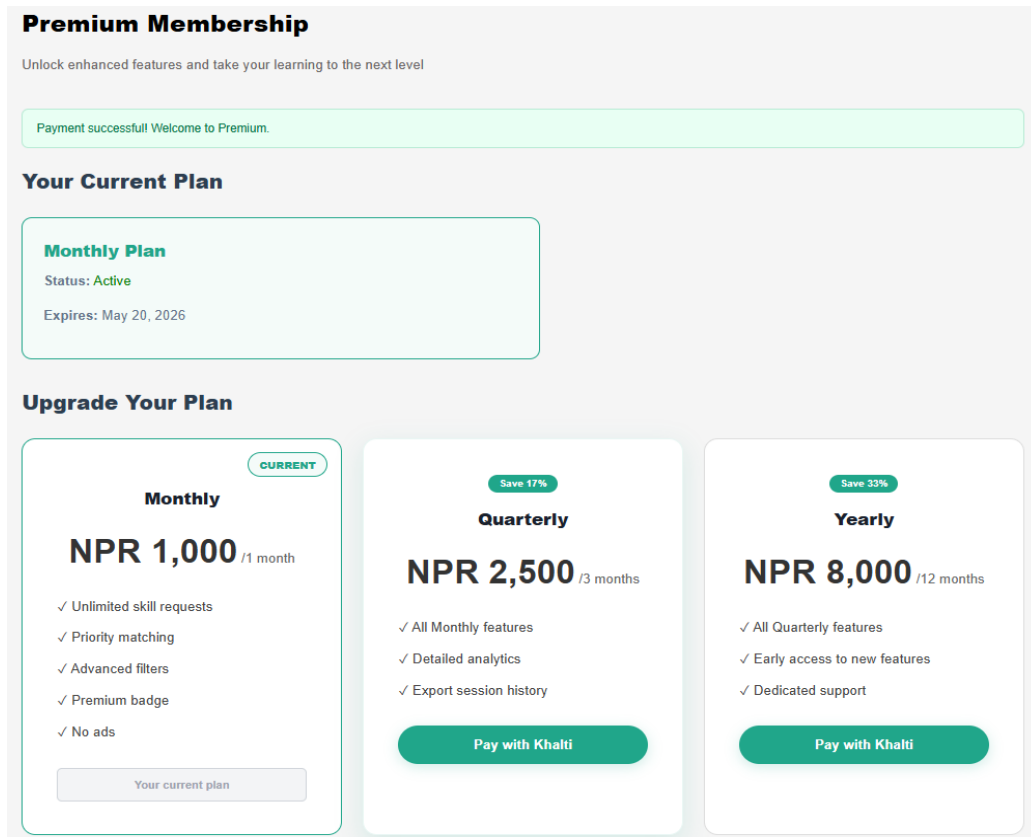


Figure 117: Successful payment

After payment, the student can see the match.

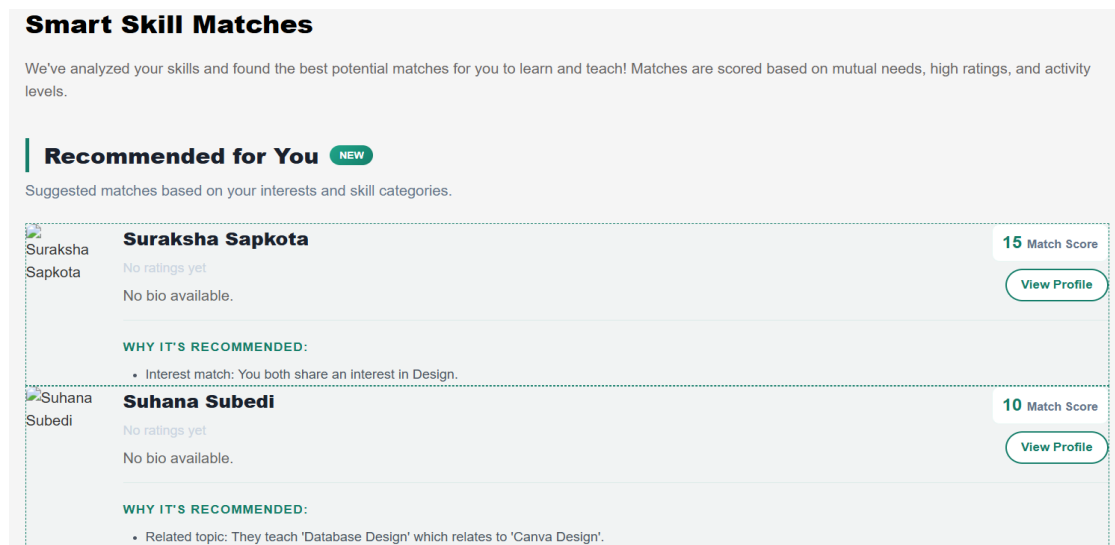
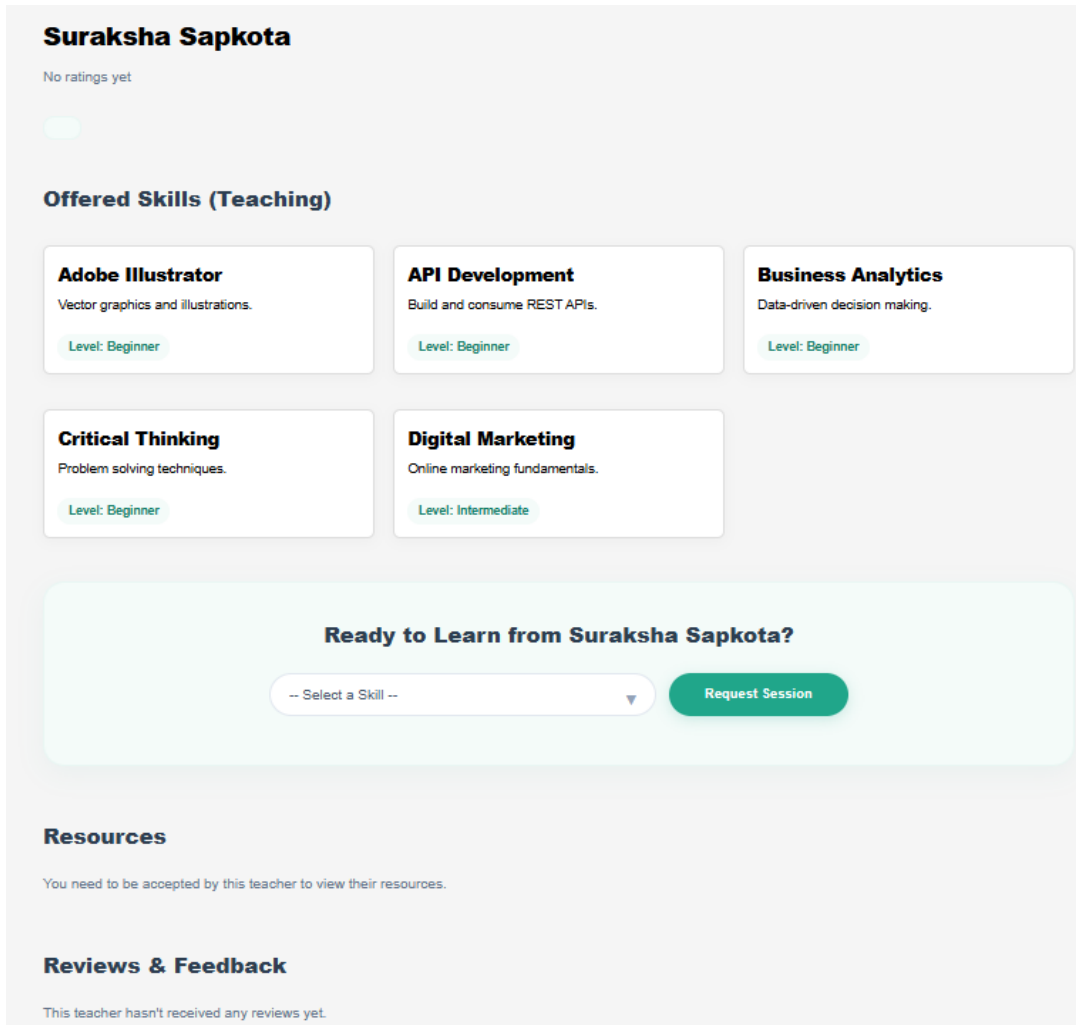


Figure 118: System showing a match for the student

Session & Notification:

From there, the student can view the teacher's profile and view the profile and can request the teacher for teaching. All the skills offered by the teacher, resources, reviews & feedback are shown in the teacher's profile, and below, students can choose the skill that they want to learn from the dropdown and request a session.



Suraksha Sapkota
No ratings yet

Offered Skills (Teaching)

Adobe Illustrator Vector graphics and illustrations. Level: Beginner	API Development Build and consume REST APIs. Level: Beginner	Business Analytics Data-driven decision making. Level: Beginner
Critical Thinking Problem solving techniques. Level: Beginner	Digital Marketing Online marketing fundamentals. Level: Intermediate	

Ready to Learn from Suraksha Sapkota?

-- Select a Skill -- [Request Session](#)

Resources
You need to be accepted by this teacher to view their resources.

Reviews & Feedback
This teacher hasn't received any reviews yet.

Figure 119: Teacher's profile

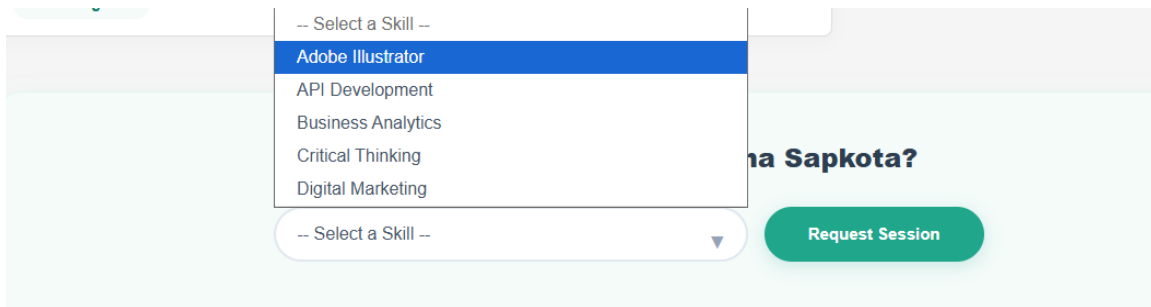
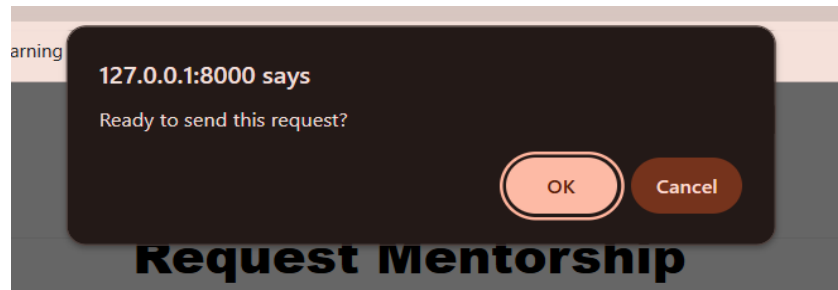


Figure 120: The dropdown feature

The student can request a teacher to fill this.

A screenshot of a 'Request Mentorship' form. The title 'Request Mentorship' is in bold black text. Below it, a subtitle says 'Connect with Suraksha Sapkota to master this skill.' The form has a light green background. A section titled 'Digital Marketing' is highlighted in a light green box, with 'Level: Intermediate' below it. Below this is a section titled 'INTRODUCE YOURSELF' with a text area for 'Describe what you want to learn or any specific goals you have...'. Below that is a section titled 'PREFERRED DATE & TIME' with a date and time picker showing 'mm/dd/yyyy --:-- --'. At the bottom are two buttons: 'Send Request' (green) and 'Cancel' (white with green border).

Figure 121: Request teacher form



The request is sent successfully, and now the teacher gets a notification of this request, and after the teacher accepts or declines.

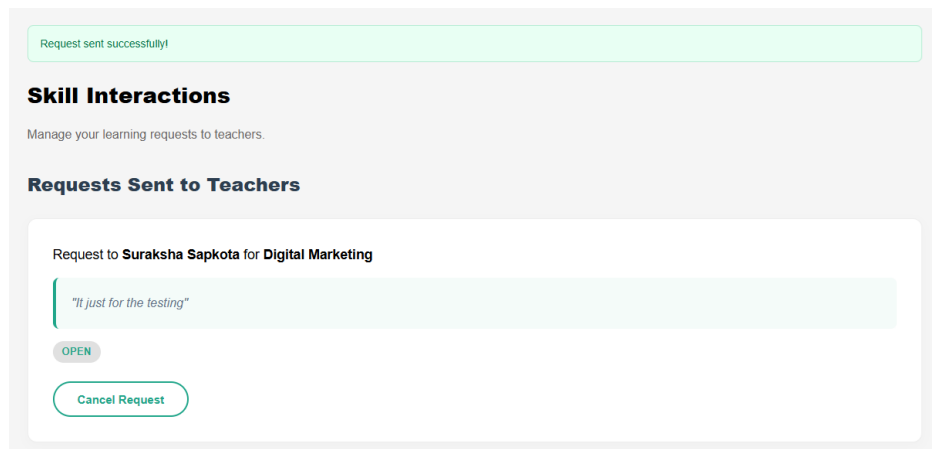


Figure 122: request sent successfully

Students get a notification after the teacher accepts the request for a session.

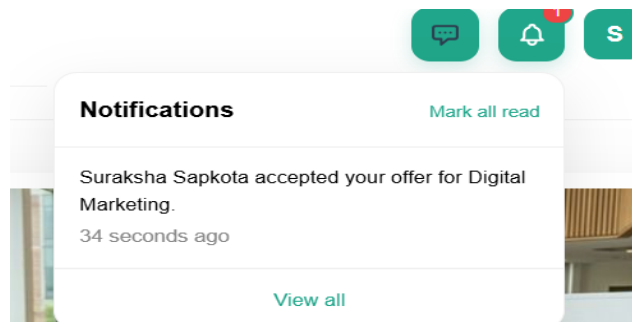


Figure 123: Notification of accepting the session request

Now, the teacher needs to create the session, and after the teacher creates the session, students get notified.

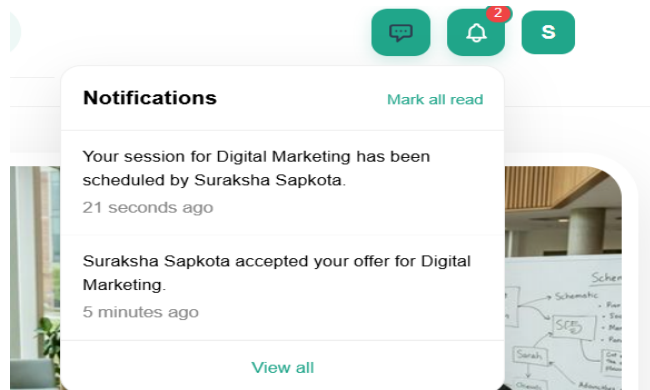
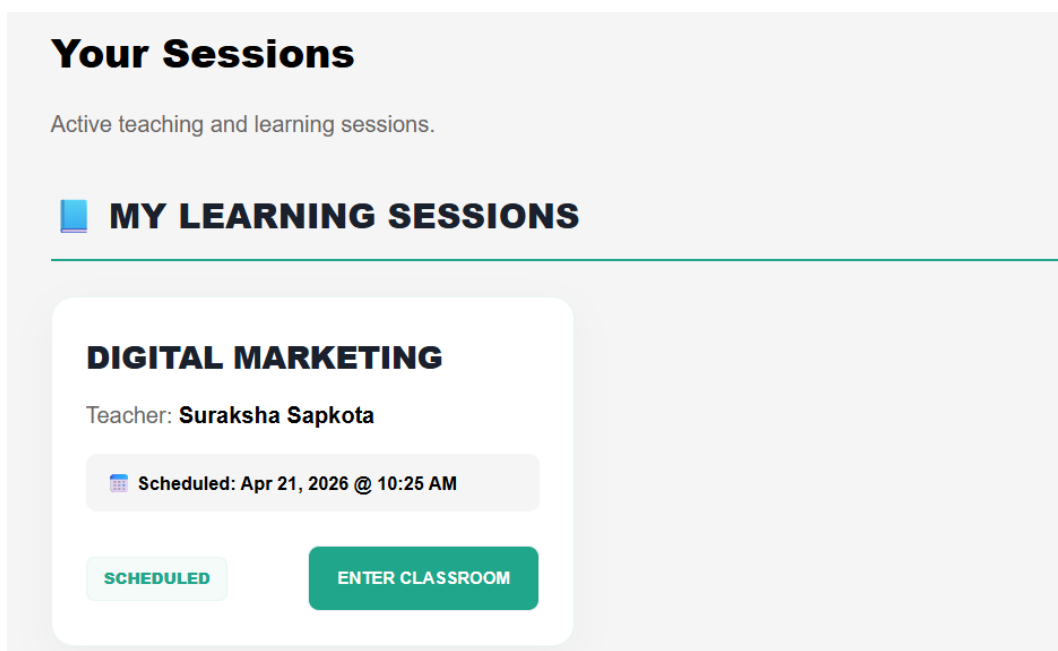


Figure 124: Notification of session

After clicking on the notification of the session created, student will be redirected to the session page and there they can see the session, and they can go to the classroom by clicking on “Enter Classroom.”



The student's classroom looks like this, and before the teacher provides the meeting link, it still looks like this.

The screenshot displays the 'CLASSROOM (STUDENT VIEW)' for a 'Digital Marketing' session. At the top, the SkillSwap logo and navigation links (Home, About, Pages, Contact) are visible, along with user icons. The classroom header includes a 'SCHEDULED' status badge. Below this, the teacher's name 'Suraksha Sapkota' is shown with a 'CHAT' button, and a note states 'No students attached to this session yet.' The 'Session Schedule' section shows the 'CURRENT TIME' as 'April 21, 2026 - 10:25 AM'. A 'REQUEST A DIFFERENT TIME' section contains a text input field with the message 'Teacher, I am not available then. Could we do Tuesday at 6 PM instead?' and a 'SEND RESCHEDULE REQUEST' button. A large section titled 'AWAITING MEETING LINK...' features an hourglass icon and a message: 'The teacher hasn't provided the live room link yet. You can relax or kindly message them a reminder.' Below this is a 'MESSAGE TEACHER' button. At the bottom, there are two side-by-side panels: 'MATERIALS' (with the note 'No materials shared yet.') and 'PRACTICE & HOMEWORK' (with the note 'No practice files submitted yet.'). The 'PRACTICE & HOMEWORK' panel includes a 'SUBMIT NEW PRACTICE' section with a 'Task Name (e.g. Design Draft v1)' input field, a 'Notes/Description to teacher...' input field, a 'Choose File' button (showing 'No file chosen'), and a 'SUBMIT WORK' button.

Figure 125: Student classroom before the teacher provides the meeting link

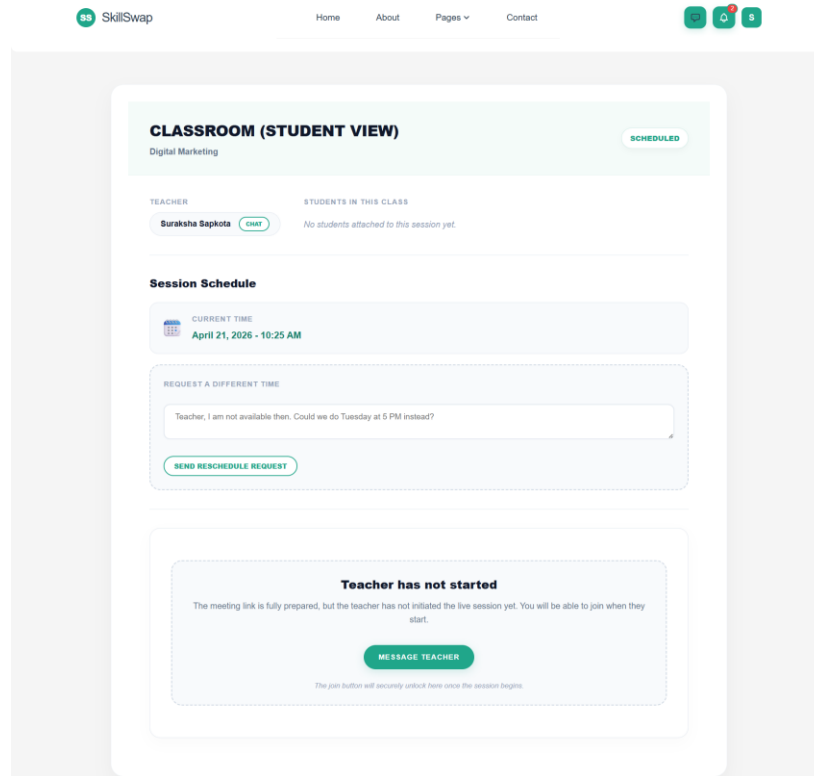


Figure 126: After the meeting link is provided by the teacher

The meeting link was provided by the teacher, but before the teacher start the session, the student cannot see the join live session.

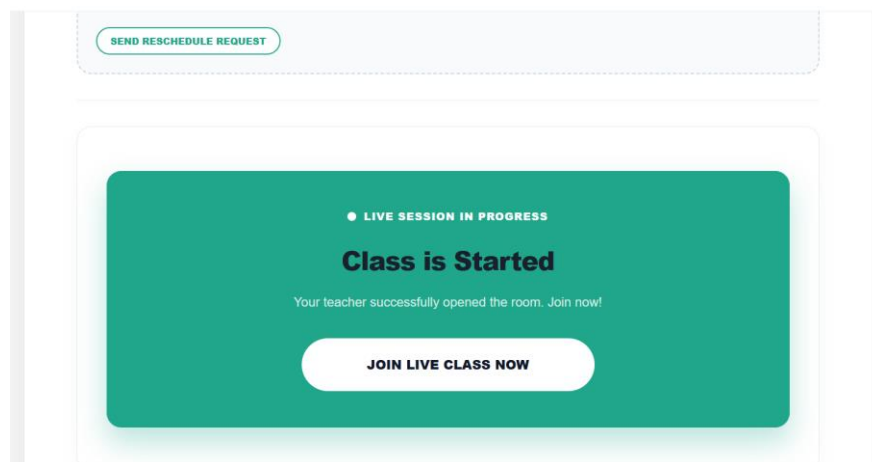


Figure 127: After the teacher started the session

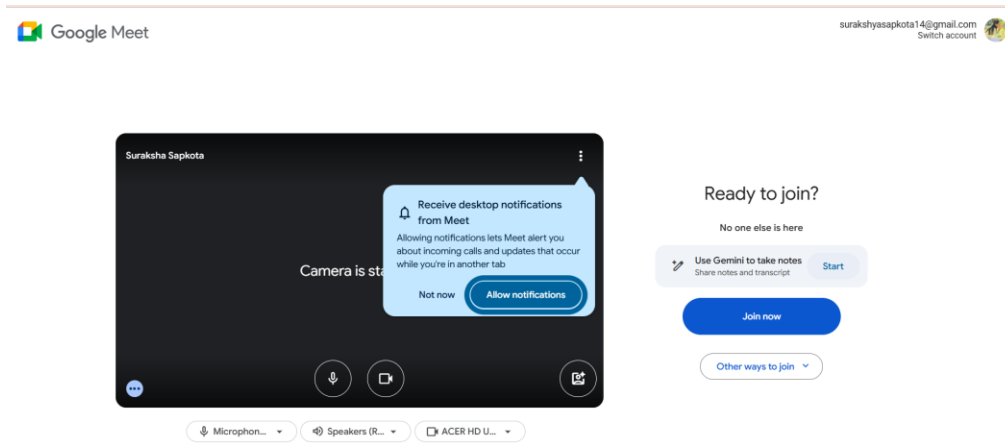


Figure 128: Redirect on google meet



Figure 129: Admit for join the live

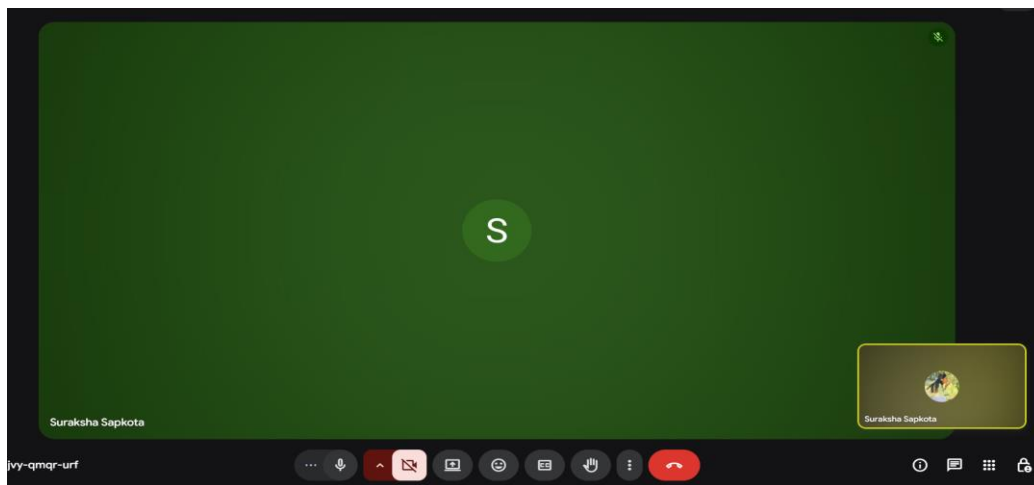


Figure 130: Live session started

Practical Upload:

Students can also ask the teacher to reschedule the class.

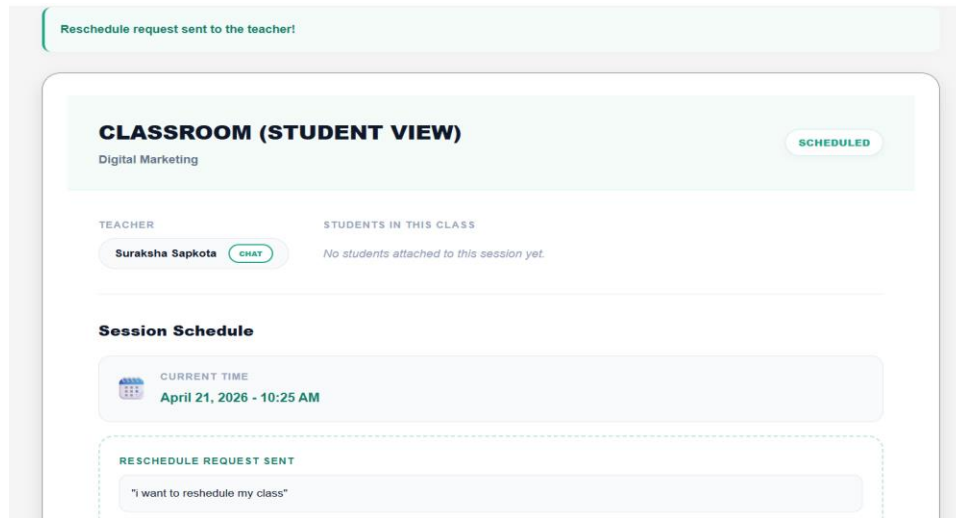


Figure 131: Sending reschedule request

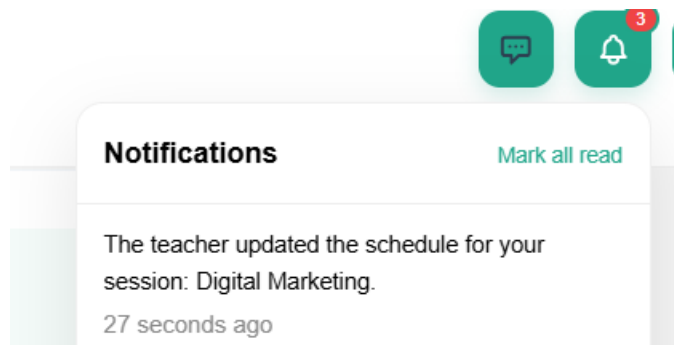


Figure 132: Get notification after teacher reschedule the class

The screenshot shows a web interface for submitting practice work. At the top, a light green header contains the text 'PRACTICE & HOMEWORK'. Below this, the section 'First day' displays a placeholder text 'bdutegfjebf jdsfbcisdgf' and a timestamp 'Submitted: 1 second ago'. A dashed line separates this from the 'SUBMIT NEW PRACTICE' section. This section includes a text input field with '@nd day', a larger text area with 'now i will upload the file', and a file upload area with a 'Choose File' button and the filename 'Log1.docx'. At the bottom of the form is a prominent green button labeled 'SUBMIT WORK'.

Figure 133: Student can also upload the practical



Figure 134: Practical submitted message

PRACTICE & HOMEWORK

First day

bdutegfjebf jdsfbcisdgf

Submitted: 40 seconds ago

@nd day

[VIEW WORK](#)

now i will upload the file

Submitted: 1 second ago

Figure 135: Practical and homework file

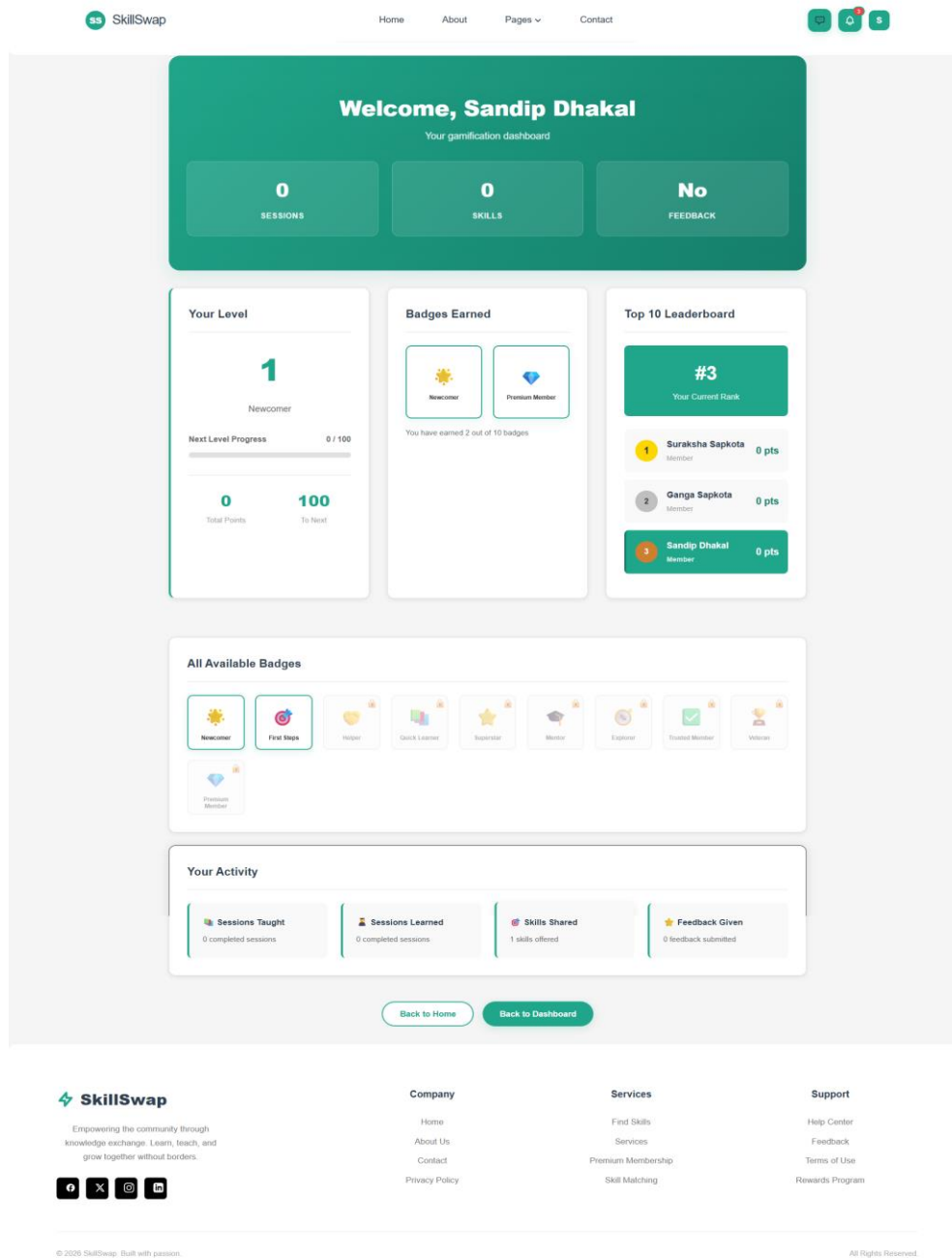
Rewards:

Figure 136: Reward Page

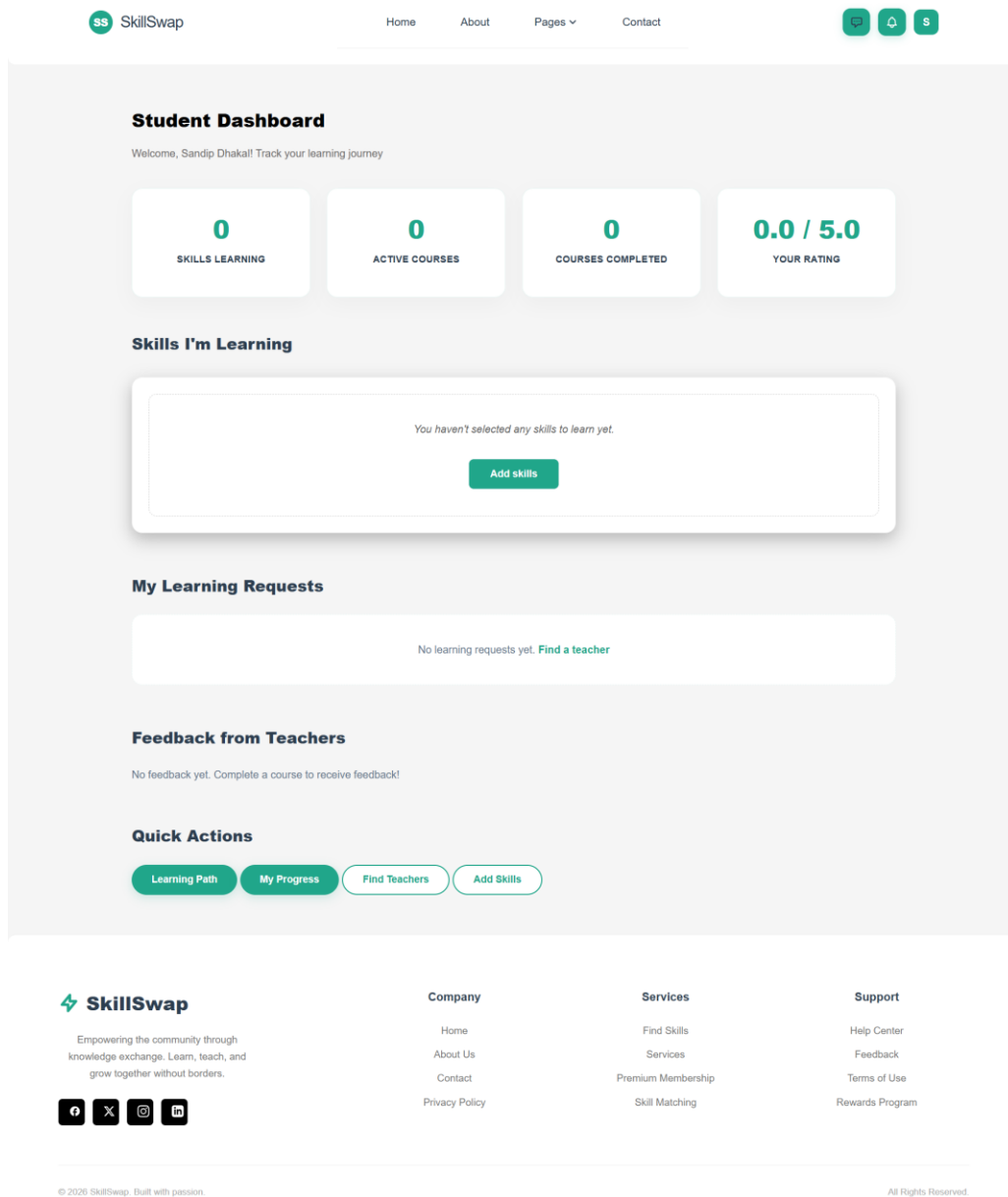
Student Dashboard:

Figure 137: Student dashboard before adding skill and session

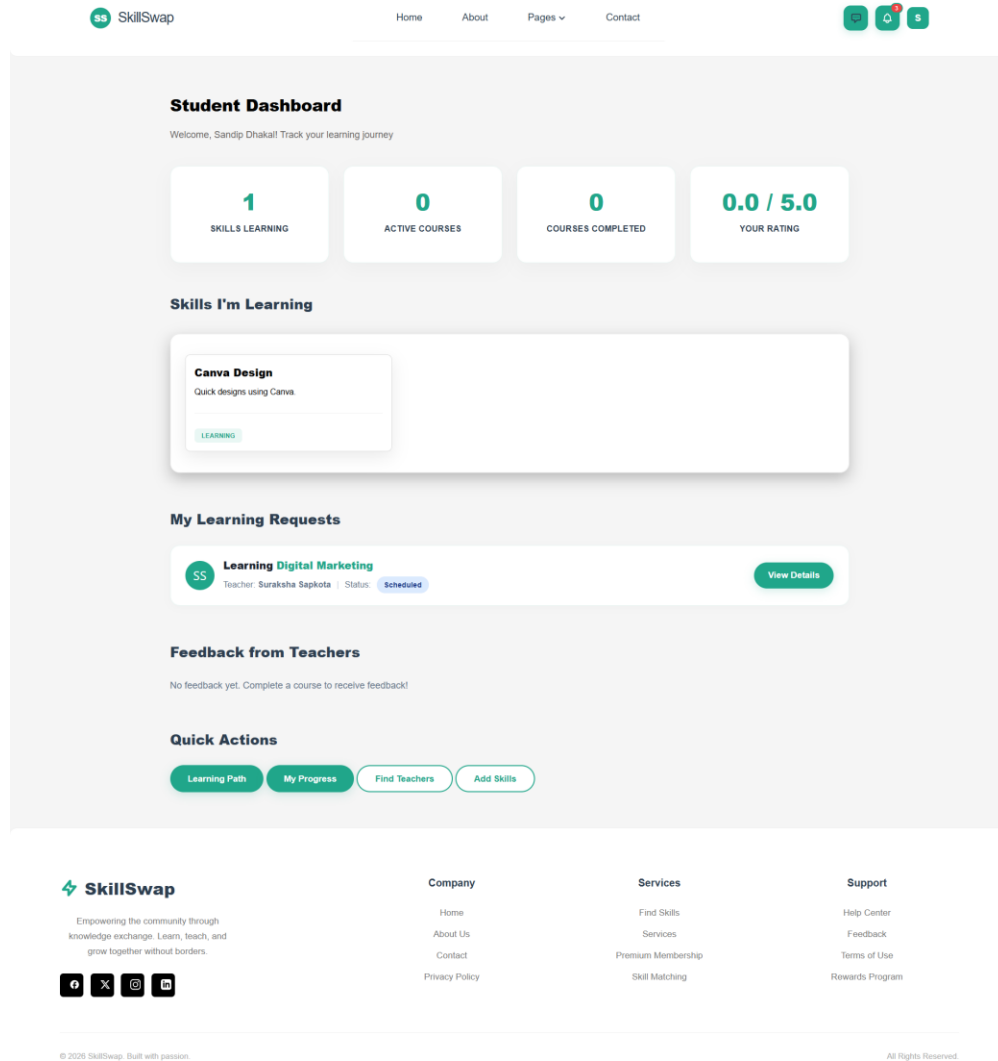


Figure 138: After adding

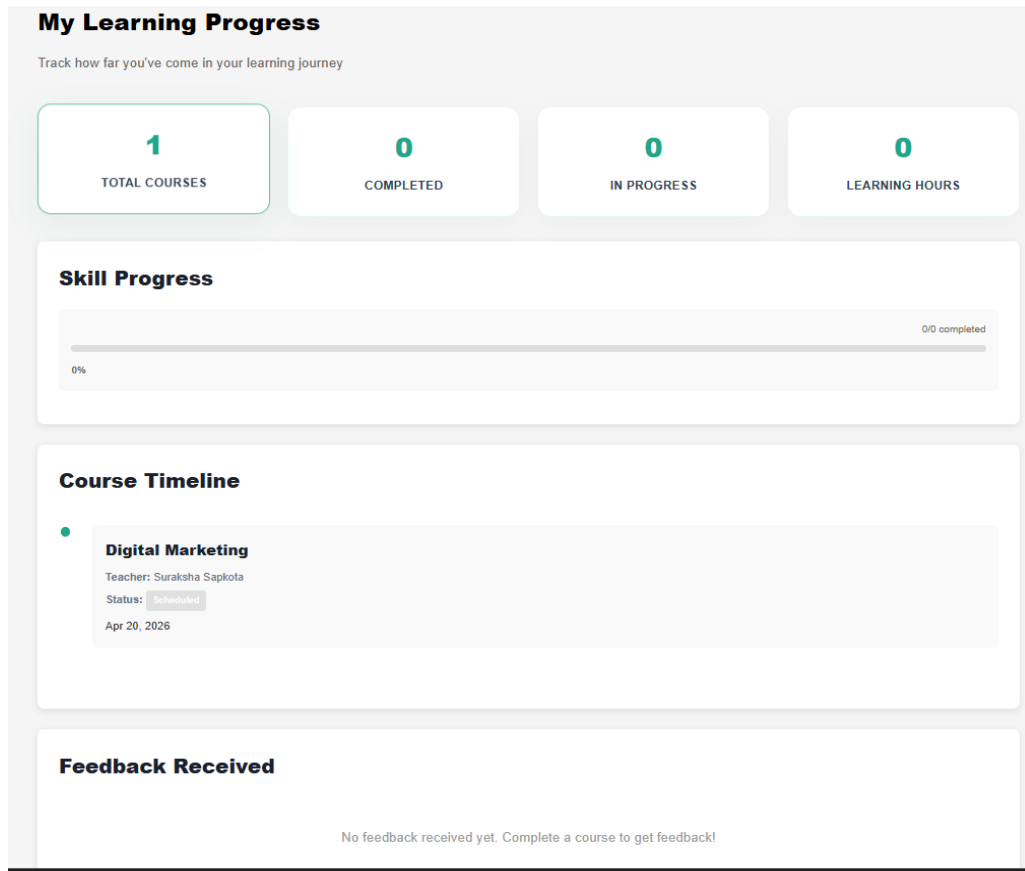


Figure 139: Learning Progress

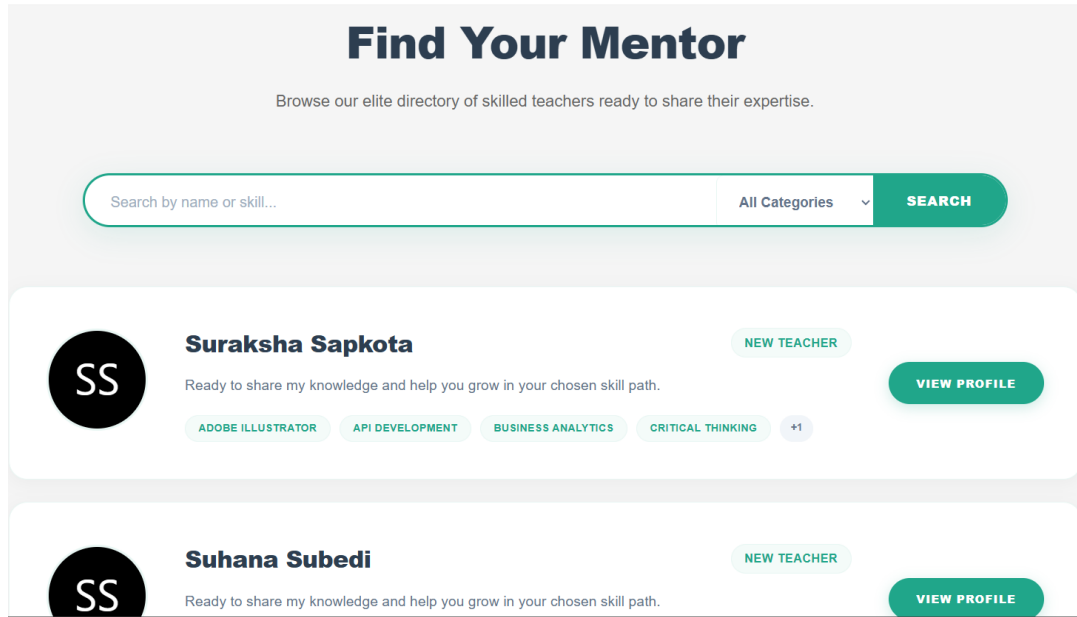


Figure 140: Find mentor page

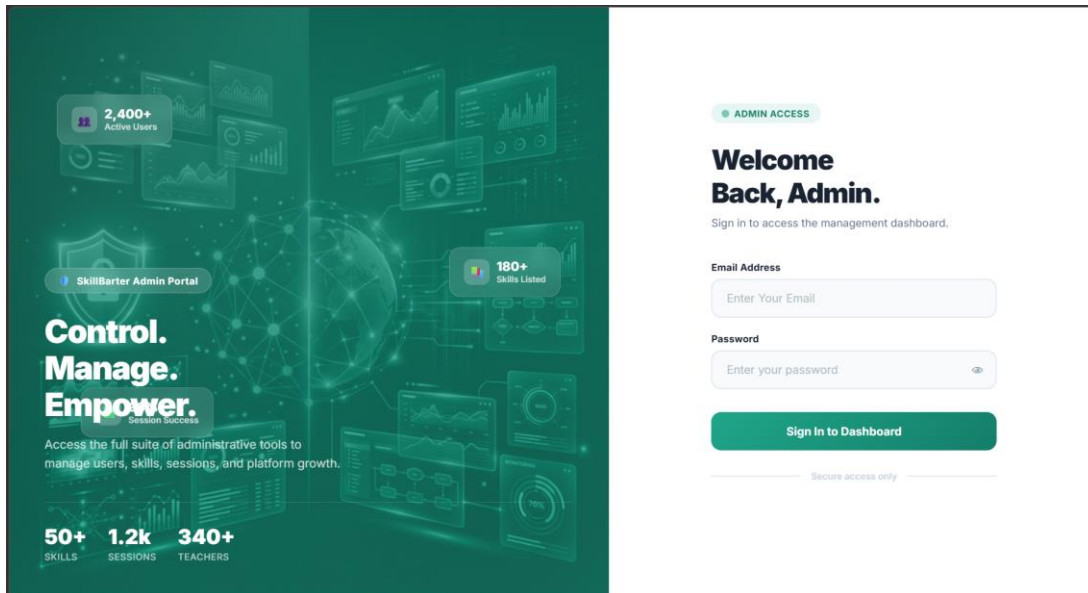
d. Admin Side**Admin Login Page:**

Figure 141: Admin login page

● ADMIN ACCESS

Welcome Back, Admin.

Sign in to access the management dashboard.

Email Address

Password

Sign In to Dashboard

Secure access only

Figure 142: Login

Admin dashboard:

Here in the dashboard shows all the details like; total users, teachers, students, how much session is going on now, total skills, total requests, premium members, and feedback.

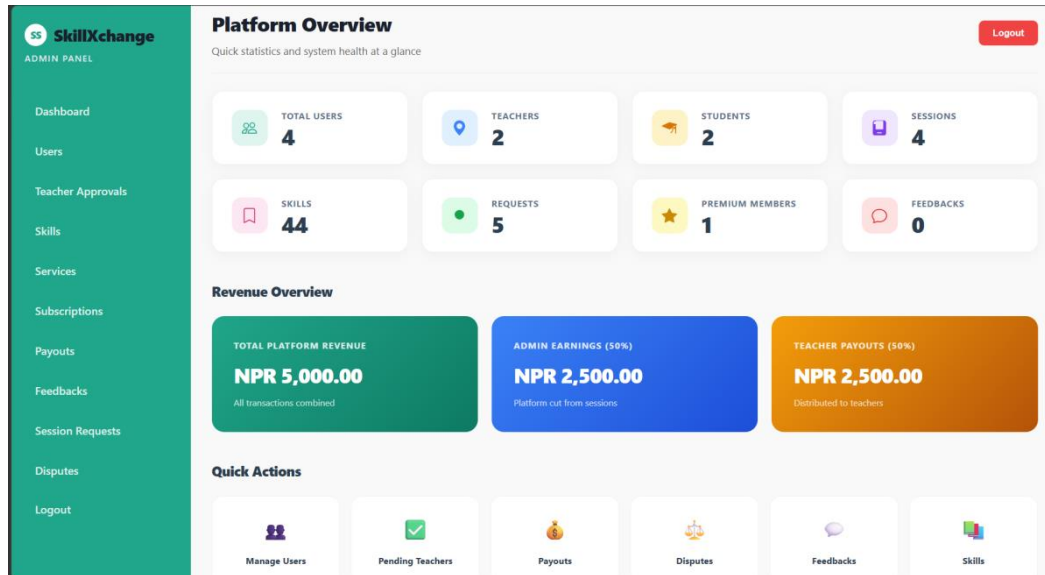


Figure 143: Admin dashboard

Teacher Approval:

Teachers need approval; without admin approval, they cannot access the dashboard, and students cannot request a non-approved teacher while searching.

The screenshot displays the SkillXchange Teacher Approvals page. The left sidebar contains the 'ADMIN PANEL' with links to Dashboard, Users, Teacher Approvals, Skills, Services, Subscriptions, Payouts, Feedbacks, Session Requests, Disputes, and Logout. The main content area is titled 'Teacher Approvals' and includes a 'Logout' button. It features a 'Pending Approvals' section with a table of pending teacher applications and an 'All Teachers' section with a table of all teachers.

NAME	EMAIL	DOCUMENTS	JOINED	ACTIONS
Ganga Sapkota	ganga@yopmail.com	View CV View Certificate View Citizenship	2026-03-21	Approve

NAME	EMAIL	APPROVED	JOINED	ACTIONS
Ganga Sapkota	ganga@yopmail.com	Pending	2026-03-21	Approve

Figure 144: Teacher approvals page

User Management:

Here in user management, we can see the total number of registered users.

NAME	EMAIL	ROLE	ACTIVE	ACTIONS
Sandip Dhakal	sandip@yopmail.com	Student	Active	Deactivate Delete
Suhana Subedi	suhana@yopmail.com	Teacher	Active	Deactivate Delete
Ganga Sapkota	ganga@yopmail.com	Student	Active	Deactivate Delete
Suraksha Sapkota	surakshyasapkota14@gmail.com	Teacher	Active	Deactivate Delete

Figure 145: User management

Skill Management:

Admin adds a skill, and all skills are shown here in skill management, and from there, the admin adds a new skill.

IMAGE	NAME	CATEGORY	ACTIONS
	Python Programming Learn Python from beginner to advanced.	Technology	Update Image Delete
	Web Development HTML, CSS, JavaScript, PHP basics.	Technology	Update Image Delete
	Laravel Framework Build modern web apps using Laravel.	Technology	Update Image Delete
	React.js Frontend development with React.	Technology	Update Image Delete
	Node.js Backend development using Node.js.	Technology	Update Image Delete
	Database Design MySQL and relational database concepts.	Technology	Update Image Delete
	API Development Build and consume REST APIs.	Technology	Update Image Delete

Skill Management
Monitor, manage and add skills available on the platform

+ Add New Skill

Skill Title *
e.g. Python Programming

Category *
e.g. Technology

Description
Brief description of this skill...

Skill Image
Choose File | No file chosen
Click to upload image (PNG, JPG, WebP)

Add Skill

IMAGE	NAME	CATEGORY	ACTIONS
-------	------	----------	---------

Figure 146: New skill adding form

Service Management:

In service management service is adding by admin and that is visible to website as service and here all service is included that I had in my platform, and also admin can add and delete service.

Service Management
Manage platform services and offerings

Add New Service

Service Title
e.g. Graphic Design Masterclass

Category
e.g. Design

Description
Briefly describe the service...

Add Service

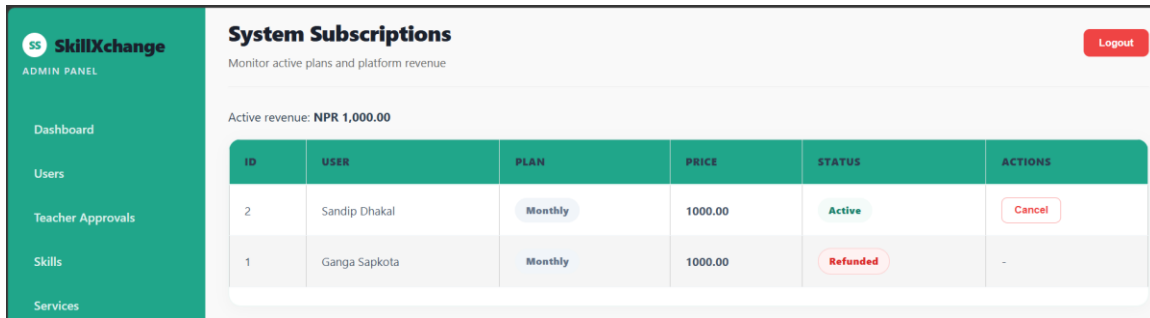
Existing Services

TITLE	CATEGORY	ACTIONS
Teaching Dashboard Manage your student requests and upcoming teaching...	N/A	
Resource Hub Upload and manage materials for your students.	N/A	
Teaching Analytics Track your ratings and teaching performance.	N/A	
Search for Mentors Connect with experts to learn new skills or broaden...	N/A	
Learning Roadmap Track your progress through your personalized learn...	N/A	
My Rewards View your earned badges and community points.	N/A	

Figure 147: Service management

System Subscription:

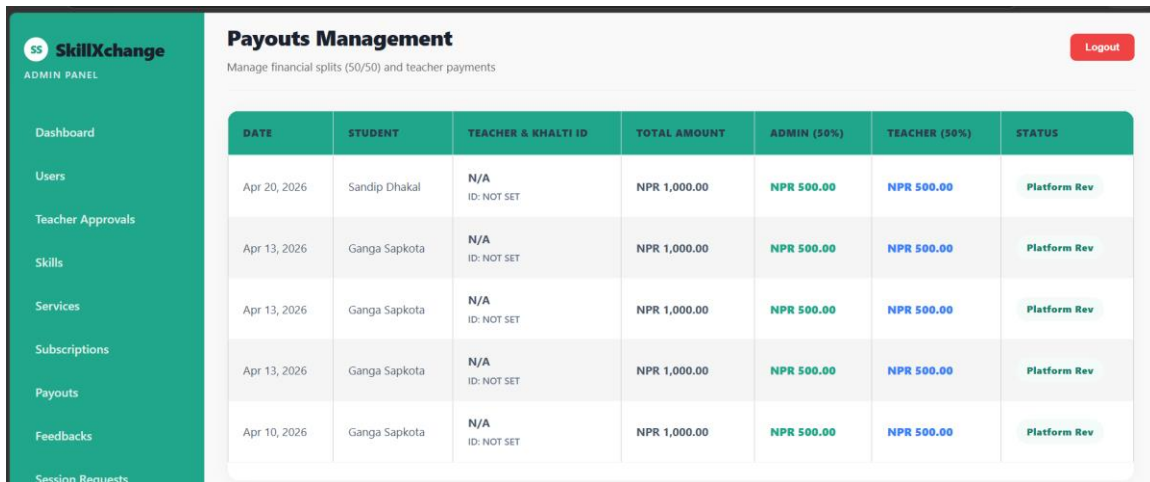
In subscription, the admin can see who used premium, and also the student will be able to report to the teacher, and the admin is also able to refund, and that shows as refunded in status.



ID	USER	PLAN	PRICE	STATUS	ACTIONS
2	Sandip Dhakal	Monthly	1000.00	Active	Cancel
1	Ganga Sapkota	Monthly	1000.00	Refunded	-

Figure 148: System Subscription

Also, the teacher will get 50% of each student's fee after the student uses the premium feature, and that is directly shown in Payouts management

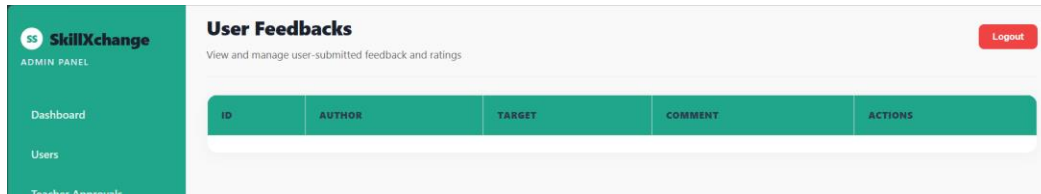


DATE	STUDENT	TEACHER & KHALTI ID	TOTAL AMOUNT	ADMIN (50%)	TEACHER (50%)	STATUS
Apr 20, 2026	Sandip Dhakal	N/A ID: NOT SET	NPR 1,000.00	NPR 500.00	NPR 500.00	Platform Rev
Apr 13, 2026	Ganga Sapkota	N/A ID: NOT SET	NPR 1,000.00	NPR 500.00	NPR 500.00	Platform Rev
Apr 13, 2026	Ganga Sapkota	N/A ID: NOT SET	NPR 1,000.00	NPR 500.00	NPR 500.00	Platform Rev
Apr 13, 2026	Ganga Sapkota	N/A ID: NOT SET	NPR 1,000.00	NPR 500.00	NPR 500.00	Platform Rev
Apr 10, 2026	Ganga Sapkota	N/A ID: NOT SET	NPR 1,000.00	NPR 500.00	NPR 500.00	Platform Rev

Figure 149: Payouts management

Feedback:

Admin can see all the feedback that is given by users (teacher and student both)

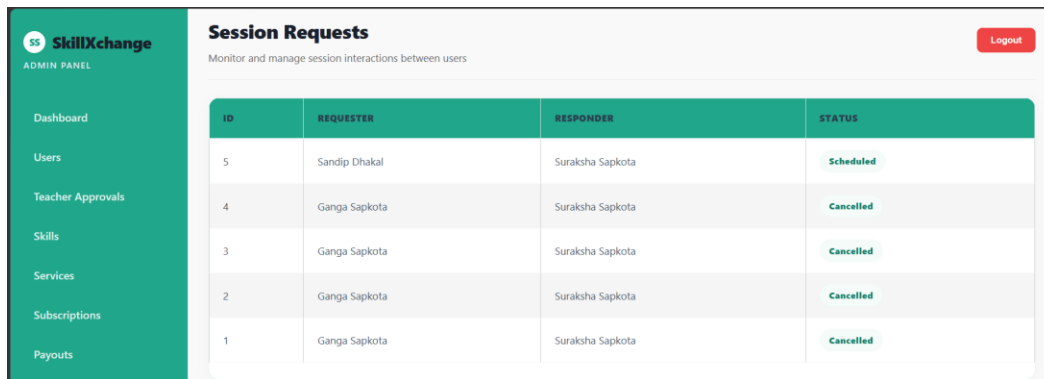


ID	AUTHOR	TARGET	COMMENT	ACTIONS

Figure 150: user Feedback

Session Request:

All session requests is shown here.

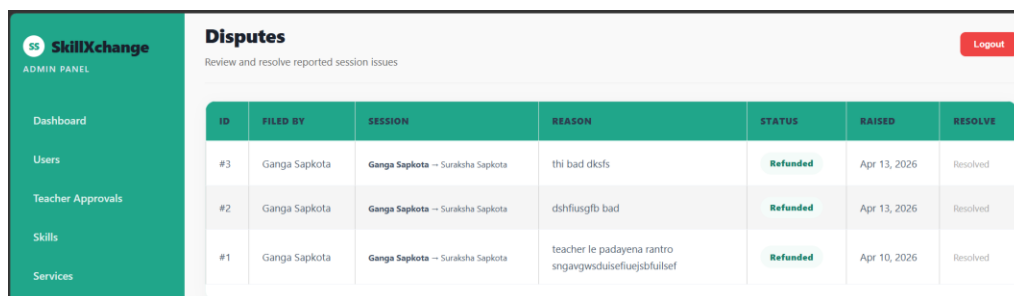


ID	REQUESTER	RESPONDER	STATUS
5	Sandip Dhakal	Suraksha Sapkota	Scheduled
4	Ganga Sapkota	Suraksha Sapkota	Cancelled
3	Ganga Sapkota	Suraksha Sapkota	Cancelled
2	Ganga Sapkota	Suraksha Sapkota	Cancelled
1	Ganga Sapkota	Suraksha Sapkota	Cancelled

Figure 151: Session Request

Disputes:

In disputes, the refunded users are shown, and who refunded whose user is shown there.

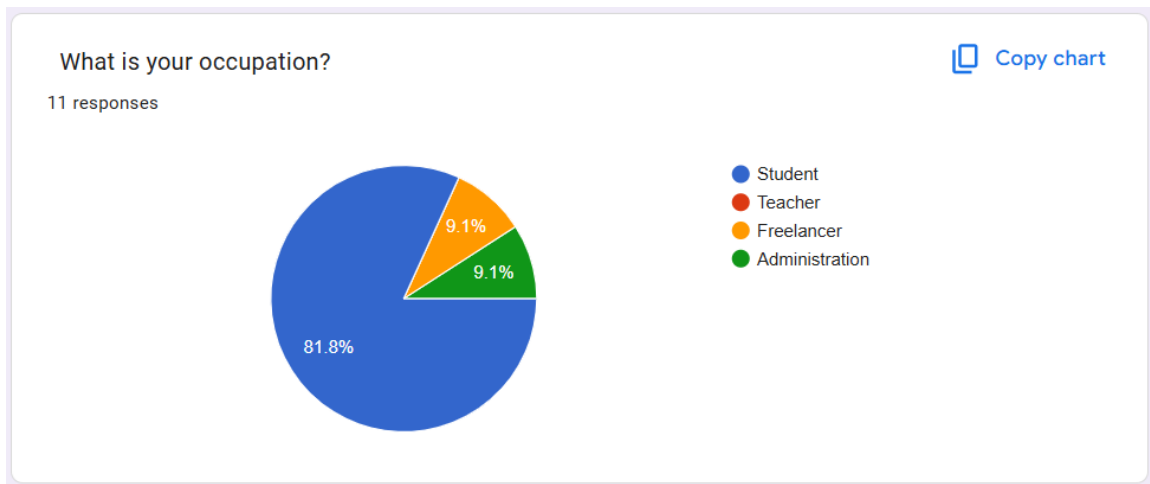
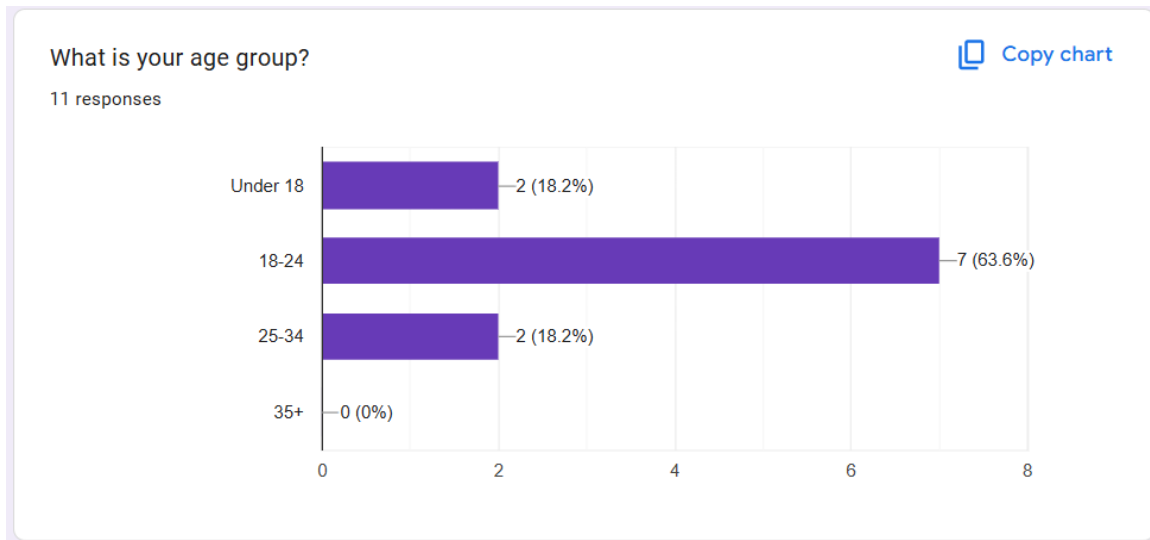


ID	FILED BY	SESSION	REASON	STATUS	RAISED	RESOLVE
#3	Ganga Sapkota	Ganga Sapkota -- Suraksha Sapkota	thi bad dksfs	Refunded	Apr 13, 2026	Resolved
#2	Ganga Sapkota	Ganga Sapkota -- Suraksha Sapkota	dshfusgfb bad	Refunded	Apr 13, 2026	Resolved
#1	Ganga Sapkota	Ganga Sapkota -- Suraksha Sapkota	teacher le padayena ranthro sngavgwdsduisefuejsbfuisef	Refunded	Apr 10, 2026	Resolved

Figure 152: Disputes

6.3 Post Survey

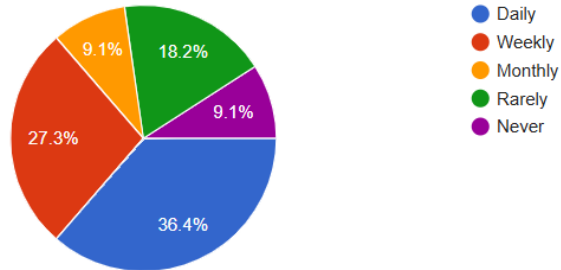
I have created a form with a certain question regarding the platform. I showcased my project to some of the end users and asked them to fill out the form. The response has been stored properly. There are suggestions that I have received, which are below:



How often do you use online learning platforms?

 [Copy chart](#)

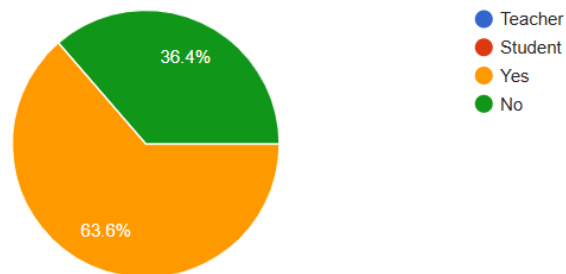
11 responses



What is your role in the system?

 [Copy chart](#)

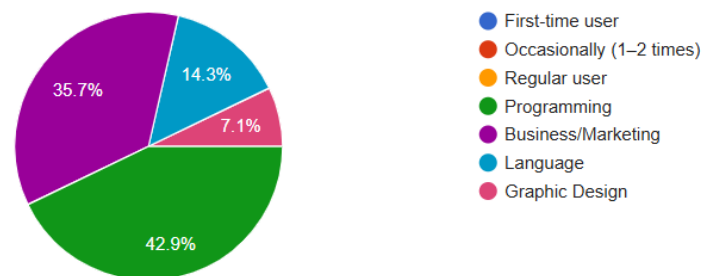
11 responses



How frequently do you use the platform?

 [Copy chart](#)

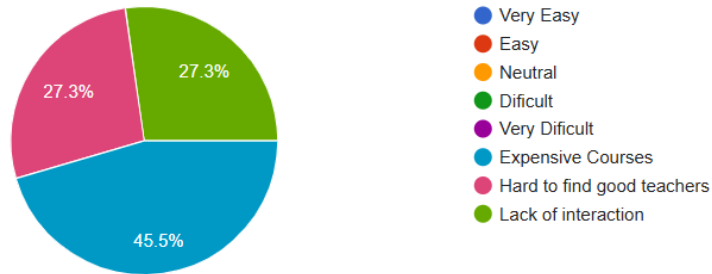
11 responses



How would you rate the ease of registration and login process?

 [Copy chart](#)

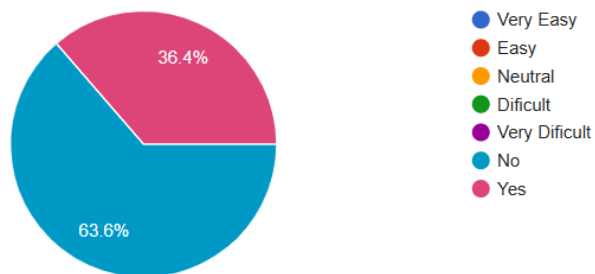
11 responses



How easy is it to navigate through different features of the system?

 [Copy chart](#)

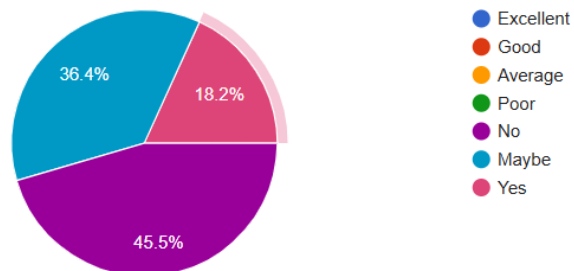
11 responses



How would you rate the overall user interface (design and layout)?

 [Copy chart](#)

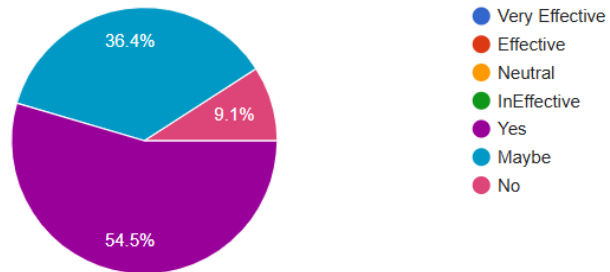
11 responses



How effective is the search functionality (finding mentors or skills)?

 [Copy chart](#)

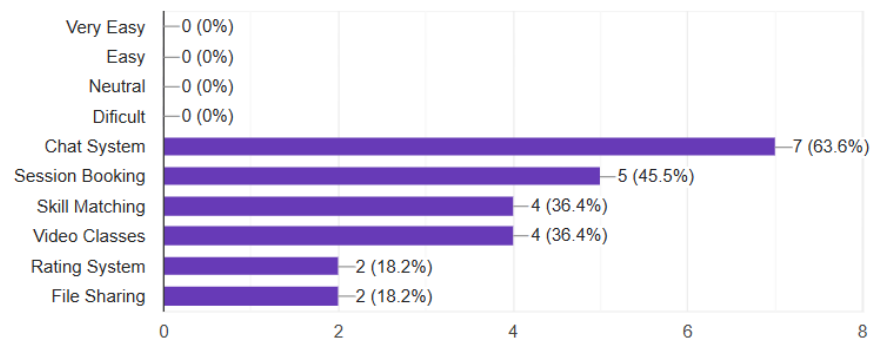
11 responses



How easy is it to send or manage session requests?

 [Copy chart](#)

11 responses



How satisfied are you with the session process (learning/teaching interaction)?

 [Copy chart](#)

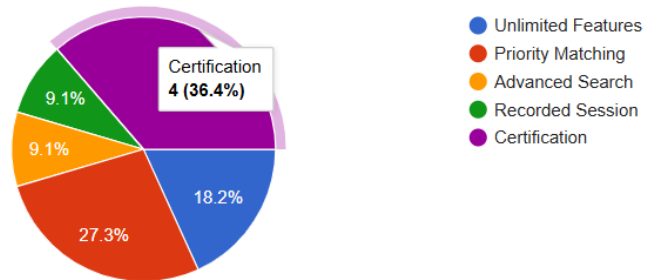
11 responses



What premium features did you like the most?

11 responses

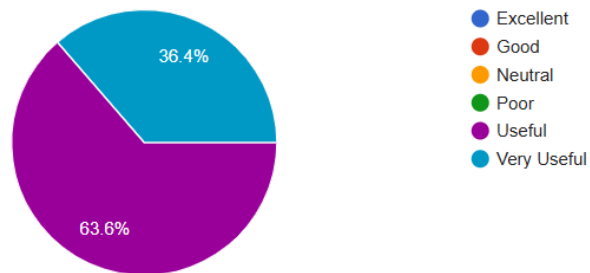
 [Copy chart](#)



How would you rate the performance of the system (speed and responsiveness)?

11 responses

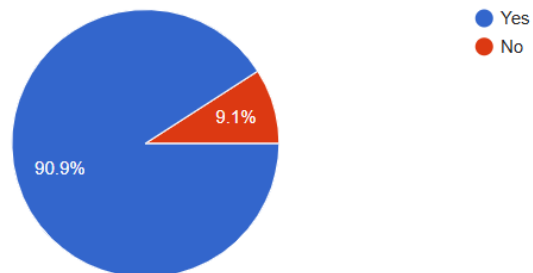
 [Copy chart](#)



Would you recommend this platform to others?

11 responses

 [Copy chart](#)



What improvements or features would you suggest for the system?

11 responses

No

Great

Yes i would use this platform

No its all good

Great!

Nothing

Nothing 🤔

Skill barter system should be introduced

haha

7 Sustainability

7.1 Business Rule

User Registration:

- Users must create an account with a valid email and personal information.
- The password must meet the validation criteria and is encoded in the database.
- The users should choose the role of Student or Teacher when registering.
- The teacher users must indicate further verification information to be given before they are granted.
- To activate the user account, email verification is required.

Access Control and login:

- Only verified users are allowed to log into the system.
- Users can access features based on their role (Student, Teacher, Admin).
- Admin can manage the systems and user activities fully.
- Unauthorized users cannot access secured pages.

Skill Management:

- Users will be able to add the skills they would like to learn or teach.
- Duplicate skill entries are not allowed in the system.
- Skills have to be chosen correctly to facilitate correct matching.
- Users can see and maintain their skill lists.

Session Request and Management.

- Learners can request sessions with teachers.
- Teachers can approve or disapprove session requests.
- Only sessions are created upon request approval.
- Status of the session should be up to date in the system.

Communication (Chat System):

- During sessions, users can send and receive messages.
- Chat messages are stored in the database.
- Previous chat history can be viewed by the users.

Resource Sharing:

- Teachers are able to post learning materials in between sessions.
- The files uploaded should be valid.
- Learners are able to download common materials.

System Notification (Session-Based):

- In-app notifications about session requests and updates are provided to the users.
- Notifications are shown inside the system interface.
- Notifications assist users to monitor activities of a session.

Email Notification:

- Only account verification is done using email.
- Password reset is done via email.
- Contact related communication is done through email.

Payment System:

- Users can upgrade to premium using Khalti payment system.
- Payment has to be confirmed prior to updating user status.
- The database is safe in storing payment records.

Admin Control:

- Admin can handle the users and data in the system.
- Teacher accounts can be approved or monitored by the Admin.
- System activities and sessions can be viewed by admin.

7.2 System Requirement Specifications (SRS)**7.2.1 Introduction**

- Purpose:

This document is a document that outlines the system requirements of SkillXChange, which is a web-based skills barter system where individuals can acquire or give skills without the use of money. The primary idea behind the system is to encourage peer-to-peer-learning, collaboration and knowledge sharing through the provision of safety and orderly environment of skills. This specification is aimed at the developers, testers, supervisors, and the academic reviewers to learn the functionality and limitations of the system.

- Scope
 - User registration and login system.
 - User profile and skill management (skills to teach and learn).
 - Browsing and matching skills between users.
 - Request and approval of skill exchange.
 - User-to-user communication during skill exchange.
 - Post-skilling feedback and rating.
- Definitions, Acronyms, and Abbreviations
 - SRS – Software Requirements Specification
 - UML – Unified Modeling Language
 - DFD – Data Flow Diagram
 - ERD – Entity Relationship Diagram

7.2.2 Overall Description

- Product Perspective:

Skill Barter System is an independent web-based application that is meant to facilitate skill exchange without any monetary transactions. The system is used by the users with the Web interface, and all the information concerning the users, skills, exchanges, and feedback is stored and managed in the central database. The system is client-server based on the MVC pattern of design.

- User Classes and Characteristics
 - Admin
 - Manage user
 - Manage skills
 - Monitor activity
 - Handle policy
 - User
 - Register and Log In
 - Manage profile
 - Adds skills to teach and learn
 - Skill exchange
 - Provide feedback
- Operating Environment
 - Web browsers
 - Internet connection
 - Server with PHP (Laravel Framework)
 - MySQL database
- Design and Implementation Constraints
 - System must be web-based
 - Internet connection required
 - Data must store
 - Only registered users can access system features

7.2.3 Functional Requirements

- User Authentication
 - The system should enable its users to register using valid information.
 - The system will enable users to log in with email/username and password.
 - The system must ensure that the credentials provided by the user are validated before access is allowed.
- Profile and Skill Management
 - The system should enable users to create and modify their profiles.
 - The system should enable users to enter, modify, and remove skills.
 - The system should be capable of storing skill information like category and description.
- Skill Browsing and Matching
 - The system should enable browsing skills by users.
 - The system should support searching for skills based on categories or keywords.
 - Users will see relevant skill correlations within the system.
- Skill Exchange Requests
 - The system will enable the sending of skill exchange requests.
 - It shall be possible in this system for a user to accept or reject requests.
 - The system should record the status of every exchange.
- Communication
 - The system should enable users to communicate within the system.
 - Messages will be associated with exchange sessions.
- Feedback and Rating
 - The system shall enable feedback from users following an exchange.
 - The system shall be capable of storing ratings based on user reliability.
 - Admin should be able to view feedback.

7.2.4 Non-functional Requirements

- Performance
 - The system will take 2-3 seconds to load pages.
 - The system must support several users at once.
- Security
 - Password must be encrypted for users
 - Role-Based Access Control must be implemented by the system.
 - Secure protocols (HTTPS) must be used for information transport.
- Usability
 - The system should be equipped with an easy-to-use interface
 - Both desktop computers and mobile phones should be able to access the system.
 - The error messages will be simple to read and comprehend.
- Reliability
 - The data integrity will be supported by the system.
 - Minor errors will be properly corrected by the system.

7.2.5 Future Enhancements

- AI-based skill matching
- Gamification and rewards system
- Premium membership
- Advanced analytics & reporting

7.3 Business Plan

7.3.1 Estimation of costs based on the COCOMO Model.

To determine the effort, the time and the cost needed to develop the Skill Barter System, COCOMO (Constructive Cost Model) is utilized. It is an Organic Project, as it is developed on the individual level with the help of familiar technologies and the complexity of the system is medium. Organic projects typically have small teams and loose requirements and are less complicated to control.

Effort Estimation Formula

Effort (Person-Months) = $a \times (\text{KLOC})^b$.

Where:

$$a = 2.4$$

$$b = 1.05$$

The estimated size of the project is around 30,000 lines of code, so:

$$\text{KLOC} = 30$$

$$\text{Effort} = 2.4 \times (30)^{1.05}$$

$$\text{Effort} = 2.4 \times 33.8$$

$$\text{Effort} \approx 81.12 \text{ Person-Months}$$

Development Time Formula

Time (months) = $c (\text{Effort})^d$.

Where:

$$c = 2.5$$

$$d = 0.38$$

$$\text{Time} = 2.5 \times (81.12)^{0.38}$$

$$\text{Time} = 2.5 \times 5.5$$

$$\text{Time} \approx 13.75 \text{ months}$$

Cost Estimation

Assuming the salary of the developers = Rs. 25,000/month

$$\text{Cost} = 13.75 \times 25,000$$

$$\text{Cost} \approx \text{Rs. } 3,43,750$$

- **Hosting Cost**

The system needs to be hosted on the web in order to be available online. Better scalability and better performance are preferred to cloud hosting. The hosting fee is approximately to be of Rs. 30,000 – 40,000 per year, depending on the service provider.

- **Domain Registration**

The platform needs a domain name in order to be available to users. Provided there is a domain of .com.np, it will be free otherwise a .com domain may be registered. It would cost approximately Rs. 1500 – 3000 per year.

- **Maintenance Cost**

Bug fixing, updates and database management are all part of maintenance. Periodic maintenance can guarantee the stability and performance of the system. The approximate price will be Rs. 10,000–20,000 per year.

7.3.2 Organic and Inorganic Growth Plan

- Organic Growth Plan

The platform can be promoted via social media and referrals by the users. Users will be encouraged to invite others because of a positive user experience. User feedback may be used to enhance system features. On-going enhancement aids in creating trust among users. This results in gradual growth of the platform.

- Inorganic Growth Plan

Paid advertisements on such websites as Google and Facebook can be used to promote the system. New users can be attracted by special offers like free premium access. Cooperation with educational institutions can raise the platform popularity. A bigger audience can be reached with the help of marketing campaigns. This helps in accelerated development of the system.

8 Testing

8.1 Incremental 1 Testing

8.1.1 User Authentication:

Register:

- Test Case 1: Role Selection Required

Table 3: Test1: Registration without role

Item	Description
Objective	To verify, the system needs role selection.
Action	Enter without selecting a role.
Expected Outcome	The system should display an error message.
Actual Result	An error message has been displayed.
Conclusion	Success

Figure 153: Role based registration



Create Account

Suhana Subedi

suhana@yopmail.com

Password

Confirm Password

Select your role:

☐ Student - I want to learn new skills

☐ Teacher - I want to teach skills

The role field is required.

Register

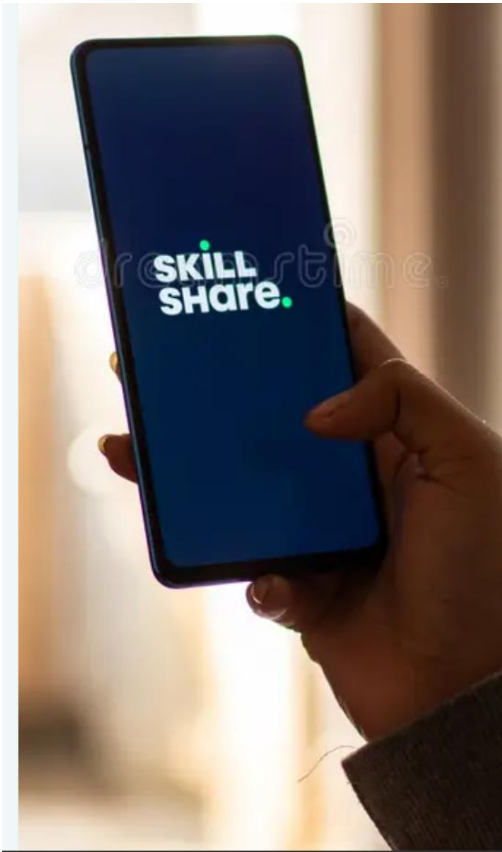
[Already have an account?](#)

Figure 154: Trying registration without selecting role

- Test Case 2: Teacher Role Displays Additional Fields

Table 4: Test2 Teacher role's fields

Item	Description
Objective	To verify that the select the Teacher role will show more document upload fields.
Action	During registration, choose Teacher - I want to teach skills
Expected Outcome	Select "Teacher – I want to teach skills" during registration.
Actual Result	Fields to upload CV, Certificate, and Citizenship should be displayed.
Conclusion	Success



Confirm Password

Select your role:

☐ Student - I want to learn new skills

☒ Teacher - I want to teach skills

Teacher Verification Documents

Khalti ID (Phone Number)

Upload CV (PDF, DOCX)

No file chosen

Upload Certificate (PDF, Image)

No file chosen

Upload Citizenship (PDF, Image)

No file chosen

Register

[Already have an account?](#)

Figure 155: Showing teacher's additional fields

- Test Case 3: Teacher Must Upload Required Documents

Table 5: Test3 Upload required documents

Item	Description
Objective	To verify that the system shows an error when the user tries to register as a teacher without uploading a document
Action	Trying to register without uploading a CV, certificate, and citizenship.
Expected Outcome	Select "Teacher – I want to teach skills" during registration.
Actual Result	An error message has been displayed.
Conclusion	Success



Select your role:

☐ Student - I want to learn new skills
☒ Teacher - I want to teach skills

Teacher Verification Documents

9827472492

Upload CV (PDF, DOCX)

Choose File No file chosen

The cv field is required when role is teacher.

Upload Certificate (PDF, Image)

Choose File No file chosen

The certificate field is required when role is teacher.

Upload Citizenship (PDF, Image)

Choose File No file chosen

The citizenship field is required when role is teacher.

Register

Figure 156: Teacher registration without documents

- Test case 4: Teacher Registration

Table 6: Test 4 Teacher registration

Item	Description
Objective	To verify that the system allows a user to register as a teacher with all the necessary documents and fields.
Action	Select "Teacher", fill in all required details & document and submit the registration form.
Expected Outcome	The system must be able to create a user as a teacher.
Actual Result	The user was created successfully and stored in the database.
Conclusion	Success



.....

.....

Select your role:

☐ Student - I want to learn new skills

☒ Teacher - I want to teach skills

Teacher Verification Documents

092738675264

Upload CV (PDF, DOCX)

CDL.pdf

Upload Certificate (PDF, Image)

SD.drawio.png

Upload Citizenship (PDF, Image)

JournalExport_20260125_211533.pdf

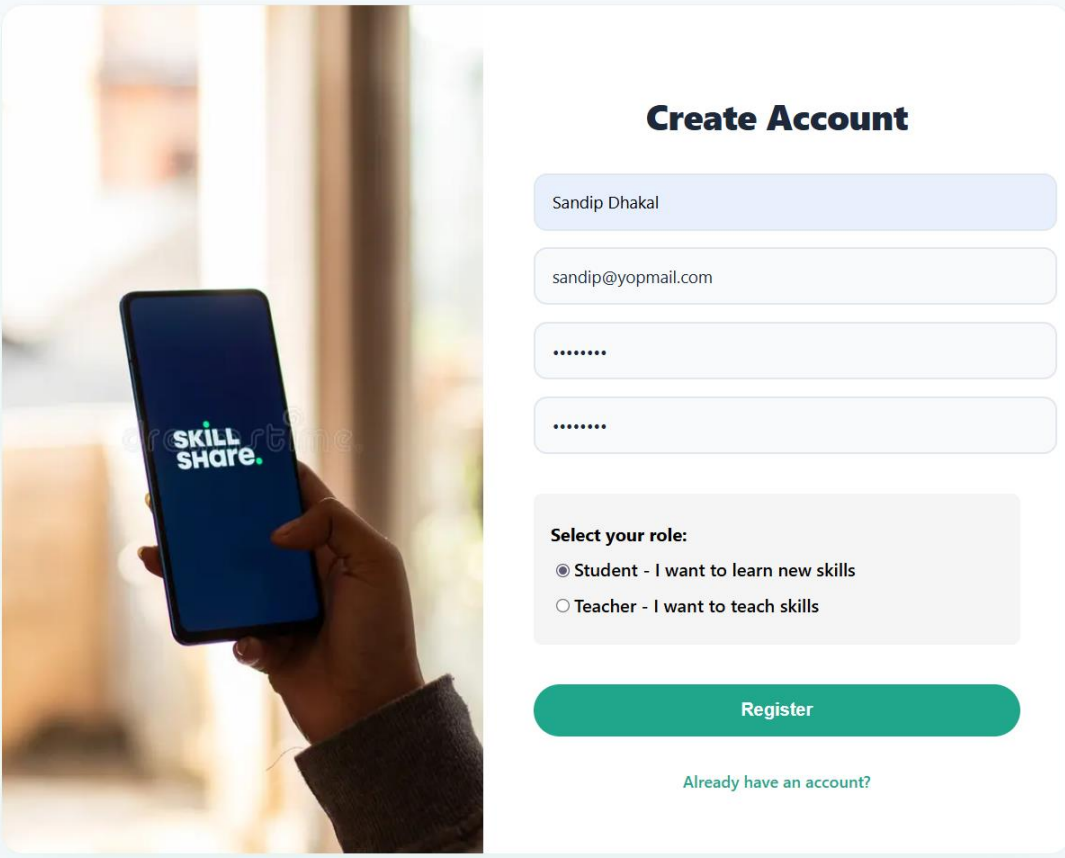
	id	name	email	phone	email_verified_at	password	provider	provider_id	provider_token	provider_type	oauth_synced_at	role	bio	avatar	remember_token
te	8	Suhana Subedi	suhana@yopmail.com	NULL	2026-04-13 17:20:27	\$2y\$10\$FZyBwxZlfl6Pi	NULL	NULL	NULL	NULL	NULL	teacher	NULL	NULL	NULL
te	7	Ganga Sapkota	ganga@yopmail.com	NULL	2026-03-22 12:37:39	\$2y\$10\$ZApnZUezeoi	NULL	NULL	NULL	NULL	NULL	student	NULL	NULL	NULL
te	6	Suraksha Sapkota	surakshyasapkota14@gmail.com	NULL	2026-03-22 12:34:03	\$2y\$10\$ROwjh7Tp1xd	NULL	NULL	NULL	NULL	NULL	teacher	NULL	NULL	NULL

Figure 157: Teacher registration

- Test case5: Student Registration

Table 7: Test 5 Student registration

Item	Description
Objective	To verify that the system allows a user to register as a student.
Action	Select "Student", fill in all required details, and submit the registration form.
Expected Outcome	The system must be able to create a user as a student.
Actual Result	The user was created successfully and stored in the database.
Conclusion	Success



Create Account

Sandip Dhakal

sandip@yopmail.com

.....

.....

Select your role:

☒ Student - I want to learn new skills

☐ Teacher - I want to teach skills

Register

[Already have an account?](#)

Figure 158: Insert value to successful registration

- Verified data storage in the database.

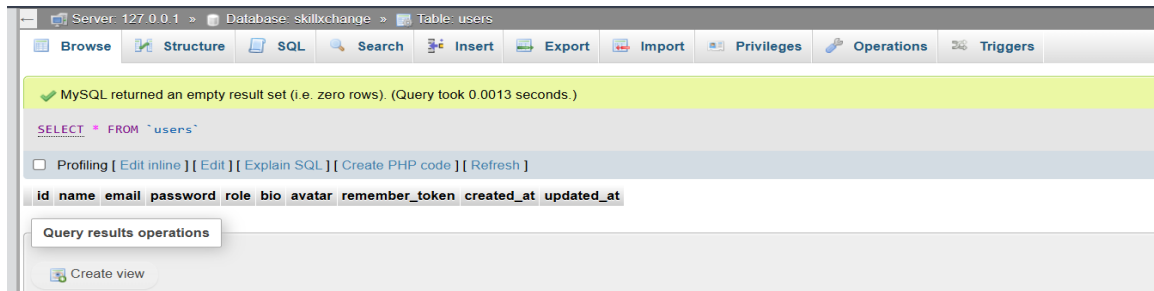


Figure 159: Before Registration

	id	name	email	phone	email_verified_at	password	provider	provider_id	provider_token	provider_ref	role	oauth_synced_at	bio	avatar
to	9	Sandip Dhakal	sandip@yopmail.com	NULL	2026-04-13 17:56:38	\$2y\$10\$8xwplaWAZZc	NULL	NULL	NULL	NULL	student	NULL	NULL	NULL
te	8	Suhana Subedi	suhana@yopmail.com	NULL	2026-04-13 17:20:27	\$2y\$10\$FZyBwxZlI6Pik	NULL	NULL	NULL	NULL	teacher	NULL	NULL	NULL
te	7	Ganga Sapkota	ganga@yopmail.com	NULL	2026-03-22 12:37:39	\$2y\$10\$ZApnZUeZeoit	NULL	NULL	NULL	NULL	student	NULL	NULL	NULL
te	6	Suraksha Sapkota	surakshyasapkota14@gmail.com	NULL	2026-03-22 12:34:03	\$2y\$10\$ROwjh7Tp1xd	NULL	NULL	NULL	NULL	teacher	NULL	NULL	NULL

Figure 160: After Registration

Email Verification:

- Test case 6: Verification required in registration

Table 8: Test 6 verification

Item	Description
Objective	To verify that the system needs email verification after registration.
Action	Fill in all registration details correctly and submit the form.
Expected Outcome	The system will require a verification code, which will be sent to the email of the user.
Actual Result	Verification code prompt successfully displayed.
Conclusion	Success

Secure Your Access

One final step to join the SkillBarter community. Verify your identity to protect your account and start trading skills.

Verify Email

We've sent a unique 6-digit code to your inbox to ensure your account security.

Code expires in 2 minutes

suhana@yopmail.com

ENTER 6-DIGIT CODE

000000

Complete Verification

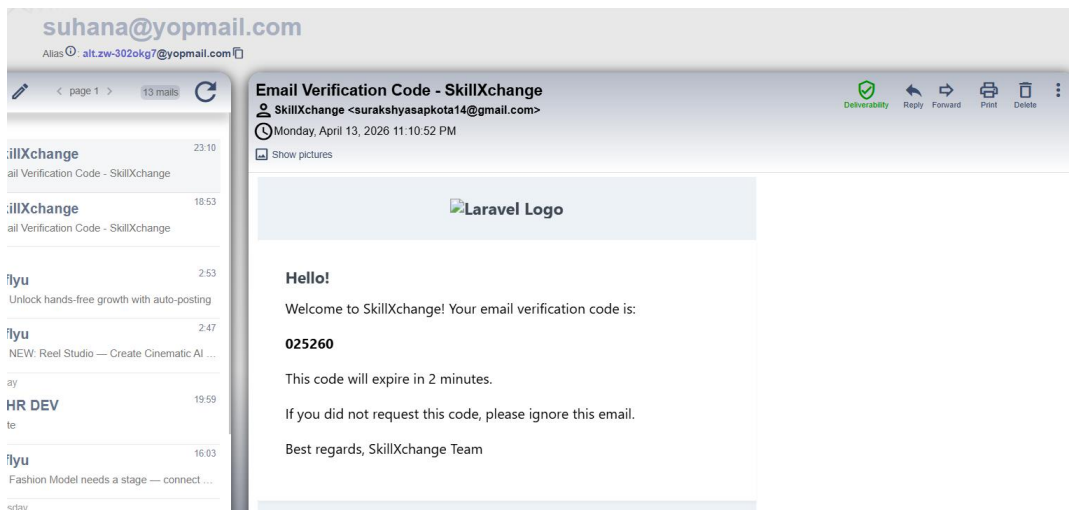
Didn't receive a code? Please check your spam folder or request a new one.

Resend New Code

- Test Case 7: Verification Code Sent to Email

Table 9: Test7 verification code received

Item	Description
Objective	To verify that a confirmation code has been sent to the registered email.
Action	Complete the registration form and check your email inbox.
Expected Outcome	A verification code is to be received by email.
Actual Result	Verification code received successfully.
Conclusion	Success



- Test Case 8: Incorrect Verification Code

Table 10: Test8Incorrect verification code

Item	Description
Objective	To verify the system's behavior in case of entering an incorrect verification code.
Action	Enter the wrong verification code.
Expected Outcome	The system should show an error message and reject verification.
Actual Result	An error message has been displayed.
Conclusion	Success

Secure Your Access

One final step to join the SkillBarter community. Verify your identity to protect your account and start trading skills.

Verify Email

We've sent a unique 6-digit code to your inbox to ensure your account security.

Code expires in 2 minutes

suhana@yopmail.com

The verification code is invalid or has expired.

ENTER 6-DIGIT CODE

000000

Complete Verification

Didn't receive a code? Please check your spam folder or request a new one.

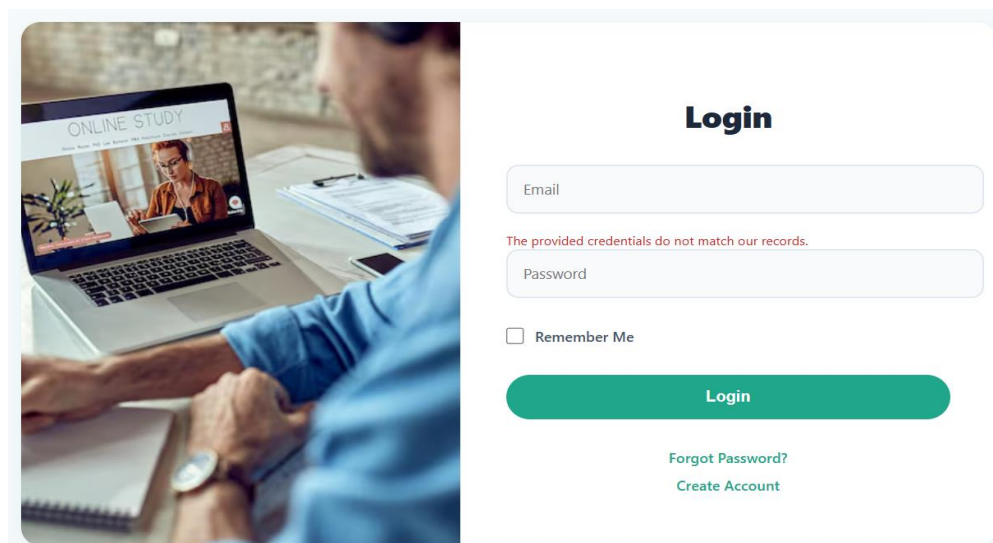
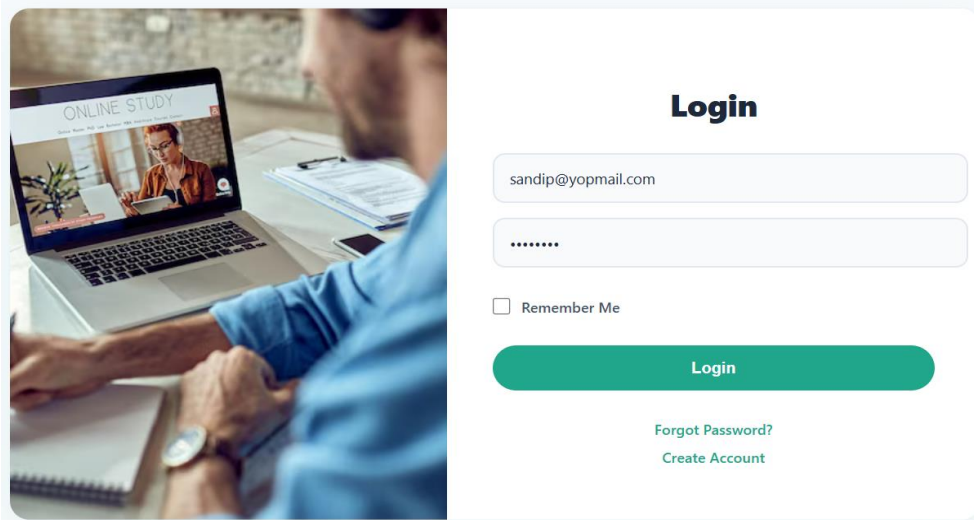
Resend New Code

Login:

- Test case 9: Login block without verification email

Table 11: Test 9 Login failed without verification email

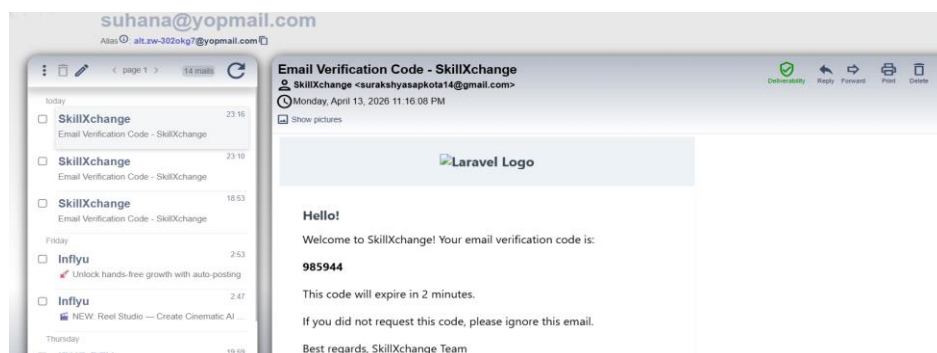
Item	Description
Objective	To verify that the system cannot be accessed by an unverified user.
Action	Users try to log in without verifying their email.
Expected Outcome	A system should be through error message.
Actual Result	An error message has been displayed.
Conclusion	Success



- Test case 10: Resend verification code received successfully

Table 12: Test 10 Resend code

Item	Description
Objective	To verify that the user can request a new verification code.
Action	Click the "Resend Code" option.
Expected Outcome	New verification code must be sent to email.
Actual Result	New code was received successfully.
Conclusion	Success



- Test case 11: Successful Email Verification and login

Table 13: Test 11 Successfully email verification

Item	Description
Objective	To verify that user can register after the correct verification code.
Action	Enter the correct verification code.
Expected Outcome	User accounts should be verified and registration completed.
Actual Result	Account verified successfully and redirect to login.
Conclusion	Success

	id	name	email	phone	email_verified_at	password	provider	provider_id	provider_token	provider_ref	role	oauth_synced_at	bio	avatar
to	9	Sandip Dhakal	sandip@yopmail.com	NULL	2026-04-13 17:56:38	\$2y\$10\$8xwptaWAZc	NULL	NULL	NULL	NULL	student	NULL	NULL	NULL
to	8	Suhana Subedi	suhana@yopmail.com	NULL	2026-04-13 17:20:27	\$2y\$10\$FZyBwxZl6Pik	NULL	NULL	NULL	NULL	teacher	NULL	NULL	NULL
to	7	Ganga Sapkota	ganga@yopmail.com	NULL	2026-03-22 12:37:39	\$2y\$10\$ZApnZUeZeoit	NULL	NULL	NULL	NULL	student	NULL	NULL	NULL
to	6	Suraksha Sapkota	surakshyasapkota14@gmail.com	NULL	2026-03-22 12:34:03	\$2y\$10\$ROwjh7Tp1xd	NULL	NULL	NULL	NULL	teacher	NULL	NULL	NULL



Login

Email

Password

☐ Remember Me

Login

[Forgot Password?](#)
[Create Account](#)

- Test case 12: login with incorrect credentials.

Table 14: Test12 Login with wrong credentials

Item	Description
Objective	To verify that the system rejects invalid credentials
Action	Enter an incorrect email or password into the login form.
Expected Outcome	The system should display an error message.
Actual Result	An error message has been shown.
Conclusion	Success

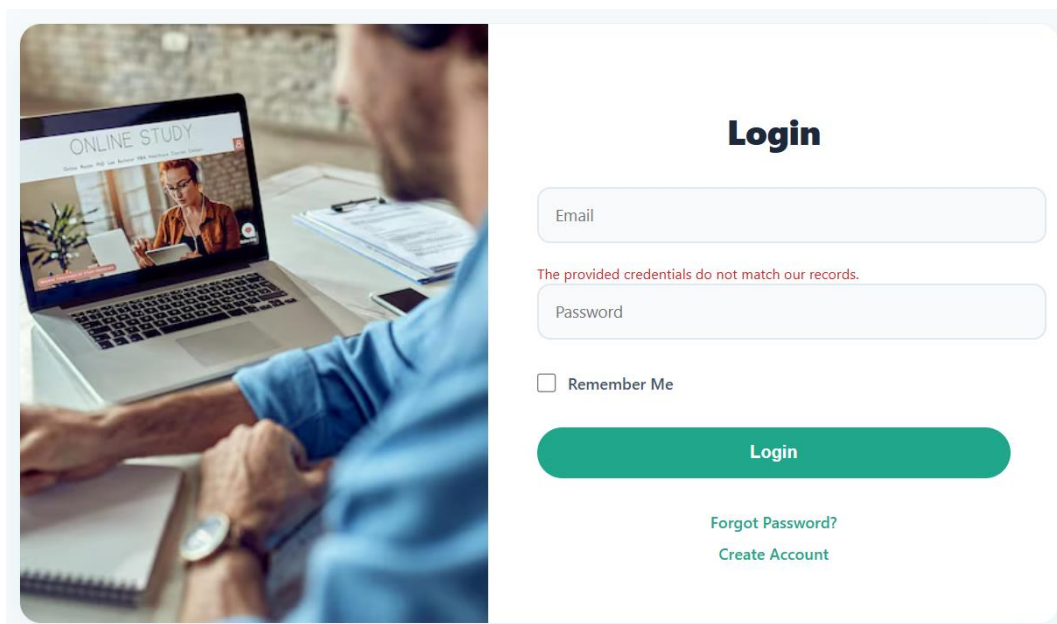


Figure 161: Testing with incorrect values

- Test case 13: login with correct credentials.

Table 15: Test13 Login with correct credentials

Item	Description
Objective	To verify that a new user can login successfully.
Action	Enter valid user details into the login form.
Expected Outcome	The system should authenticate the user and redirect it to the homepage.
Actual Result	The user is logged in.
Conclusion	Success

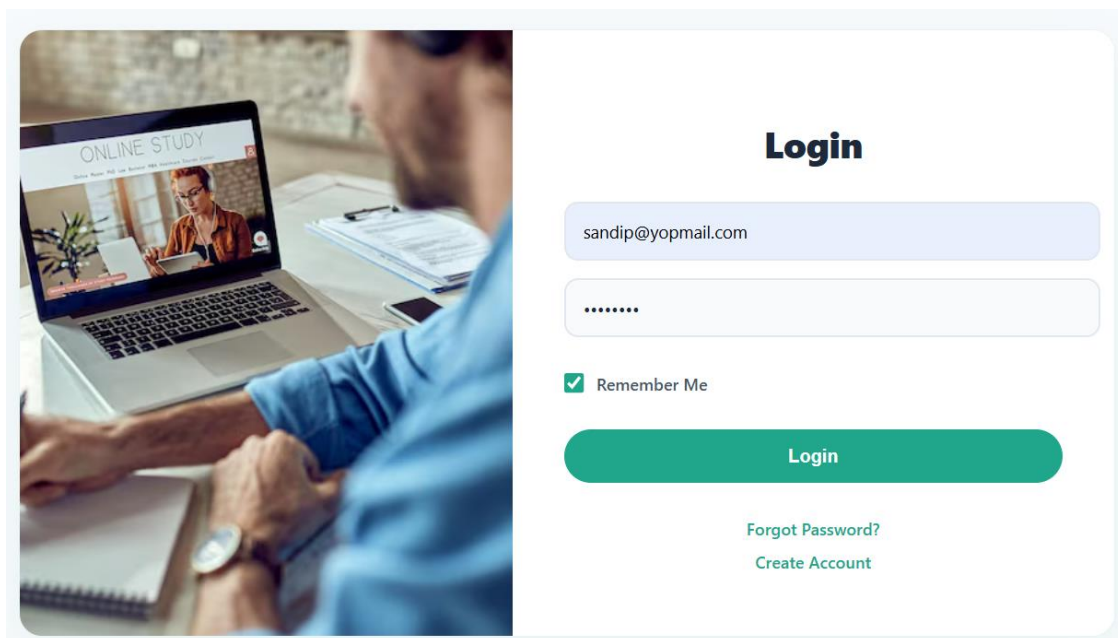


Figure 162: Insert correct value for testing

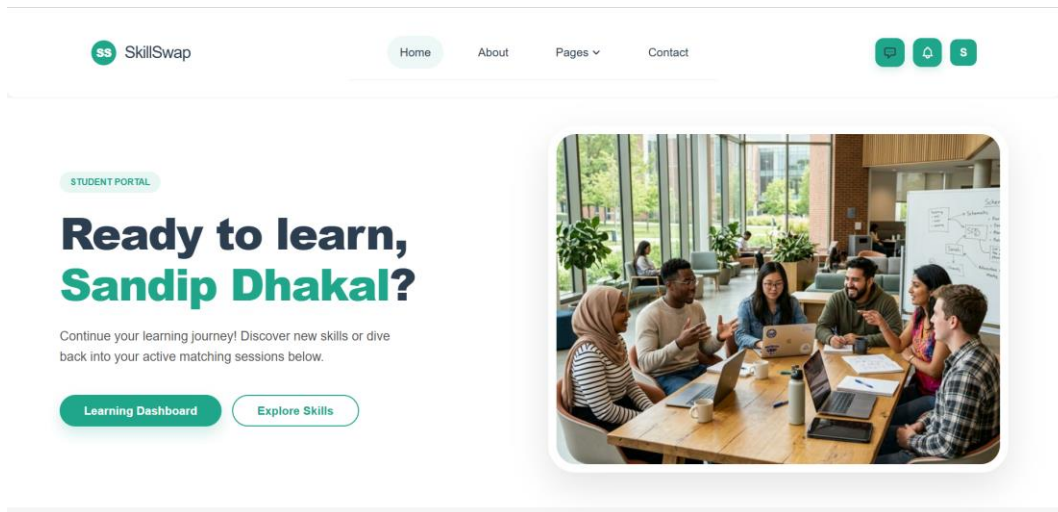


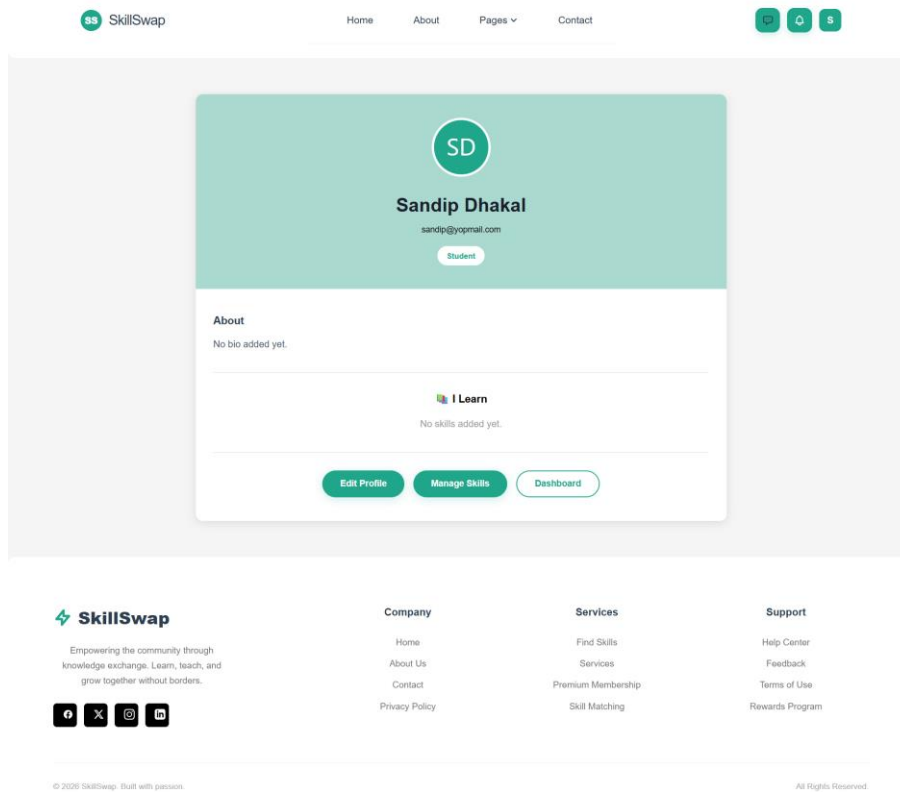
Figure 163: Home page after login

8.1.2 Profile Management Testing

- Test Case 14: View Profile

Table 16: Test14 View profile

Item	Description
Objective	To verify that users are able to see their profile information
Action	Navigate to the user's profile page.
Expected Outcome	The user information must be displayed properly.
Actual Result	The profile information is displayed.
Conclusion	Success



- Test case 15: Update profile

Table 17: Test15 Updating user details

Item	Description
Objective	To verify that users are able to update their profile information
Action	Edit profile details and save changes.
Expected Outcome	Updated information should be saved and displayed.
Actual Result	The profile information was updated successfully.
Conclusion	Success

Edit Profile

Name
Sandp Dhakal

Email
sandp@yopmail.com

Phone Number
9800000000
This number will be used for payments.

Bio (optional)

Profile Picture (Avatar)
Choose File | No file chosen

[Save Changes](#) [Cancel](#)

SkillSwap
Empowering the community through knowledge exchange. Learn, teach, and grow together without borders.

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Services
Premium Membership
Skill Matching

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Help Center
Feedback
Terms of Use
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Figure 164: Edit profile page

SkillSwap Home About Pages Contact

Edit Profile

Name
Sandip Dhakal

Email
sandip@vpmat.com

Phone Number
9827472492
This number will be used for payments.

Bio (optional)
this is just for the testing

Profile Picture (Avatar)
Choose File No file chosen

Save Changes Cancel

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Figure 165: Updating User details

SkillSwap Home About Pages Contact

✓ Profile updated successfully!

SD

Sandip Dhakal
sandip@vpmat.com
Student

About
this is just for the testing

I Learn
No skills added yet.

Edit Profile Manage Skills Dashboard

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Figure 166: Successfully updated

- Test Case 16: User's profile Picture (Avatar) Upload

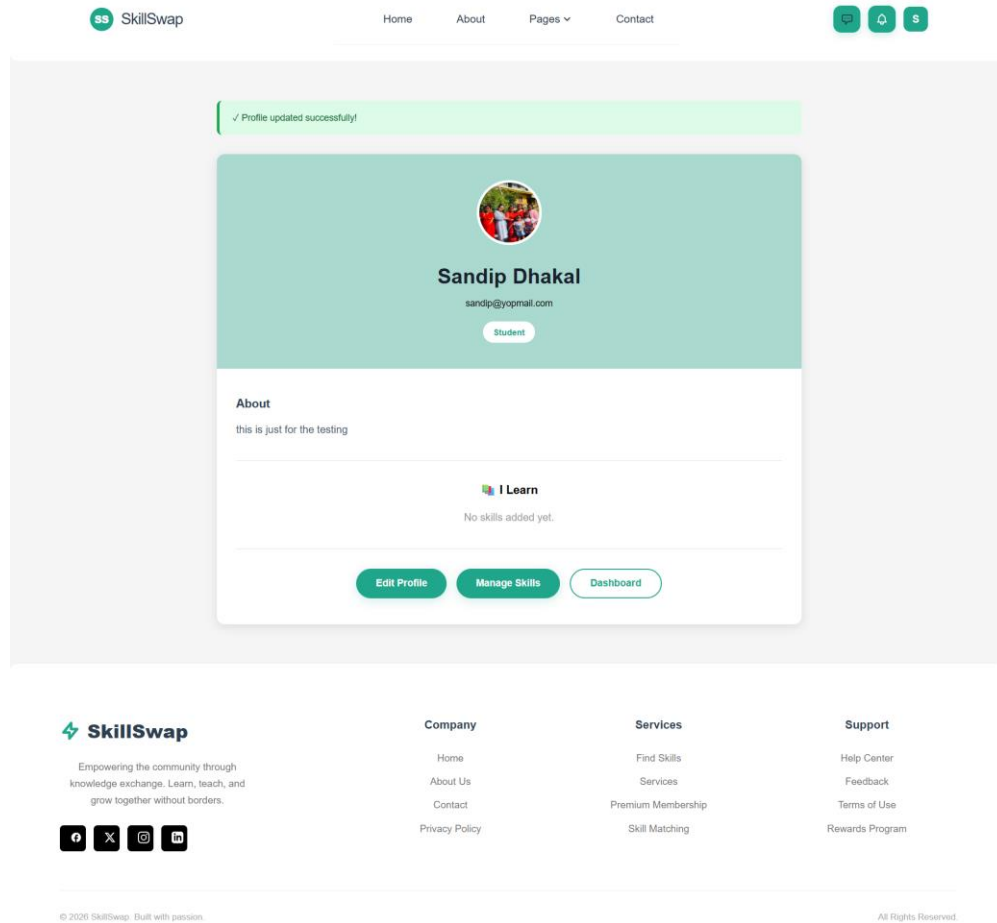
Table 18: Test16 updating user's profile picture

Item	Description
Objective	To verify that users are able to upload their profile picture
Action	upload profile picture and save changes.
Expected Outcome	Profile picture should be saved and displayed.
Actual Result	The profile picture was uploaded successfully.
Conclusion	Success

The screenshot shows the 'Edit Profile' form on the SkillSwap website. The form is centered on a light gray background. At the top, there is a navigation bar with the SkillSwap logo, links for Home, About, Pages, and Contact, and three circular icons on the right. The form itself has a white background and a rounded border. It contains the following fields:

- Name:** A text input field with the value 'Sandip Dhakal'.
- Email:** A text input field with the value 'sandip@yopmail.com'.
- Phone Number:** A text input field with the value '9827472492'. Below this field is a small note: 'This number will be used for payments'.
- Bio (optional):** A text input field with the value 'this is just for the testing'.
- Profile Picture (Avatar):** A section with a 'Choose File' button and a file path: 'hsMuGm6XLCV1rTcoymoyxgF0h...plaintext_638338112829797990.jpg'.

At the bottom of the form are two buttons: 'Save Changes' (a green button) and 'Cancel' (a white button with a green border). Below the form, the footer of the website is visible, featuring the SkillSwap logo, a tagline 'Empowering the community through knowledge exchange. Learn, teach, and grow together without borders.', social media icons, and four columns of links: Company (Home, About Us, Contact, Privacy Policy), Services (Find Skills, Services, Premium Membership, Skill Matching), and Support (Help Center, Feedback, Terms of Use, Rewards Program).



8.2 Increment 3

8.2.1 Skill Management Testing

- Test Case 17: Add Teaching Skill

Table 19: Test 17 Adding skill by the teacher

Item	Description
Objective	To verify that teachers can add skills they would like to teach
Action	Select skill and choose "Skills I Teach".
Expected Outcome	Skills should be added to the teaching list.
Actual Result	Added Skill successfully.
Conclusion	Success

wap Home About Pages ▾ Contact

My Skills

Manage the skills you teach

+ Add a Skill

Select Skill *

-- Choose Skill -- ▾

Type

Skills I Teach

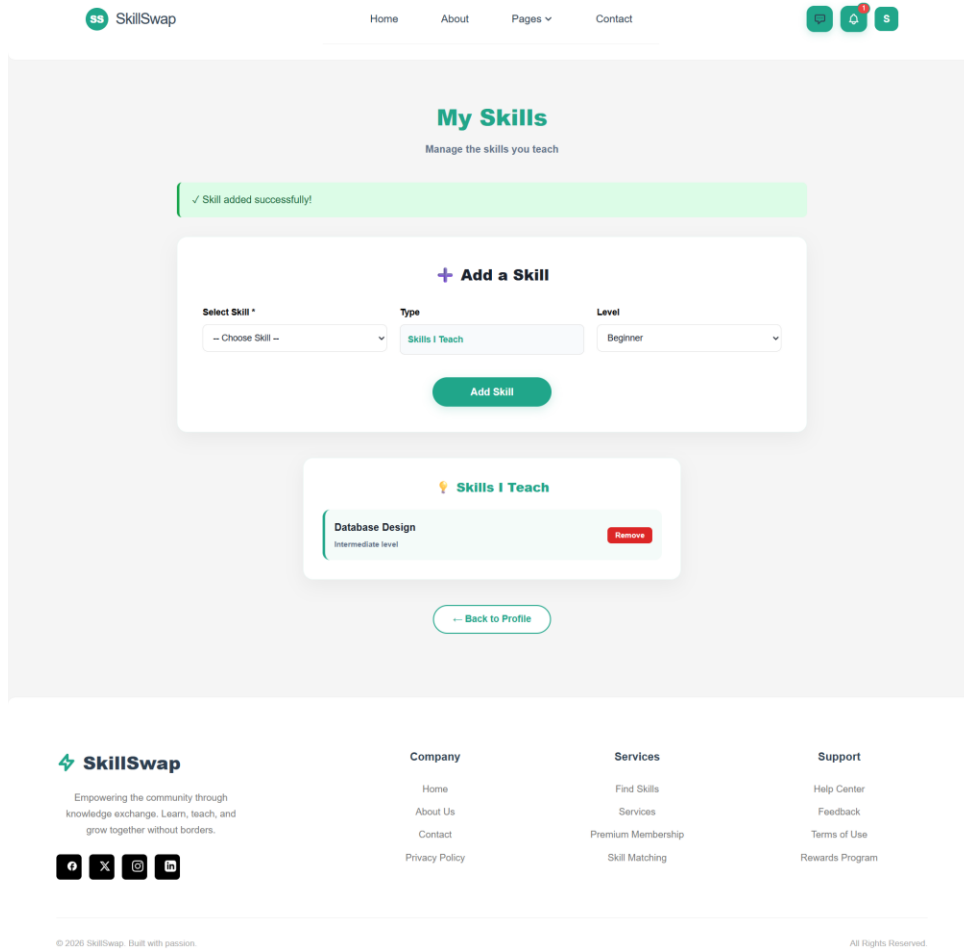
Level

Beginner ▾

Add Skill

💡 Skills I Teach

No teaching skills added yet.



- Test Case 18: Add Learning Skill

Table 20: Test 18 Adding skill by the student

Item	Description
Objective	To verify that students can add skills they would like to teach
Action	Select a skill and choose "Skills I want to learn".
Expected Outcome	Skills should be added to the student list.
Actual Result	Added Skill successfully.
Conclusion	Success

The screenshot shows the SkillSwap website's 'My Skills' page. At the top, there's a navigation bar with 'Home', 'About', 'Pages', and 'Contact'. Below this, the 'My Skills' section is titled 'Manage the skills you want to learn'. A central white card contains the '+ Add a Skill' form. This form has three dropdown menus: 'Select Skill *' (showing '-- Choose Skill --'), 'Type' (showing 'Skills I Want to Learn'), and 'Level' (showing 'Beginner'). Below these is a green 'Add Skill' button. Underneath the form is a section titled 'Skills I Want to Learn' with the text 'No learning skills added yet.' and a green button labeled '-- Back to Profile'. The footer contains the SkillSwap logo, a mission statement, social media icons, and four columns of links: Company (Home, About Us, Contact, Privacy Policy), Services (Find Skills, Services, Premium Membership, Skill Matching), and Support (Help Center, Feedback, Terms of Use, Rewards Program).

My Skills
Manage the skills you want to learn

✓ Skill added successfully!

+ Add a Skill

Select Skill *
-- Choose Skill --

Type
Skills I Want to Learn

Level
Beginner

Add Skill

Skills I Want to Learn

Canva Design
Beginner level

Remove

-- Back to Profile

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- Test Case 19: Prevent Duplicate Skill

Table 21: Test 19 Trying to add same skill again

Item	Description
Objective	To verify that duplicate skills are not allowed.
Action	Add same skill again.
Expected Outcome	Error should be shown in system.
Actual Result	An error message has been shown.
Conclusion	Success.

Manage the skills you teach

+ Add a Skill

Select Skill *

Adobe Illustrator ▾

Type

Skills I Teach

Level

Beginner ▾

Add Skill

 Skills I Teach

Adobe Illustrator

Beginner level

Remove

API Development

Beginner level

Remove

My Skills

Manage the skills you teach

 You already have this skill listed.**+ Add a Skill**

Select Skill *

-- Choose Skill -- ▾

Type

Skills I Teach

Level

Beginner ▾

Add Skill

8.2.2 Search Testing

Search Mentor:

- Test Case 20: Search Mentor by Name

Table 22: Test 20 Search Mentor by name

Item	Description
Objective	To verify that students can search for the teacher by their name.
Action	Enter the teacher's name and enter.
Expected Outcome	The matching teacher profile should be shown.
Actual Result	The matching teacher profile displayed successfully.
Conclusion	Success.

Find Your Mentor

Browse our elite directory of skilled teachers ready to share their expertise.

All Categories ▾

SEARCH



Suhana Subedi

NEW TEACHER

Ready to share my knowledge and help you grow in your chosen skill path.

DATABASE DESIGN

VIEW PROFILE

- Test Case 21: Search Mentor by Name with Invalid Input

Table 23: Test 21 Search by invalid name

Item	Description
Objective	To verify behavior when searching for an invalid teacher's name.
Action	Enter the teacher's name that does not exist.
Expected Outcome	The system should show no teacher matching your search.
Actual Result	The message was displayed successfully.
Conclusion	Success.

Find Your Mentor

Browse our elite directory of skilled teachers ready to share their expertise.

All Categories ▾ SEARCH



No teachers matching your search

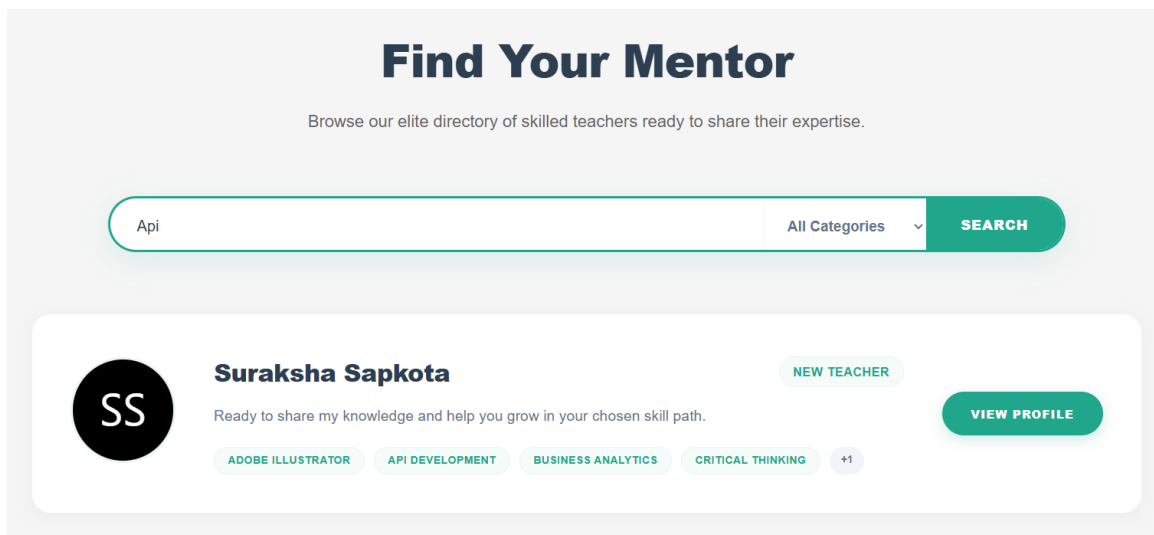
Try different keywords or browse all teachers.

Show All Teachers

- Test case 22: Search mentor by skills

Table 24: Test 22 Search mentor by skill

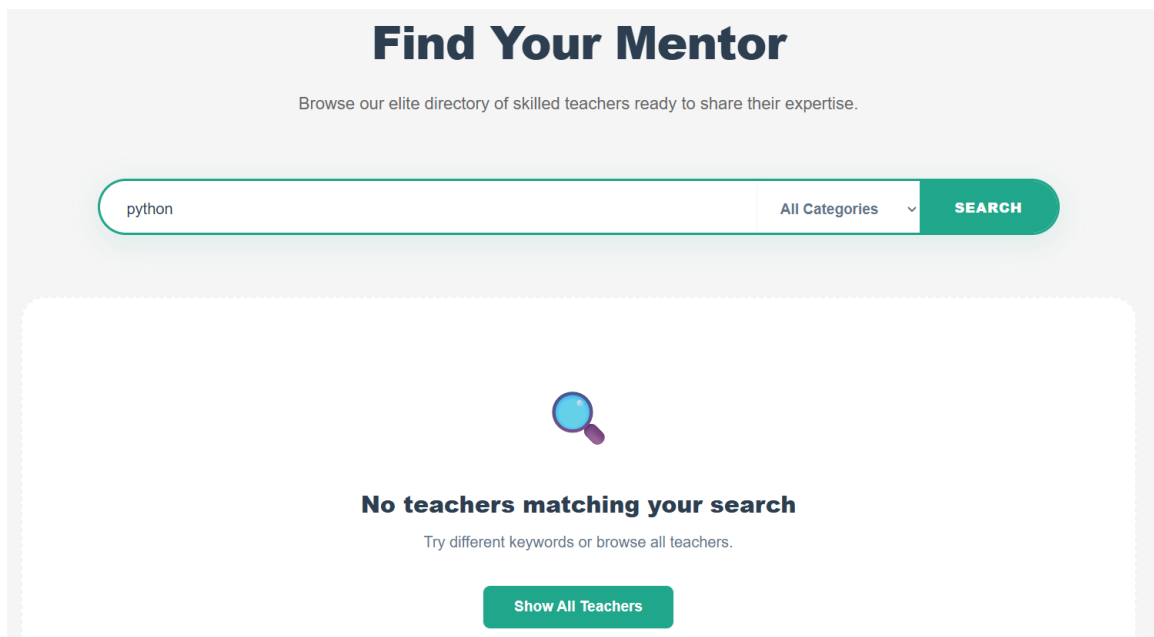
Item	Description
Objective	To verify that students can search for the teacher by their skills.
Action	Enter the skill and search.
Expected Outcome	The teacher who matches that skill should be shown.
Actual Result	The matching teacher displayed successfully.
Conclusion	Success.



- Test Case 23: Search Mentor by Skill with No Available Teacher

Table 25: Test 23 Search mentor by invalid skill

Item	Description
Objective	To verify behavior, when no teacher has the selected skills.
Action	Enter the skill that is not available in the teacher profile.
Expected Outcome	The system should show no teacher matching your search.
Actual Result	The message displayed successfully.
Conclusion	Success.



- Test Case 24: Search mentor by category

Table 26: Test 24 Search mentor by category

Item	Description
Objective	To verify that students can search for the teacher by their skills.
Action	Enter the skill and enter.
Expected Outcome	The teacher who matches that skill should be shown.
Actual Result	The matching teacher displayed successfully.
Conclusion	Success.

Find Your Mentor

Browse our elite directory of skilled teachers ready to share their expertise.

Business

SEARCH



Suraksha Sapkota

Ready to share my knowledge and help you grow in your chosen skill path.

ADOBE ILLUSTRATOR

API DEVELOPMENT

BUSINESS ANALYTICS

CRITICAL THINKING

+1

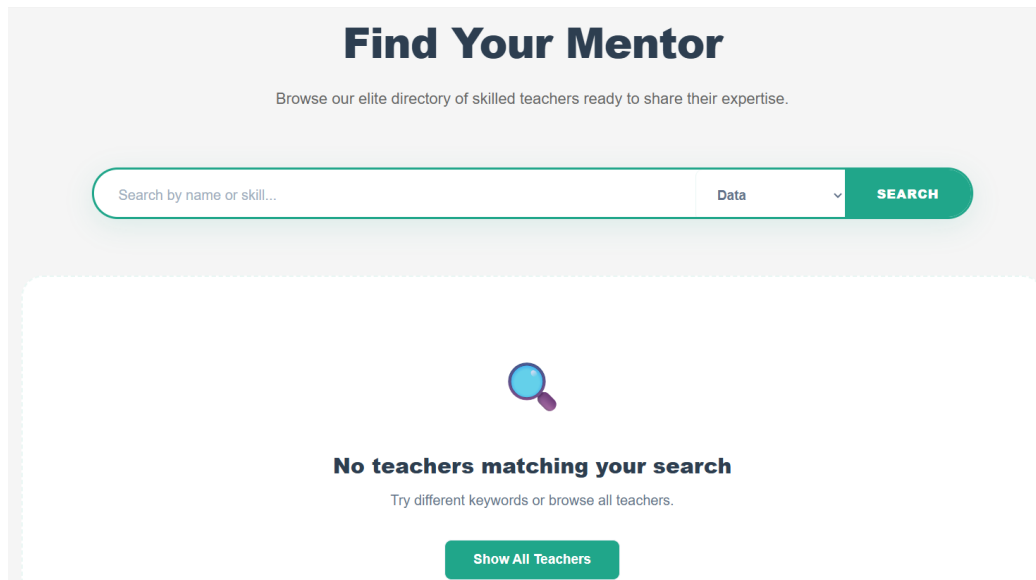
NEW TEACHER

VIEW PROFILE

- Test Case 25: Search Mentor by Category with No Available Teacher

Table 27: Test 25 Search by invalid category

Item	Description
Objective	To verify behavior when the selected category had no teacher.
Action	Select the category with no teacher data.
Expected Outcome	The system should show no teacher matching your search.
Actual Result	The message was shown successfully.
Conclusion	Success.

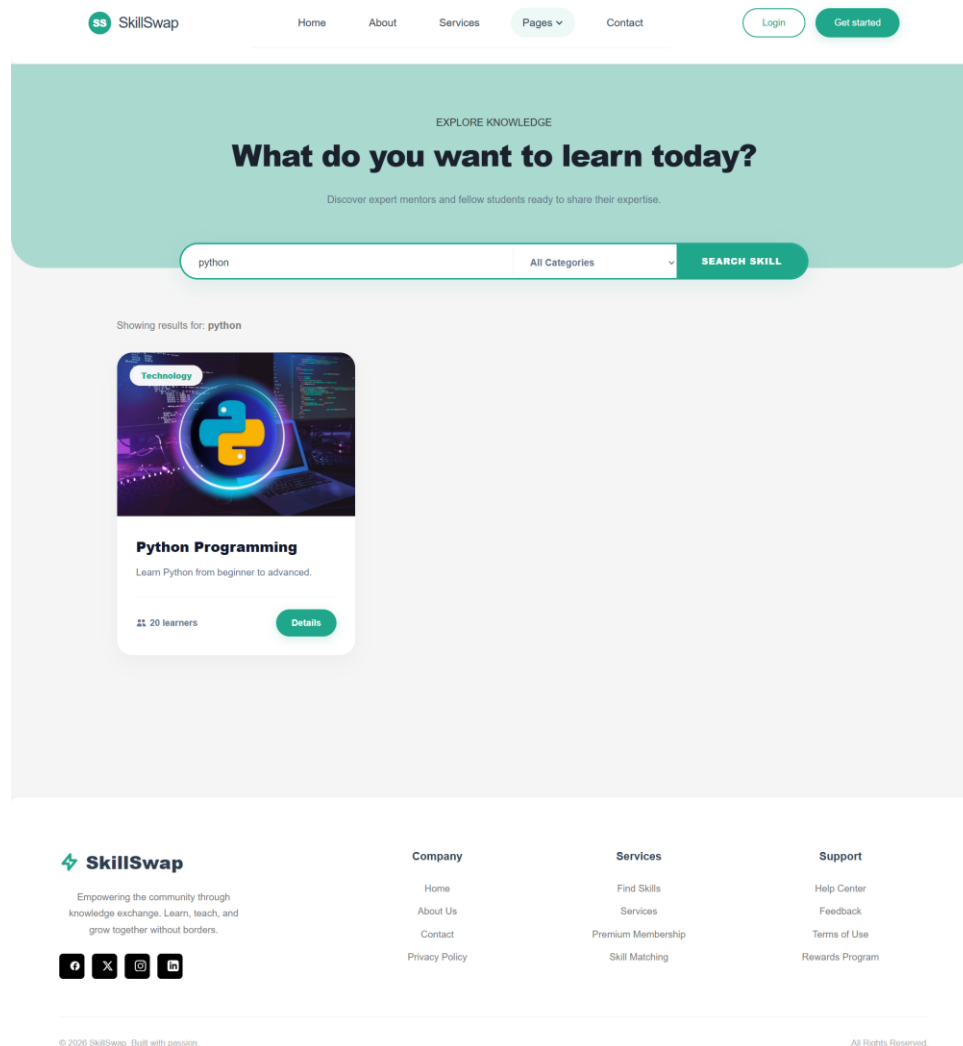


8.2.3 Find-Skill Testing

- Test Case 26: Search skill

Table 28: Test 26 Search skill

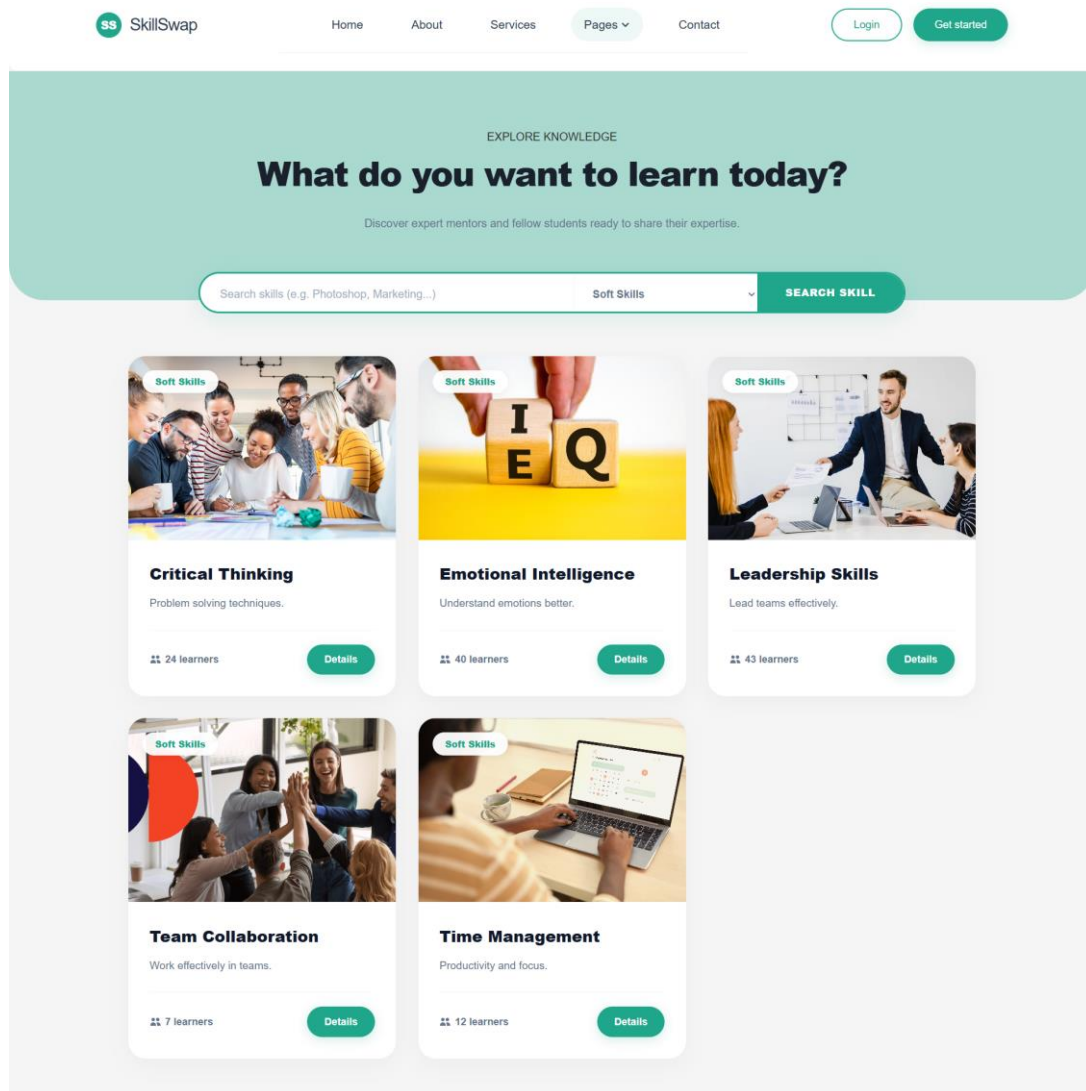
Item	Description
Objective	To verify that the user can search for the skill.
Action	Enter the skill and search.
Expected Outcome	The system should show the related skill.
Actual Result	The result shows successfully.
Conclusion	Success.



- Test Case 27: Search Skill with Category Filter

Table 29: Test 27 Search skills by category

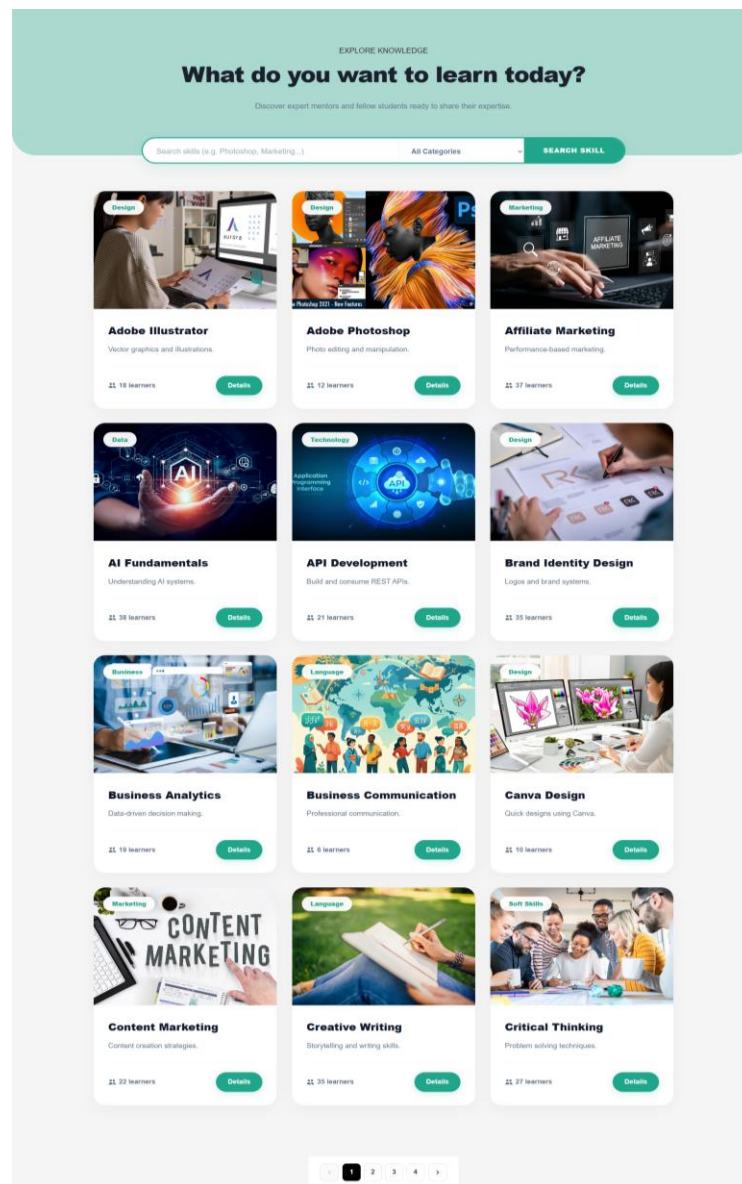
Item	Description
Objective	To verify that the user can search for skills by category.
Action	Click on the category and search.
Expected Outcome	The system should show the skills in the related category.
Actual Result	The result was shown successfully.
Conclusion	Success.



- Test Case 28: Search skills Without Input

Table 30: Test 28 Search skill without input

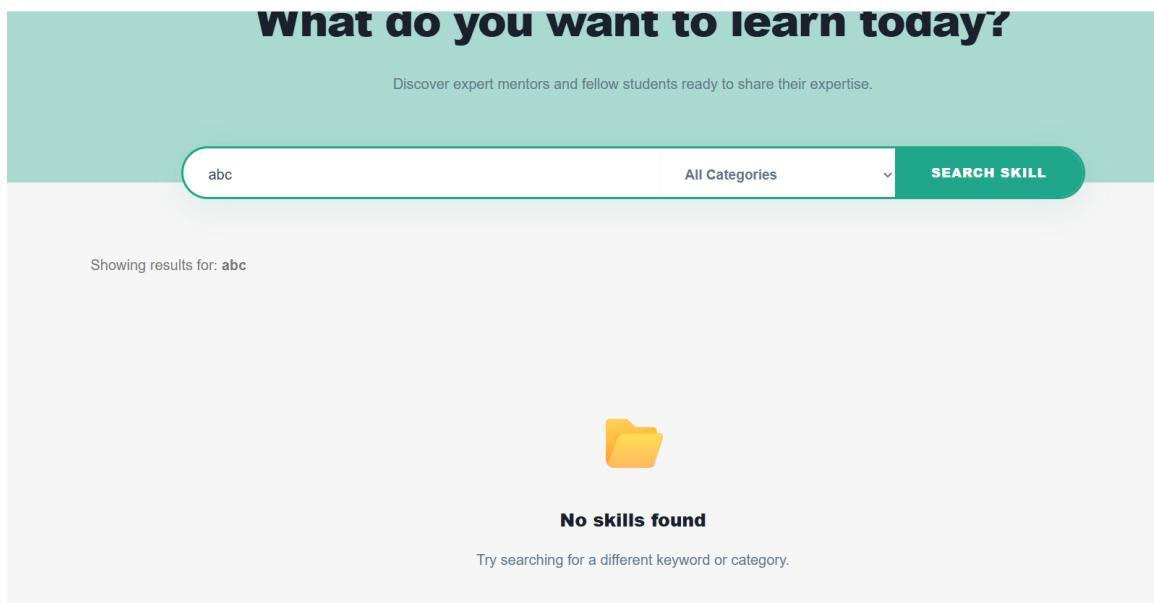
Item	Description
Objective	To verify how the system handles an empty search.
Action	Click search without entering a keyword.
Expected Outcome	The system should show the all skills.
Actual Result	The result was shown successfully.
Conclusion	Success.



- Test Case 29: Search invalid skill

Table 31: Test 29 Search by non-existing skill

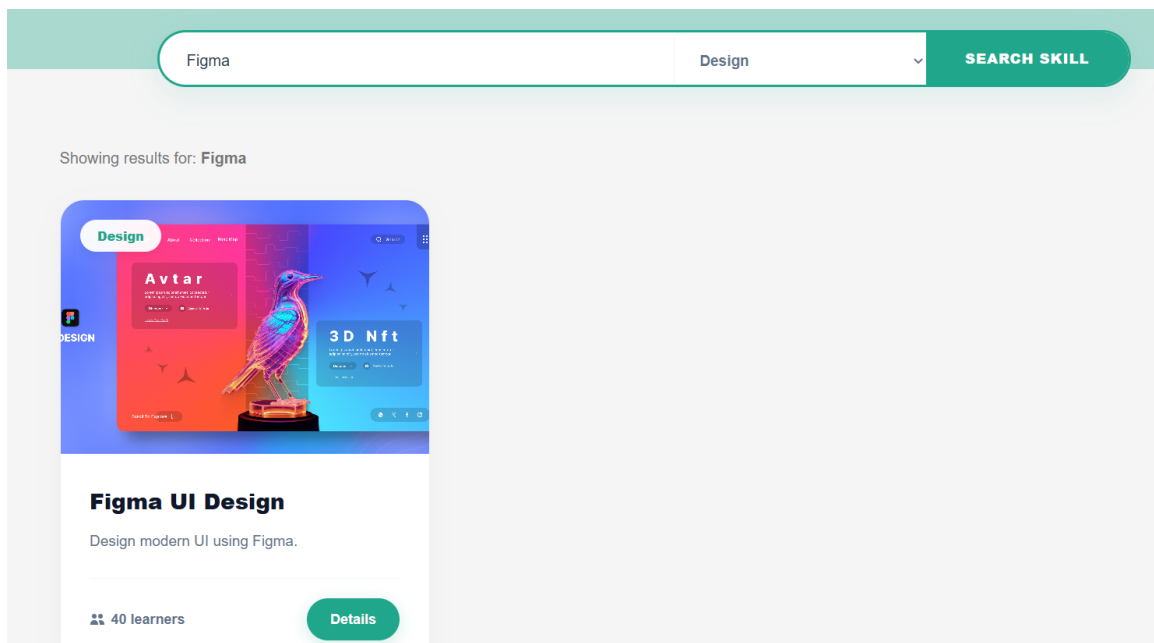
Item	Description
Objective	To verify how the system handles an invalid search.
Action	Enter a nonexistent skill and search.
Expected Outcome	The system should show a valid error message for skills.
Actual Result	The result was shown successfully.
Conclusion	Success.



- Test Case 30: Combined Search (Name + Category)

Table 32: Test 30 Search by skill & category both

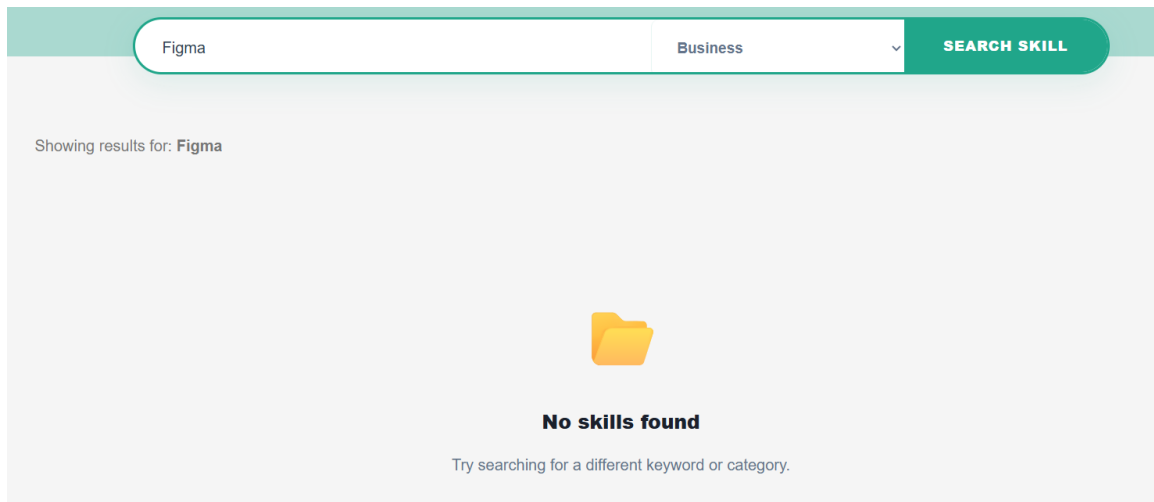
Item	Description
Objective	To verify how the system handles combine filtering
Action	Enter the keyword and category both.
Expected Outcome	Refined results displayed.
Actual Result	The result was shown successfully.
Conclusion	Success.



- Test Case 31: Search Skill with Keyword and Category Mismatch

Table 33: Test 31 Search skill with Mismatch keyword and category

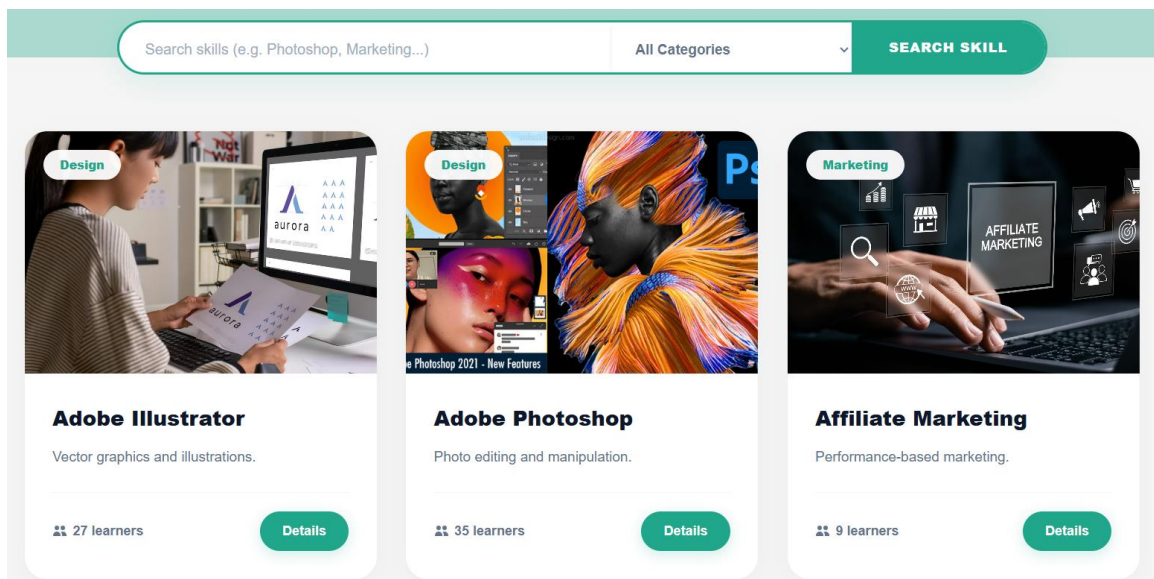
Item	Description
Objective	To verify system behavior when the keyword and category do not match.
Action	Enter skill and unrelated category and search.
Expected Outcome	The system should display the " No skills found" message.
Actual Result	The error message is shown successfully.
Conclusion	Success.



- Test Case 32: View skill details

Table 34: Test 32 Skill details

Item	Description
Objective	To verify that user can view skill details.
Action	Click on details button from any skill card.
Expected Outcome	Skill details page should be open with correct details.
Actual Result	Skill details page displayed successfully.
Conclusion	Success.



 SkillSwap
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DESIGN

Adobe Illustrator

 1 Experts 4.9 Rating Quality Verified
 [Join to Start Learning](#)


[About Skill](#) [Mentors \(1\)](#)

Skill Overview

Vector graphics and illustrations.

What you'll achieve

- ✓ Solidify core concepts of Adobe Illustrator
- ✓ Build real-world portfolio projects
- ✓ Learn industry-standard best practices
- ✓ 1-on-1 personalized feedback
- ✓ Master advanced optimization techniques
- ✓ Prepare for technical interviews

Learning Path Syllabus

A structured breakdown of what you can expect to cover with your mentor.

Phase 1: Fundamentals & Core Concepts

Establish a strong foundation by understanding the basic terminology, tools, and theoretical principles required for Adobe Illustrator. You will set up your environment and complete your first guided exercise.

3 Modules Est. 2 Weeks

Phase 2: Intermediate Techniques & Workflows

Move beyond the basics into standard industry workflows. Learn how professionals structure their projects, handle common problems, and utilize advanced tools efficiently.

4 Modules Est. 3 Weeks

Phase 3: Advanced Projects & Optimization

Apply everything you've learned to build a complex, capstone-level project. Receive intensive code/design reviews from your mentor to optimize performance, clean up architecture, and polish the final result.

2 Modules Est. 3 Weeks

Skill Profile

Category	Design
Experience Level	Beginner - Advanced
Prerequisites	None Required
Format	1-on-1 Mentorship
Language	English
Avg. Response	< 24 Hours

Satisfaction Guarantee

Not satisfied with your mentor match? Our dispute resolution team will easily refund your Skill Tokens.

8.3 Increment 4

8.3.1 Session Testing

- Test Case 33: Send Session Request

Table 35: Test 33 Send a session request to the teacher

Item	Description
Objective	To verify that the system should allow students to send session requests to the teacher.
Action	Search for the teacher who has matching skills and send the request.
Expected Outcome	The request should be sent successfully, and the system sends a notification to the teacher.
Actual Result	The request was sent successfully.
Conclusion	Success.

Request Mentorship

Connect with **Suraksha Sapkota** to master this skill.

Digital Marketing

Level: Intermediate

INTRODUCE YOURSELF

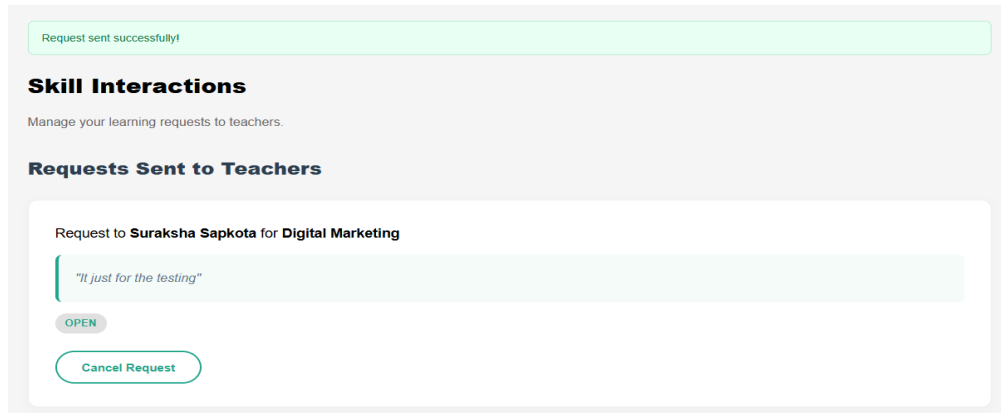
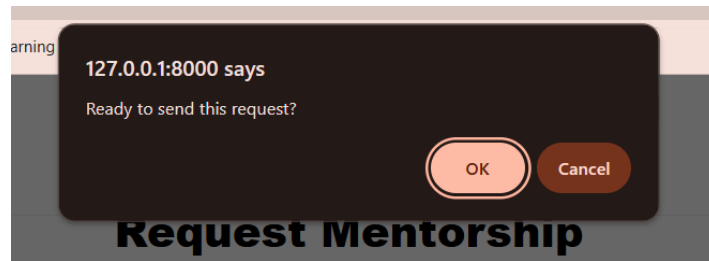
Describe what you want to learn or any specific goals you have...

PREFERRED DATE & TIME

mm/dd/yyyy --:-- --

Send Request

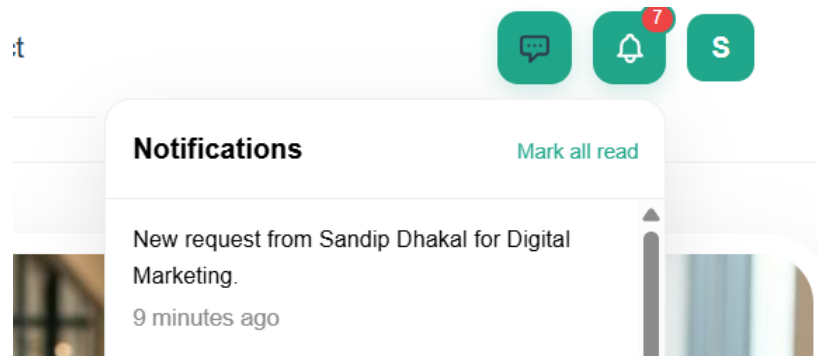
Cancel



- Test Case 34: Teacher Receives Request Notification

Table 36: Test 34 Teacher receives request notification

Item	Description
Objective	To verify that the teacher gets a notification for a new request.
Action	The student sent a request.
Expected Outcome	The teacher should get the notification.
Actual Result	The notification was shown successfully.
Conclusion	Success.



- Test Case 35: Reject Session Request

Table 37: Test 35 Reject Session request

Item	Description
Objective	To verify how the system can reject the students' requests.
Action	The teacher clicks on decline.
Expected Outcome	The system should change the status to decline.
Actual Result	The result was shown successfully.
Conclusion	Success.

Request Details

Requester: Sandip Dhakal

Responder: Suraksha Sapkota

Skill: **Adobe Illustrator**

Message: jhdgjsfs

Status: **OPEN**Offer Type: **TEACHING OFFER**

You offered to teach this skill.

Accept Offer

Decline

Request Details

Requester: Sandip Dhakal

Responder: Suraksha Sapkota

Skill: **Adobe Illustrator**

Message: jhdgjsfs

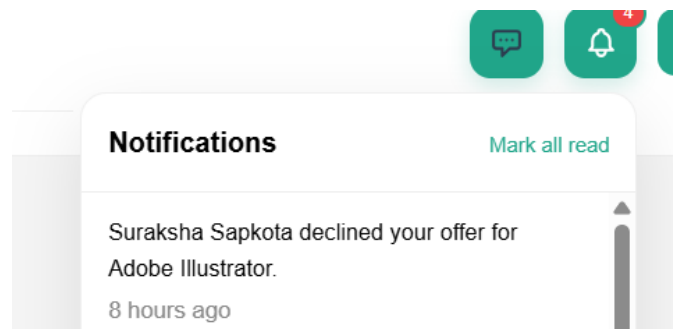
Status: **DECLINED**Offer Type: **TEACHING OFFER**

You offered to teach this skill.

- Test Case 36: Student Receives Rejection Notification

Table 38: Test 36 Student receives rejection notification

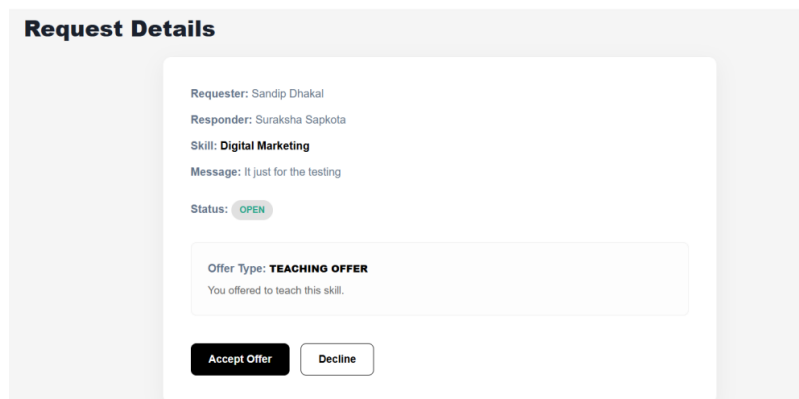
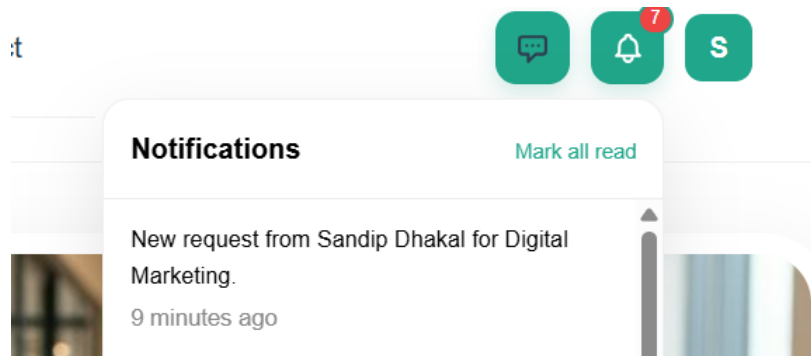
Item	Description
Objective	To verify that the student gets a notification after being rejected.
Action	The teacher clicks on decline request.
Expected Outcome	The system should show the decline notification to the student.
Actual Result	The notification was shown successfully.
Conclusion	Success.



- Test Case 37: Accept Session Request

Table 39: Test 37 accept session request

Item	Description
Objective	To verify that the teacher can accept student requests.
Action	The teacher clicks on Accept Request.
Expected Outcome	The system should show the request accepted message.
Actual Result	The result was shown successfully.
Conclusion	Success.



Request accepted! Now you can schedule the session below.

Skill Interactions

Manage your inbound student requests.

Inbound Student Requests

Sandip Dhakal offered to teach you **Digital Marketing**

"It just for the testing"

ACCEPTED

Schedule this Session

Set the date and time for your exchange with Sandip Dhakal.

START DATE & TIME

mm/dd/yyyy --:-- --

MEETING LINK (OPTIONAL)

https://zoom.us/j/...

Confirm Schedule & Create Session

- Test Case 38: Student Cannot View Session Before Creation

Table 40: Test 38 Cannot view session before creation

Item	Description
Objective	To verify that the student cannot see the session before the teacher creates it.
Action	The teacher accepted the request but did not create the session.
Expected Outcome	The students should see the session details.
Actual Result	No session was shown before created by the teacher.
Conclusion	Success.

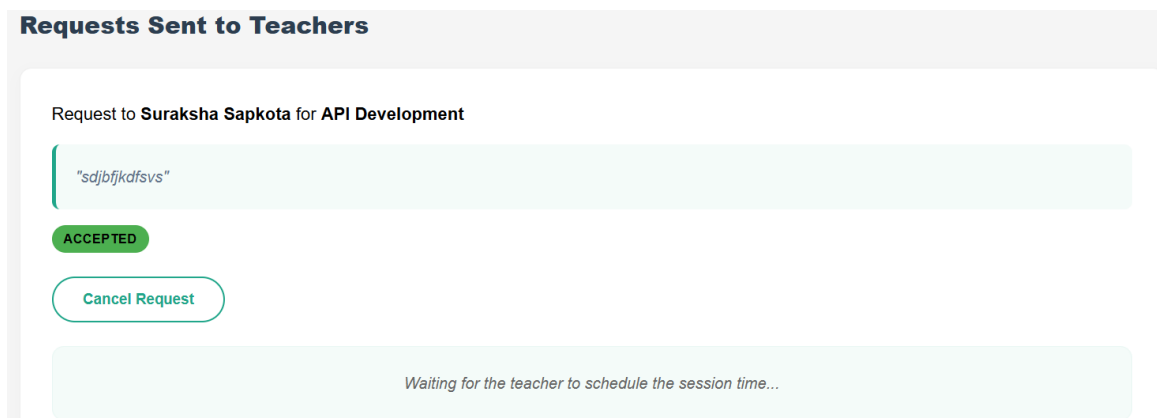


Figure 167: Waiting for session schedule

- Test Case 39: Create Session Schedule (Valid Input)

Table 41: Test 39 Create session schedule

Item	Description
Objective	To verify how the teacher can create the session after accepting the student's request.
Action	Accept the student request, and the system redirects the teacher to the request page.
Expected Outcome	The system should show a valid error message for skills.
Actual Result	The result was shown successfully.
Conclusion	Success.

Request accepted! Now you can schedule the session below.

Skill Interactions

Manage your inbound student requests.

Inbound Student Requests

Sandip Dhakal offered to teach you Digital Marketing

"It just for the testing"

ACCEPTED

Schedule this Session

Set the date and time for your exchange with Sandip Dhakal.

START DATE & TIME

mm/dd/yyyy --:-- --

MEETING LINK (OPTIONAL)

https://zoom.us/j/...

Confirm Schedule & Create Session

Your Sessions

Active teaching and learning sessions.

Session scheduled successfully and student notified.

MY TEACHING SESSIONS

Digital Marketing

Student: Sandip Dhakal

SCHEDULED

MANAGE CLASS

API Development

Student: Ganga Sapkota

SCHEDULED

MANAGE CLASS

Critical Thinking

Student: Ganga Sapkota

SCHEDULED

MANAGE CLASS

Business Analytics

Student: Ganga Sapkota

SCHEDULED

MANAGE CLASS

- Test Case 40: Prevent Session Without Date & Time

Table 42: Test 40 Cannot create session without data and time

Item	Description
Objective	For verification of schedule fields.
Action	Enter a nonexistent skill and search.
Expected Outcome	The system should show a valid error message for skills.
Actual Result	The result was shown successfully.
Conclusion	Success.

Sandip Dhakal offered to teach you **Digital Marketing**

"It just for the testing"

ACCEPTED

Schedule this Session

Set the date and time for your exchange with Sandip Dhakal.

START DATE & TIME

MEETING LINK (OPTIONAL)

! Please fill out this field.

Confirm Schedule & Create Session

- Test Case 41: Student Receives Session Created Notification

Table 43: Test 41 Student receives a session notification after the teacher creates

Item	Description
Objective	To verify that students get a notification after the teacher creates the session.
Action	The teacher creates a session.
Expected Outcome	The student should get a notification.
Actual Result	The notification was received successfully.
Conclusion	Success.



- Test Case 42: Open Session Page from Notification

Table 44: Test 42 Open session notification

Item	Description
Objective	To verify that clicking the notification redirects the student to the session request/classroom page.
Action	The student clicks on the notification.
Expected Outcome	The system should open the session page showing “Enter Classroom” and “Report Issue”.
Actual Result	Session page opened correctly.
Conclusion	Success.



- Test Case 43: Create Session Without Meeting Link

Table 45: Test 43 Created session without link

Item	Description
Objective	To verify that the session is created without a meeting link.
Action	Submit the session without the meeting link.
Expected Outcome	The session should be created successfully.
Actual Result	Session created.
Conclusion	Success.

Sandip Dhakal offered to teach you API Development

"sdjbfjkdsvs"

ACCEPTED

Schedule this Session

Set the date and time for your exchange with Sandip Dhakal.

START DATE & TIME

04/22/2026 08:56 PM

MEETING LINK (OPTIONAL)

https://zoom.us/j/...

Confirm Schedule & Create Session

Your Sessions

Active teaching and learning sessions.

Session scheduled successfully and student notified.

MY TEACHING SESSIONS

API Development

Student: Sandip Dhakal

SCHEDULED

MANAGE CLASS

Digital Marketing

Student: Sandip Dhakal

SCHEDULED

MANAGE CLASS

API Development

Student: Ganga Sapkota

SCHEDULED

MANAGE CLASS

Live Class Controls

MEETING LINK (ZOOM, GOOGLE MEET, ETC.)

https://zoom.us/j/...

SAVE

Please save a meeting link first to unlock live class controls.

- Test Case 44: Student View Session After Creation

Table 46: Test 44 Student views the session after creation

Item	Description
Objective	To verify students, they can view the session after creation.
Action	Teacher create session.
Expected Outcome	The students should see the session details.
Actual Result	Session visible successfully.
Conclusion	Success.

Your Sessions

Active teaching and learning sessions.

MY LEARNING SESSIONS

API DEVELOPMENT

Teacher: **Suraksha Sapkota**

 Scheduled: Apr 22, 2026 @ 08:56 PM

SCHEDULED **ENTER CLASSROOM**

DIGITAL MARKETING

Teacher: **Suraksha Sapkota**

 Scheduled: Apr 21, 2026 @ 11:27 AM

SCHEDULED **JOIN LIVE**

- Test Case 45: No Meeting Link Message

Table 47: Test 45 In session shown no meeting link message

Item	Description
Objective	To verify that students can view the session after creation.
Action	Teacher create session.
Expected Outcome	The students should see the session details.
Actual Result	Session visible successfully.
Conclusion	Success.

CLASSROOM (STUDENT VIEW)
SCHEDULED

API Development

TEACHER

Suraksha Sapkota

CHAT

STUDENTS IN THIS CLASS

No students attached to this session yet.

Session Schedule

CURRENT TIME
April 22, 2026 - 8:56 PM

REQUEST A DIFFERENT TIME

Teacher, I am not available then. Could we do Tuesday at 5 PM instead?

SEND RESCHEDULE REQUEST

AWAITING MEETING LINK...
The teacher hasn't provided the live room link yet. You can relax or kindly message them a reminder.

- Test Case 46: Teacher Upload Meeting Link Later

Table 48: Test 46 teacher upload meeting link

Item	Description
Objective	To verify how the system handles an invalid search.
Action	Enter a nonexistent skill and search.
Expected Outcome	The system should show a valid error message for skills.
Actual Result	The result was shown successfully.
Conclusion	Success.

Live Class Controls

MEETING LINK (ZOOM, GOOGLE MEET, ETC.)

SAVE

Meeting link updated successfully.

CLASSROOM (TEACHER VIEW)

SCHEDULED

API Development

- Test Case 47: Student Cannot Join Before Session Starts

Table 49: Test 47 Student cannot join the meeting before the teacher starts the session

Item	Description
Objective	To verify how the system handles an invalid search.
Action	Enter a nonexistent skill and search.
Expected Outcome	The system should show a valid error message for skills.
Actual Result	The result was shown successfully.
Conclusion	Success.

CLASSROOM (STUDENT VIEW)
SCHEDULED

API Development

TEACHER

Suraksha Sapkota

CHAT

STUDENTS IN THIS CLASS

No students attached to this session yet.

Session Schedule

CURRENT TIME

April 22, 2026 - 8:56 PM

REQUEST A DIFFERENT TIME

Teacher, I am not available then. Could we do Tuesday at 5 PM instead?

SEND RESCHEDULE REQUEST

Teacher has not started

The meeting link is fully prepared, but the teacher has not initiated the live session yet. You will be able to join when they start.

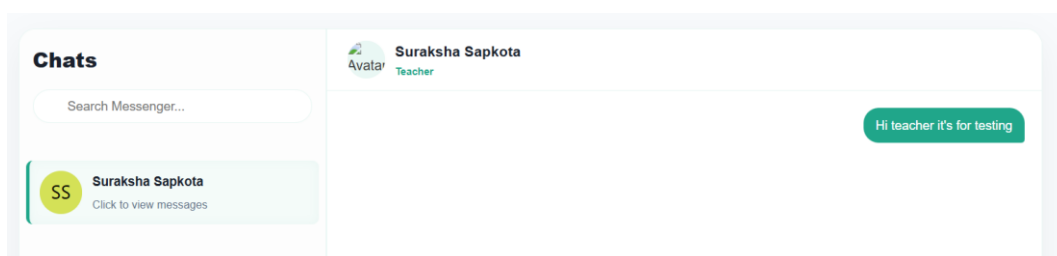
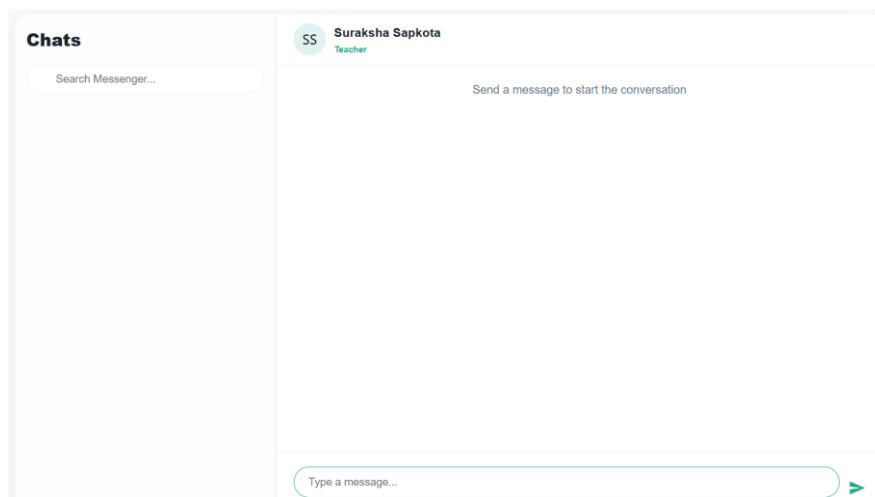
MESSAGE TEACHER

The join button will securely unlock here once the session begins.

- Test Case 48: Student Can Message Teacher Before Session

Table 50: Test 48 Student can message teacher before session

Item	Description
Objective	To verify how the system handles an invalid search.
Action	Enter a nonexistent skill and search.
Expected Outcome	The system should show a valid error message for skills.
Actual Result	The result was shown successfully.
Conclusion	Success.



- Test Case 49: Teacher Starts Live Session

Table 51: Test 49 Teacher start session

Item	Description
Objective	To verify how the system handles an invalid search.
Action	Enter a nonexistent skill and search.
Expected Outcome	The system should show a valid error message for skills.
Actual Result	The result was shown successfully.
Conclusion	Success.

Live session started!

CLASSROOM (TEACHER VIEW)
SCHEDULED

TEACHER

Suraksha Sapkota

STUDENTS IN THIS CLASS

No students attached to this session yet.

ADD ANOTHER ACCEPTED STUDENT:

Select Student...

ADD TO CLASS

Session Schedule

CURRENT TIME

April 22, 2026 - 8:56 PM

UPDATE START TIME

04/22/2026 08:56 PM

UPDATE END TIME

mm/dd/yyyy --:--

UPDATE SCHEDULE

Live Class Controls

MEETING LINK (ZOOM, GOOGLE MEET, ETC.)

https://meet.google.com/vkm-rbqg-wkr

UPDATE

• SESSION IS LIVE

OPEN MEETING ROOM

STOP LIVE SESSION

- Test Case 50: Student Sees Join Live Option

Table 52: Test 50 Student sees the option to join live

Item	Description
Objective	To verify how the system handles an invalid search.
Action	Enter a nonexistent skill and search.
Expected Outcome	The system should show a valid error message for skills.
Actual Result	The result was shown successfully.
Conclusion	Success.

CLASSROOM (STUDENT VIEW)

API Development

SCHEDULED

TEACHER

STUDENTS IN THIS CLASS

Suraksha Sapkota

CHAT

No students attached to this session yet.

Session Schedule

 CURRENT TIME

April 22, 2026 - 8:56 PM

REQUEST A DIFFERENT TIME

Teacher, I am not available then. Could we do Tuesday at 5 PM instead?

SEND RESCHEDULE REQUEST

LIVE SESSION IN PROGRESS

Class is Started

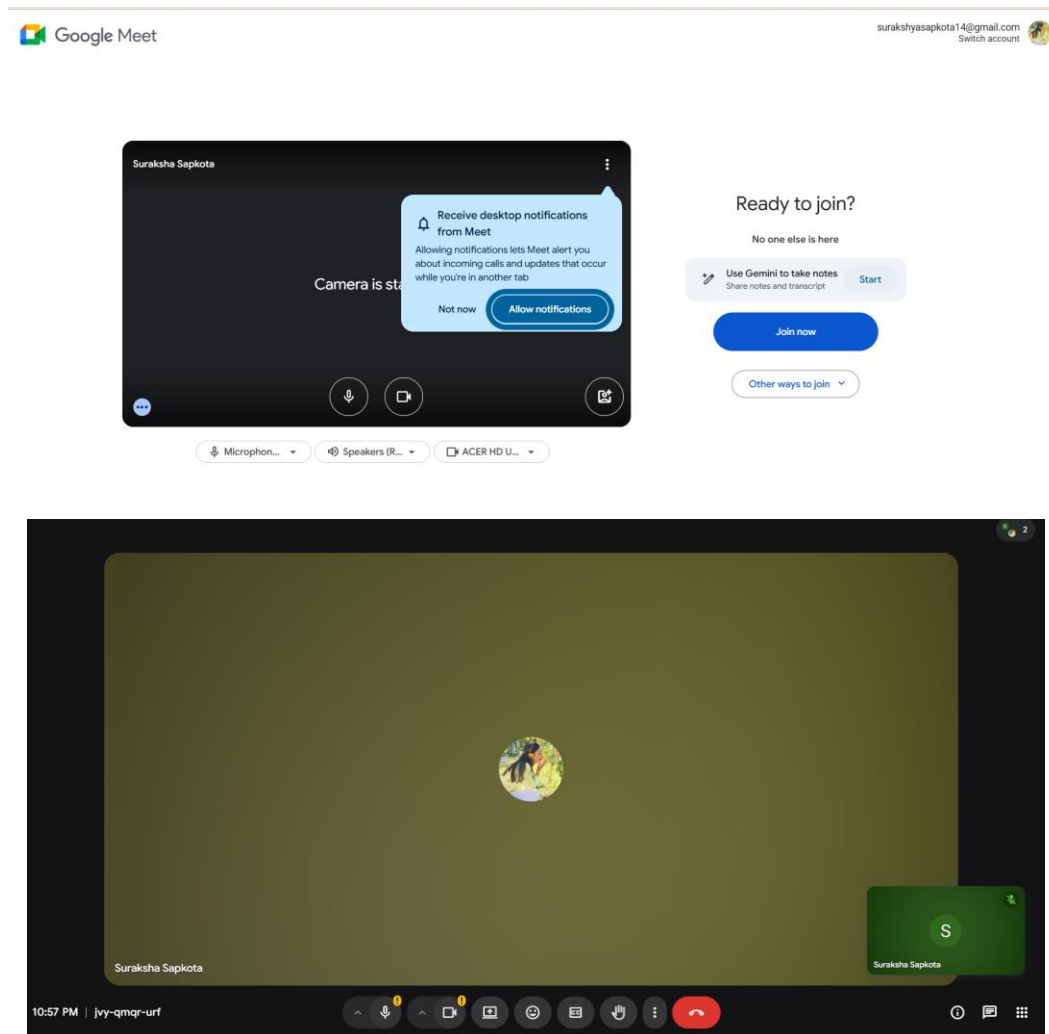
Your teacher successfully opened the room. Join now!

JOIN LIVE CLASS NOW

- Test Case 51: Student Joins Live Session

Table 53: Test 51 Student join session

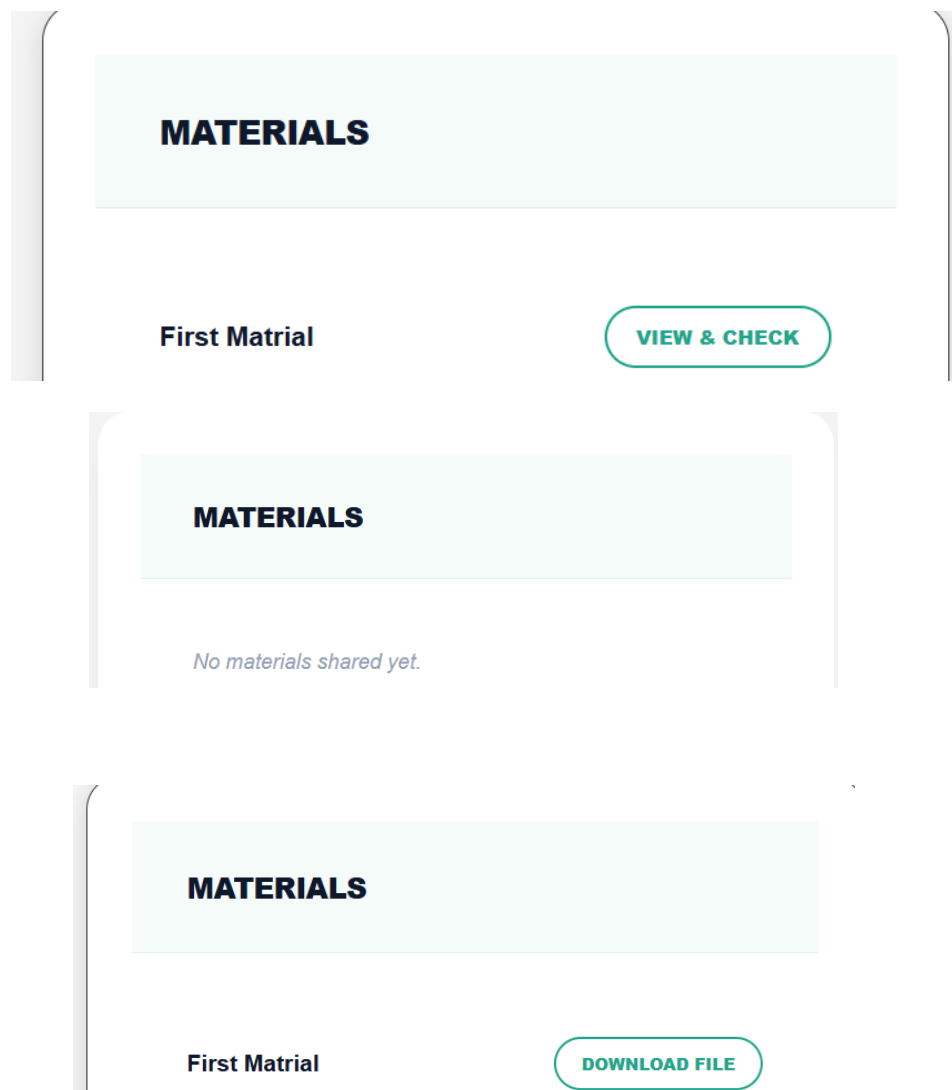
Item	Description
Objective	To verify how the system handles an invalid search.
Action	Enter a nonexistent skill and search.
Expected Outcome	The system should show a valid error message for skills.
Actual Result	The result was shown successfully.
Conclusion	Success.



- Test Case 52: Teacher Upload Learning Material

Table 54: Test 52 Teacher upload learning

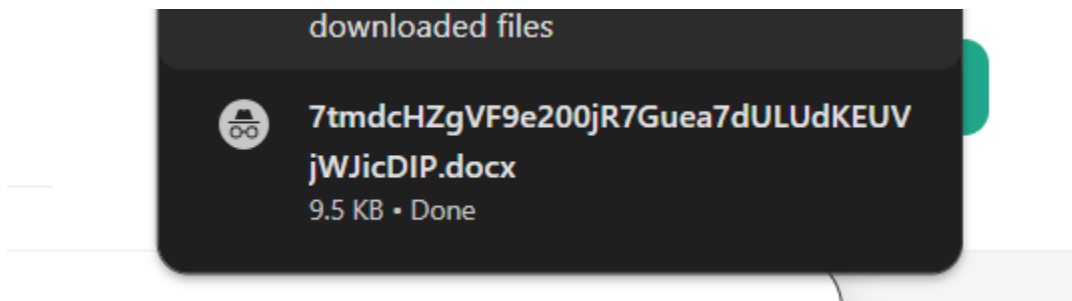
Item	Description
Objective	To verify teacher can upload material.
Action	Upload the file to the classroom.
Expected Outcome	File should be available to the student.
Actual Result	The file uploading successfully.
Conclusion	Success.



- Test Case 53: Student Download Learning Material

Table 55: Test 53 Downloading learning material

Item	Description
Objective	To verify that the student can get the material.
Action	Click the download file
Expected Outcome	File should be download.
Actual Result	The file was downloaded successfully.
Conclusion	Success.



- Test Case 54: Student Upload Assignment

Table 56: Test 54 Uploading assignment

Item	Description
Objective	To verify students can upload assignment.
Action	Upload the file to the classroom.
Expected Outcome	File should be available to the teacher to view.
Actual Result	The file uploading successfully.
Conclusion	Success.

PRACTICE & HOMEWORK

No practice files submitted yet.

PRACTICE & HOMEWORK

First day assignment

[VIEW WORK](#)

For Testing

Submitted: 1 second ago

PRACTICE & HOMEWORK

First day assignment

[VIEW WORK](#)

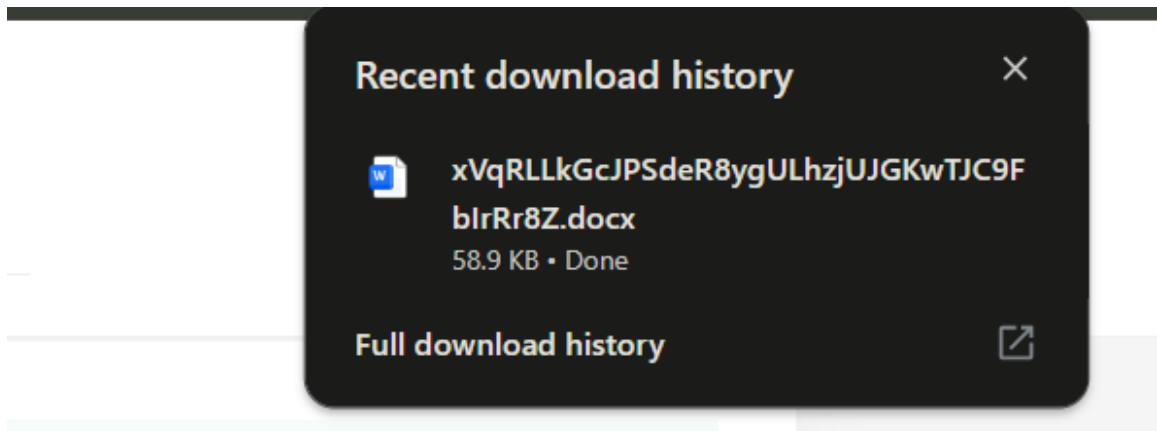
For Testing

Submitted: 31 seconds ago

- Test Case 55: Teacher View Student Assignment

Table 57: Test 55 View student assignment

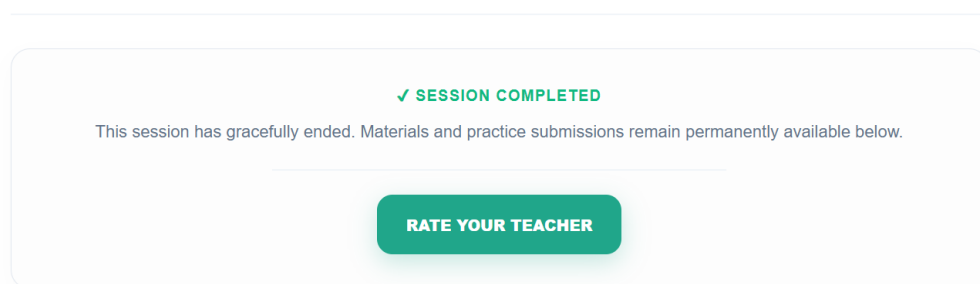
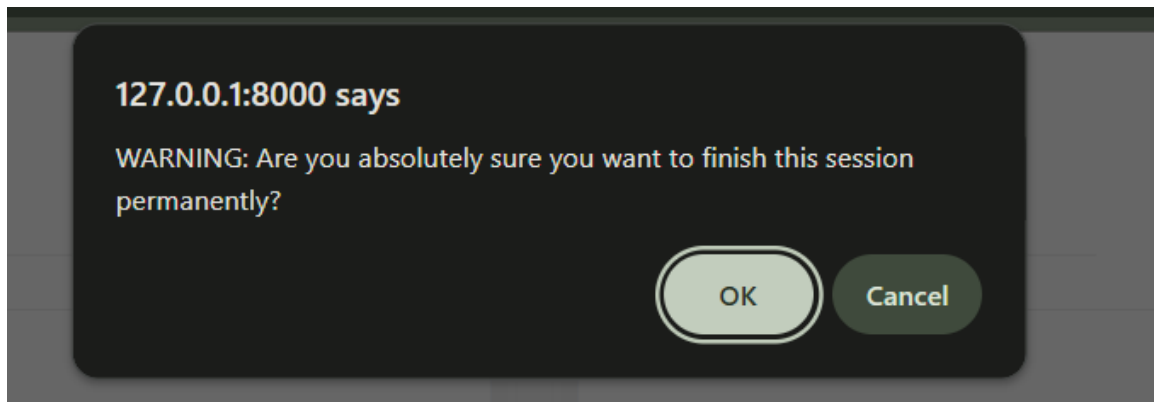
Item	Description
Objective	To verify that the teacher can download the assignment.
Action	Click the download file
Expected Outcome	File should be download.
Actual Result	The file was downloaded successfully.
Conclusion	Success.



- Test Case 56: Mark completed the session

Table 58: Test 56 session completed

Item	Description
Objective	To verify that the teacher can make session complete.
Action	Click on marks complete button in the classroom and make the session complete
Expected Outcome	The system should show a pop-up message, and after the teacher clicks on OK, the system should mark that session completed.
Actual Result	The session should be completed.
Conclusion	Success.



8.3.2 Feedback & Review Testing

- Test Case 57: submit feedback only without rating

Table 59: Test 57 Feedback form without rating

Item	Description
tries	To verify that the system shows an error when the user try to submit the feedback form without rating.
Action	Click submit without selecting a rating.
Expected Outcome	The system should display the error "The rating field is required."
Actual Result	The error was displayed successfully.
Conclusion	Success.

Leave Feedback

Rate your experience with Sandip Dhakal

Session Details

Skill: API Development

With: Sandip Dhakal

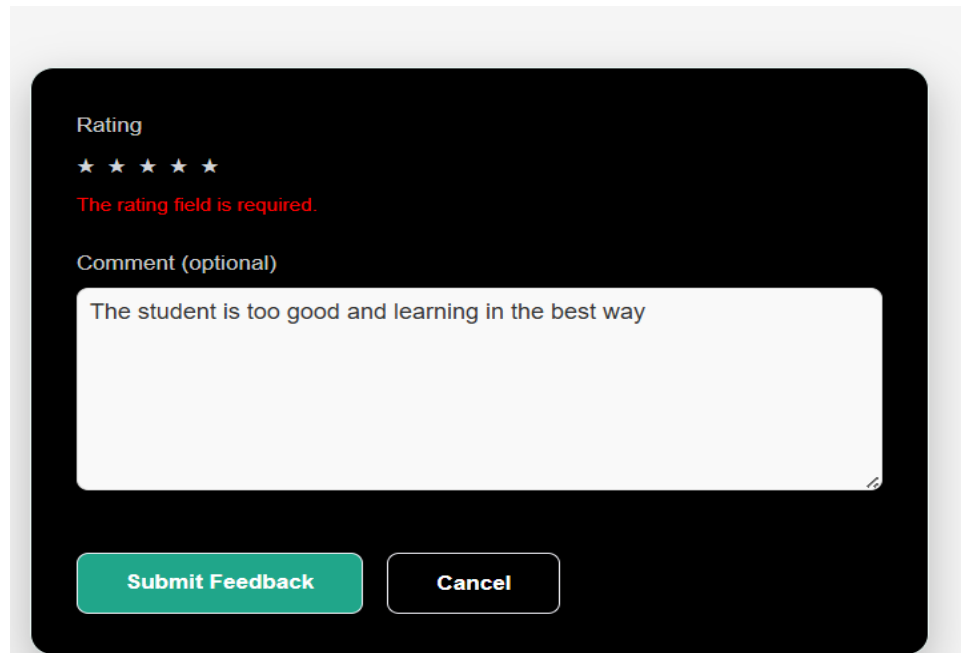
Rating

★ ★ ★ ★ ★

Comment (optional)

The student is too good and learning in the best way

Submit Feedback Cancel

A dark-themed feedback form modal with rounded corners. It contains a 'Rating' section with five stars, a red error message 'The rating field is required.', a 'Comment (optional)' section with a text area containing the text 'The student is too good and learning in the best way', and two buttons at the bottom: 'Submit Feedback' (green) and 'Cancel' (white with a grey border).

Rating

★ ★ ★ ★ ★

The rating field is required.

Comment (optional)

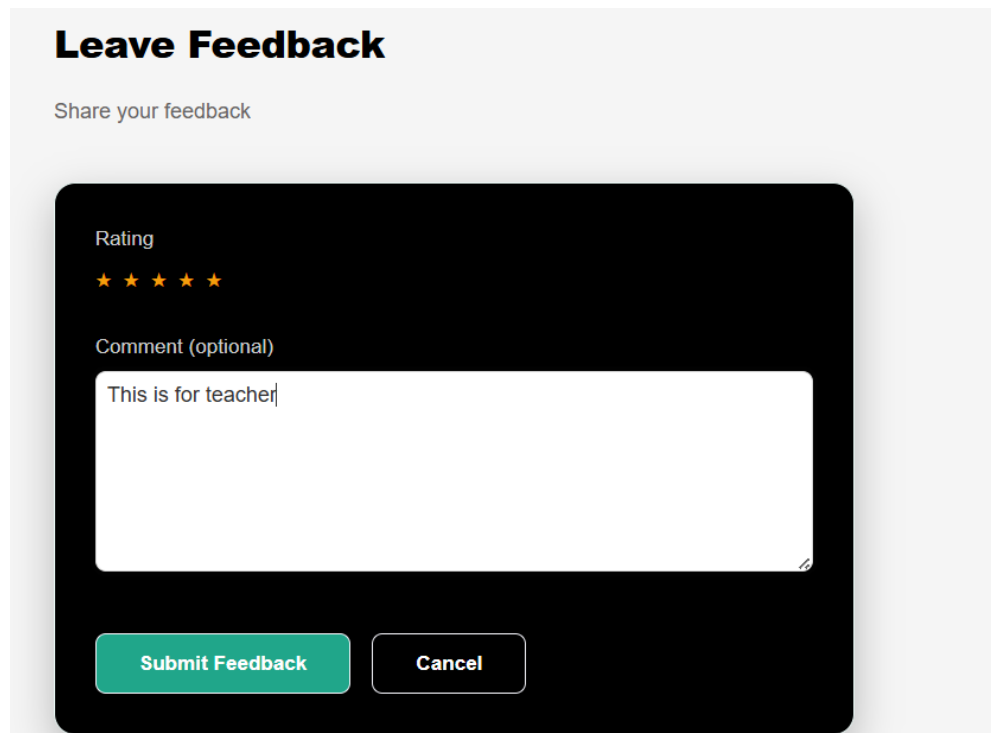
The student is too good and learning in the best way

Submit Feedback Cancel

- Test Case 58: Submit the feedback form correctly

Table 60: Test 58 Submit the feedback form correctly

Item	Description
tries	To verify that system submit feedback correctly with rating and command.
Action	Click submit with all write info.
Expected Outcome	The feedback should be shown in user feedback page.
Actual Result	The feedback was displayed successfully.
Conclusion	Success.



Leave Feedback

Share your feedback

Rating

★ ★ ★ ★ ★

Comment (optional)

This is for teacher

Submit Feedback Cancel

Feedback

View feedback you've received and given

N/A
AVERAGE RATING

0
TOTAL REVIEWS

Feedback Received

- No feedback received yet.

Feedback Given

- ★ ★ ★ ★ ★

The student is too good and learning in the best way

30 minutes ago

Feedback

View feedback you've received and given

5.0
AVERAGE RATING

1
TOTAL REVIEWS

Feedback Received

- ★ ★ ★ ★ ★ by Suraksha Sapkota

The student is too good and learning in the best way

41 minutes ago

Feedback Given

- You haven't given any feedback yet.

8.3.3 Premium Membership & Payment Testing

- Test Case 59: View Premium Plans

Table 61: Test 59 View Premium plan

Item	Description
tries	To verify that user can view all available premium plans.
Action	Open premium membership page.
Expected Outcome	Monthly, Quarterly, and Yearly plans should be displayed.
Actual Result	The plans displayed correctly
Conclusion	Success.

Premium Membership

Unlock enhanced features and take your learning to the next level

Choose a Plan

Monthly

NPR 1,000 /1 month

- ✓ Unlimited skill requests
- ✓ Priority matching
- ✓ Advanced filters
- ✓ Premium badge
- ✓ No ads

Pay with Khalti

Save 17%

Quarterly

NPR 2,500 /3 months

- ✓ All Monthly features
- ✓ Detailed analytics
- ✓ Export session history

Pay with Khalti

Save 33%

Yearly

NPR 8,000 /12 months

- ✓ All Quarterly features
- ✓ Early access to new features
- ✓ Dedicated support

Pay with Khalti

- Test Case 60: Select Premium Plan

Table 62: Test 60 Select plan

Item	Description
Objective	To verify users can select a plan
Action	Click “Pay with Khalti” on any plan.
Expected Outcome	Users should be redirected to payment page.
Actual Result	The System redirected successfully.
Conclusion	Success.

Payment Details

This payment will expire on Apr 20, 2026 22:13 PM

Billed To:
Sandip Dhakal
9827472492 , sandip@yopmail.com

BSD Apple Care

Product Summary:

Monthly Premium
(x1)
Rs 1000.00

Amount Summary:

Subscription Fee	Rs 1000.00
Total Payable Amount	Rs 1000.00

PAYMENT POWERED BY

← Pay via Khalti Wallet

Enter Khalti ID

Nepal

Khalti Mobile Number

Khalti Password / MPIN

Forgot your password ? [Reset Password](#)

[Cancel Payment](#)

- Test Case 61: Display Payment Details

Table 63: Test 61 Payment details

Item	Description
Objective	To verify correct billing details.
Action	Open payment page.
Expected Outcome	User info, plan name, and amount should be correct.
Actual Result	The details was displayed successfully.
Conclusion	Success.

← Pay via Khalti Wallet

Enter OTP

 Nepal

Khalti Mobile Number
9800000005

Khalti Password / MPIN
..... 

Enter OTP sent to your Mobile Number/Email Address

|

Pay Rs. 1000.00

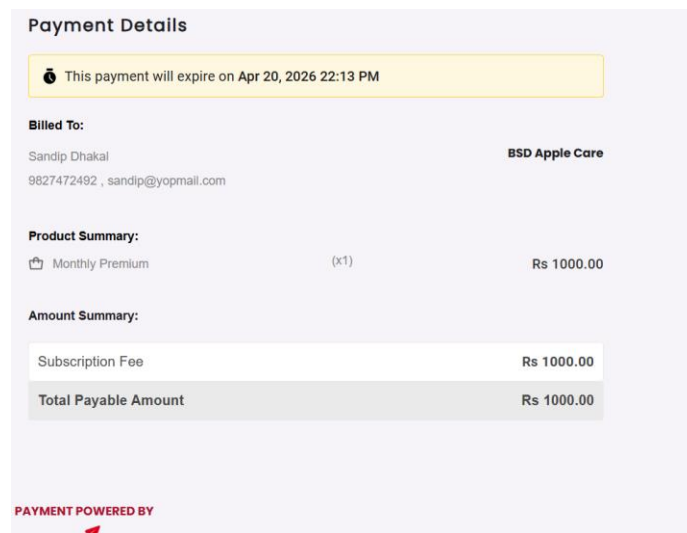
Forgot your password ? [Reset Password](#)

Cancel Payment

- Test Case 62: Payment Expiry Timer

Table 64: Test 62 payment expiry time

Item	Description
Objective	To verify that the payment expiry message is shown.
Action	Open payment page.
Expected Outcome	Expiry time should be displayed.
Actual Result	The message shown successfully.
Conclusion	Success.



- Test Case 63: Enter Khalti Credentials

Table 65: Enter Khalti credentials

Item	Description
Objective	To verify that the Khalti login is working
Action	Enter Khalti ID and password.
Expected Outcome	The system should allow proceeding to the OTP step.
Actual Result	The Proceeded successfully.
Conclusion	Success.

Payment Details

This payment will expire on Apr 20, 2026 22:13 PM

Billed To:
Sandip Dhakal
9827472492 , sandip@yopmail.com

Product Summary:
 Monthly Premium (x1) Rs 1000.00

Amount Summary:

Subscription Fee	Rs 1000.00
Total Payable Amount	Rs 1000.00

PAYMENT POWERED BY

Pay via Khalti Wallet

Enter OTP

Nepal

Khalti Mobile Number
9800000005

Khalti Password / MPIN

••••

Enter OTP sent to your Mobile Number/Email Address

Pay Rs. 1000.00

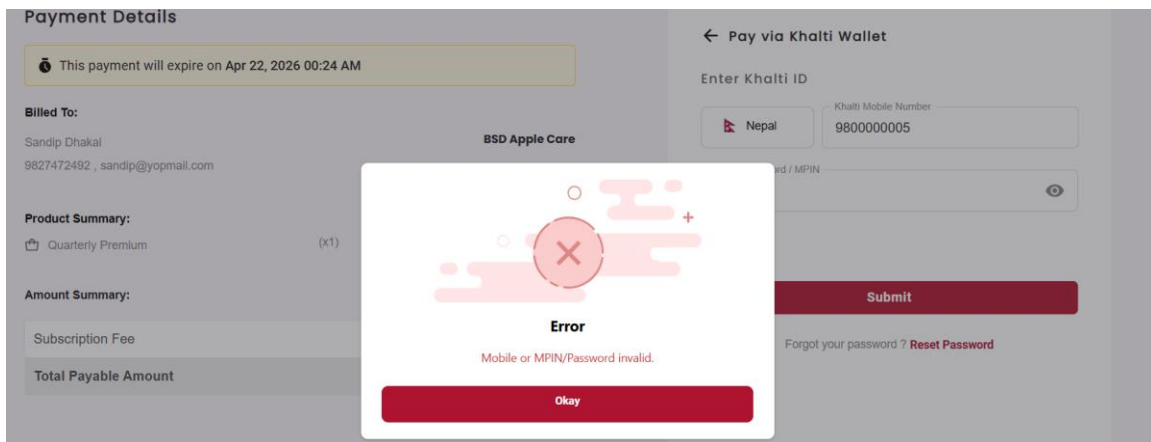
Forgot your password ? [Reset Password](#)

Cancel Payment

- Test Case 64: Invalid Khalti Credentials (Error)

Table 66: Test 64 Invalid khalti login

Item	Description
Objective	To verify invalid error handling.
Action	Enter wrong Khalti ID or password.
Expected Outcome	Error messages should be shown.
Actual Result	The error was shown successfully.
Conclusion	Success.



- Test Case 65: Enter OTP

Table 67: Test 65 Enter OTP

Item	Description
Objective	To verify OTP validation
Action	Enter correct OTP.
Expected Outcome	Payment should be proceed.
Actual Result	The OTP was accepted successfully.
Conclusion	Success.

← Pay via Khalti Wallet

Enter OTP

Nepal

Khalti Mobile Number 9800000005

Khalti Password / MPIN

Enter OTP sent to your Mobile Number/Email Address

9 8 7 6 5 4

Pay Rs. 1000.00

Forgot your password ? [Reset Password](#)

- Test Case 66: Successful Payment and Premium Activation

Table 68: Test 66 Successfully payment and activate premium plan

Item	Description
Objective	To verify successful payment, premium activation, and current plan display.
Action	Complete payment by entering valid OTP and open premium page.
Expected Outcome	Payment success message should be displayed, premium plan should be activated, and current plan with expiry date should be visible.
Actual Result	"Payment successful! Welcome to Premium." shown, plan activated, and current plan displayed correctly.
Conclusion	Success.

Premium Membership

Unlock enhanced features and take your learning to the next level

Payment successful! Welcome to Premium.

Your Current Plan

Monthly Plan

Status: Active

Expires: May 20, 2026

Upgrade Your Plan

Monthly

NPR 1,000 /1 month

✓ Unlimited skill requests

✓ Priority matching

✓ Advanced filters

✓ Premium badge

✓ No ads

Your current plan

Save 17%

Quarterly

NPR 2,500 /3 months

✓ All Monthly features

✓ Detailed analytics

✓ Export session history

Pay with Khalti

Save 33%

Yearly

NPR 8,000 /12 months

✓ All Quarterly features

✓ Early access to new features

✓ Dedicated support

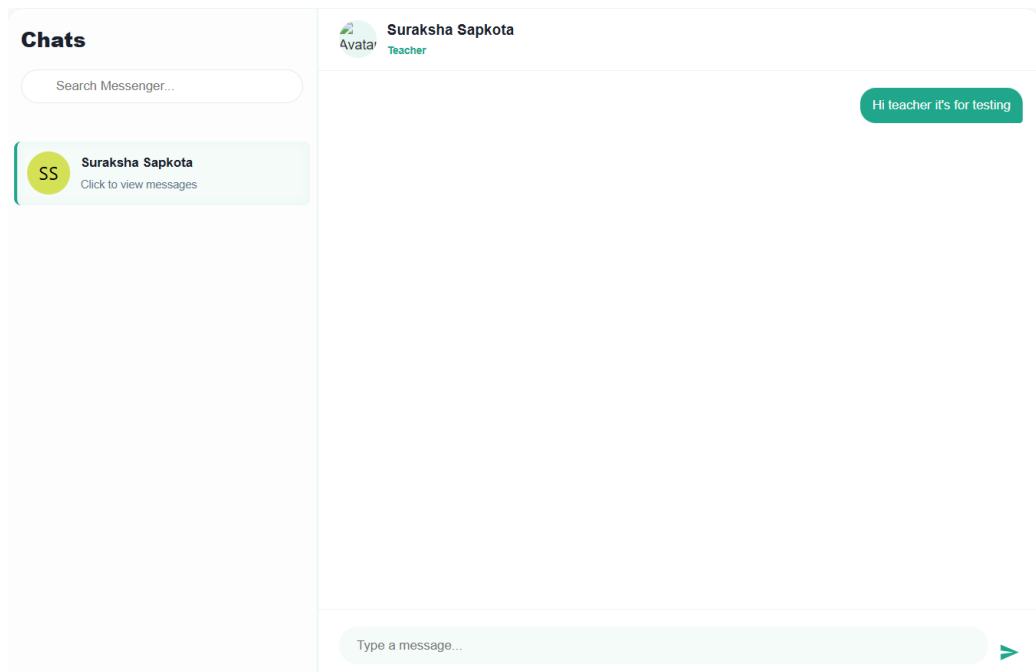
Pay with Khalti

8.3.4 Chat System Testing

- Test Case 67: Send Message

Table 69: Test 67 Send message to teacher

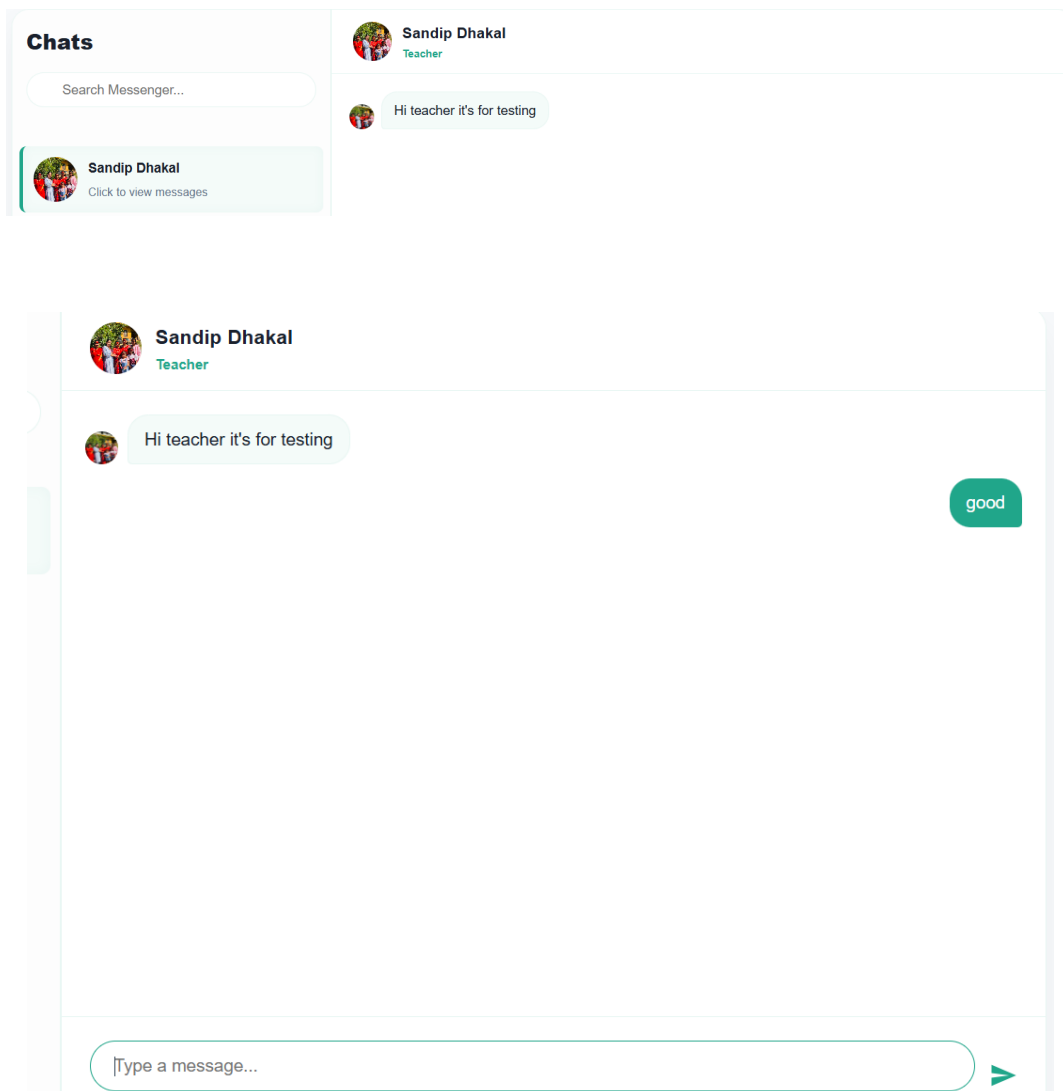
Item	Description
Objective	To verify that the student can send message to the teacher
Action	Send the message to the teacher.
Expected Outcome	The message is delivered.
Actual Result	The message was delivered successfully.
Conclusion	Success.



- Test Case 69: Receive Message and Reply

Table 70: Test 69 Receive and reply the message

Item	Description
Objective	To verify how the system handles an invalid search.
Action	Enter a nonexistent skill and search.
Expected Outcome	The system should show a valid error message for skills.
Actual Result	The result was shown successfully.
Conclusion	Success.



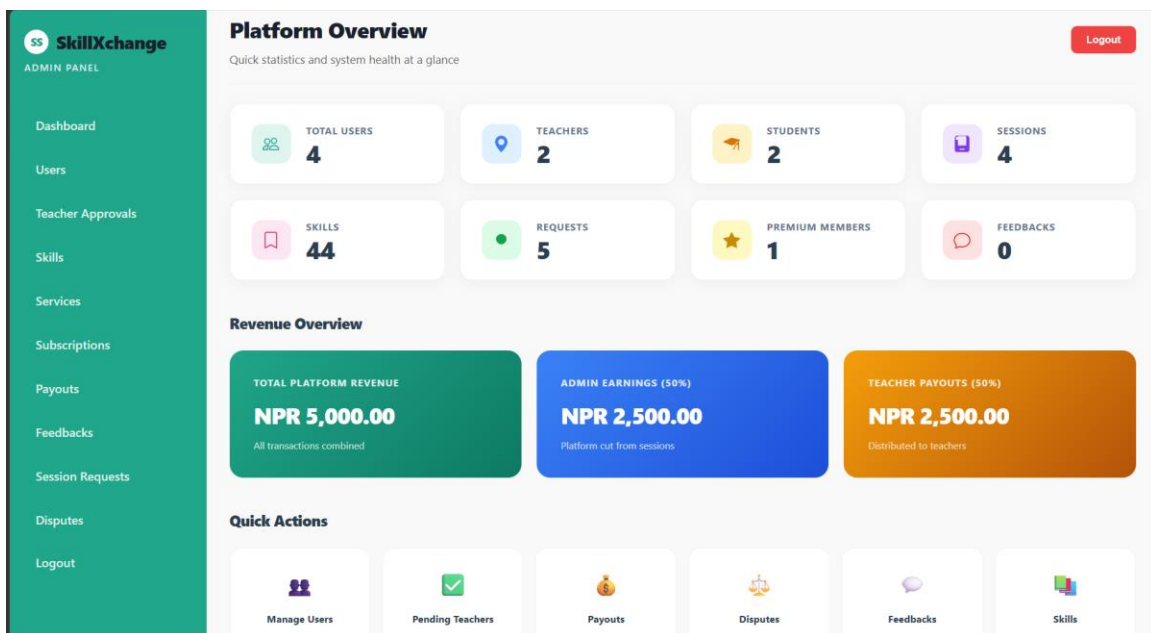
8.4 Increment 5

8.4.1 Admin Module Testing:

- Test Case 70: Admin Dashboard Access

Table 71: Test 70 Admin dashboard access

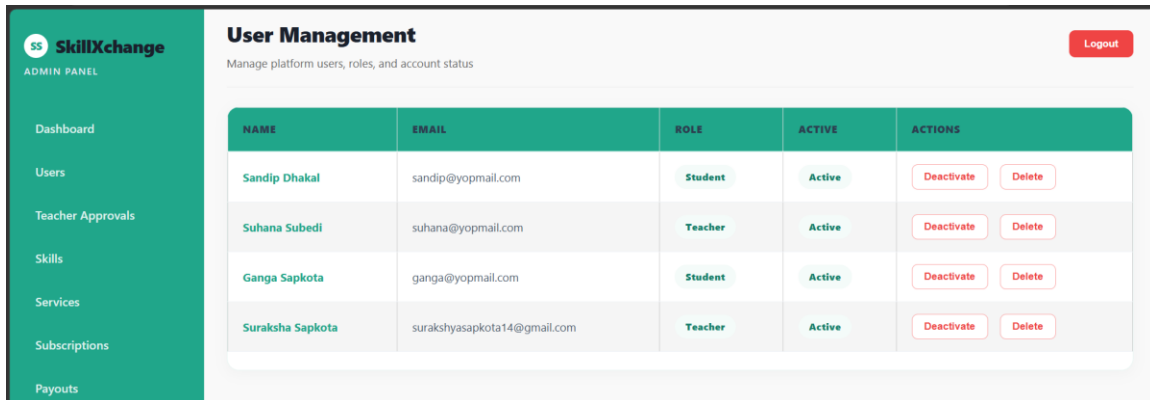
Item	Description
Objective	To verify that admin can access dashboard after login.
Action	Login as admin and open dashboard.
Expected Outcome	Dashboard should load with all statistics and panels.
Actual Result	Dashboard loaded successfully.
Conclusion	Success.



- Test Case 71: View User List

Table 72: Test 71 User management

Item	Description
Objective	To verify admin can view total users.
Action	Navigate to Users section
Expected Outcome	User list with name, email, role, status displayed
Actual Result	Users displayed correctly.
Conclusion	Success.



SkillXchange
ADMIN PANEL

User Management
Manage platform users, roles, and account status

Logout

NAME	EMAIL	ROLE	ACTIVE	ACTIONS
Sandip Dhakal	sandip@yopmail.com	Student	Active	Deactivate Delete
Suhana Subedi	suhana@yopmail.com	Teacher	Active	Deactivate Delete
Ganga Sapkota	ganga@yopmail.com	Student	Active	Deactivate Delete
Suraksha Sapkota	surakshyasapkota14@gmail.com	Teacher	Active	Deactivate Delete

Dashboard
Users
Teacher Approvals
Skills
Services
Subscriptions
Payouts

- Test Case 72: Add New Skill

Table 73: Test 72 add new skill by admin

Item	Description
Objective	To verify admin can add new skill
Action	Enter skill details and submit
Expected Outcome	Skill added to list
Actual Result	Skill added successfully
Conclusion	Success.

+ Add New Skill

Skill Title * Category *

testing Design

Description

Just for testing the add skill

Skill Image

Choose File Screenshot 2026-03-21 215358.png

Screenshot 2026-03-21 215358.png

Add Skill

Skill Management Logout

Monitor, manage and add skills available on the platform

Skill "testing" added successfully.

Skill "testing" added successfully.

+ Add New Skill

IMAGE	NAME	CATEGORY	ACTIONS
	testing Just for testing the add skill	Design	Update Image

- Test Case 73: View Subscriptions

Table 74: Test 73 Subscription view by admin

Item	Description
Objective	To verify admin can view subscription plan.
Action	Open subscriptions page
Expected Outcome	Plan, price, status displayed
Actual Result	Data displayed correctly.
Conclusion	Success.

System Subscriptions
Monitor active plans and platform revenue

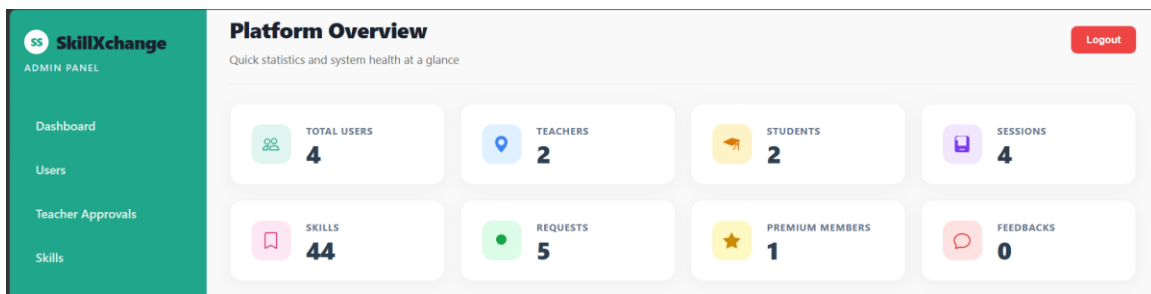
Active revenue: NPR 1,000.00

ID	USER	PLAN	PRICE	STATUS	ACTIONS
2	Sandip Dhakal	Monthly	1000.00	Active	Cancel
1	Ganga Sapkota	Monthly	1000.00	Refunded	-

- Test Case 74: Display Teachers and Students Count

Table 75: Test 73 Display total student and teacher

Item	Description
Objective	To verify teacher and student count accuracy.
Action	Open dashboard
Expected Outcome	Correct number of teachers and students shown.
Actual Result	Data displayed correctly.
Conclusion	Success.



- Test Case 74: View Payouts

Table 76: Test 74 view payouts

Item	Description
Objective	To verify admin can view revenue split
Action	Open payouts page.
Expected Outcome	Admin and teacher share displayed (50/50)
Actual Result	Data displayed correctly.
Conclusion	Success.

SkillXchange
ADMIN PANEL

Payouts Management
Manage financial splits (50/50) and teacher payments

Logout

DATE	STUDENT	TEACHER & KHALTI ID	TOTAL AMOUNT	ADMIN (50%)	TEACHER (50%)	STATUS
Apr 20, 2026	Sandip Dhakal	N/A ID: NOT SET	NPR 1,000.00	NPR 500.00	NPR 500.00	Platform Rev
Apr 13, 2026	Ganga Sapkota	N/A ID: NOT SET	NPR 1,000.00	NPR 500.00	NPR 500.00	Platform Rev
Apr 13, 2026	Ganga Sapkota	N/A ID: NOT SET	NPR 1,000.00	NPR 500.00	NPR 500.00	Platform Rev
Apr 13, 2026	Ganga Sapkota	N/A ID: NOT SET	NPR 1,000.00	NPR 500.00	NPR 500.00	Platform Rev
Apr 10, 2026	Ganga Sapkota	N/A ID: NOT SET	NPR 1,000.00	NPR 500.00	NPR 500.00	Platform Rev

Dashboard
Users
Teacher Approvals
Skills
Services
Subscriptions
Payouts
Feedbacks
Session Requests

- Test Case 75: View Feedback

Table 77: Test 75 Feedback

Item	Description
Objective	To verify admin can view user feedback.
Action	Open feedback section.
Expected Outcome	Feedback list displayed
Actual Result	Feedback shown correctly.
Conclusion	Success.

SkillXchange
ADMIN PANEL

User Feedbacks
View and manage user-submitted feedback and ratings

Logout

ID	AUTHOR	TARGET	COMMENT	ACTIONS
1	Suraksha Sapkota	User #9	The student is too good and learning in the best way	Delete

- Test Case 76: View Session Requests

Table 78: Test 76 View session request

Item	Description
Objective	To verify admin can monitor session requests.
Action	Open session requests page.
Expected Outcome	Requester, responder, status shown
Actual Result	Data displayed correctly.
Conclusion	Success.

SkillXchange
ADMIN PANEL

Session Requests
Monitor and manage session interactions between users

Logout

ID	REQUESTER	RESPONDER	STATUS
5	Sandip Dhakal	Suraksha Sapkota	Scheduled
4	Ganga Sapkota	Suraksha Sapkota	Cancelled
3	Ganga Sapkota	Suraksha Sapkota	Cancelled
2	Ganga Sapkota	Suraksha Sapkota	Cancelled
1	Ganga Sapkota	Suraksha Sapkota	Cancelled

- Test Case 77: Add New Service

Table 79: Test 77 Add new service

Item	Description
Objective	To verify admin can add new service and that added service is shown in user service page.
Action	Enter service details and submit.
Expected Outcome	Service added and shown in user's page.
Actual Result	Service added correctly.
Conclusion	Success.

Service Management
Manage platform services and offerings

Logout

Service added successfully.

Service added successfully.

Add New Service

Service Title
e.g. Graphic Design Masterclass

Category

Existing Services

TITLE	CATEGORY	ACTIONS
Testing Just for testing, i will delete it later	Technology	

SkillSwap Home About Services Pages ▾ Contact Login Get started

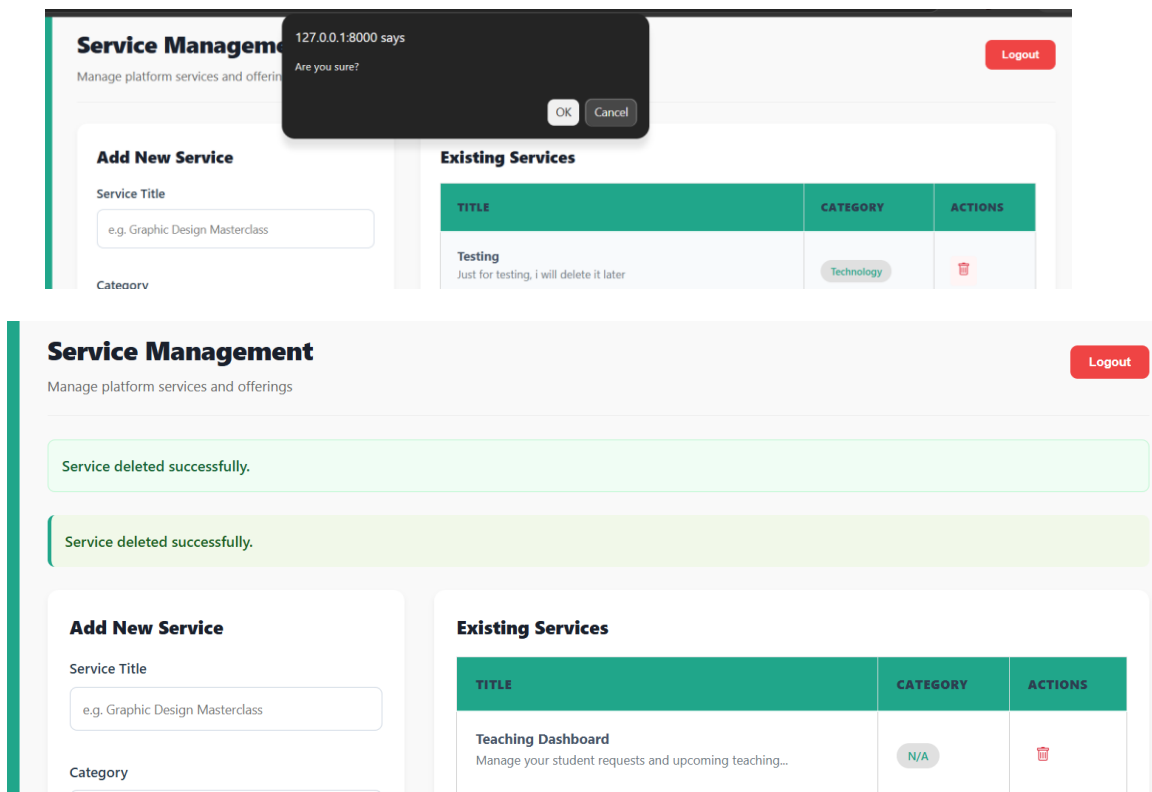
swapping classes. skill-bartering journey. amazing community.

Testing
Just for testing, i will delete it later

- Test Case 78: Delete Service

Table 80: Test 78 Delete service by admin

Item	Description
Objective	To verify admin can delete service and that the deleted service is gone from the user service page.
Action	Click on the delete icon.
Expected Outcome	Service removed
Actual Result	Service deleted successfully.
Conclusion	Success.



- Test Case 79: Forgot Password Flow

Table 81: Test 79 Forgot password flow

Item	Description
Objective	To verify that the user can reset their password successfully.
Action	Click “Forgot Password”, enter email, submit request, reset password via email link
Expected Outcome	The password reset link should be sent, and the user should be able to set a new password
Actual Result	Email sent successfully, and password reset completed
Conclusion	Success.

Forgot Password?

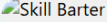
Enter your email address and we will send you a password reset link.

We have emailed your password reset link!

Email

Send Password Reset Link

[Back to Login](#)

Skill Barter

Forgot Password?

Enter your email address and we will send you a password reset link.

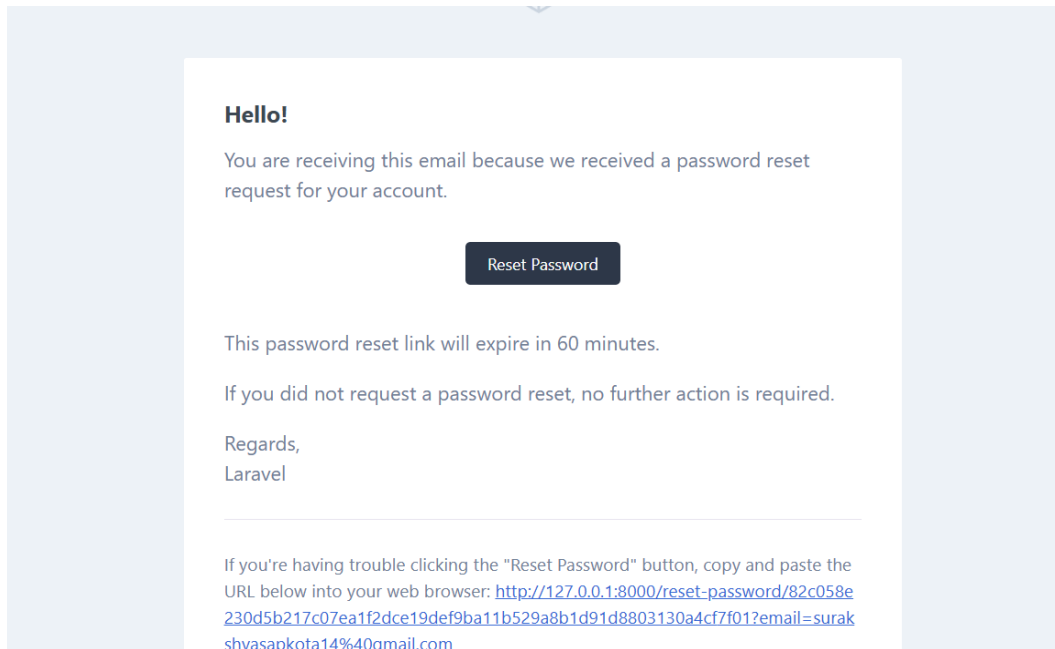
[Back to Login](#)



Login

☐ Remember Me

[Forgot Password?](#)
[Create Account](#)



Reset Password

Create a new password for your account

Email

New Password

Confirm Password

[Reset Password](#)

9 Analysis

9.1 SWOT Analysis

Table 82: SWOT Analysis

Strength	• The system is simple to deal with a user-friendly interface. • It is adaptable and convenient because it enables users to learn and teach skills. • The site offers a live communication platform via sessions and communication. • Safe payments (Khalti) is a safe payment integration.
Weakness	• It requires user involvement and therefore with a smaller number of users, it can be less effective. • Teachers can have delays in accepting sessions or scheduling them. • The platform needs constant internet connectivity to be functional.
Opportunity	• Increase in the demand of online learning platforms makes it have more potential. • Has the opportunity to grow and introduce new options such as certifications or group learning. • Potential collaborations with schools or organizations.
Threats	• Security risks like hacking, or breach of data can impact on user trust. • Existing learning platforms such as Udemy or Coursera. • Transactions may be affected by changes in payment regulations or policies.

9.2 PEST Analysis

Table 83: PEST Analysis

Political	• Government policies in favor of digital platforms and online education. • Laws governing online payment systems and data security shall be adhered to.
Economical	• •Saves users money by sharing skills rather than charging them a lot of money. • Offers the possibility of the users earning by teaching skills.
Social	• Growing popularity of online learning and skill development. • Individuals like online and distance learning.
Technical	• Availability of internet and Smartphones facilitate system use. • Web technologies contribute to the enhancement of performance and capabilities of the system.

9.3 Risk Analysis

Table 84: Risk Analysis

Risk	Description	Mitigation Strategy
SQL injection	Hackers may try to access or manipulate database data	Use input validation and prepared statements
Payment Failure	Failure of payment process can be caused by the network or gateway problems	Retry mechanism and error handling should be implemented.
Session Mismatch	Users can demand sessions with unmatched skills	Strict skill matching validation.
Server Overload	Increased number of users can slow down the system	Optimized queries and scalable hosting.
Data Breach	Sensitive user information can be disclosed	Encrypt and authenticate with security.
Notification Failure	Users might not get session updates	Create effective notification system and logs.

10 CONCLUSION

The Skill Barter System is an effective web-based platform which has been designed and developed successfully to allow users to trade skills in a well-organized and efficient way. The primary aim of the project was to develop a system in which users can be both learners and teachers and enable them to share knowledge without necessarily having to depend on monetary transactions. This has been accomplished by adopting basic functionalities like user authentication, skill management, session management, feedback system and premium subscription.

The incremental approach was used through the whole process of development and this ensured that the system was developed in phases and stability and correct validation was achieved at one stage before the next step was undertaken. The first point was the essential functions, i.e. registration, login, and profile management. Over time, the more advanced features like skill search, session requests, scheduling, communication and feedback systems were incorporated into the system. This strategy assisted in minimizing mistakes and consistency within various modules.

The session management feature is highly featured in the system as one is able to request, accept, schedule and hold learning sessions. The presence of dynamic elements of interaction (chat, resources sharing and assignment submission) makes the overall user experience of the platform more practical and interactive. Moreover, the feedback and rating system provides trust and transparency as it enables the user to rate each other on the basis of the completed sessions. Besides the main functions, the platform is also integrated with a premium subscription system based on Khalti payment gateway, which contributes value to the platform, providing advanced functions and monetization. The system provides safety in the processing of payments and appropriate validation prior to the activation of the premium features which enhances reliability and confidence of the user.

Technically, the project is also an example of successful application of such modern web technologies like Laravel to develop the backend or MySQL to manage the database. Appropriate validation, authentication and safe data handling procedures are used so as to maintain system reliability and data integrity.

However, there are aspects that can be enhanced like any system. Future development can involve better matching algorithms, real-time video integration, support of mobile applications, and greater scalability to support more users. The resulting gains will aid in the system being more robust and competitive within the changing digital learning landscape.

To sum up, the Skill Barter System offers an effective approach towards collaborative learning and skills exchange. It effectively addresses the disconnect between learners and teachers by providing a secure, flexible and user-friendly platform. The system does not only fulfill the original project goals but also gives a good base towards future development and growth.

11 FURTHER WORK

This section will describe the remaining tasks and upcoming upgrades to the Skill Barter System. Due to time constraints, academic focus considerations, and the need to establish a foundational system before exploring additional enhancements, certain features included in the first project proposal submission had to be modified. As a result, the development process will be progressive, starting with system design, user interface components, and authentication.

11.1 Reflection on Changes from the Original Proposal

A full skill trading system with real-time chat, credit transaction systems, rating facilities, and full skill trading was to be built in a single development cycle, according to the original design. However, it has been reduced to a single topic because of academic time constraints and the complexity in developing multiple interacting systems.

This section will tell about the remaining task and future improvements in the Skill Barter System:

- Limited Development Time in the academic semester.
- The need to focus on basic functionalities and avoid sophistication.
- Ensure stability in a system before adding financial/credit-based modules.
- System logic and user flow validation by increment.

The project is currently concentrating on user authentication, system modeling, user interface design, and basic backend implementation. The more complex tasks of skill exchange and messaging have been planned for implementation during the later stages of the project.

11.2 Remedial Plans Adopted

To overcome these limitations, an incrementally developed system design methodology has been employed in this project. The major functionalities like user registration, user login, identification of roles, and validation processes are included in the project design initially. The system design concepts like detailed UML diagram, DFD diagram, ER diagram, flow chart, sequence diagram, and collaboration diagrams are employed to provide proper understanding to the system design.

A user interface design was completed using wireframes and prototypes based on usability principles that came out of research. Backend validation was incorporated with a focus on avoiding submitting data incorrectly. It is important since it ensures that whenever a new module is added in the future, integration is smooth.

11.3 Further Work to be Done Using Incremental Development

Development in the future will be carried out incrementally so that each module can be designed independently.

- Increment 1: Authentication and access control

This phase has been completed as planned. It includes session handling for user authentication, login, and registration.

- Increment 1: Management of Skills

Skills can be added, changed, and removed by users. The talents are tied to user profiles, searchable, and arranged. To guarantee accurate description and classification, the skills are validated.

- Increment 3: Skill Matching and Barter Logic

This update will focus on matching users according to the talents that can be provided and those that are needed. Barter requests, acceptances, rejections, and tracking of ongoing exchanges will all be managed by the system.

- Increment 4: Credit / Point-Based Exchange System

There would be a credit system to enable fair value trading. Credits would be earned, spent, and documented on a secure log under this system.

- Increment 5: Communication and Collaboration Features

To allow users to communicate while exchanging talents, this phase will build message and student communication functionality. The student will be informed of the various actions carried out by user notifications.

- Increment 6: Rating, Feedback, and Trust Management

After completing a skills exchange, users will be able to review and rate. This will have an impact on the platform's display and trust levels.

➤ Increment 7: Testing, Security, and Performance Optimization

There will be lots of testing, including user experience, authentication, and validation testing. Access control, safe data management, and encrypted credentials are some of the security features that will be enhanced.

11.4 Deployment and Documentation

The Skill Barter System will run on a web server environment that supports Laravel and an Relational Database. The documentation will cover user manuals, administrator manuals, explanations of the system architecture, and testing reports. There will be screenshots and guides that facilitate ease of use.

11.5 Alignment with Proposed Gantt Chart

To show progressive progression over a linear temporal approach, the Gantt diagram has been revised. This is due to the sequential design of each increment in incremental development, which permits overlap between design, implementation, and testing.

11.6 Expected Outcomes

The Skill Barter System will provide a safe and welcoming environment for skill trade after the remaining tasks are finished. Users will be able to register, manage talents, exchange services fairly through credits, communicate clearly, and gain the trust of others through reviews and ratings. Scalability and improvement will be made possible by the gradual approach.

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